

Canonical GHZ configurations - cluster P2 - helicity flipping

species: electron
polarization cluster: P2
source cluster configurations: 97
canonical configurations collected: 46

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

helicity-mechanism subset: helicity flipping
classification: complete-chain R_flip/conserved > 1
per-page labels: isolated flip/conserved and T/L powers; coherent cross terms are reported separately

cluster retained-component count distribution:

- 2 components: 56 configurations
- 4 components: 7 configurations
- 5 components: 8 configurations
- 6 components: 21 configurations
- 7 components: 4 configurations
- 8 components: 1 configurations

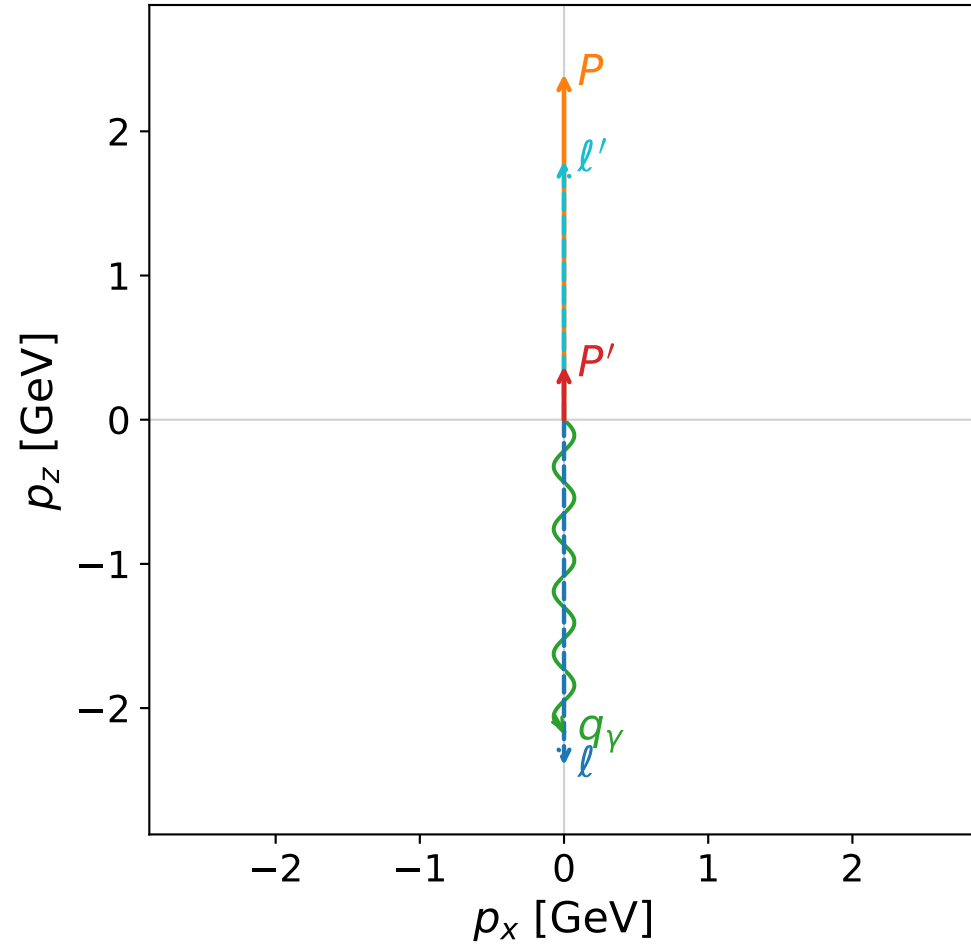
CSV index:

Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_02\combined\
helicity_flipping\canonical_dghz_polarization_cluster_02_helicity_flipping_configurati
ons_electron.csv

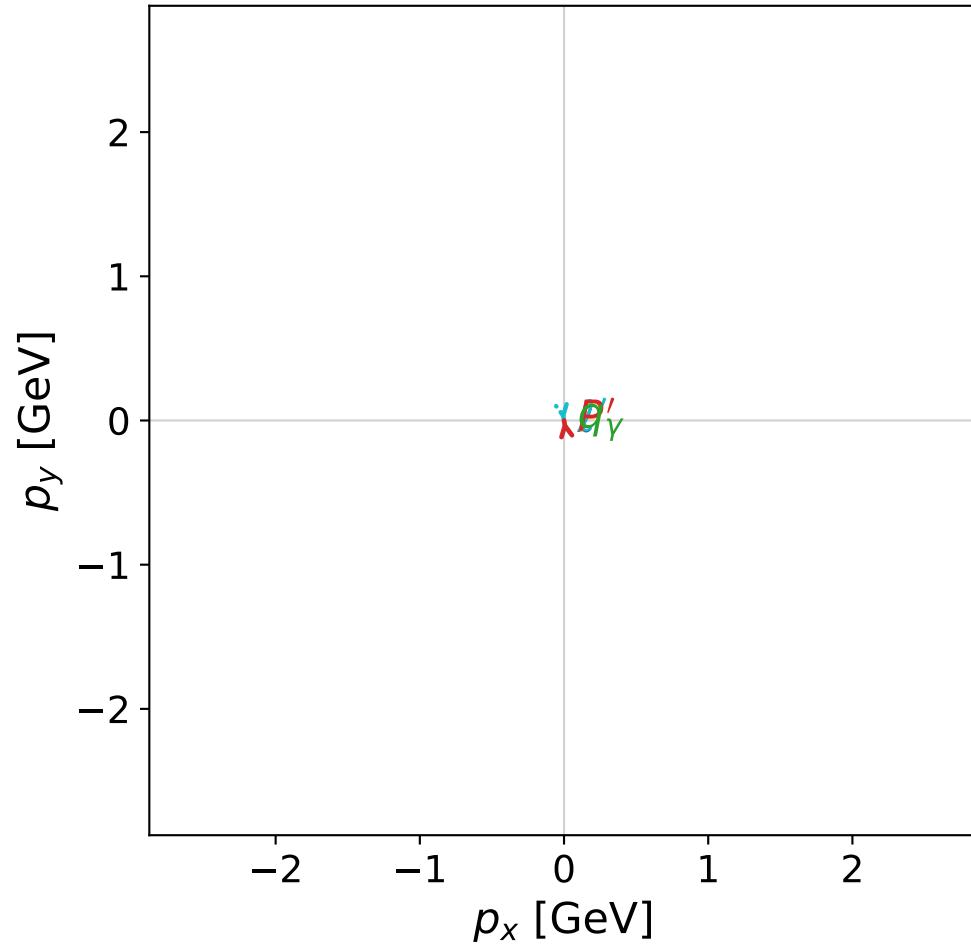
Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

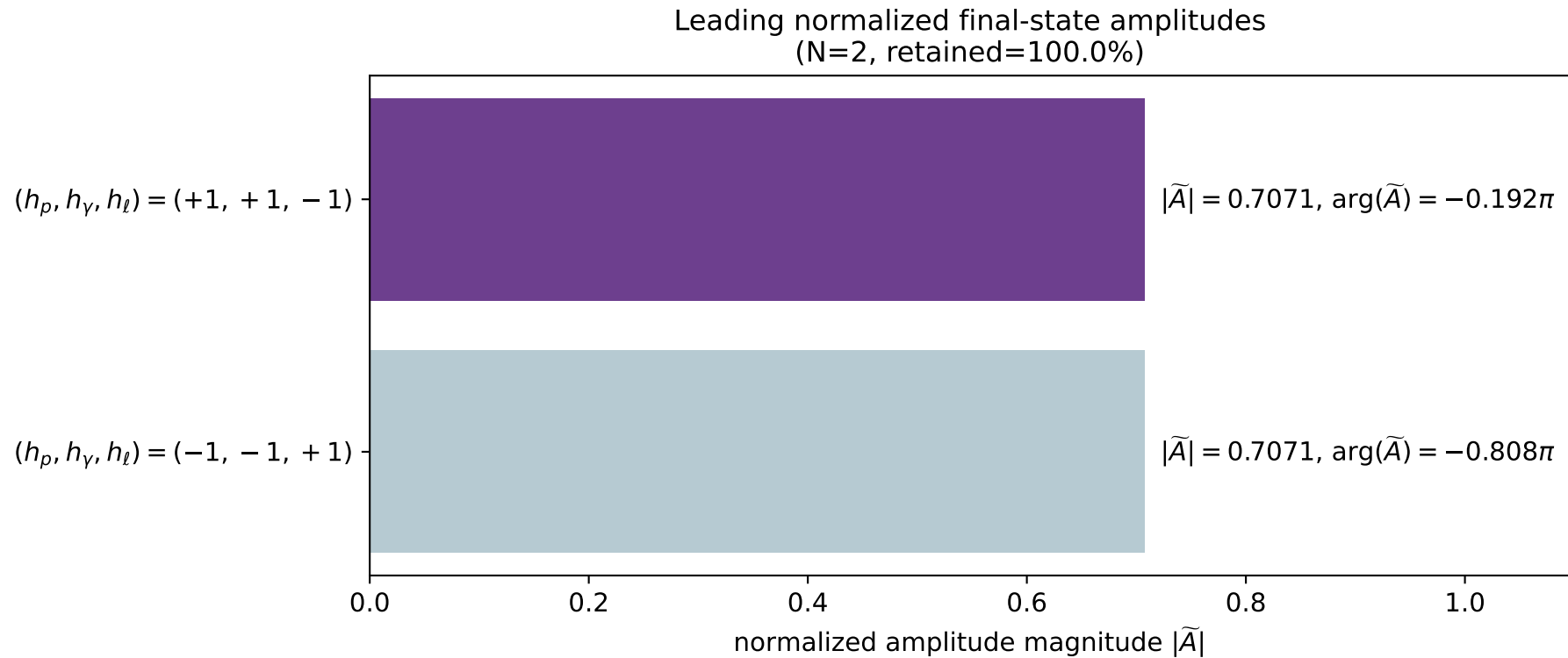


dghz_mixing_angles_polarization_02_local_minimum_0
 $d_{\text{GHZ}}=2.22835\text{e-}08$
 region: polarization_cluster_02_local_minimum_0
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0000
 lepton mixing angle: $\alpha_e=0.7579928$ rad
 incoming lepton: $0.726217|+\rangle +0.687465|-\rangle$
 proton mixing angle: $\alpha_p=2.3835959$ rad
 incoming proton: $-0.726215|+\rangle +0.687468|-\rangle$

$s=24.7062$, $\sqrt{s}=4.97053$, $|\vec{P}|=2.39676$, $|\vec{P}'|=0.371823$
 $\theta_{P'}=1.30111\text{e-}07$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.74218$, $\phi_\gamma=3.74627$
 $E_\gamma=2.16667$, $\theta_\ell=2.58096\text{e-}08$, $\phi_\ell=4.88378$
 $Q^2=17.2073$, $x_B=0.741984$, $t=-1.65188$, $W^2=6.86346$, $y=0.973331$

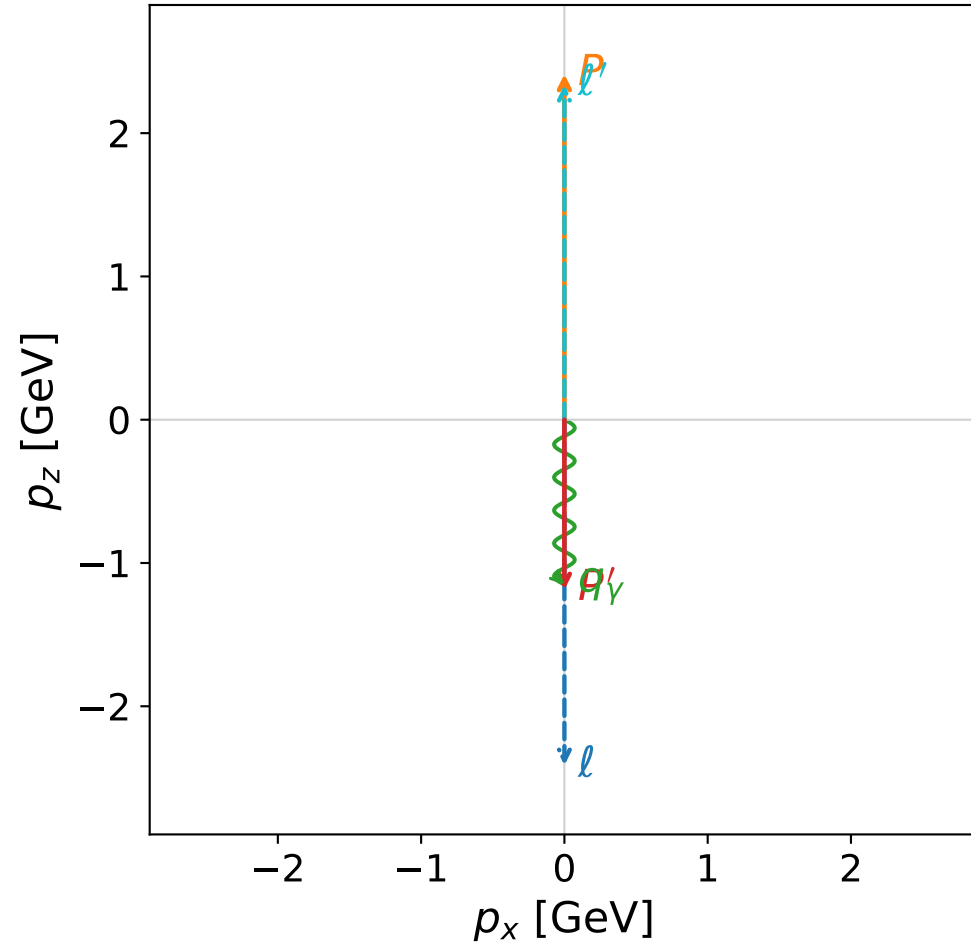
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3968$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3968$
 P $E= 2.5738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3968$
 ℓ' $E= 1.7948$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7948$
 P' $E= 1.0090$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3718$
 q_γ $E= 2.1667$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1667$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=9.890\text{e+}22$; $I_h=-2.423\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.174\text{e-}18$; $I_{TL}=-3.283\text{e-}11$

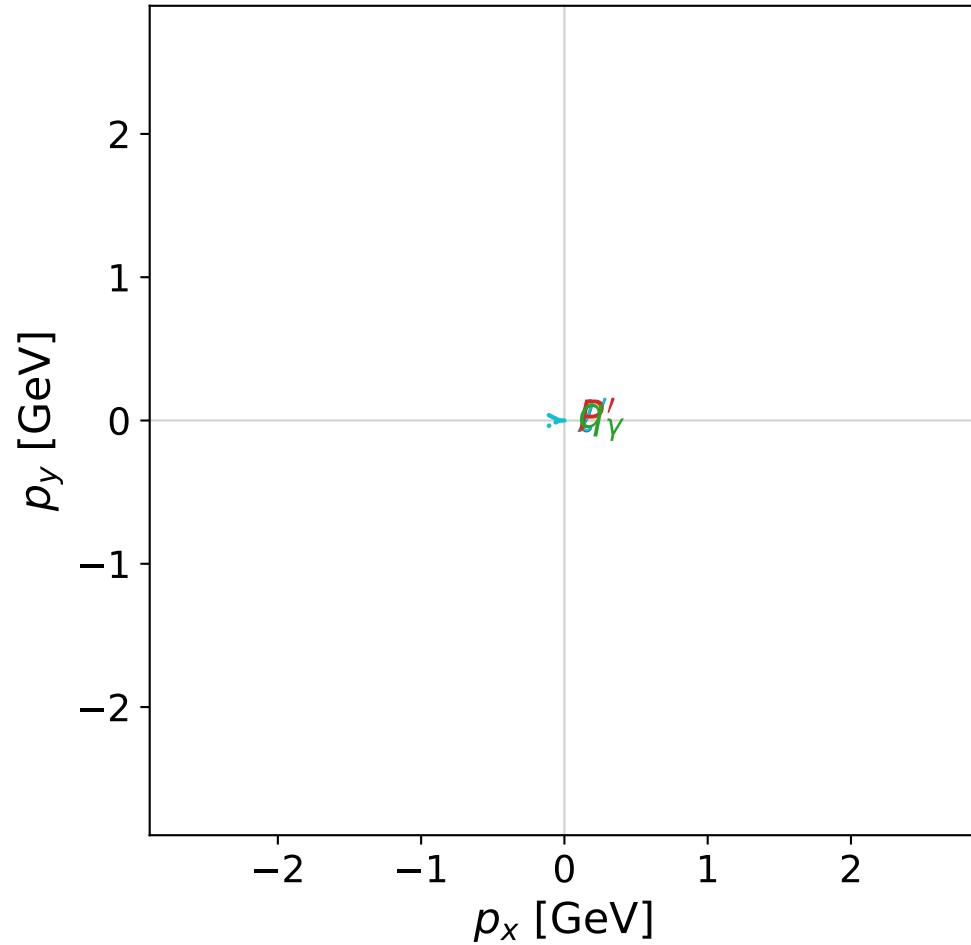


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

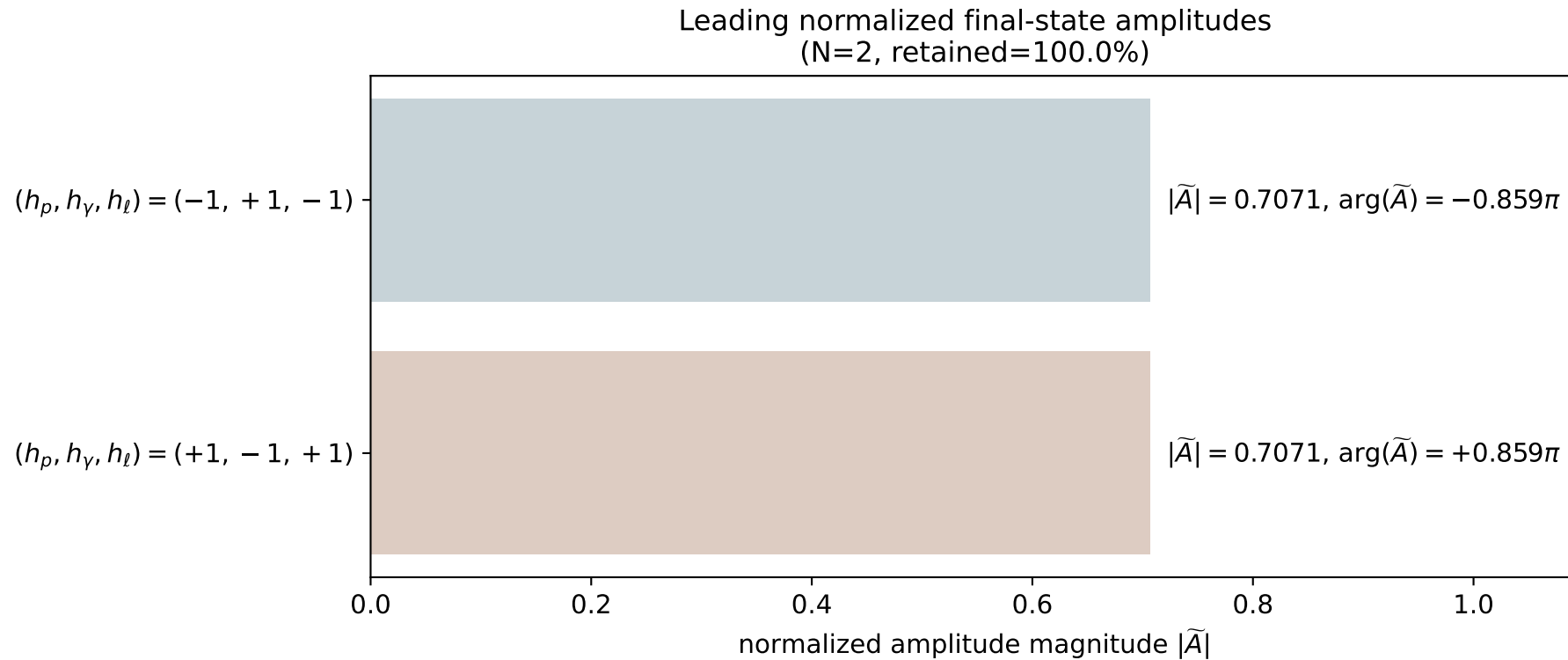


dghz_mixing_angles_polarization_02_local_minimum_3
 $d_{\text{GHZ}}=2.32433\text{e-}08$
 region: polarization_cluster_02_local_minimum_3
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0003
 lepton mixing angle: $\alpha_e=0.82685667$ rad
 incoming lepton: $0.677192|+\rangle + 0.735806|-\rangle$
 proton mixing angle: $\alpha_p=2.3147292$ rad
 incoming proton: $-0.677187|+\rangle + 0.735811|-\rangle$

$s=24.9996$, $\sqrt{s}=4.99996$, $|\vec{P}|=2.412$, $|\vec{P}'|=1.18707$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.98106$, $\phi_\gamma=2.69995$
 $E_\gamma=1.14997$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.98435$
 $Q^2=22.5478$, $x_B=0.96783$, $t=-11.7976$, $W^2=1.62932$, $y=0.965899$

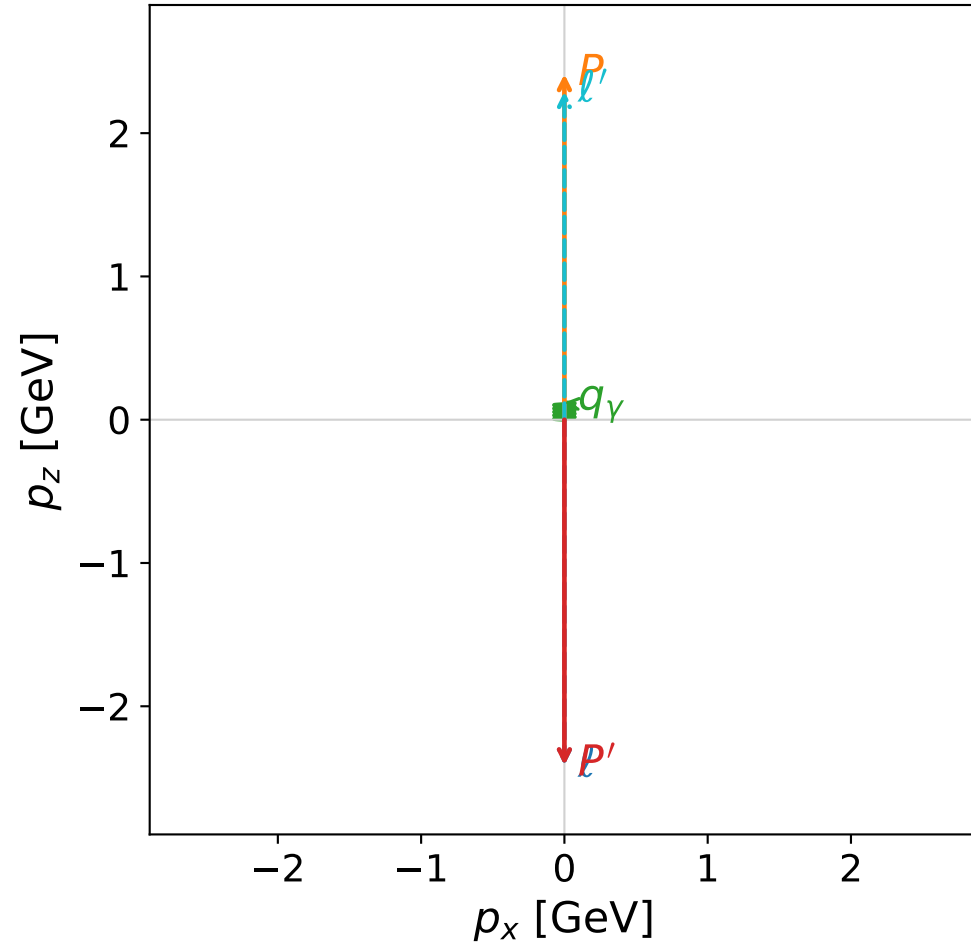
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3370$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.3370$
 P' $E= 1.5129$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.1871$
 q_γ $E= 1.1500$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.1500$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.268\text{e+}23$; $I_h=-9.534\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.654\text{e-}34$; $I_{TL}=+4.661\text{e-}20$

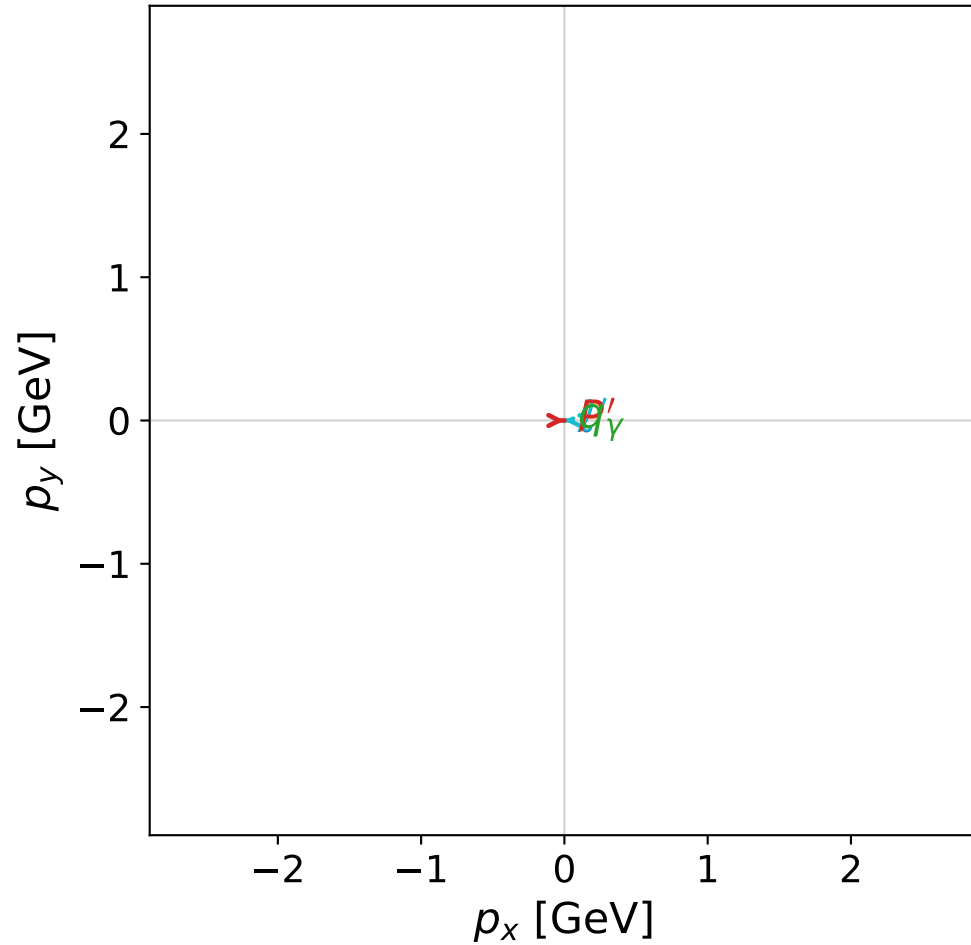


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

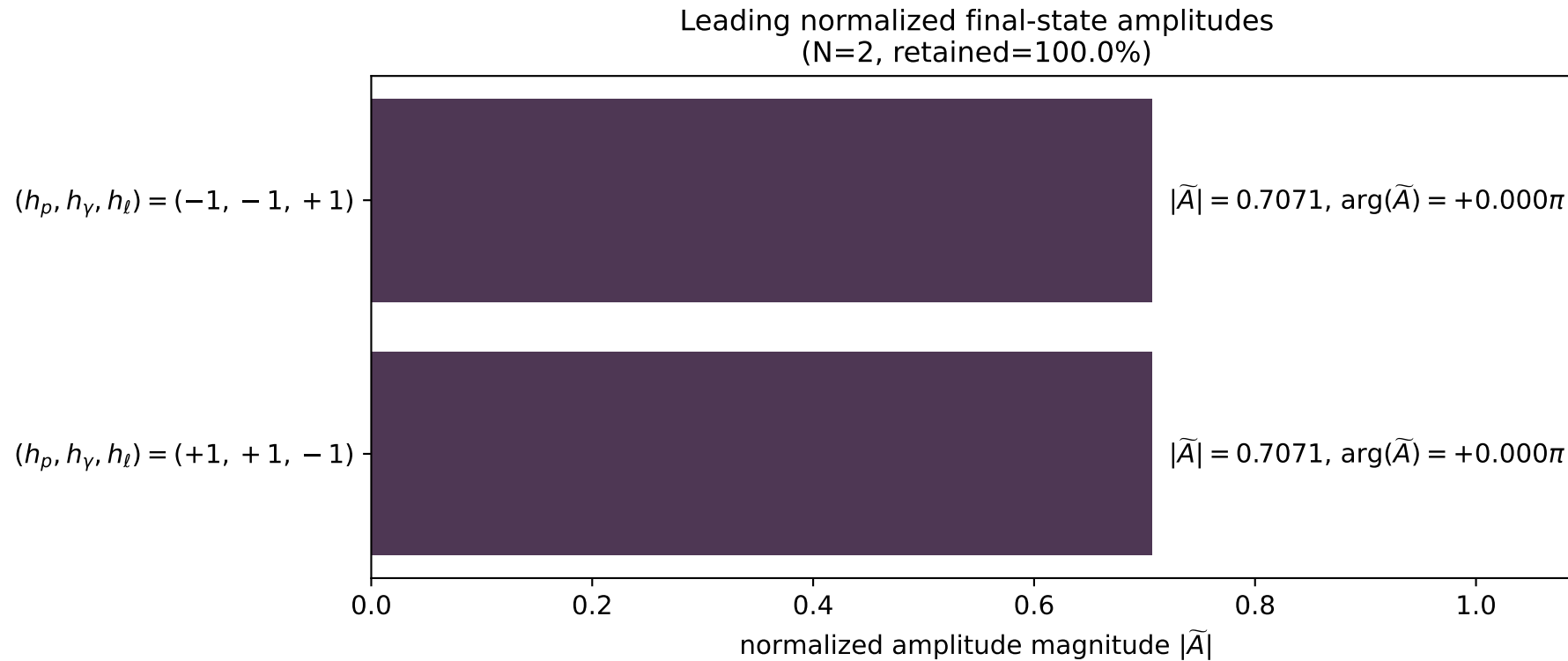


```
dghz_mixing_angles_polarization_02_local_minimum_4
d_{\rm GHZ}=2.36714e-08
region: polarization_cluster_02_local_minimum_4
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0004
lepton mixing angle: alpha_e=0.81232036 rad
incoming lepton: 0.687816|+> +0.725885|->
proton mixing angle: alpha_p=2.3831047 rad
incoming proton: -0.725877|+> +0.687825|->
```

```
s=25, sqrt(s)=5, |P|=2.41202, |P'|=2.41202
theta_{p'}=3.14159, theta_gamma=0, phi_{p'}=0, phi_gamma=0.437635
E_gamma=0.120596, theta_{l'}=0, phi_{l'}=3.14159
Q^2=22.1078, x_B=0.948273, t=-23.2713, W^2=2.0858, y=0.966566
```

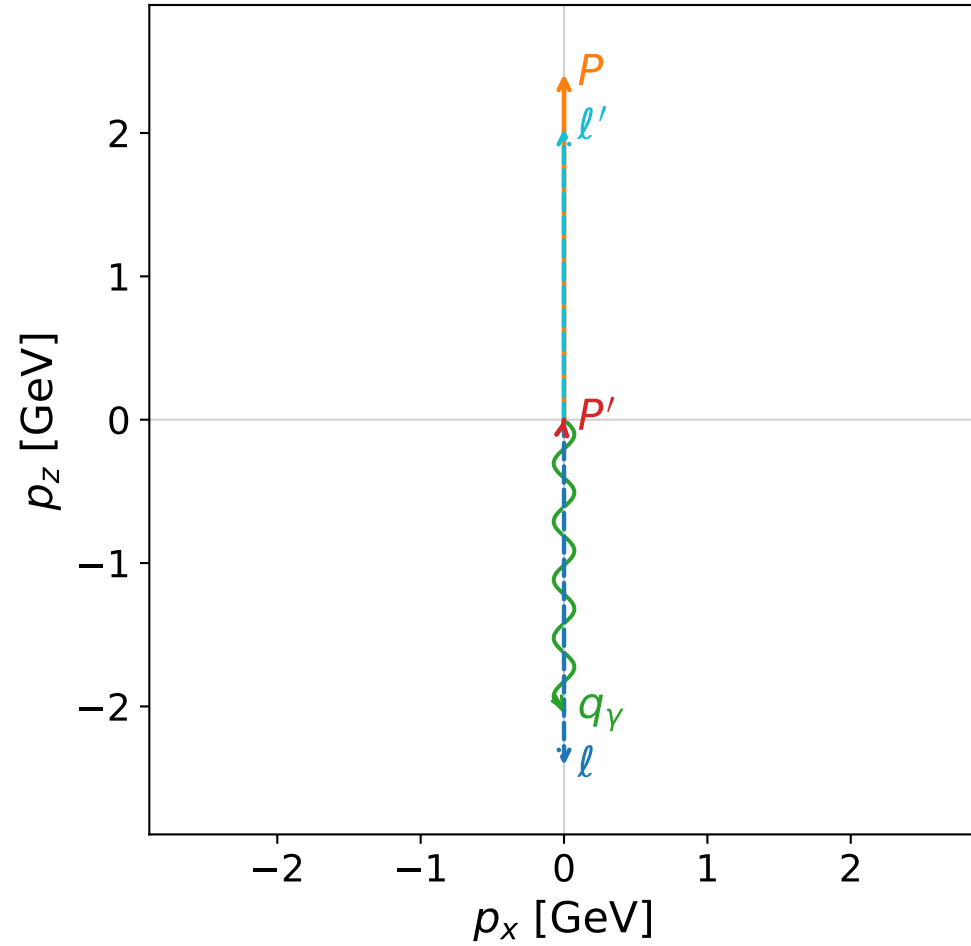
```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  2.4120 p_x=  0.0000 p_y=  0.0000 p_z= -2.4120
P      E=  2.5880 p_x=  0.0000 p_y=  0.0000 p_z=  2.4120
l'     E=  2.2914 p_x= -0.0000 p_y= -0.0000 p_z=  2.2914
P'     E=  2.5880 p_x=  0.0000 p_y=  0.0000 p_z= -2.4120
q_gamma E=  0.1206 p_x=  0.0000 p_y=  0.0000 p_z=  0.1206
```

```
GHZ mechanism: helicity flipping (full H G C)
helicity powers: flip 100.000%, conserve 0.000%
R_flip/conserve=3.932e+21; I_h=-2.607e-14
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=3.408e-28; I_TL=-2.608e-14
```

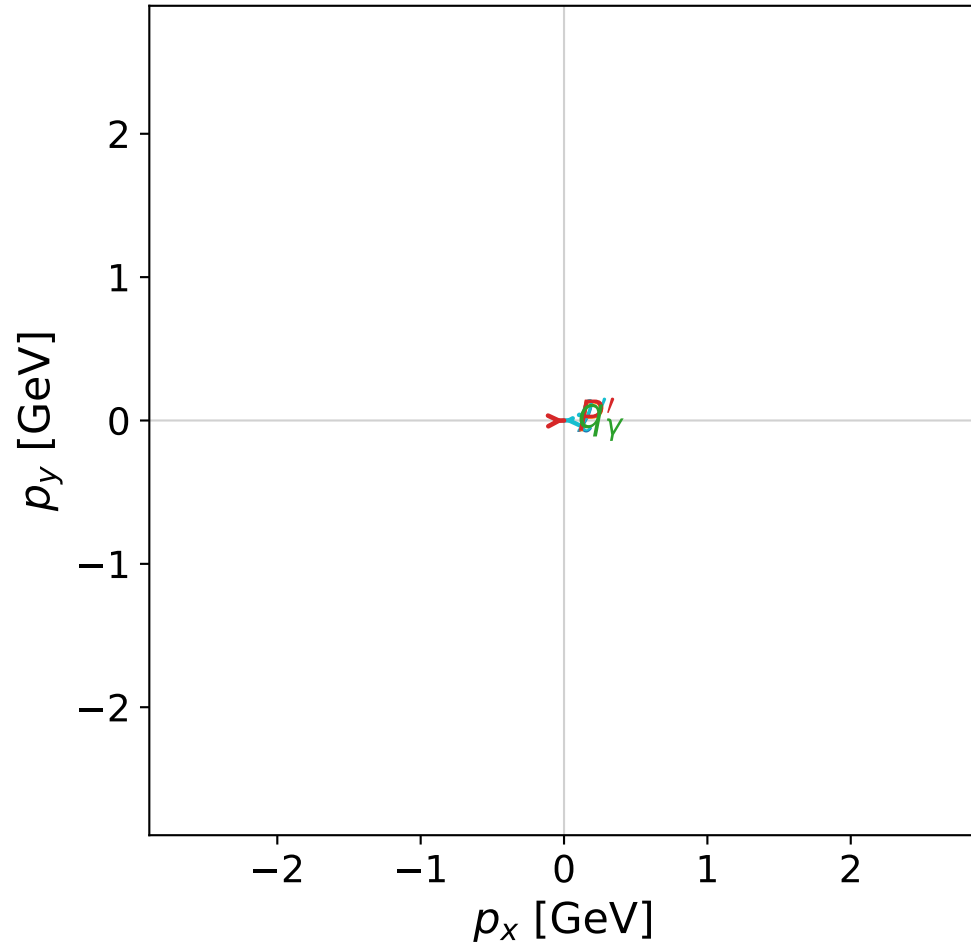


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

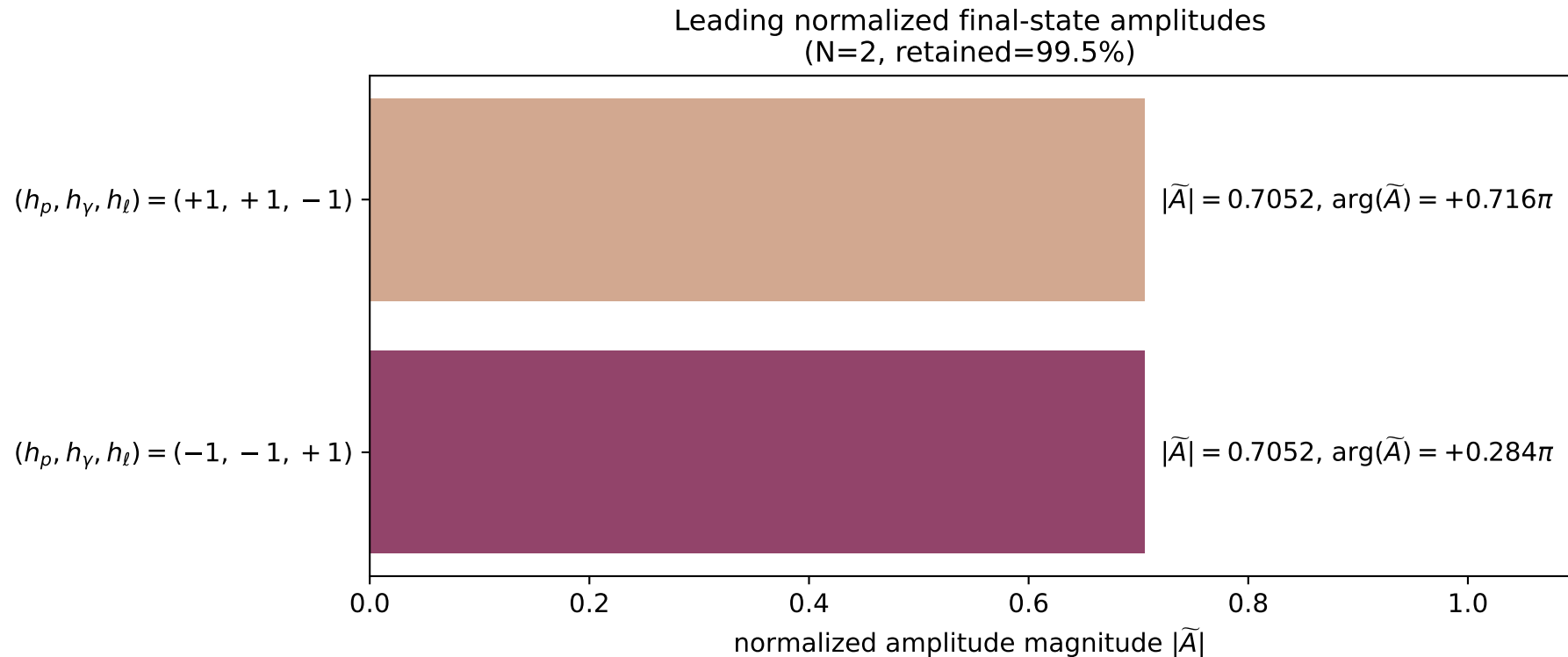


dghz_mixing_angles_polarization_02_local_minimum_6
 $d_{\text{GHZ}}=2.39329\text{e-}08$
 region: polarization_cluster_02_local_minimum_6
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0006
 lepton mixing angle: $\alpha_e=0.81040845$ rad
 incoming lepton: $0.689203|+\rangle + 0.724569|-\rangle$
 proton mixing angle: $\alpha_p=2.331173$ rad
 incoming proton: $-0.689194|+\rangle + 0.724576|-\rangle$

$s=24.9736$, $\sqrt{s}=4.99736$, $|\vec{P}|=2.41065$, $|\vec{P}'|=0.000556214$
 $\theta_{P'}=0.146926$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0285838$, $\phi_\gamma=0.891756$
 $E_\gamma=2.02996$, $\theta_{\ell'}=4.01243\text{e-}05$, $\phi_{\ell'}=3.17018$
 $Q^2=19.5688$, $x_B=0.837016$, $t=-3.09033$, $W^2=4.69028$, $y=0.970341$

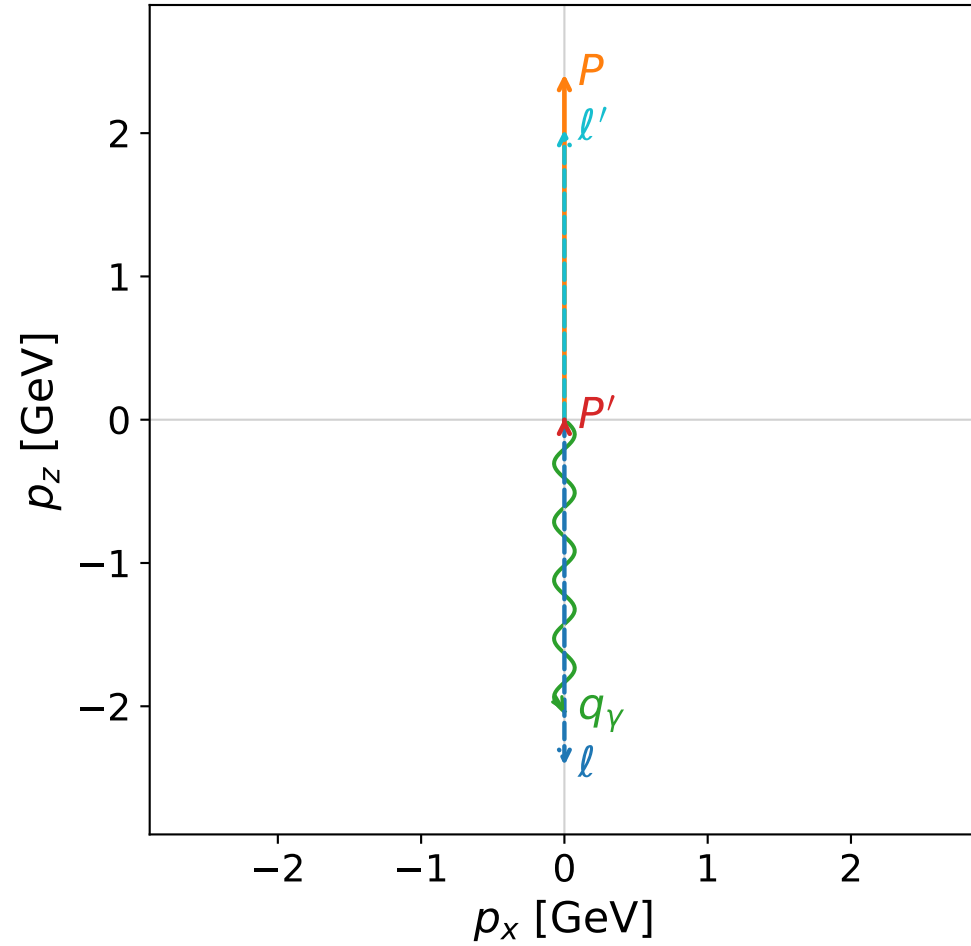
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4107$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4107$
 P $E= 2.5867$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4107$
 ℓ' $E= 2.0294$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0294$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0006$
 q_γ $E= 2.0300$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0300$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.210\text{e+}16$; $I_h=-2.527\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.044\text{e-}12$; $I_{TL}=-1.728\text{e-}07$

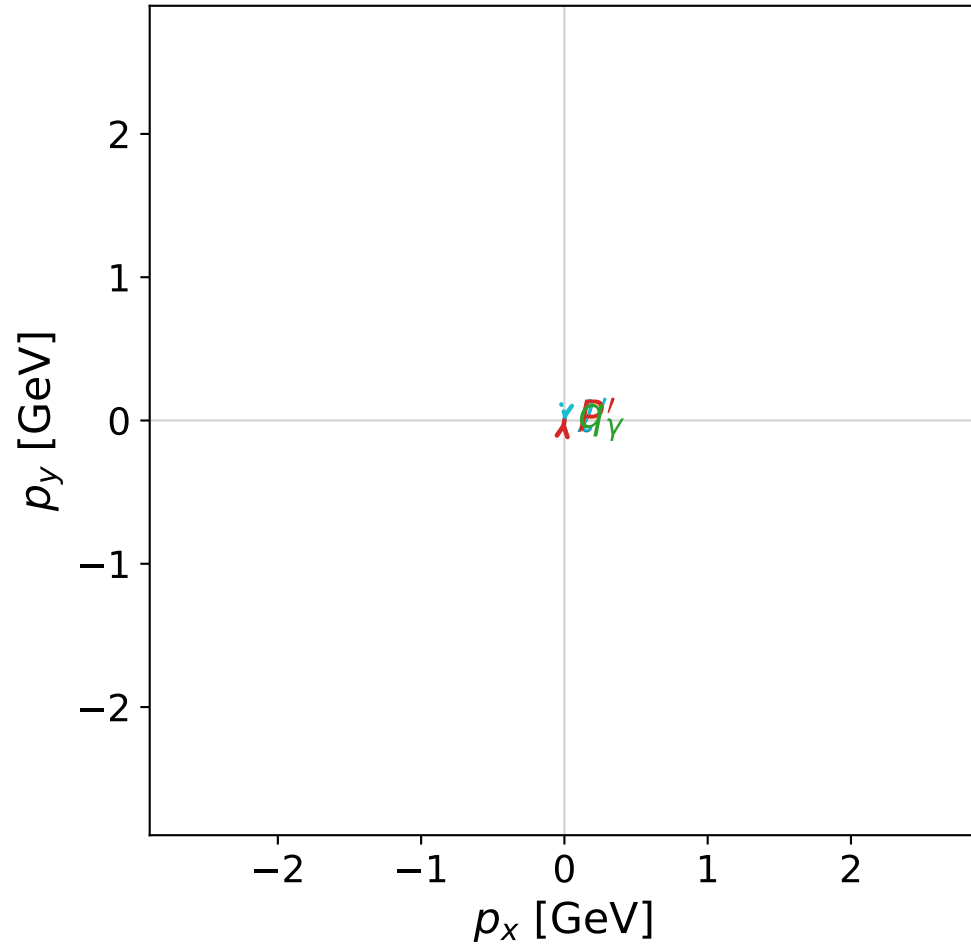


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

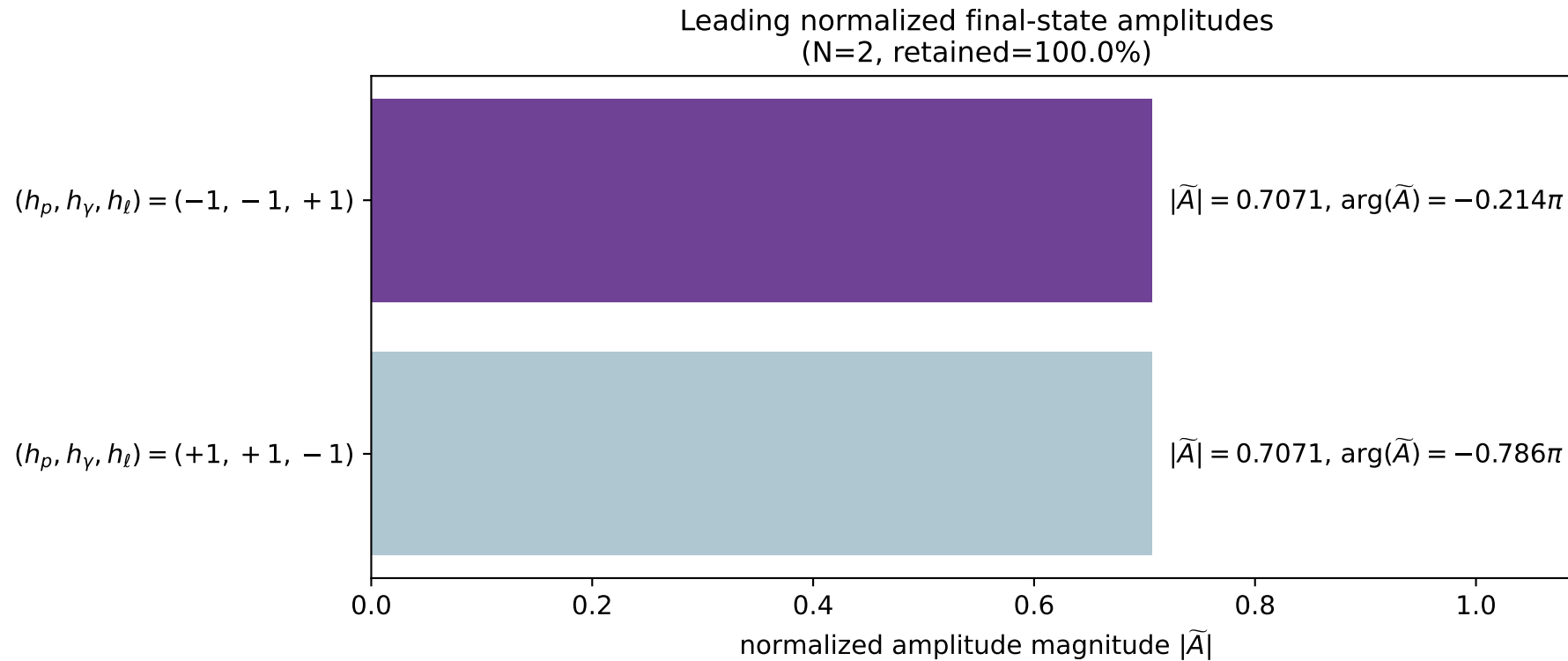


dghz_mixing_angles_polarization_02_local_minimum_7
 $d_{\text{GHZ}}=2.41149\text{e-}08$
 region: polarization_cluster_02_local_minimum_7
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0007
 lepton mixing angle: $\alpha_e=0.78796497$ rad
 incoming lepton: $0.705289|+\rangle + 0.708919|-\rangle$
 proton mixing angle: $\alpha_p=2.3536584$ rad
 incoming proton: $-0.705311|+\rangle + 0.708898|-\rangle$

$s=24.9993$, $\sqrt{s}=4.99993$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.0126281$
 $\theta_{P'}=7.54042\text{e-}07$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.42848$, $\phi_\gamma=5.61149$
 $E_\gamma=2.03724$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.57008$
 $Q^2=19.5333$, $x_B=0.834508$, $t=-3.03483$, $W^2=4.7535$, $y=0.970458$

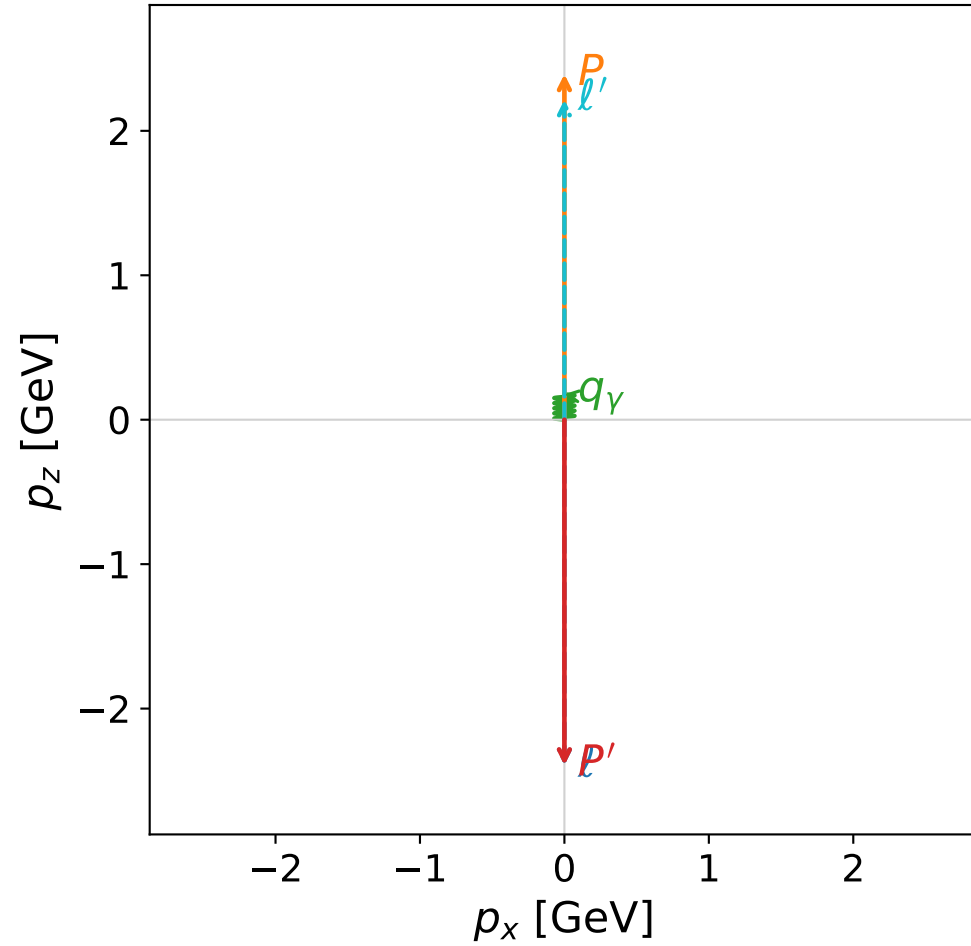
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0246$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0246$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0126$
 q_γ $E= 2.0372$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0372$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.021\text{e+}24$; $I_h=+4.200\text{e-}16$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=8.521\text{e-}20$; $I_{TL}=-3.006\text{e-}13$

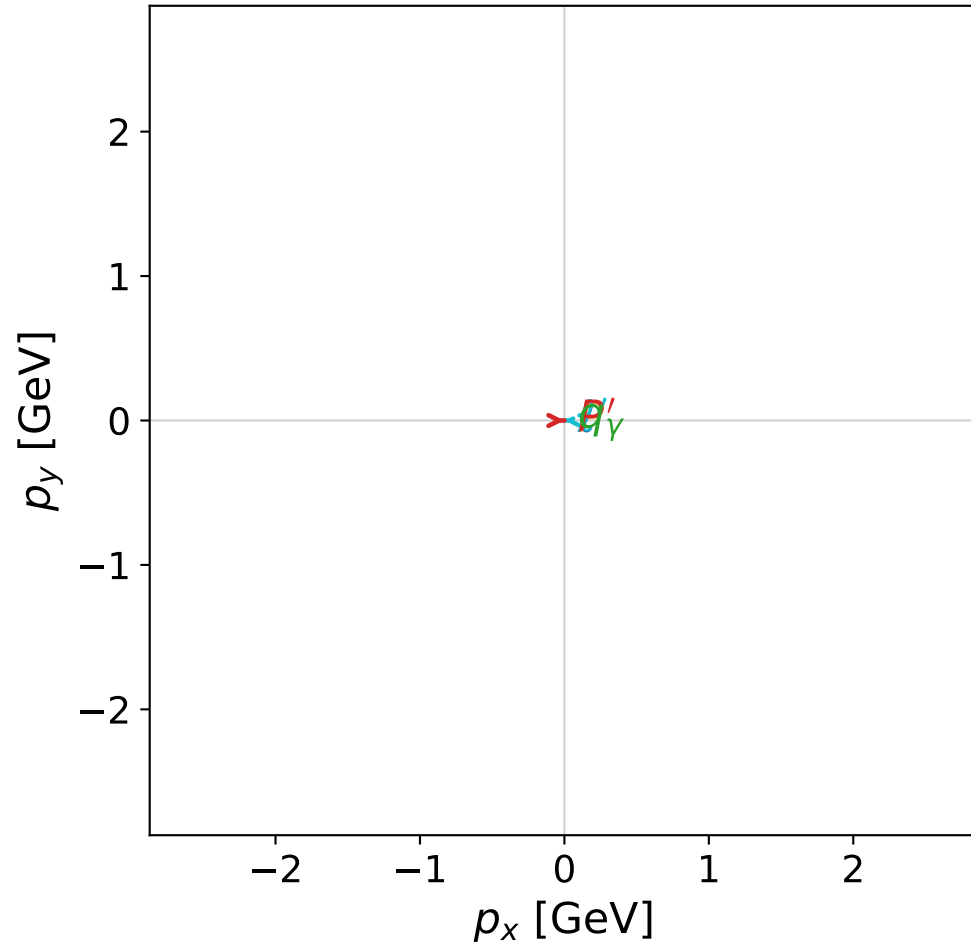


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



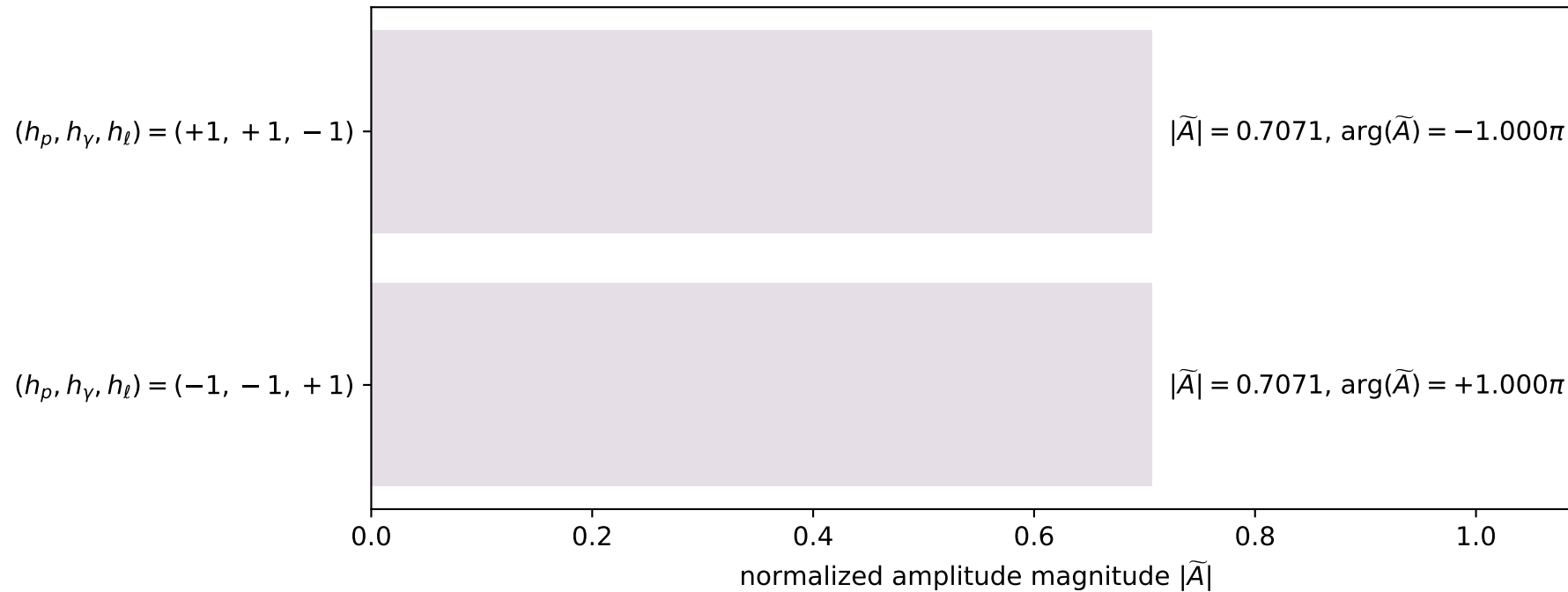
dghz_mixing_angles_polarization_02_local_minimum_14
 $d_{\text{GHZ}}=2.46768\text{e-}08$
 region: polarization_cluster_02_local_minimum_14
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0014
 lepton mixing angle: $\alpha_e=0.75308403$ rad
 incoming lepton: $0.729583|+\rangle + 0.683892|-\rangle$
 proton mixing angle: $\alpha_p=2.3238894$ rad
 incoming proton: $-0.683899|+\rangle + 0.729577|-\rangle$

$s=24.6277$, $\sqrt{s}=4.96263$, $|\vec{P}|=2.39267$, $|\vec{P}'|=2.39267$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.225183$, $\phi_\gamma=3.93532$
 $E_\gamma=0.176937$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.36678$
 $Q^2=21.2061$, $x_B=0.92352$, $t=-22.8995$, $W^2=2.63599$, $y=0.966916$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3927$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3927$
 P $E= 2.5700$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3927$
 ℓ' $E= 2.2157$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2157$
 P' $E= 2.5700$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3927$
 q_γ $E= 0.1769$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1769$

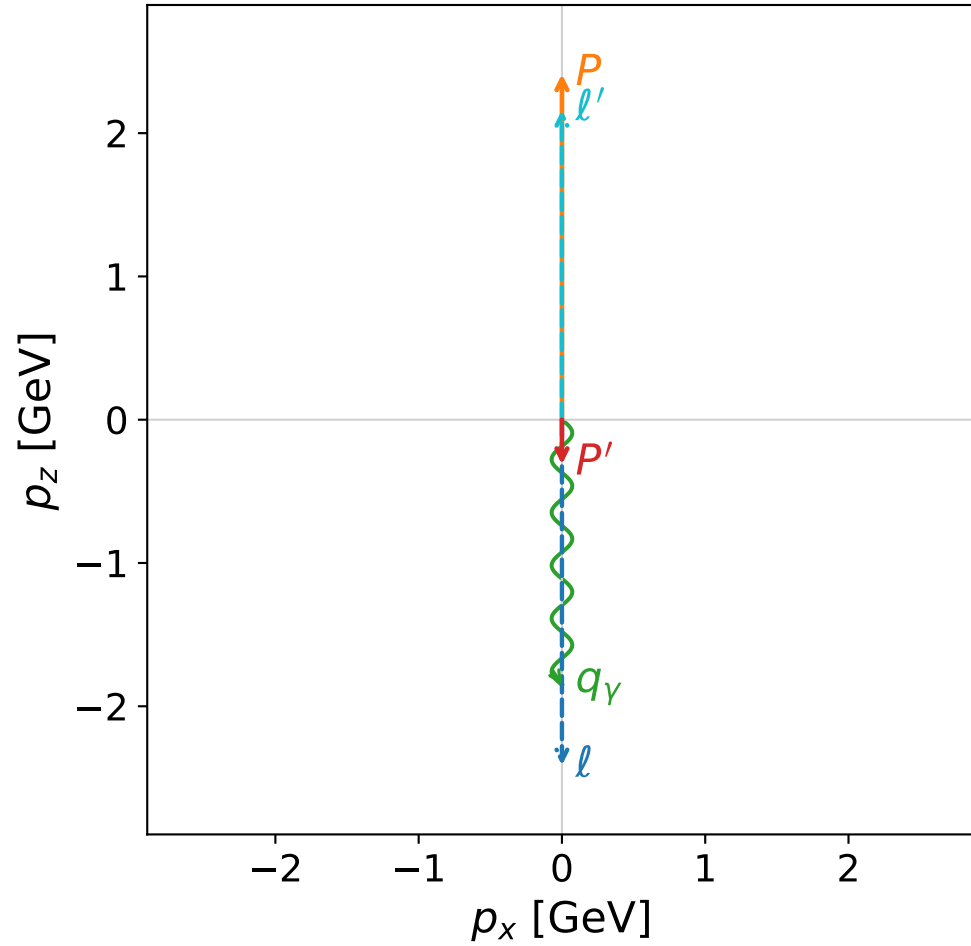
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=8.960\text{e+}21$; $I_h=+6.226\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.946\text{e-}25$; $I_{TL}=+6.226\text{e-}13$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

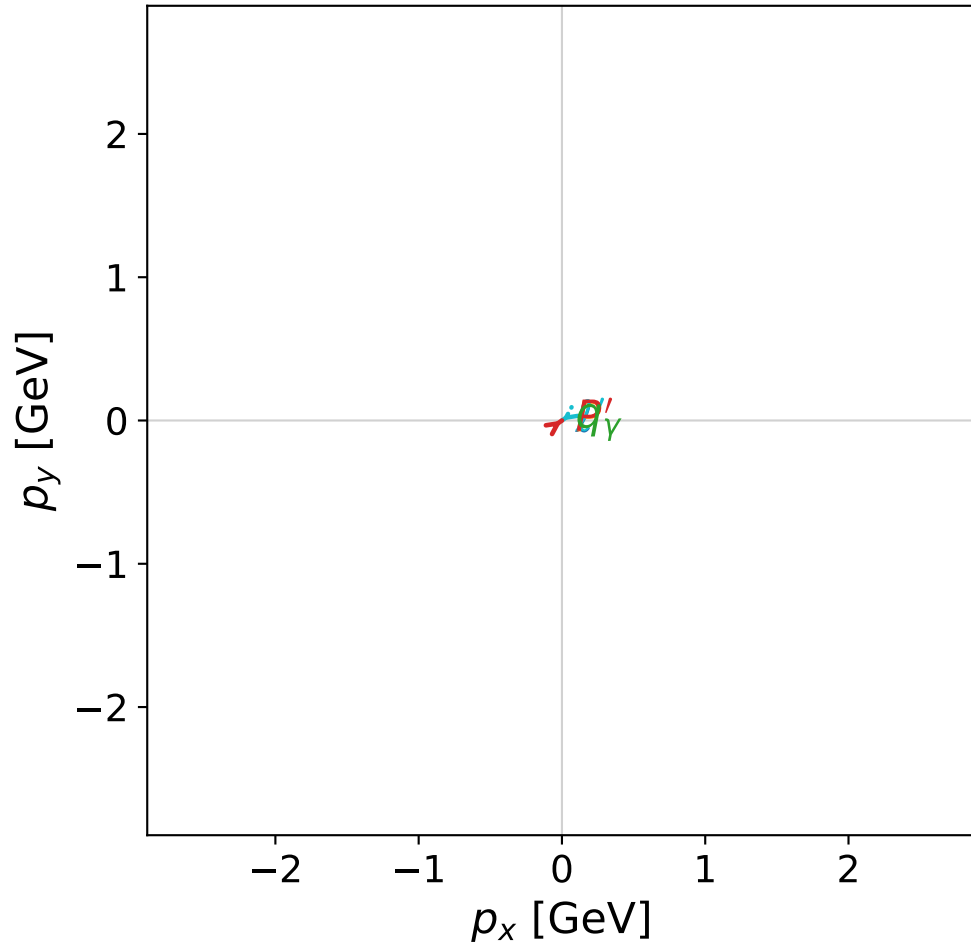


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



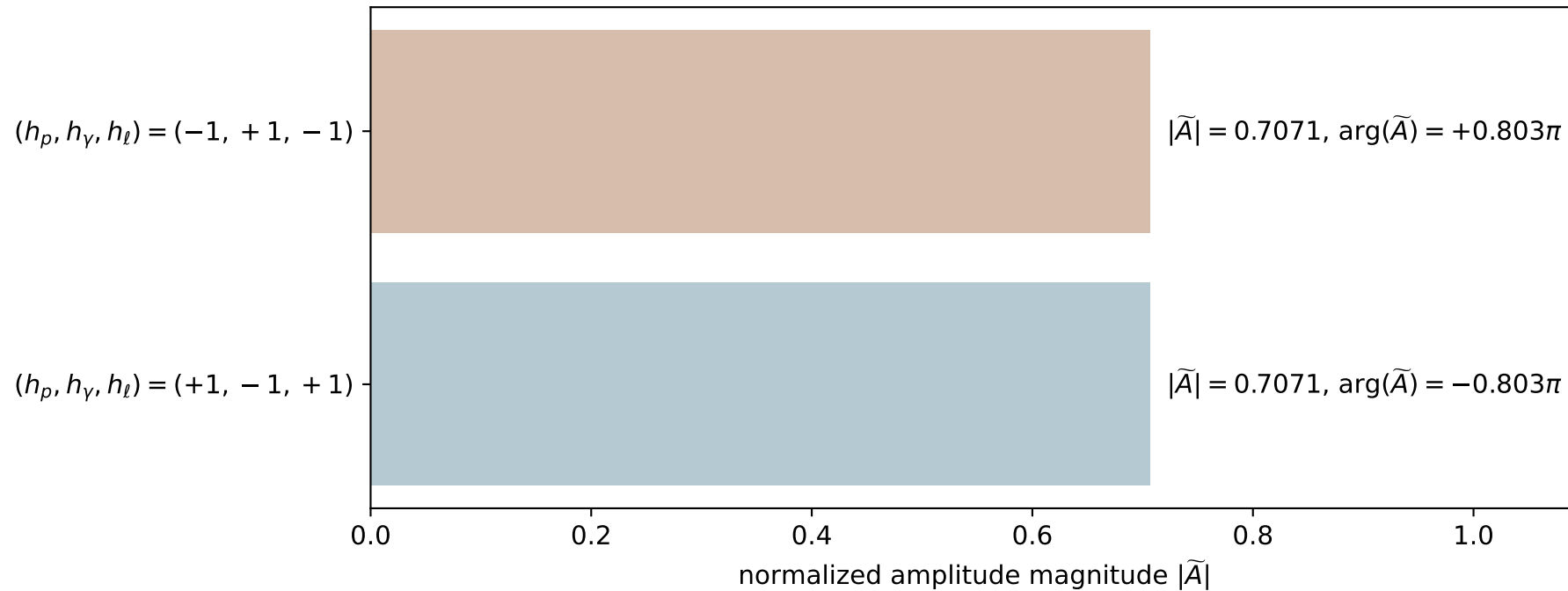
dghz_mixing_angles_polarization_02_local_minimum_21
 $d_{\text{GHZ}}=2.55996\text{e-}08$
 region: polarization_cluster_02_local_minimum_21
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0021
 lepton mixing angle: $\alpha_e=0.88921427$ rad
 incoming lepton: $0.630022|+\rangle + 0.776577|-\rangle$
 proton mixing angle: $\alpha_p=2.2523645$ rad
 incoming proton: $-0.630012|+\rangle + 0.776586|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.311681$
 $\theta_{P'}=3.14153$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.615032$, $\phi_\gamma=4.37445$
 $E_\gamma=1.84995$, $\theta_{\ell'}=8.59128\text{e-}06$, $\phi_{\ell'}=3.75662$
 $Q^2=20.8555$, $x_B=0.89281$, $t=-4.85994$, $W^2=3.38373$, $y=0.96846$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1616$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1616$
 P' $E= 0.9884$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3117$
 q_γ $E= 1.8499$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8499$

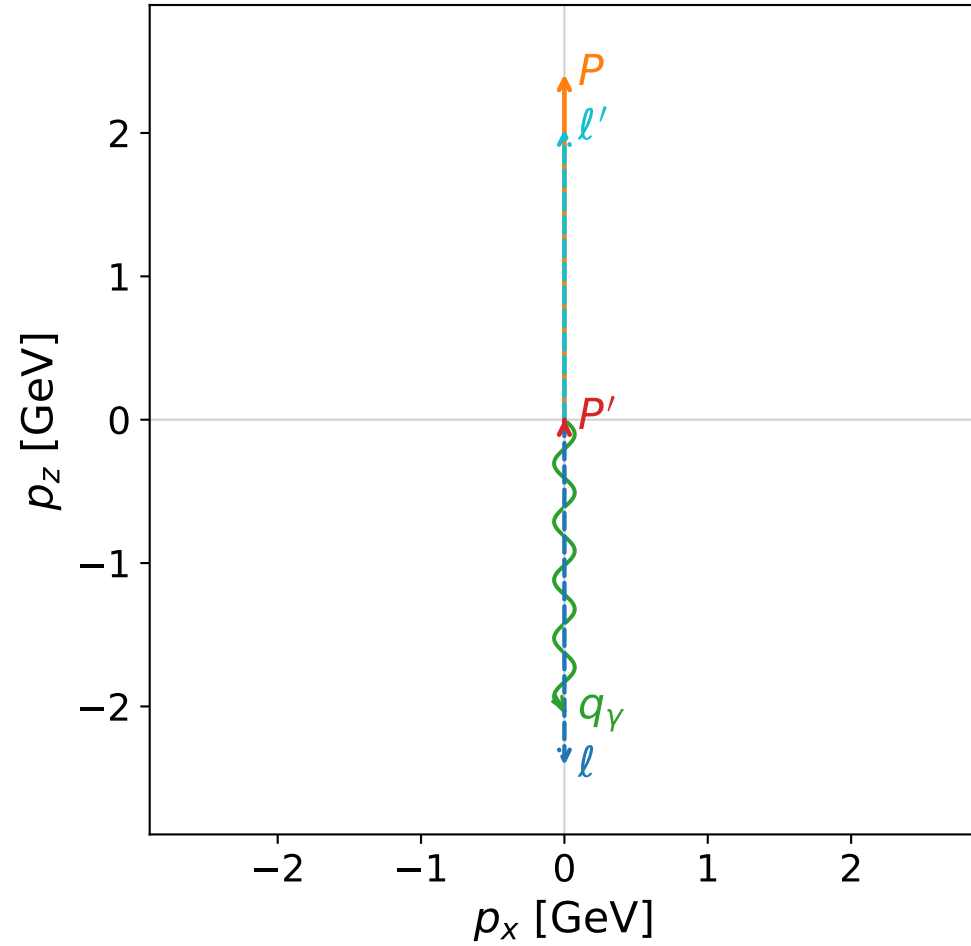
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=9.754\text{e+}17$; $I_h=-2.042\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.154\text{e-}13$; $I_{TL}=-8.084\text{e-}08$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

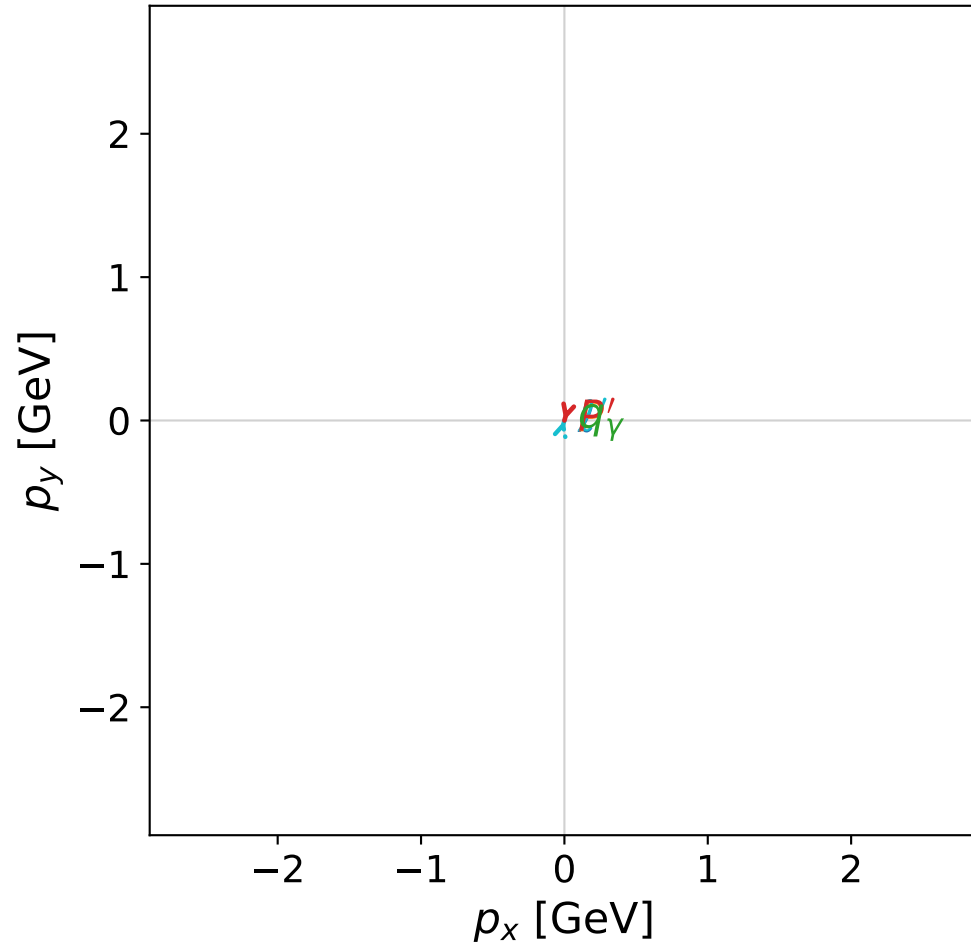


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

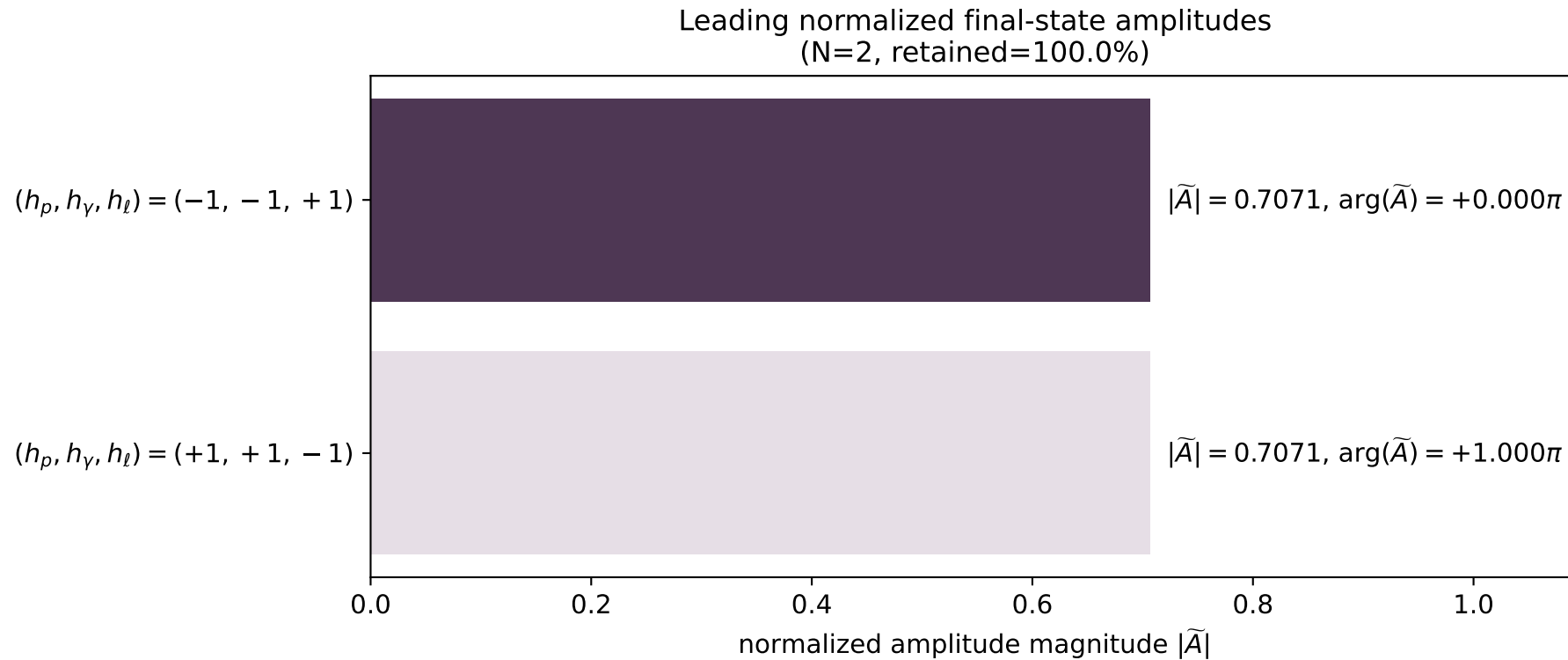


dghz_mixing_angles_polarization_02_local_minimum_26
 $d_{\text{GHZ}}=2.59712\text{e-}08$
 region: polarization_cluster_02_local_minimum_26
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0026
 lepton mixing angle: $\alpha_e=0.77423168$ rad
 incoming lepton: $0.714958|+\rangle + 0.699167|-\rangle$
 proton mixing angle: $\alpha_p=2.3673692$ rad
 incoming proton: $-0.714964|+\rangle + 0.699161|-\rangle$

$s=24.9729$, $\sqrt{s}=4.99729$, $|\vec{P}|=2.41061$, $|\vec{P}'|=0.00645061$
 $\theta_{P'}=0.00979874$, $\theta_Y=3.14159$, $\phi_P=4.43638$, $\phi_Y=6.38506\text{e-}05$
 $E_Y=2.03286$, $\theta_{\ell'}=3.11916\text{e-}05$, $\phi_{\ell'}=1.29479$
 $Q^2=19.5395$, $x_B=0.835755$, $t=-3.06193$, $W^2=4.7198$, $y=0.970383$

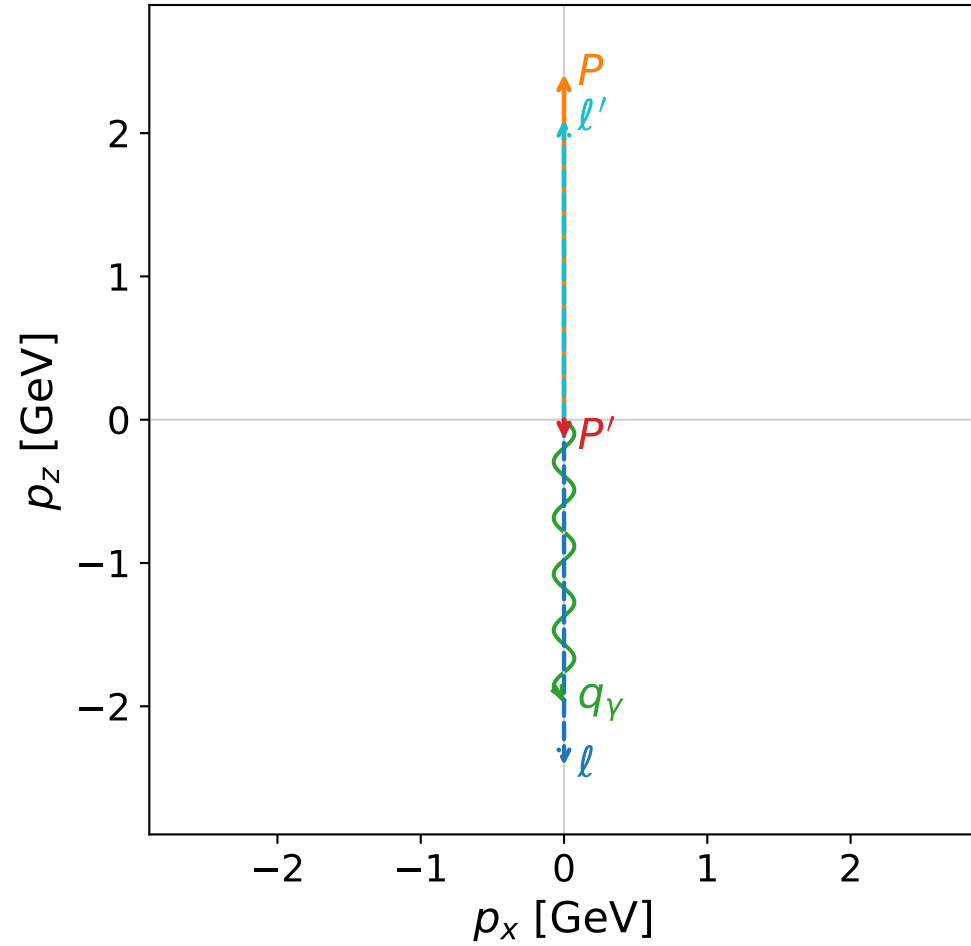
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4106$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4106$
 P $E= 2.5867$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4106$
 ℓ' $E= 2.0264$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 2.0264$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 0.0065$
 q_Y $E= 2.0329$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0329$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=8.651\text{e+}16$; $I_h=-2.388\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.699\text{e-}12$; $I_{TL}=-1.596\text{e-}08$

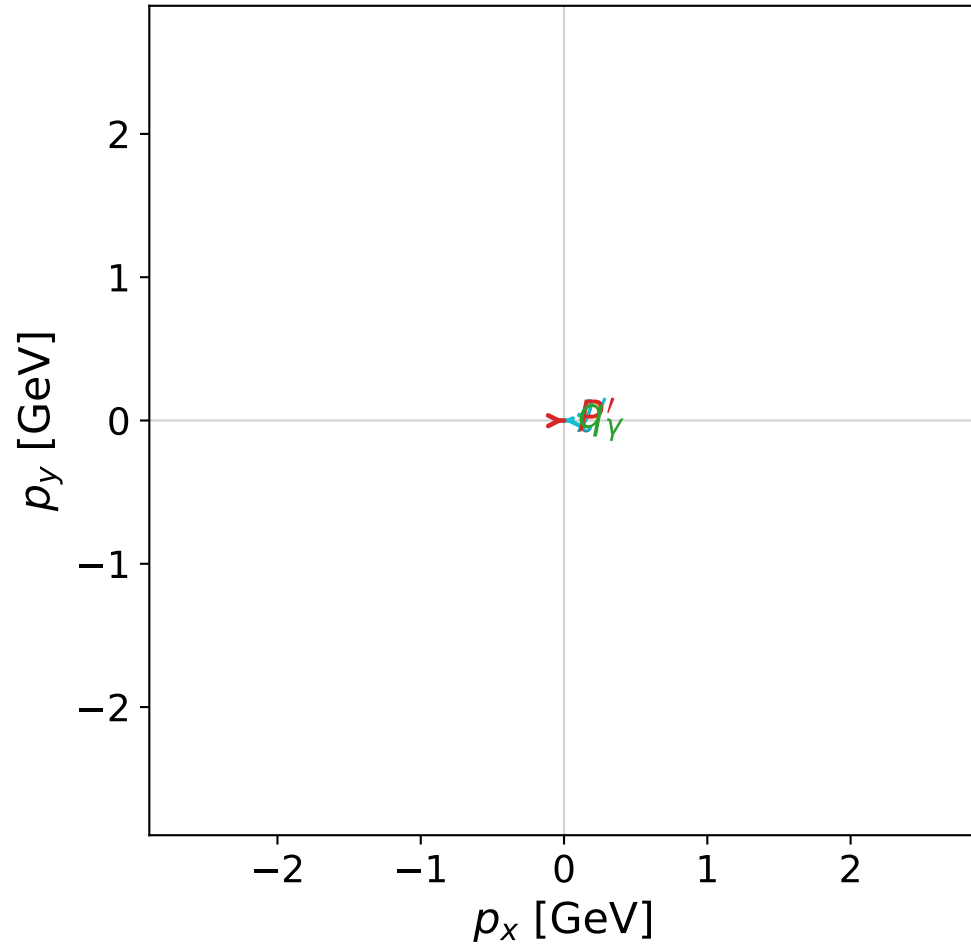


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

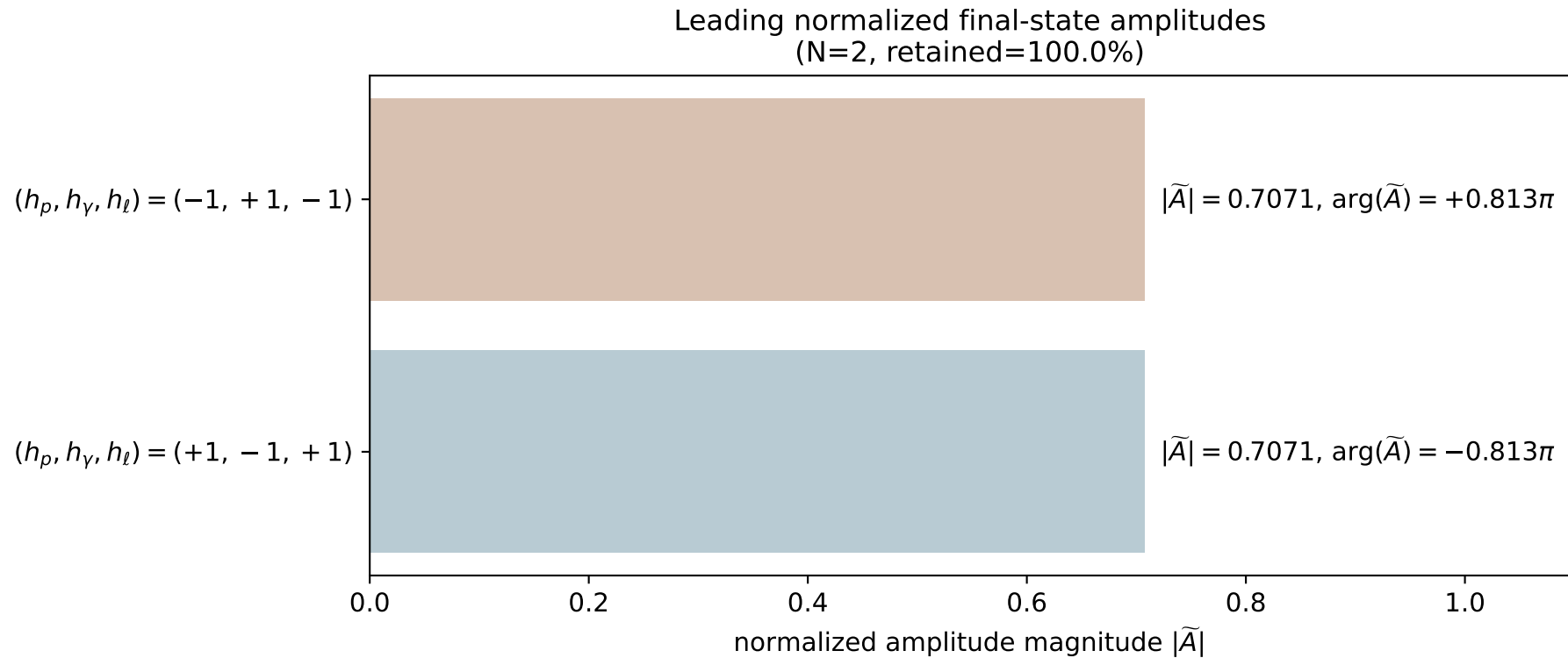


dghz_mixing_angles_polarization_02_local_minimum_31
 $d_{\text{GHZ}}=2.62572\text{e-}08$
 region: polarization_cluster_02_local_minimum_31
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0031
 lepton mixing angle: $\alpha_e=0.72294087$ rad
 incoming lepton: $0.749863|+\rangle +0.661593|-\rangle$
 proton mixing angle: $\alpha_p=2.4186407$ rad
 incoming proton: $-0.749856|+\rangle +0.661601|-\rangle$

$s=24.9982$, $\sqrt{s}=4.99982$, $|\vec{P}|=2.41192$, $|\vec{P}'|=0.133473$
 $\theta_{p'}=3.14093$, $\theta_Y=3.14159$, $\phi_{p'}=0.0144773$, $\phi_Y=3.74298$
 $E_Y=1.95945$, $\theta_{\ell'}=4.19601\text{e-}05$, $\phi_{\ell'}=3.15607$
 $Q^2=20.1919$, $x_B=0.863574$, $t=-3.78797$, $W^2=4.06974$, $y=0.969459$

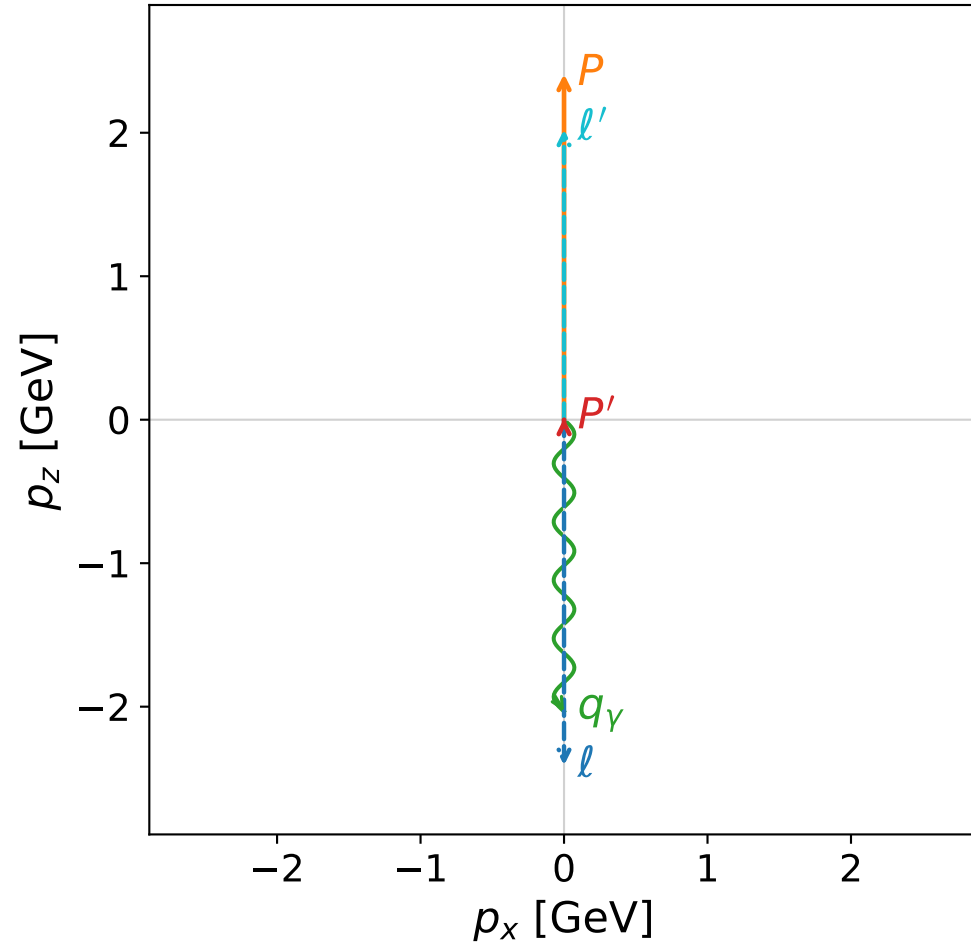
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0929$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0929$
 P' $E= 0.9474$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.1335$
 q_Y $E= 1.9595$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9595$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.545\text{e+}16$; $I_h=+6.870\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.668\text{e-}12$; $I_{TL}=+3.816\text{e-}07$

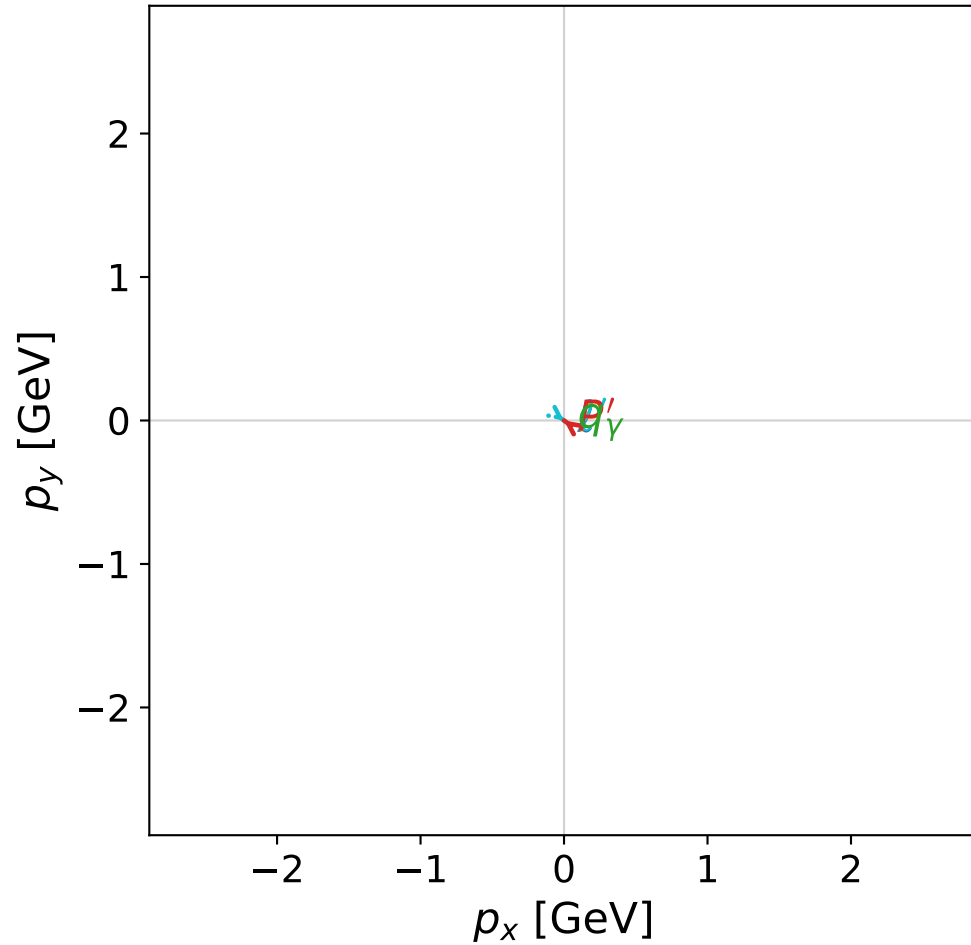


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

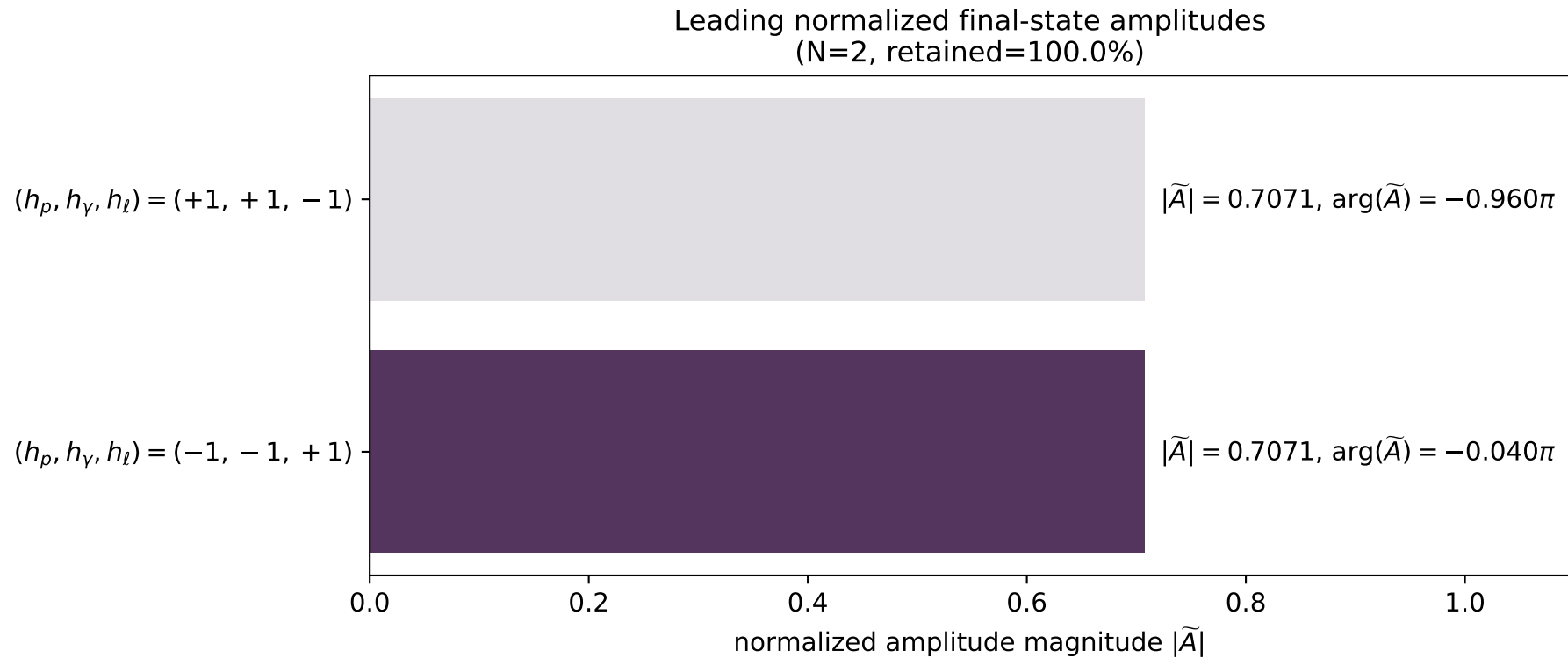


dghz_mixing_angles_polarization_02_local_minimum_33
 $d_{\text{GHZ}}=2.63319\text{e-}08$
 region: polarization_cluster_02_local_minimum_33
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0033
 lepton mixing angle: $\alpha_e=0.79116621$ rad
 incoming lepton: $0.703016|+\rangle +0.711174|-\rangle$
 proton mixing angle: $\alpha_p=2.3504057$ rad
 incoming proton: $-0.703002|+\rangle +0.711188|-\rangle$

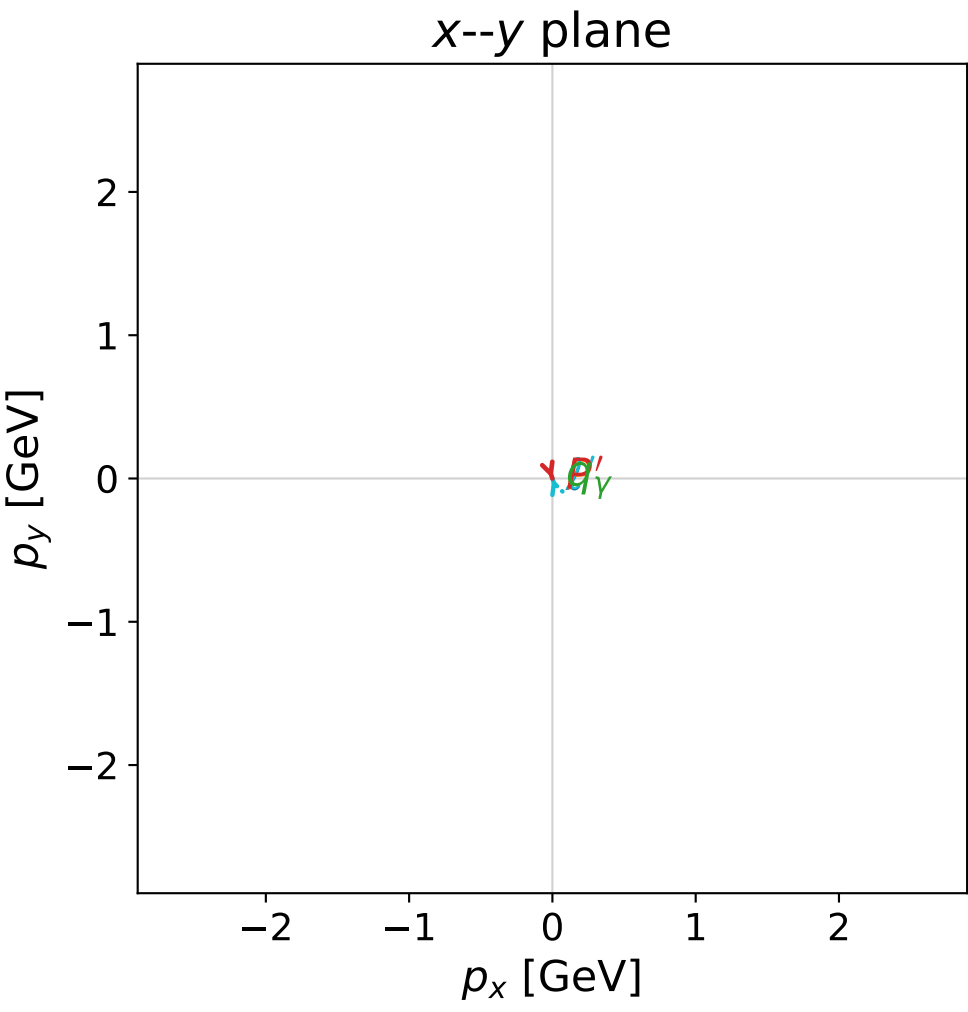
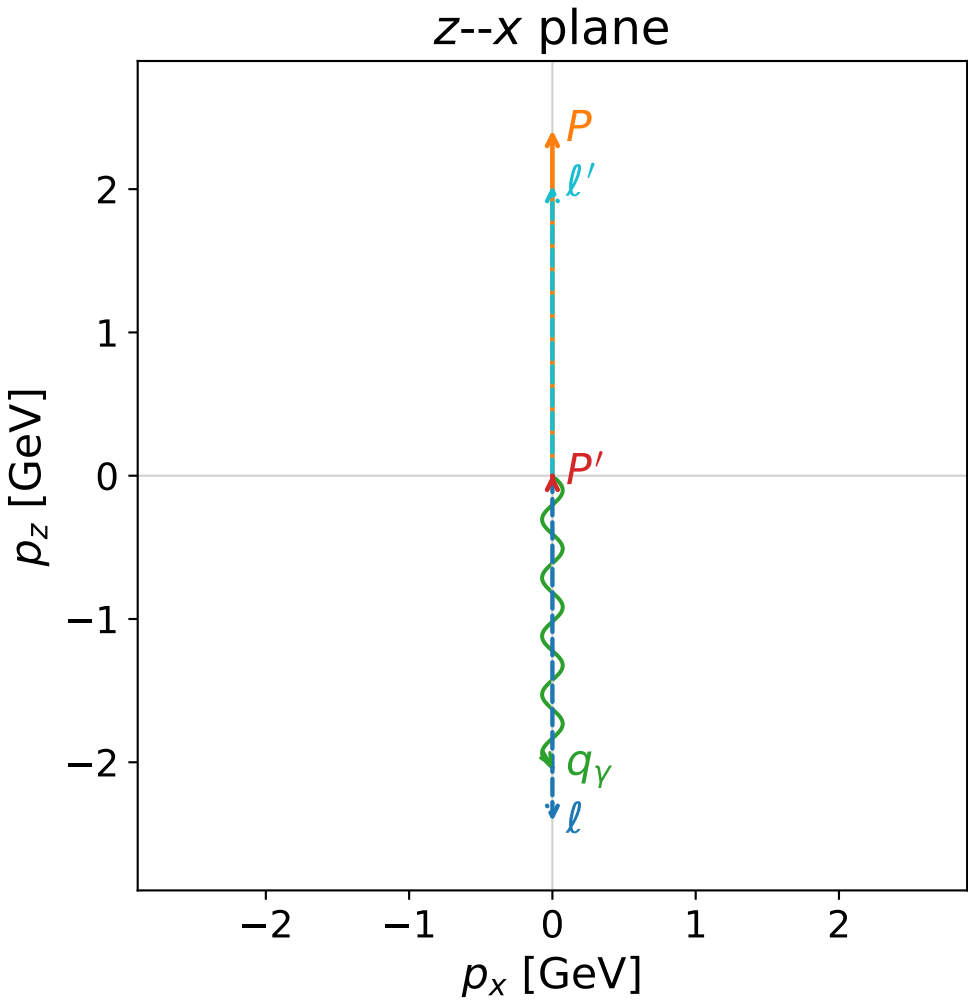
$s=24.934$, $\sqrt{s}=4.9934$, $|\vec{P}|=2.4086$, $|\vec{P}'|=0.0106855$
 $\theta_{P'}=0.0073264$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.51128$, $\phi_\gamma=6.15755$
 $E_\gamma=2.03301$, $\theta_{\ell'}=3.87108\text{e-}05$, $\phi_{\ell'}=5.65288$
 $Q^2=19.4839$, $x_B=0.834731$, $t=-3.03824$, $W^2=4.73747$, $y=0.970372$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4086$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4086$
 P $E= 2.5848$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4086$
 ℓ' $E= 2.0223$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 2.0223$
 P' $E= 0.9381$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 0.0107$
 q_γ $E= 2.0330$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0330$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.612\text{e+}16$; $I_h=+4.556\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.773\text{e-}12$; $I_{TL}=+3.258\text{e-}08$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

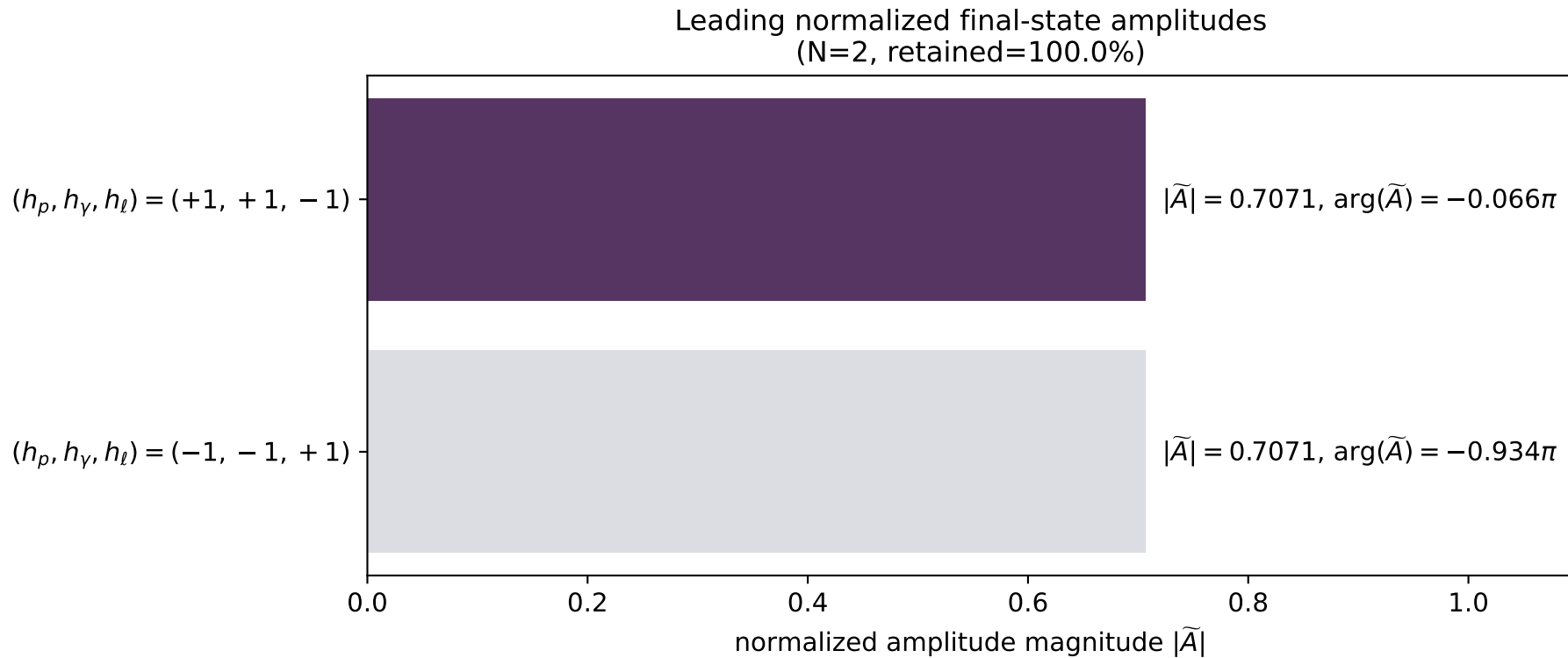


dghz_mixing_angles_polarization_02_local_minimum_34
 $d_{\{\rm GHZ\}}=2.63343\text{e-}08$
region: polarization_cluster_02_local_minimum_34
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0034
lepton mixing angle: $\alpha_e=0.76007598$ rad
incoming lepton: $0.724784|+\rangle +0.688977|-\rangle$
proton mixing angle: $\alpha_p=2.3815115$ rad
incoming proton: $-0.72478|+\rangle +0.68898|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0121954$
 $\theta_{p'}=0.00245437$, $\theta_\gamma=3.14159$, $\phi_{p'}=5.04048$, $\phi_\gamma=3.34859$
 $E_\gamma=2.03706$, $\theta_{\ell'}=1.47823\text{e-}05$, $\phi_{\ell'}=1.89889$
 $Q^2=19.536$, $x_B=0.834603$, $t=-3.03695$, $W^2=4.75137$, $y=0.970455$

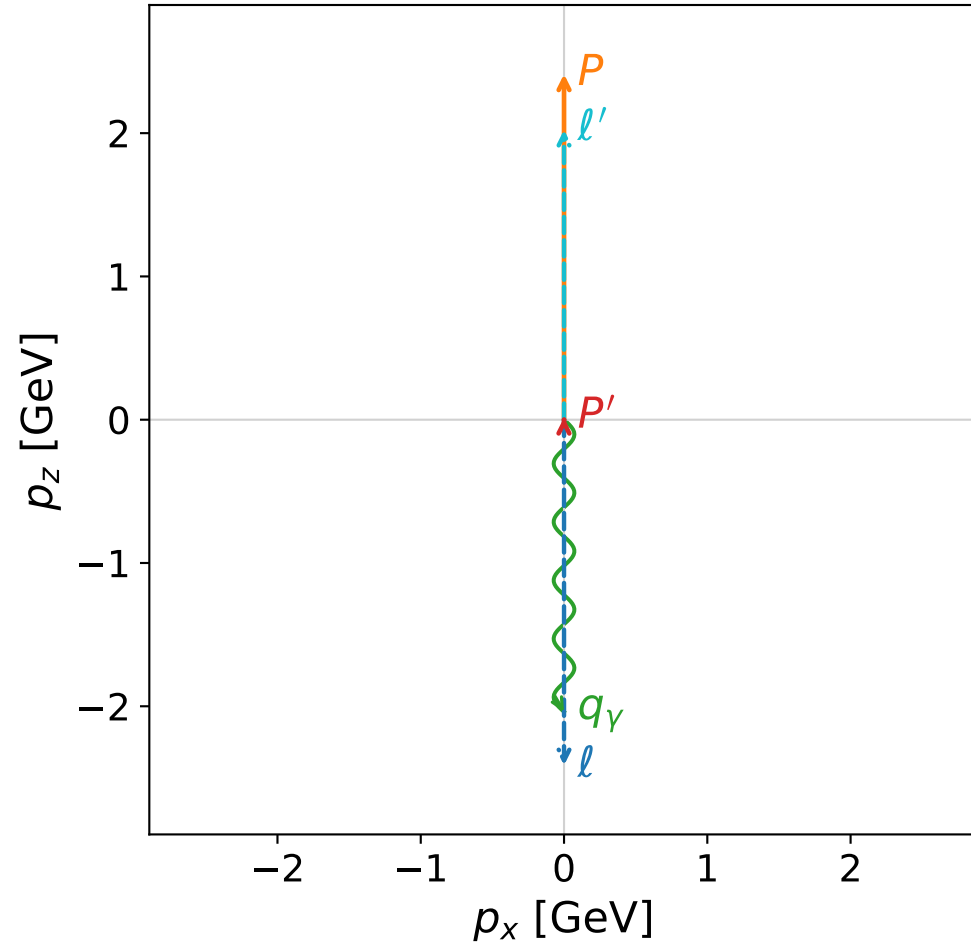
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0249$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0249$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0122$
 q_γ $E= 2.0371$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0371$

GHZ mechanism: helicity flipping (full H G C)
helicity powers: flip 100.000%, conserve 0.000%
R_flip/conserve=3.850e+17; I_h=+3.029e-11
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=8.429e-13; I_TL=+2.131e-08

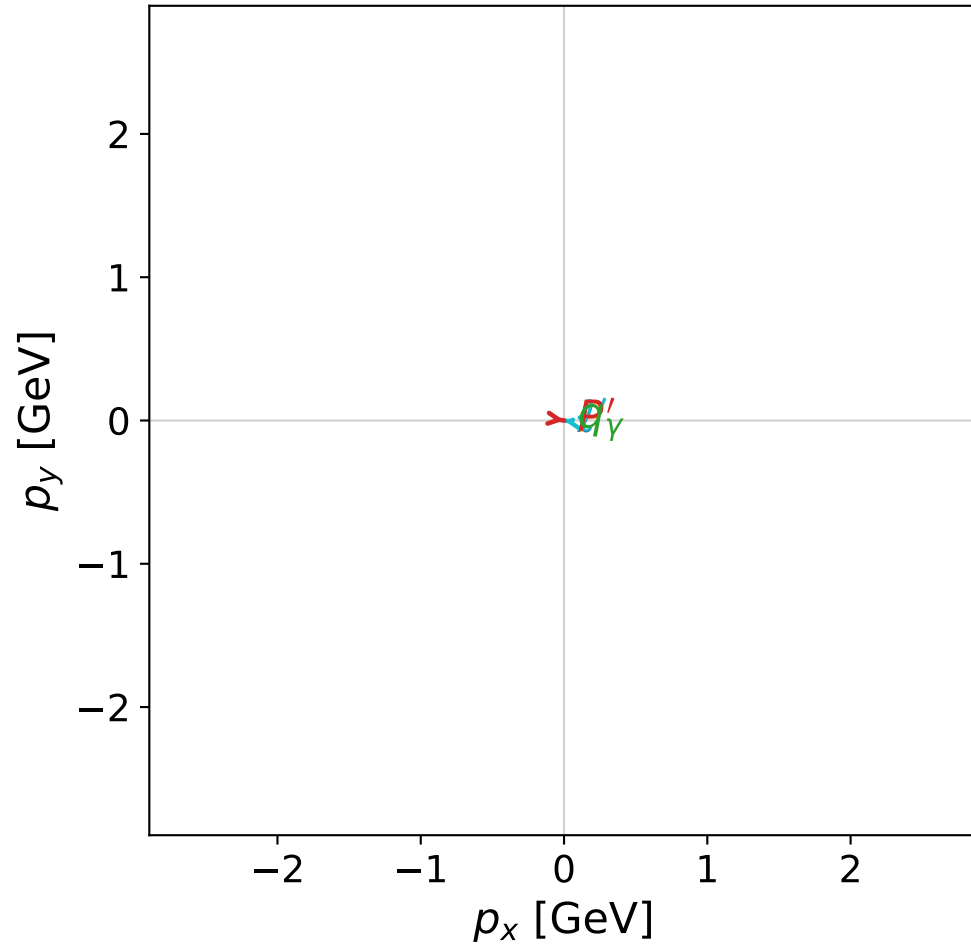


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

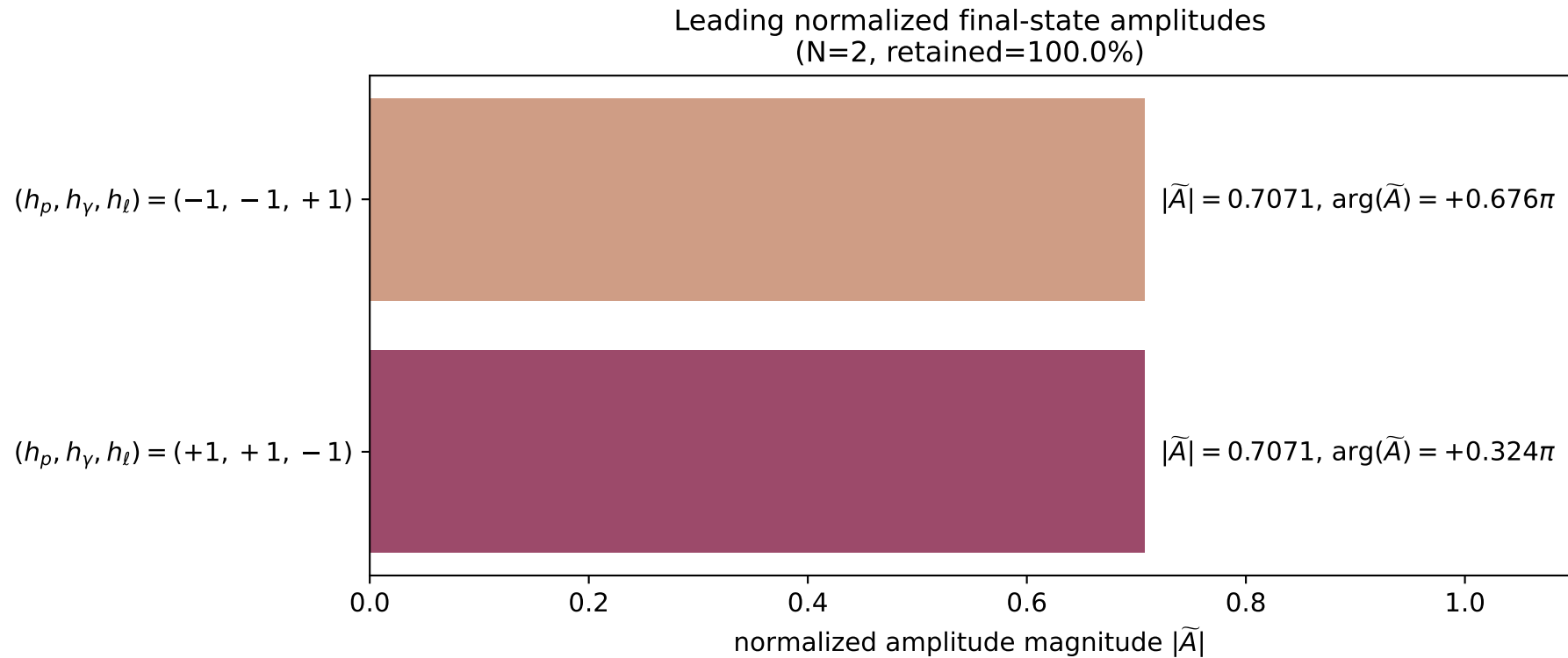


dghz_mixing_angles_polarization_02_local_minimum_36
 $d_{\text{GHZ}}=2.63908\text{e-}08$
 region: polarization_cluster_02_local_minimum_36
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0036
 lepton mixing angle: $\alpha_e=0.80546213$ rad
 incoming lepton: $0.692778|+\rangle +0.721151|-\rangle$
 proton mixing angle: $\alpha_p=2.3361295$ rad
 incoming proton: $-0.692777|+\rangle +0.721152|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0152919$
 $\theta_{P'}=0.00490874$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.14104$, $\phi_\gamma=2.12467$
 $E_\gamma=2.03858$, $\theta_{\ell'}=3.70997\text{e-}05$, $\phi_{\ell'}=2.99944$
 $Q^2=19.5208$, $x_B=0.833936$, $t=-3.02225$, $W^2=4.76708$, $y=0.970478$

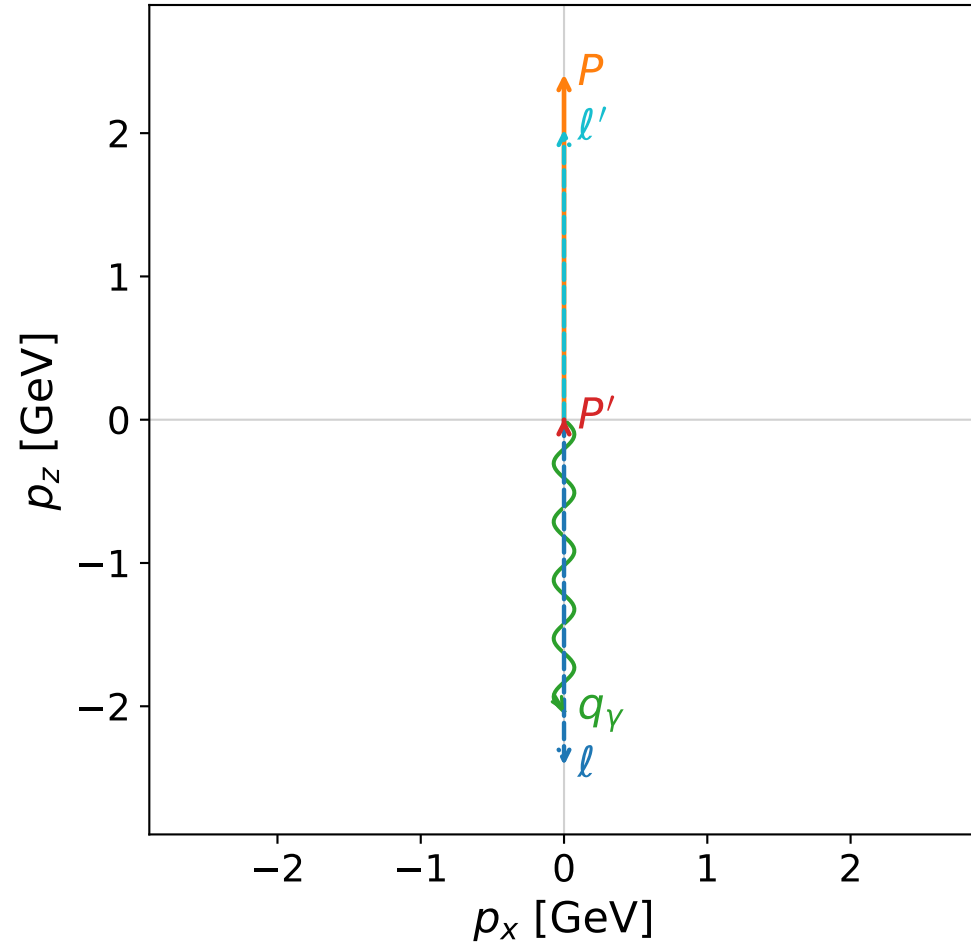
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0233$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0233$
 P' $E= 0.9381$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0153$
 q_γ $E= 2.0386$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0386$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.122\text{e+}16$; $I_h=-1.847\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.345\text{e-}12$; $I_{TL}=-1.290\text{e-}07$

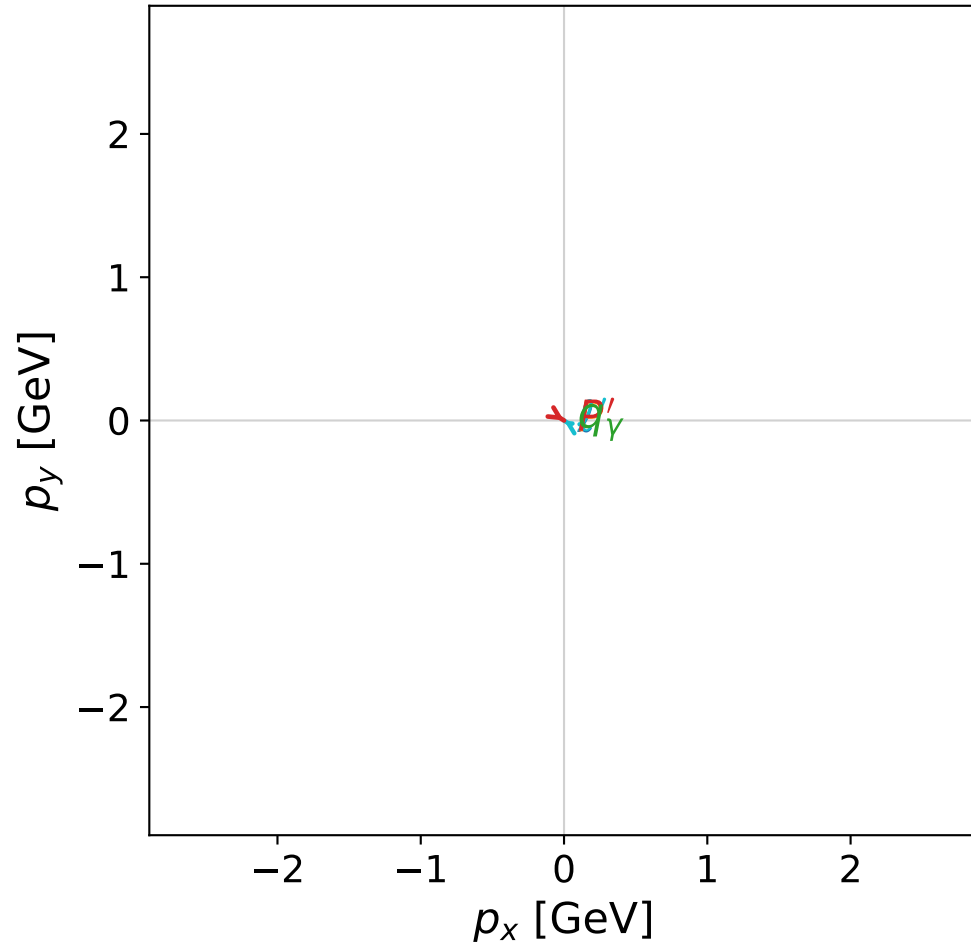


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

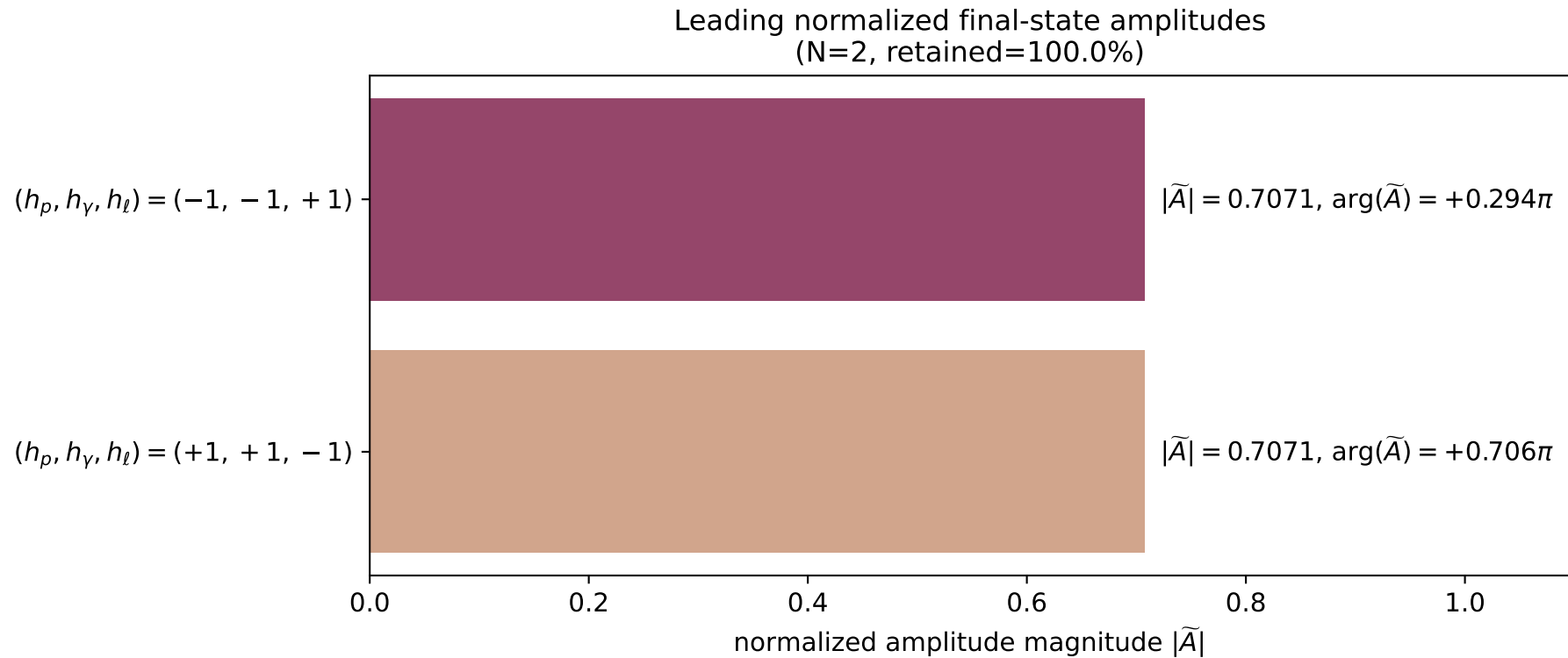


dghz_mixing_angles_polarization_02_local_minimum_37
 $d_{\text{GHZ}}=2.64099\text{e-}08$
 region: polarization_cluster_02_local_minimum_37
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0037
 lepton mixing angle: $\alpha_e=0.81766347$ rad
 incoming lepton: $0.683928|+\rangle +0.72955|-\rangle$
 proton mixing angle: $\alpha_p=2.3239276$ rad
 incoming proton: $-0.683927|+\rangle +0.729551|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0096556$
 $\theta_{P'}=0.00245437$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.7193$, $\phi_\gamma=0.92409$
 $E_\gamma=2.0358$, $\theta_{\ell'}=1.16963\text{e-}05$, $\phi_{\ell'}=2.5777$
 $Q^2=19.5484$, $x_B=0.835149$, $t=-3.04905$, $W^2=4.73853$, $y=0.970436$

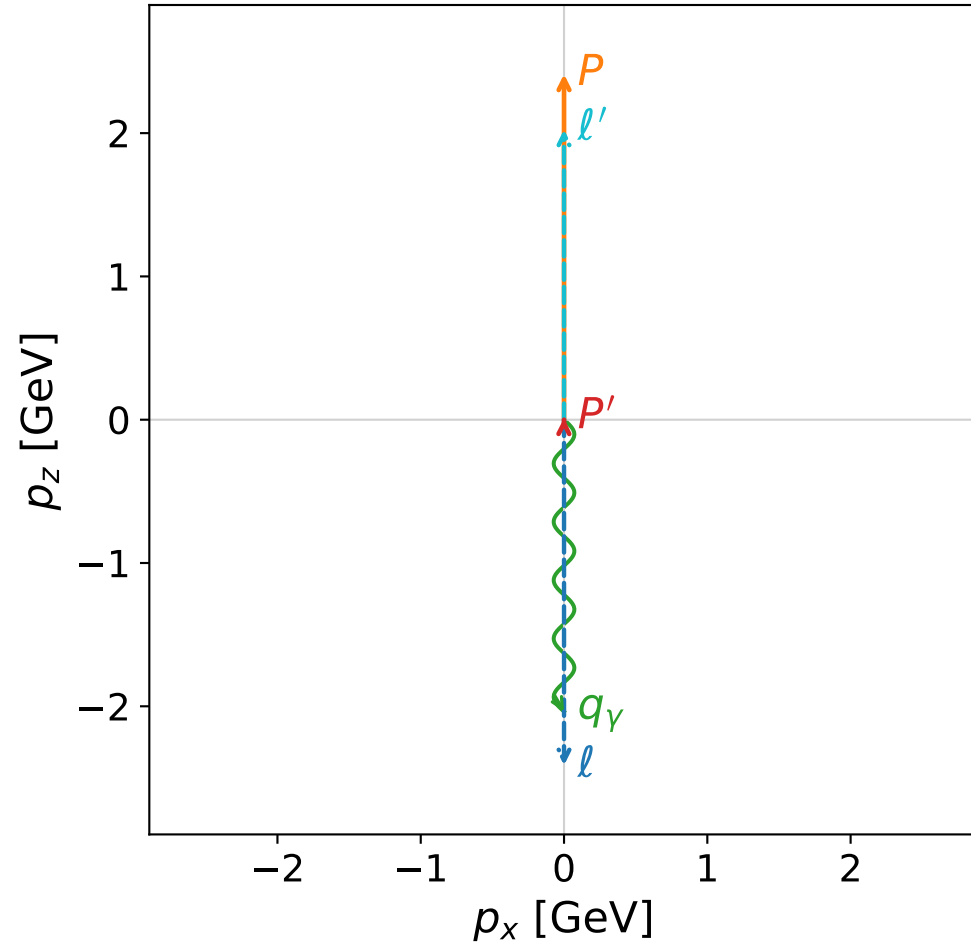
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0261$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0261$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0097$
 q_γ $E= 2.0358$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0358$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.134\text{e+}17$; $I_h=-8.016\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.252\text{e-}13$; $I_{TL}=-5.578\text{e-}08$

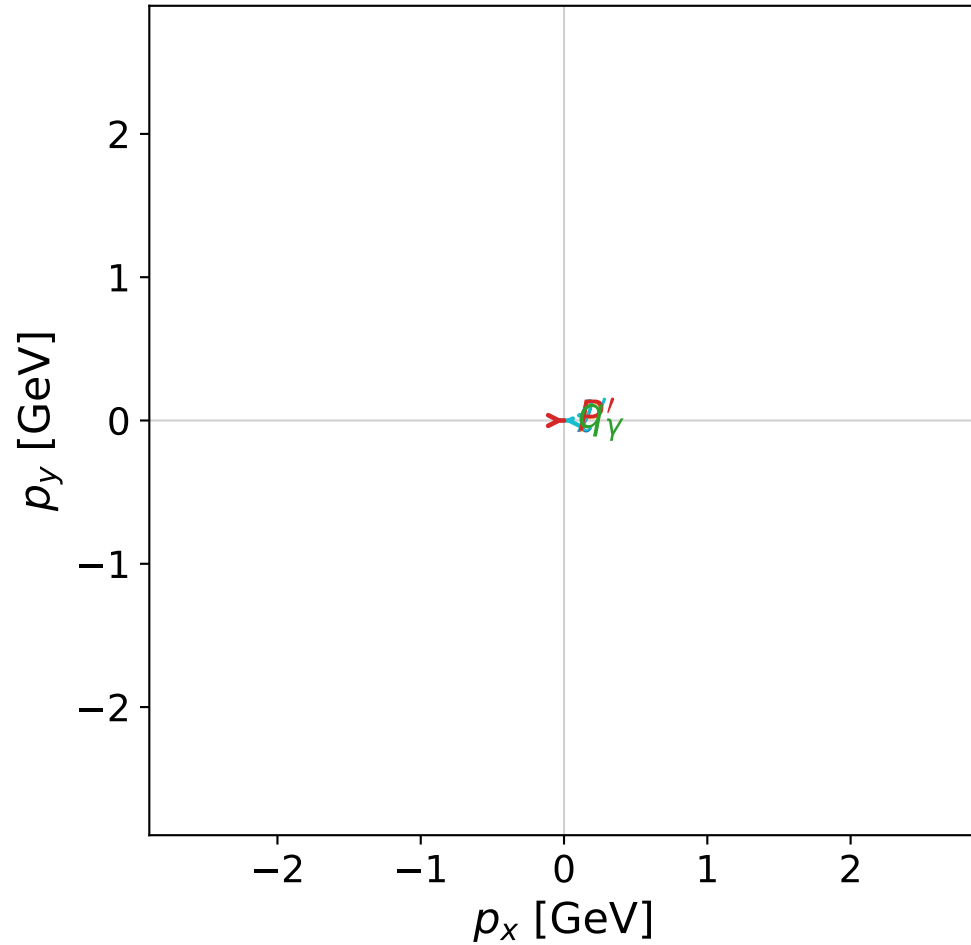


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

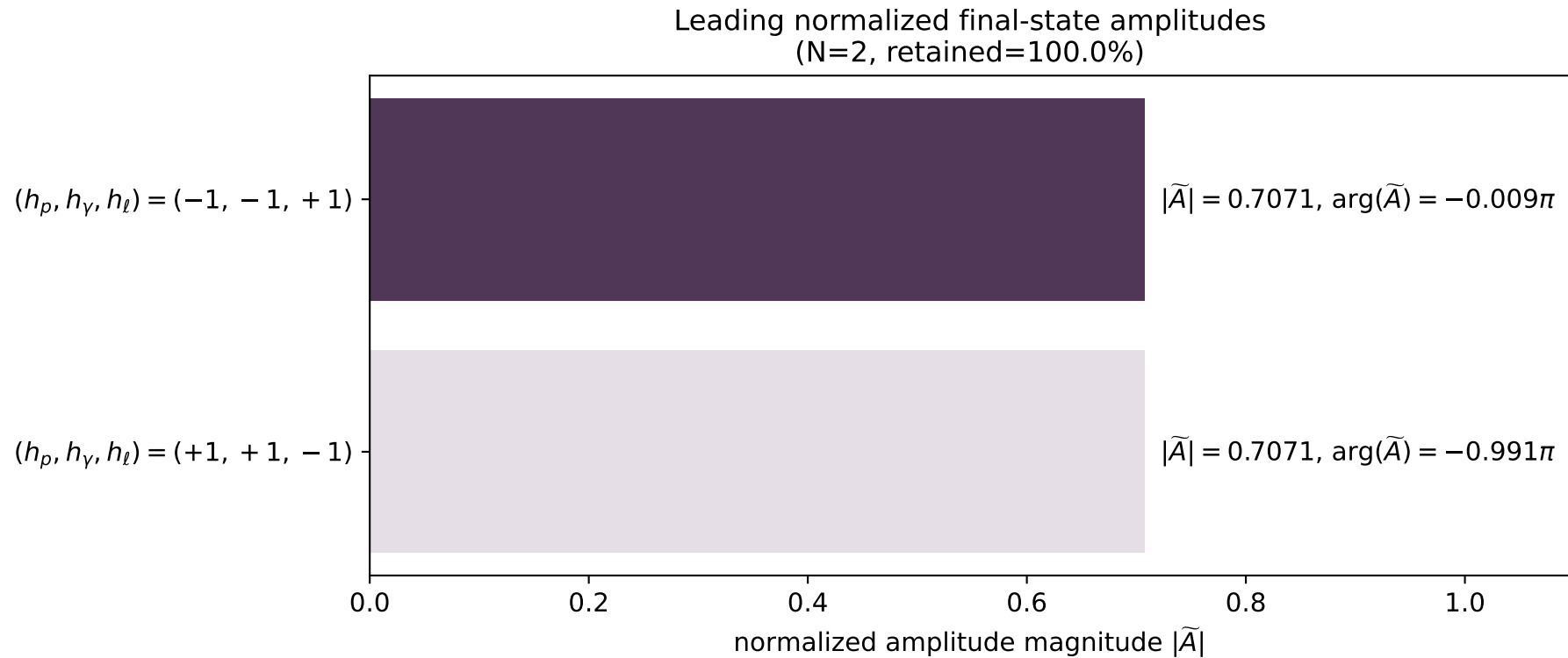


dghz_mixing_angles_polarization_02_local_minimum_40
 $d_{\text{GHZ}}=2.64788\text{e-}08$
 region: polarization_cluster_02_local_minimum_40
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0040
 lepton mixing angle: $\alpha_e=0.76726665$ rad
 incoming lepton: $0.719811|+\rangle +0.69417|-\rangle$
 proton mixing angle: $\alpha_p=2.3743383$ rad
 incoming proton: $-0.719819|+\rangle +0.694161|-\rangle$

$s=24.9961$, $\sqrt{s}=4.99961$, $|\vec{P}|=2.41181$, $|\vec{P}'|=0.0116328$
 $\theta_{P'}=0.00959617$, $\theta_Y=3.14159$, $\phi_{P'}=6.28284$, $\phi_Y=6.2538$
 $E_Y=2.03658$, $\theta_{\ell'}=5.51265\text{e-}05$, $\phi_{\ell'}=3.14125$
 $Q^2=19.5352$, $x_B=0.834713$, $t=-3.03928$, $W^2=4.74815$, $y=0.970447$

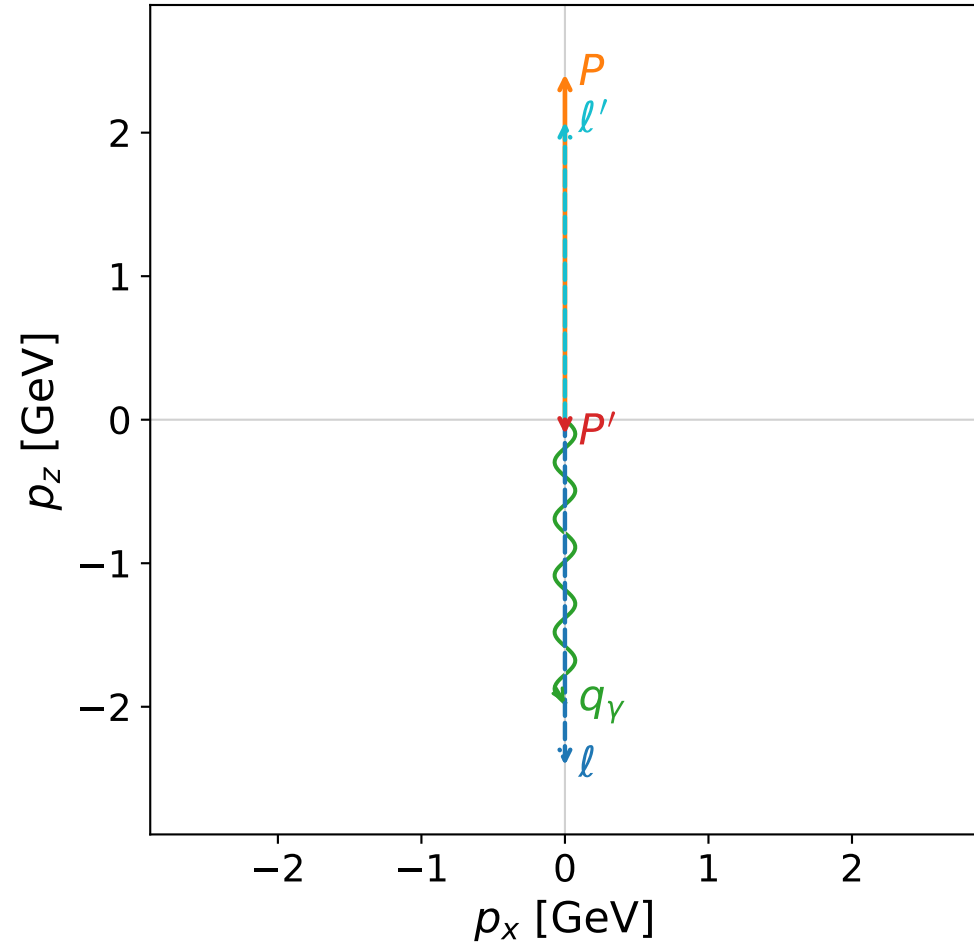
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 ℓ' $E= 2.0250$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0250$
 P' $E= 0.9381$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0116$
 q_Y $E= 2.0366$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0366$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.773\text{e+}16$; $I_h=+2.512\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.170\text{e-}11$; $I_{TL}=+1.772\text{e-}07$

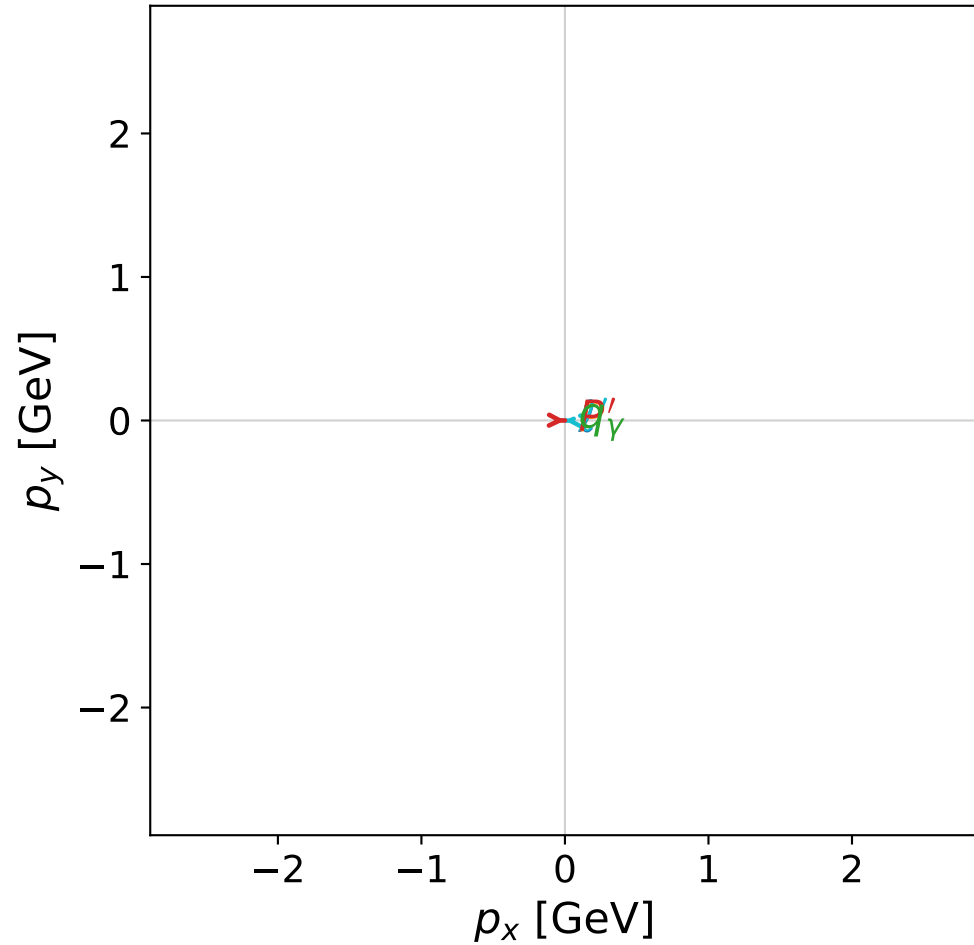


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



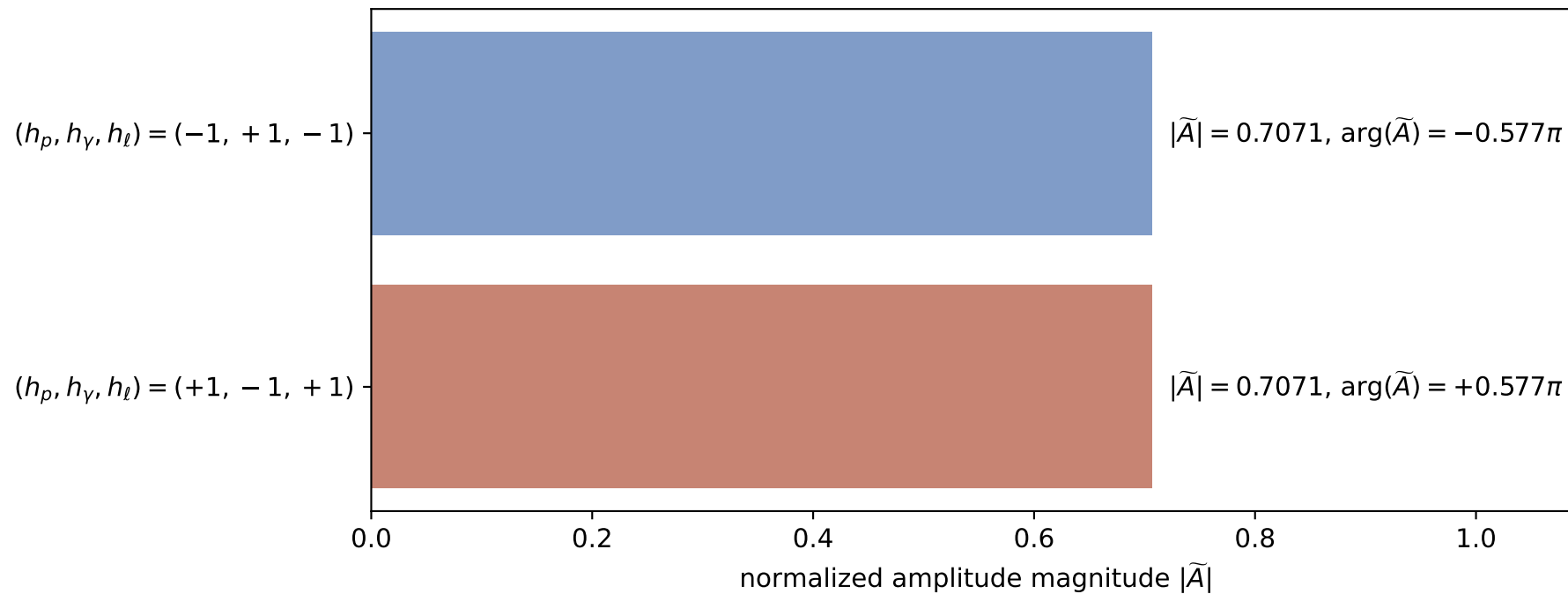
dghz_mixing_angles_polarization_02_local_minimum_41
 $d_{\text{GHZ}}=2.65018\text{e-}08$
 region: polarization_cluster_02_local_minimum_41
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0041
 lepton mixing angle: $\alpha_e=0.70648598$ rad
 incoming lepton: $0.760648|+\rangle + 0.649165|-\rangle$
 proton mixing angle: $\alpha_p=2.4350875$ rad
 incoming proton: $-0.760635|+\rangle + 0.649179|-\rangle$

$s=24.9206$, $\sqrt{s}=4.99205$, $|\vec{P}|=2.4079$, $|\vec{P}'|=0.101522$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.26287$, $\phi_\gamma=1.79274$
 $E_\gamma=1.97352$, $\theta_{\ell'}=3.12924\text{e-}07$, $\phi_{\ell'}=3.12128$
 $Q^2=19.986$, $x_B=0.857428$, $t=-3.6054$, $W^2=4.20309$, $y=0.969575$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4079$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4079$
 P $E= 2.5841$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4079$
 ℓ' $E= 2.0750$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0750$
 P' $E= 0.9435$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.1015$
 q_γ $E= 1.9735$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9735$

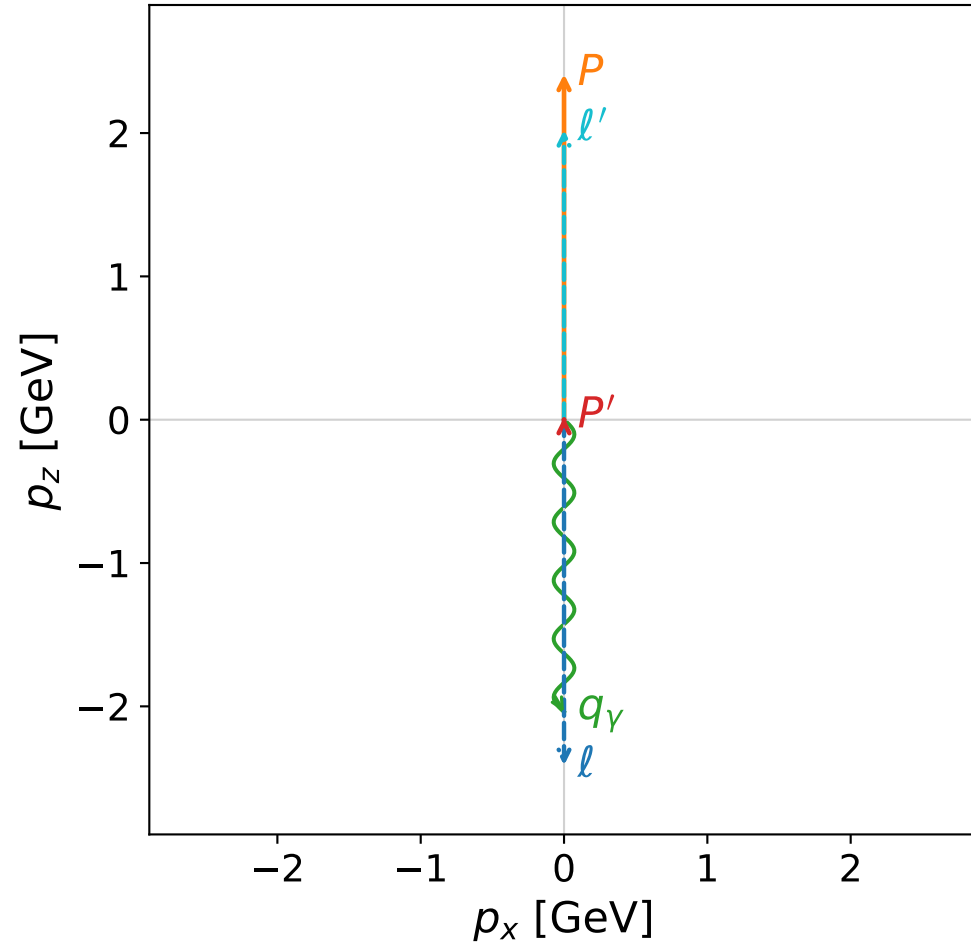
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=8.070\text{e+}20$; $I_h=+6.445\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.877\text{e-}16$; $I_{TL}=+3.769\text{e-}09$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

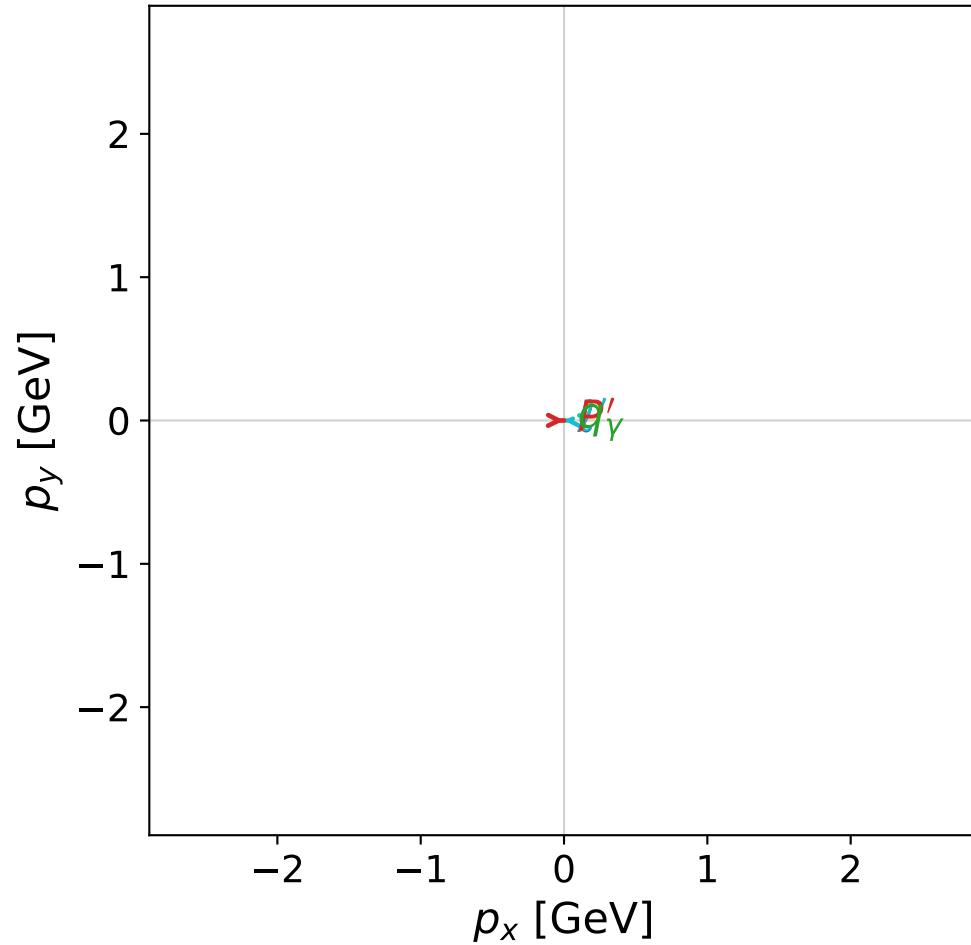


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

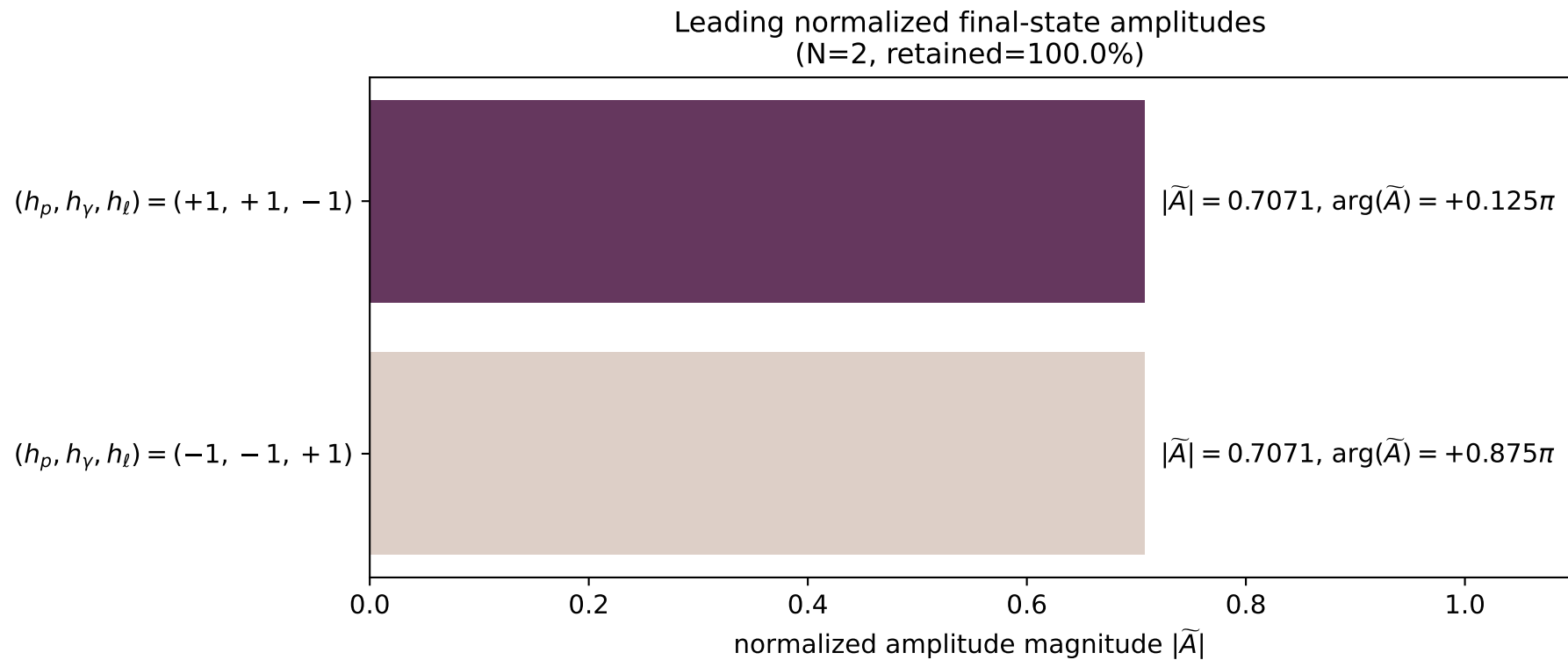


dghz_mixing_angles_polarization_02_local_minimum_42
 $d_{\text{GHZ}}=2.65173\text{e-}08$
 region: polarization_cluster_02_local_minimum_42
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0042
 lepton mixing angle: $\alpha_e=0.7472291$ rad
 incoming lepton: $0.733575|+\rangle +0.679609|-\rangle$
 proton mixing angle: $\alpha_p=2.394356$ rad
 incoming proton: $-0.73357|+\rangle +0.679614|-\rangle$

$s=24.9854$, $\sqrt{s}=4.99854$, $|\vec{P}|=2.41126$, $|\vec{P}'|=0.0171201$
 $\theta_{P'}=3.14159\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.27545$, $\phi_\gamma=2.74997$
 $E_\gamma=2.03875$, $\theta_{\ell'}=2.66143\text{e-}07$, $\phi_{\ell'}=3.13386$
 $Q^2=19.4987$, $x_B=0.833497$, $t=-3.01229$, $W^2=4.77498$, $y=0.970476$

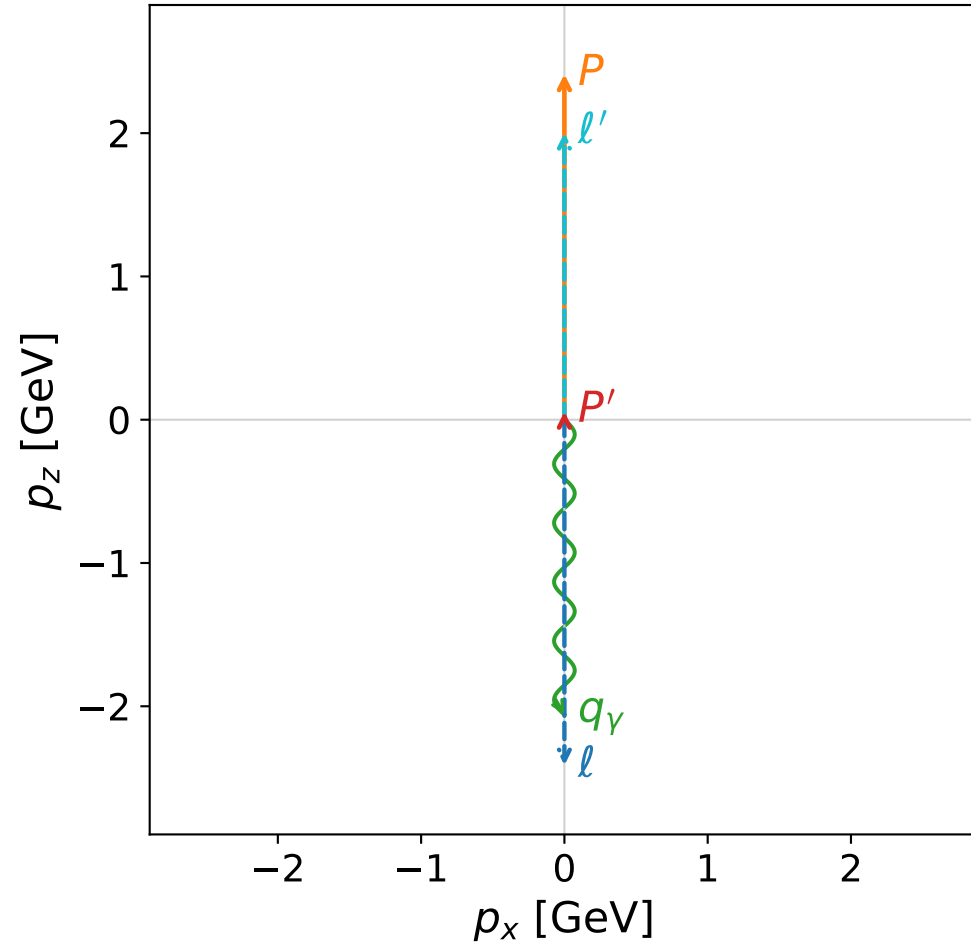
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5873$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 2.0216$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0216$
 P' $E= 0.9382$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0171$
 q_γ $E= 2.0388$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0388$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.149\text{e+}21$; $I_h=+2.564\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.776\text{e-}16$; $I_{TL}=+1.797\text{e-}09$

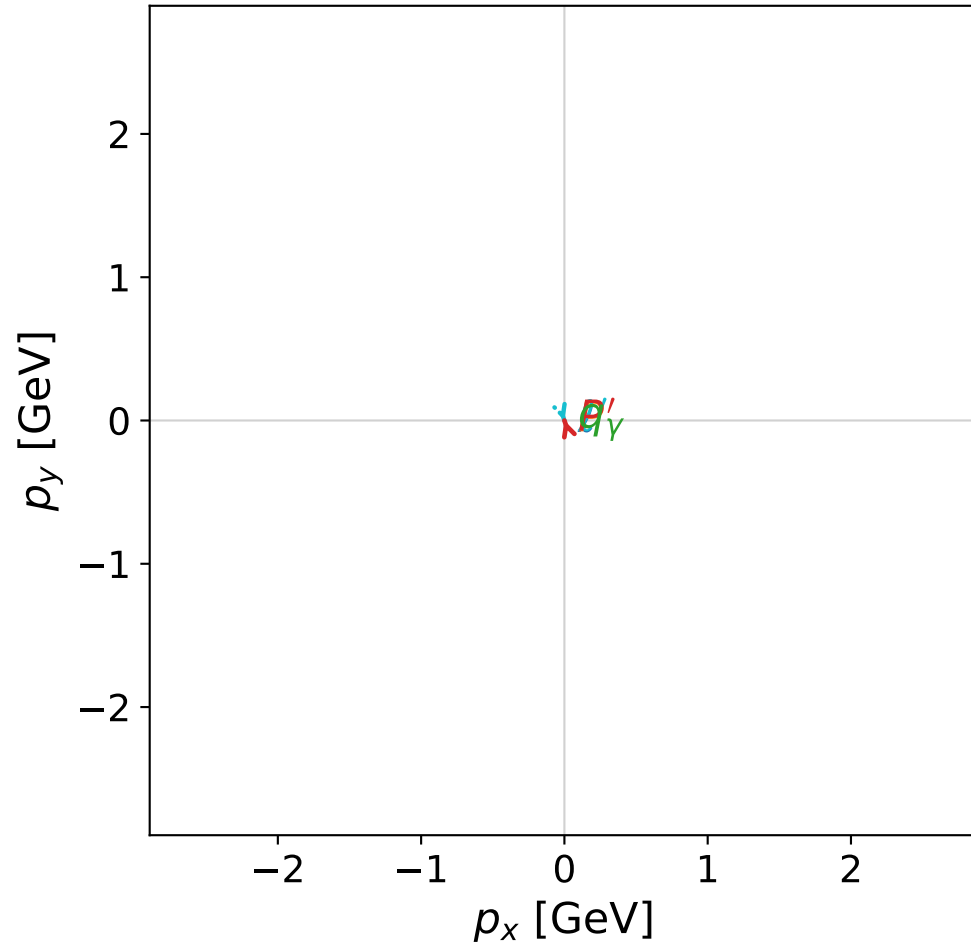


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

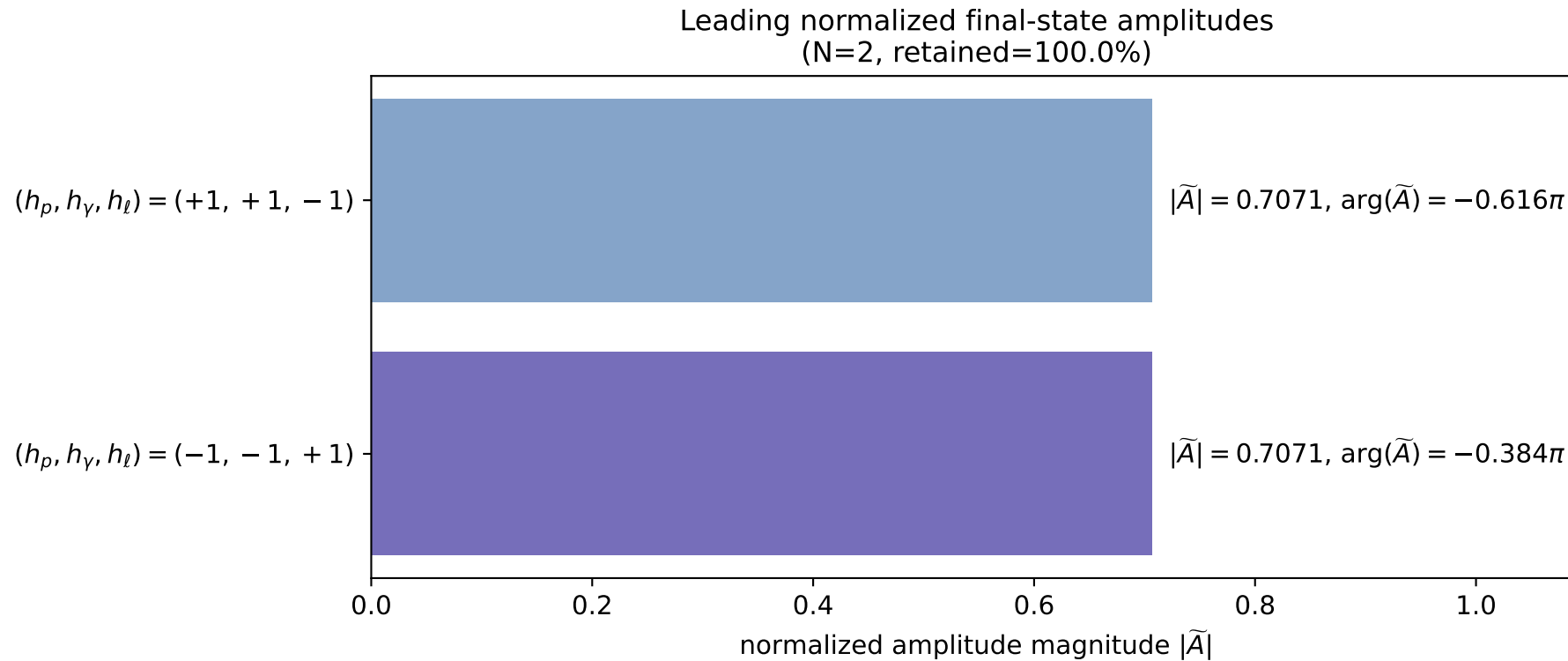


dghz_mixing_angles_polarization_02_local_minimum_49
 $d_{\text{GHZ}}=2.67844\text{e-}08$
 region: polarization_cluster_02_local_minimum_49
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0049
 lepton mixing angle: $\alpha_e=0.81607917$ rad
 incoming lepton: $0.685083|+\rangle + 0.728465|-\rangle$
 proton mixing angle: $\alpha_p=2.3255025$ rad
 incoming proton: $-0.685075|+\rangle + 0.728473|-\rangle$

$s=24.9993$, $\sqrt{s}=4.99993$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.0563059$
 $\theta_{P'}=2.80434\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.88915$, $\phi_\gamma=5.07566$
 $E_\gamma=2.05828$, $\theta_{\ell'}=7.88636\text{e-}07$, $\phi_{\ell'}=5.03075$
 $Q^2=19.3148$, $x_B=0.824895$, $t=-2.83243$, $W^2=4.9799$, $y=0.970788$

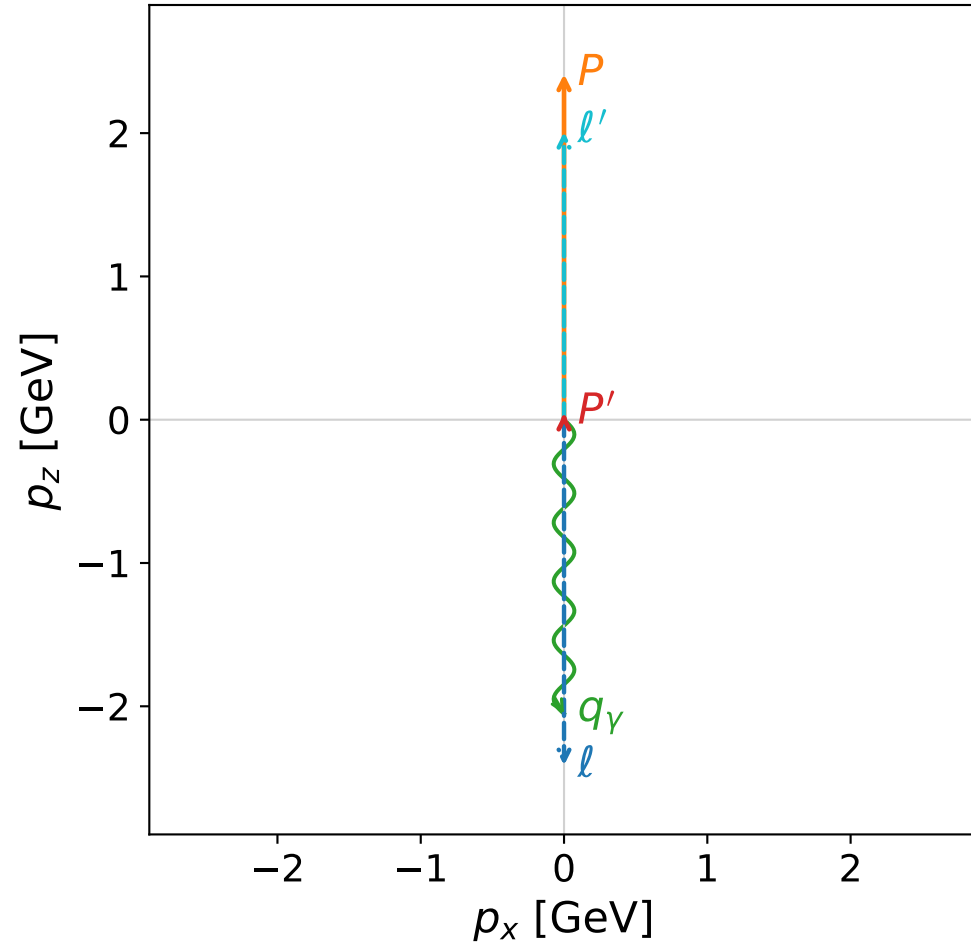
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0020$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0020$
 P' $E= 0.9397$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0563$
 q_γ $E= 2.0583$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0583$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.344\text{e+}20$; $I_h=+1.888\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.675\text{e-}15$; $I_{TL}=+1.404\text{e-}09$

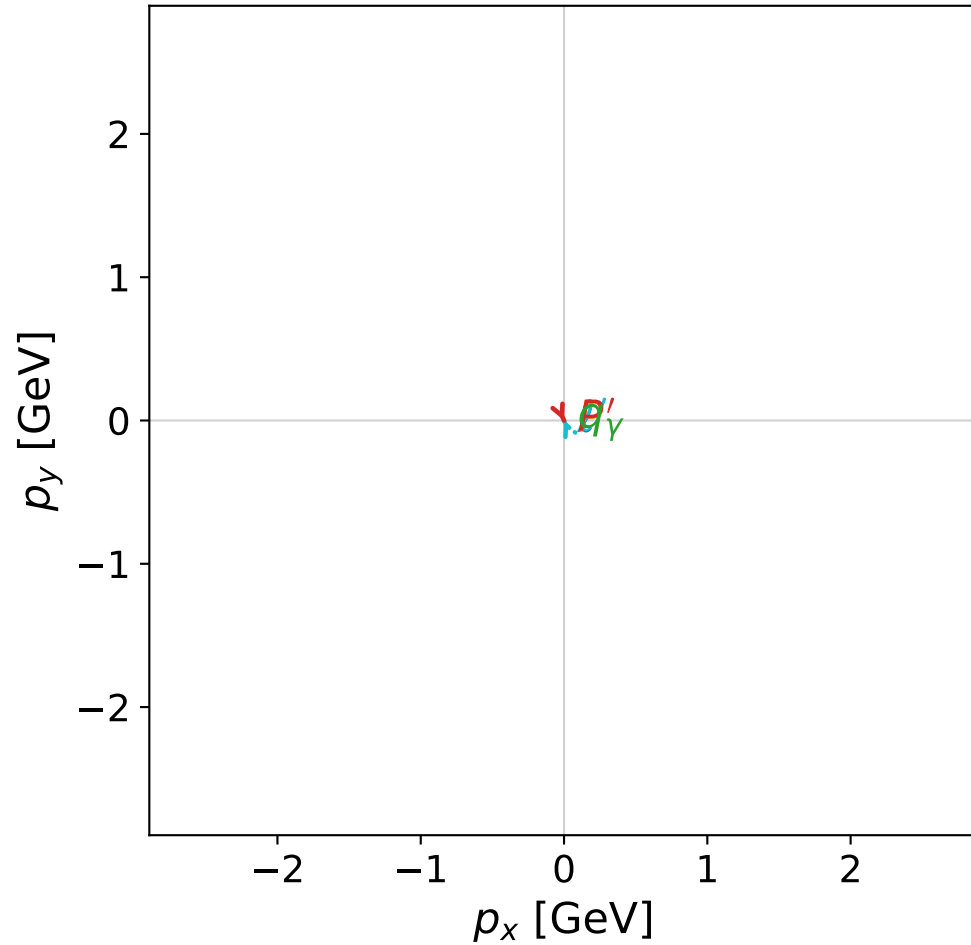


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

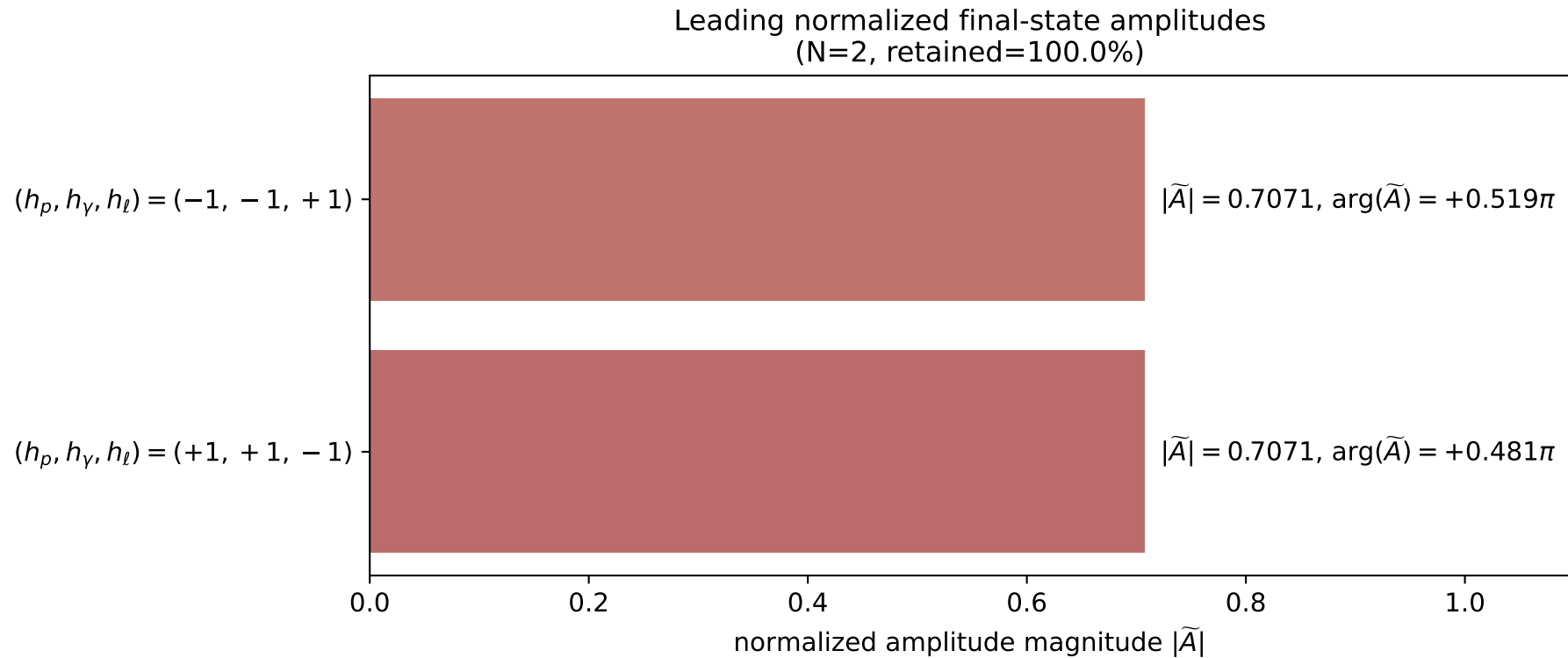


dghz_mixing_angles_polarization_02_local_minimum_54
 $d_{\text{GHZ}}=2.70595\text{e-}08$
 region: polarization_cluster_02_local_minimum_54
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0054
 lepton mixing angle: $\alpha_e=0.85741662$ rad
 incoming lepton: $0.654393|+\rangle + 0.756155|-\rangle$
 proton mixing angle: $\alpha_p=2.2841843$ rad
 incoming proton: $-0.654399|+\rangle + 0.756149|-\rangle$

$s=24.9965$, $\sqrt{s}=4.99965$, $|\vec{P}|=2.41184$, $|\vec{P}'|=0.0441066$
 $\theta_{P'}=0.00122325$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.12862$, $\phi_\gamma=1.63085$
 $E_\gamma=2.05236$, $\theta_\ell=2.68658\text{e-}05$, $\phi_\ell=1.98702$
 $Q^2=19.3743$, $x_B=0.827614$, $t=-2.88766$, $W^2=4.91537$, $y=0.970691$

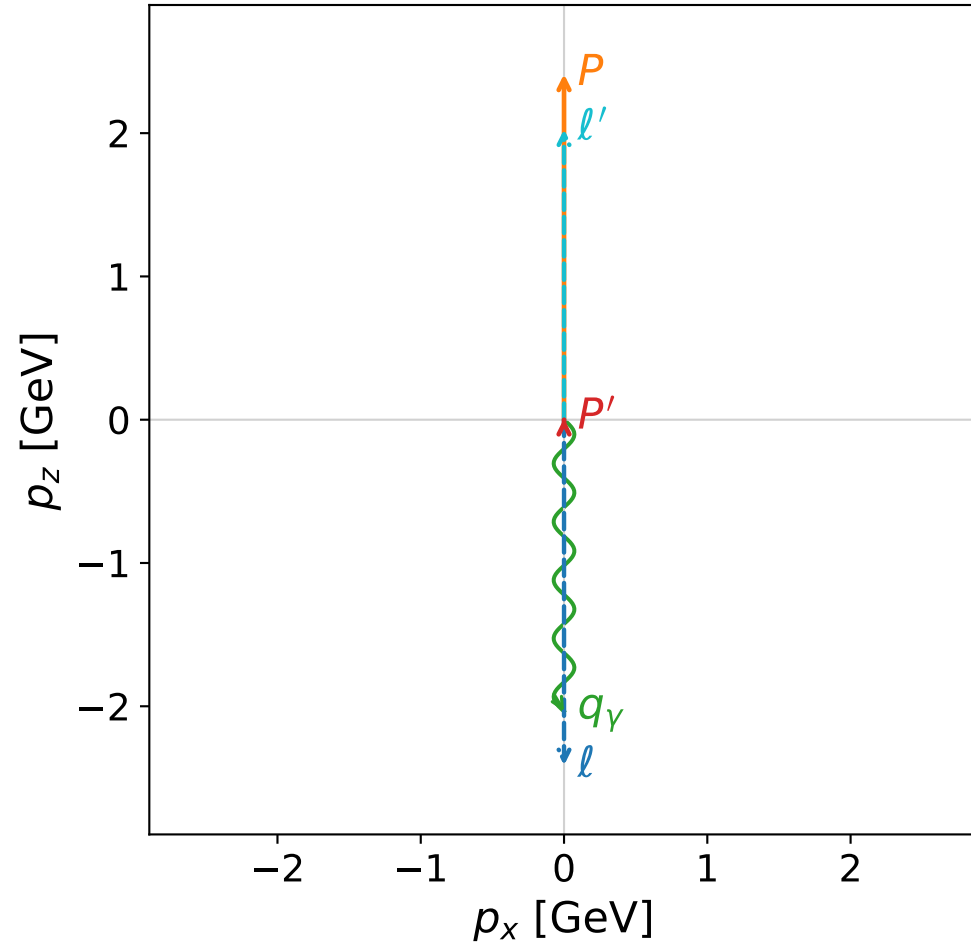
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 ℓ' $E= 2.0083$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0083$
 P' $E= 0.9390$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0441$
 q_γ $E= 2.0524$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0524$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.143\text{e+}17$; $I_h=-1.961\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.067\text{e-}12$; $I_{TL}=-1.433\text{e-}07$

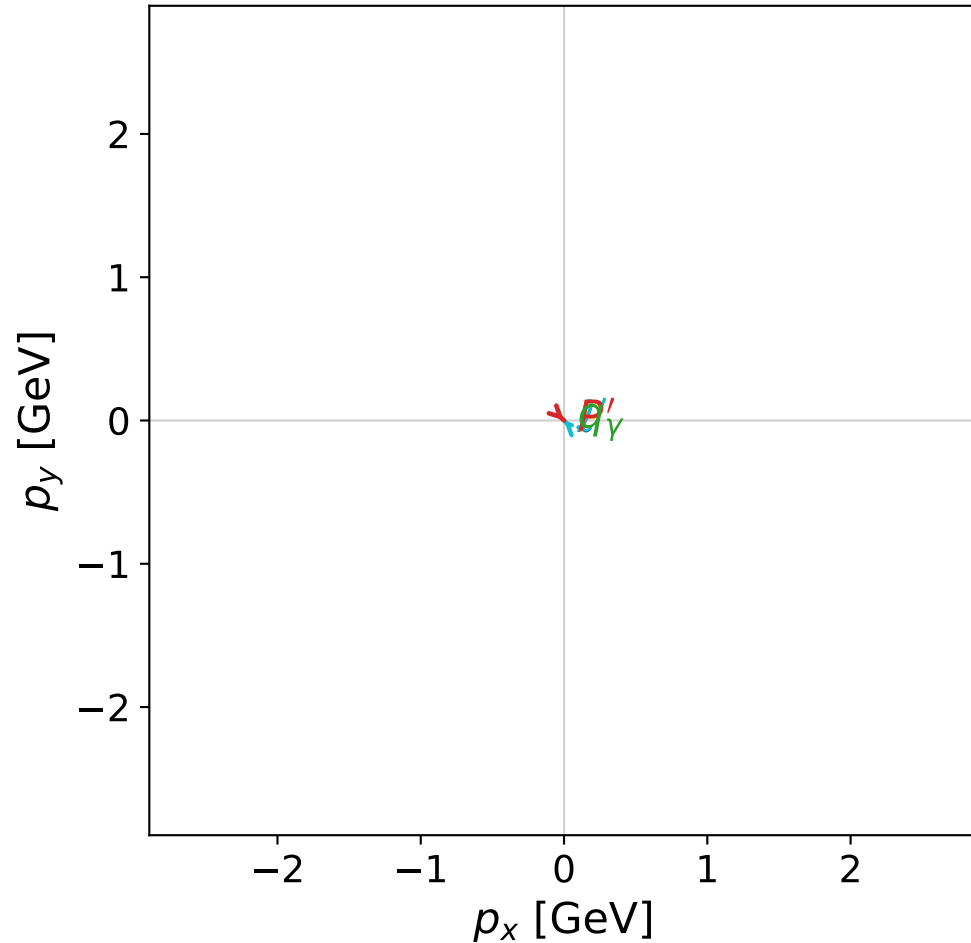


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

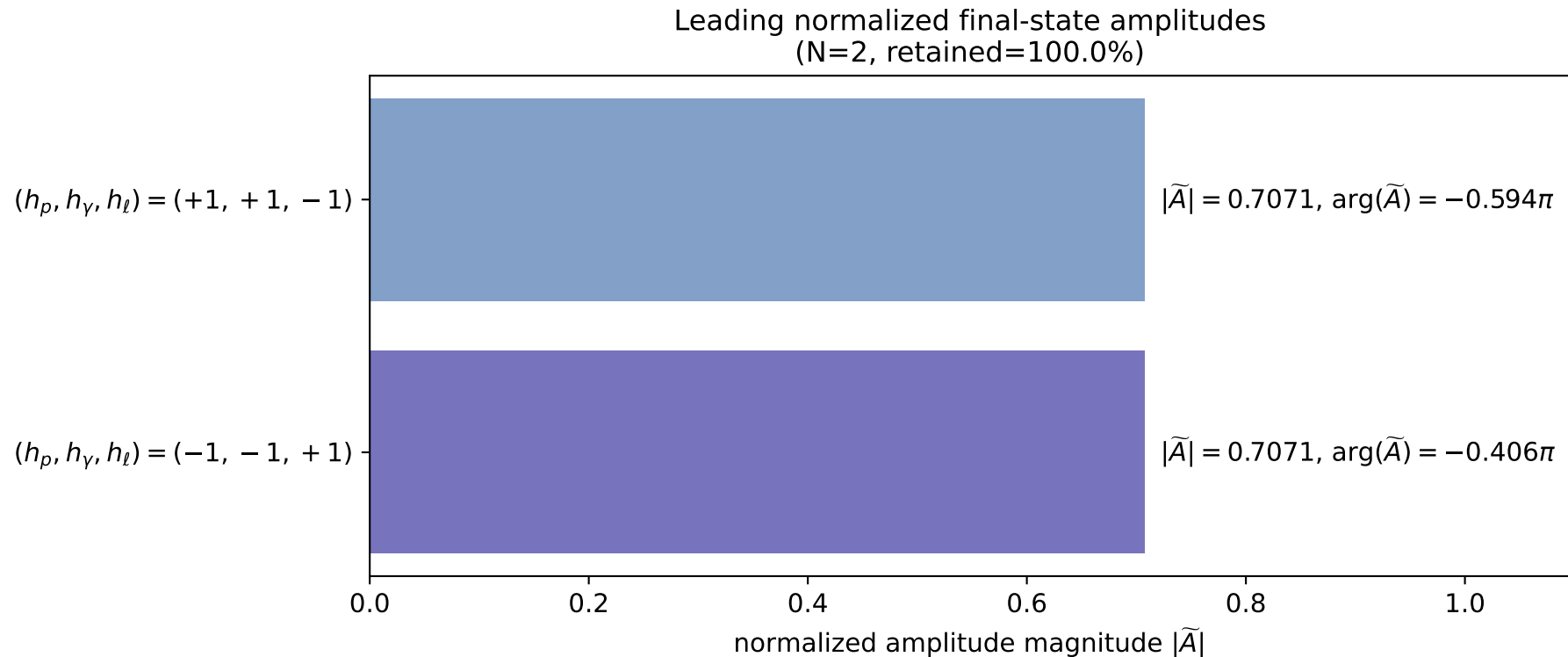


dghz_mixing_angles_polarization_02_local_minimum_62
 $d_{\text{GHZ}}=2.75303\text{e-}08$
 region: polarization_cluster_02_local_minimum_62
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0062
 lepton mixing angle: $\alpha_e=0.65340217$ rad
 incoming lepton: $0.79402|+\rangle +0.607891|-\rangle$
 proton mixing angle: $\alpha_p=2.4881891$ rad
 incoming proton: $-0.794019|+\rangle +0.607892|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00901078$
 $\theta_{P'}=0.000613592$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.51618$, $\phi_\gamma=5.00919$
 $E_\gamma=2.03548$, $\theta_\ell=2.72834\text{e-}06$, $\phi_\ell=2.37459$
 $Q^2=19.5515$, $x_B=0.835287$, $t=-3.05213$, $W^2=4.73527$, $y=0.970432$

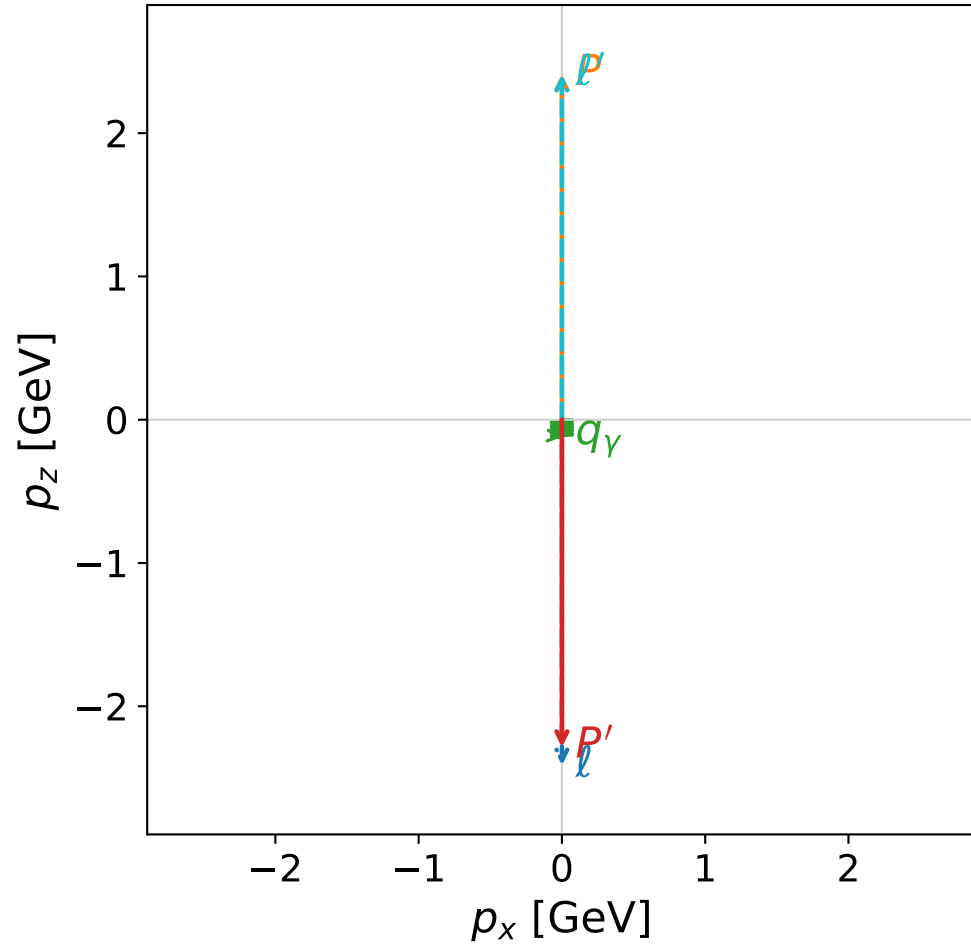
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0265$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0265$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0090$
 q_γ $E= 2.0355$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0355$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.036\text{e+}19$; $I_h=+6.663\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.049\text{e-}14$; $I_{TL}=+4.640\text{e-}08$

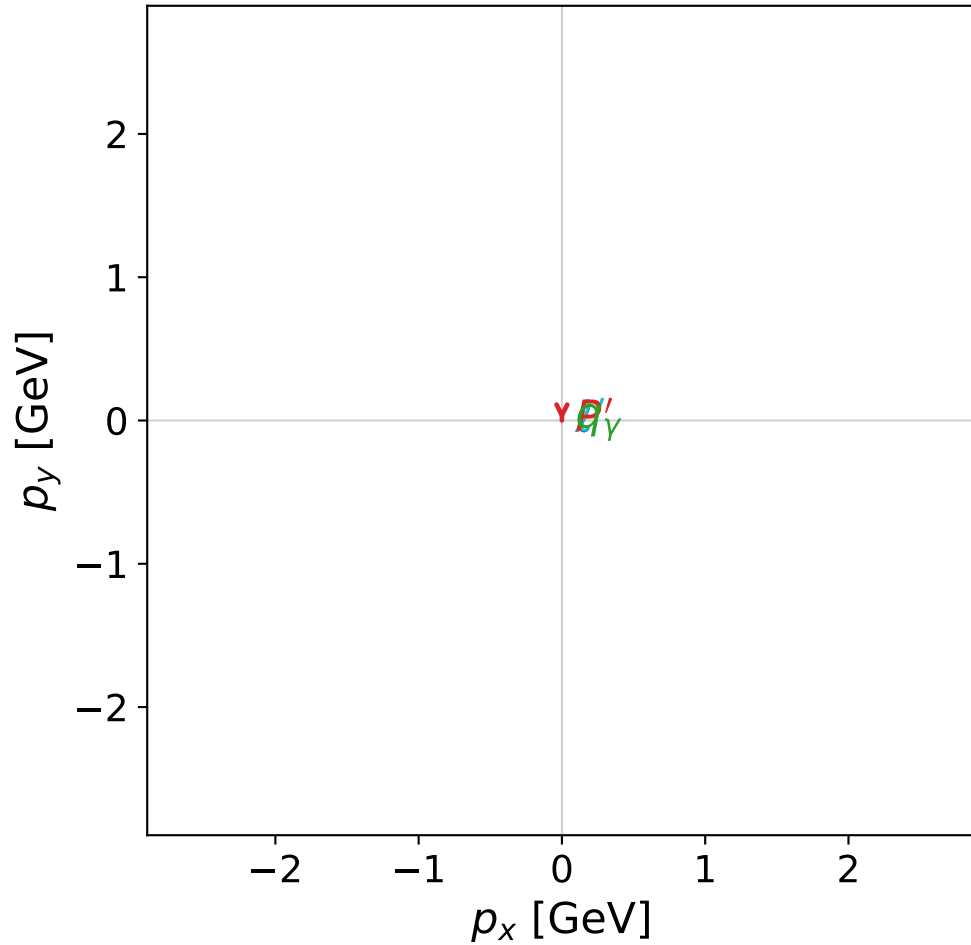


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

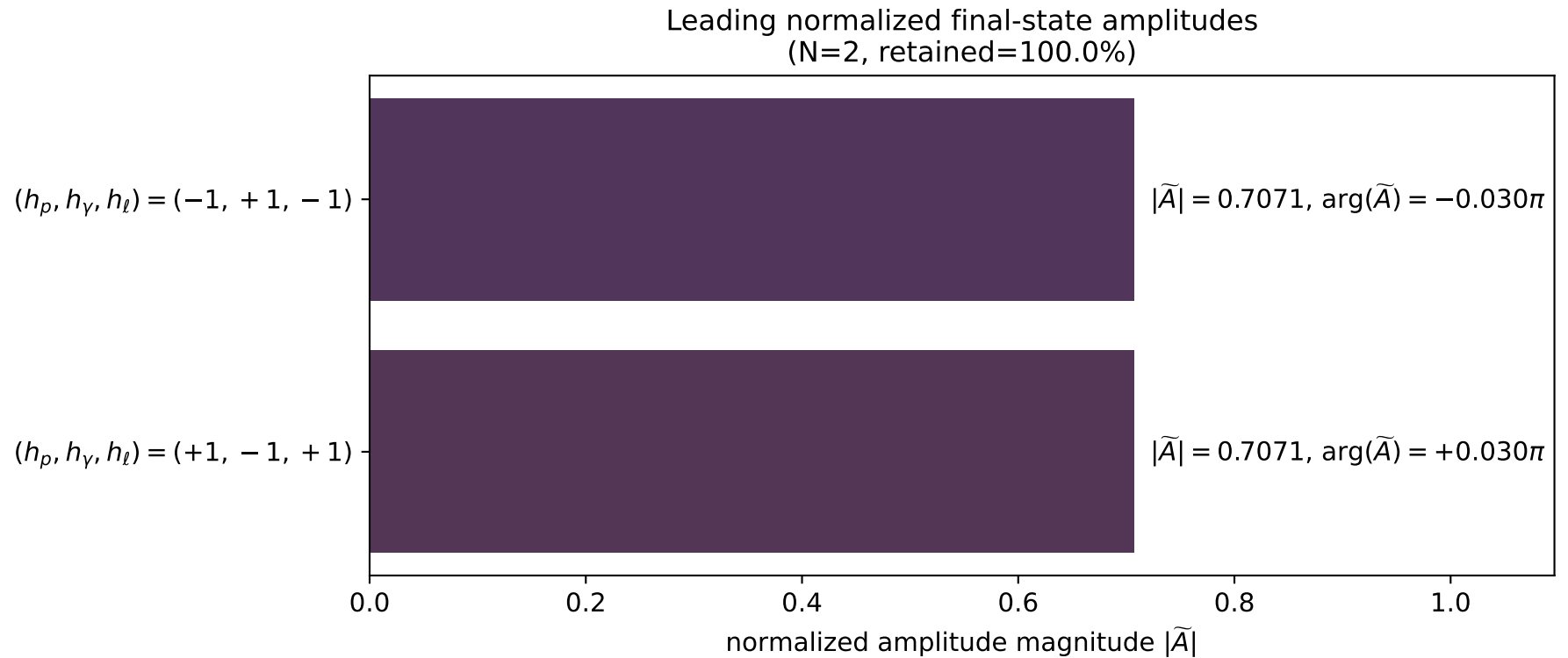


dghz_mixing_angles_polarization_02_local_minimum_77
 $d_{\{\rm GHZ\}}=2.96256\text{e-}08$
 region: polarization_cluster_02_local_minimum_77
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0077
 lepton mixing angle: $\alpha_e=1.1396606$ rad
 incoming lepton: $0.417903|+\rangle +0.908492|-\rangle$
 proton mixing angle: $\alpha_p=2.0019206$ rad
 incoming proton: $-0.417892|+\rangle +0.908497|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28693$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=4.37912$, $\phi_\gamma=0.0937313$
 $E_\gamma=0.120627$, $\theta_{l'}=0$, $\phi_{l'}=1.28657$
 $Q^2=23.2282$, $x_B=0.998083$, $t=-22.0666$, $W^2=0.924449$, $y=0.964871$

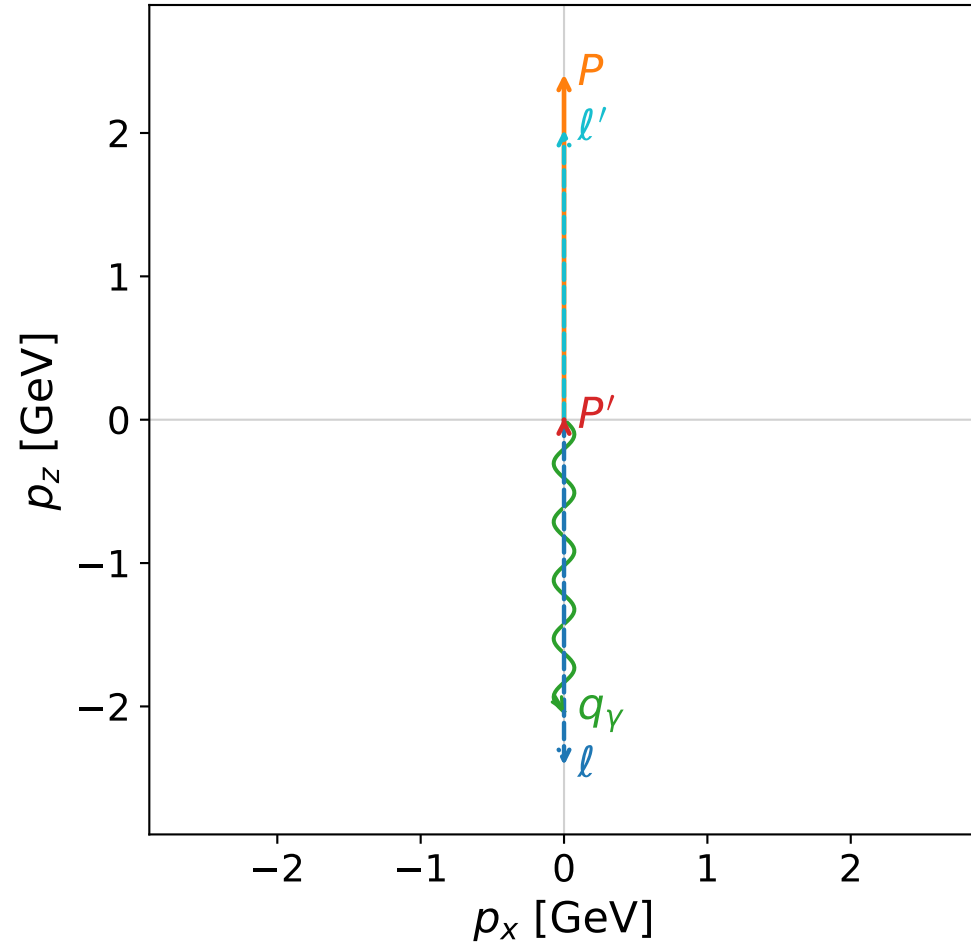
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.4076$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4076$
 P' $E= 2.4718$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.2869$
 q_γ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1206$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.579\text{e}+21$; $I_h=+1.874\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{L/T}=2.723\text{e-}25$; $I_{TL}=-1.080\text{e-}16$

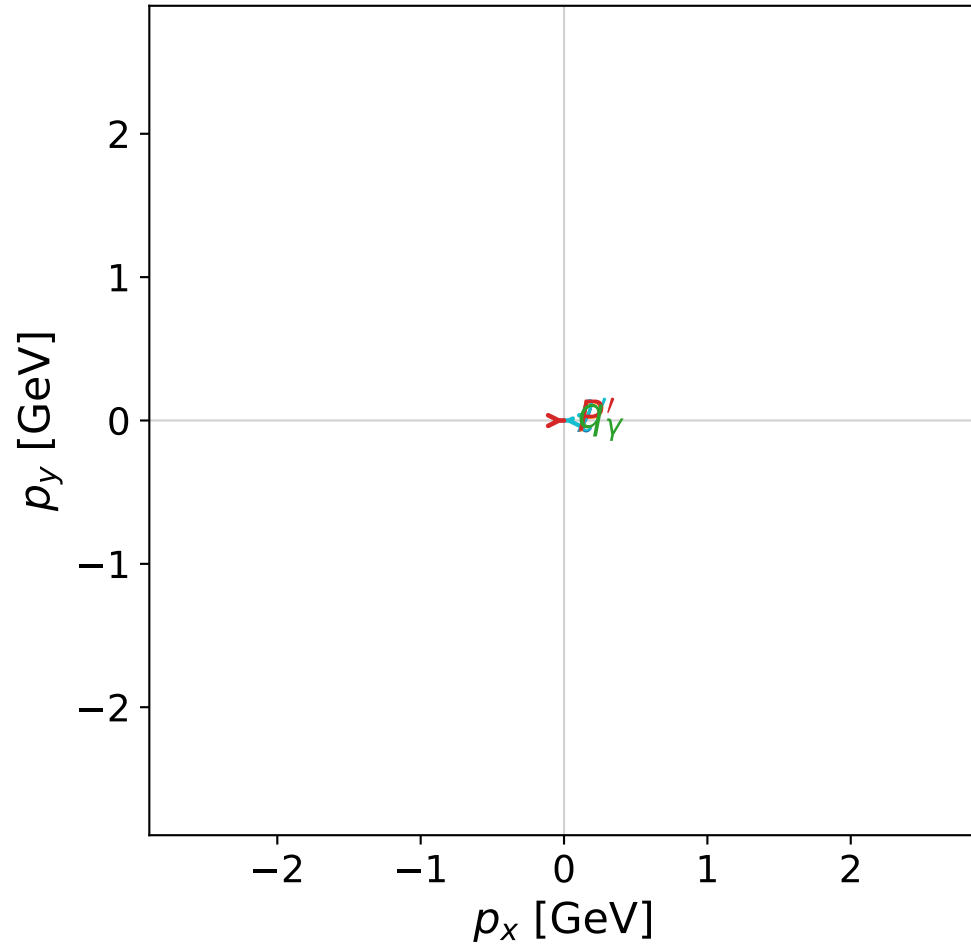


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

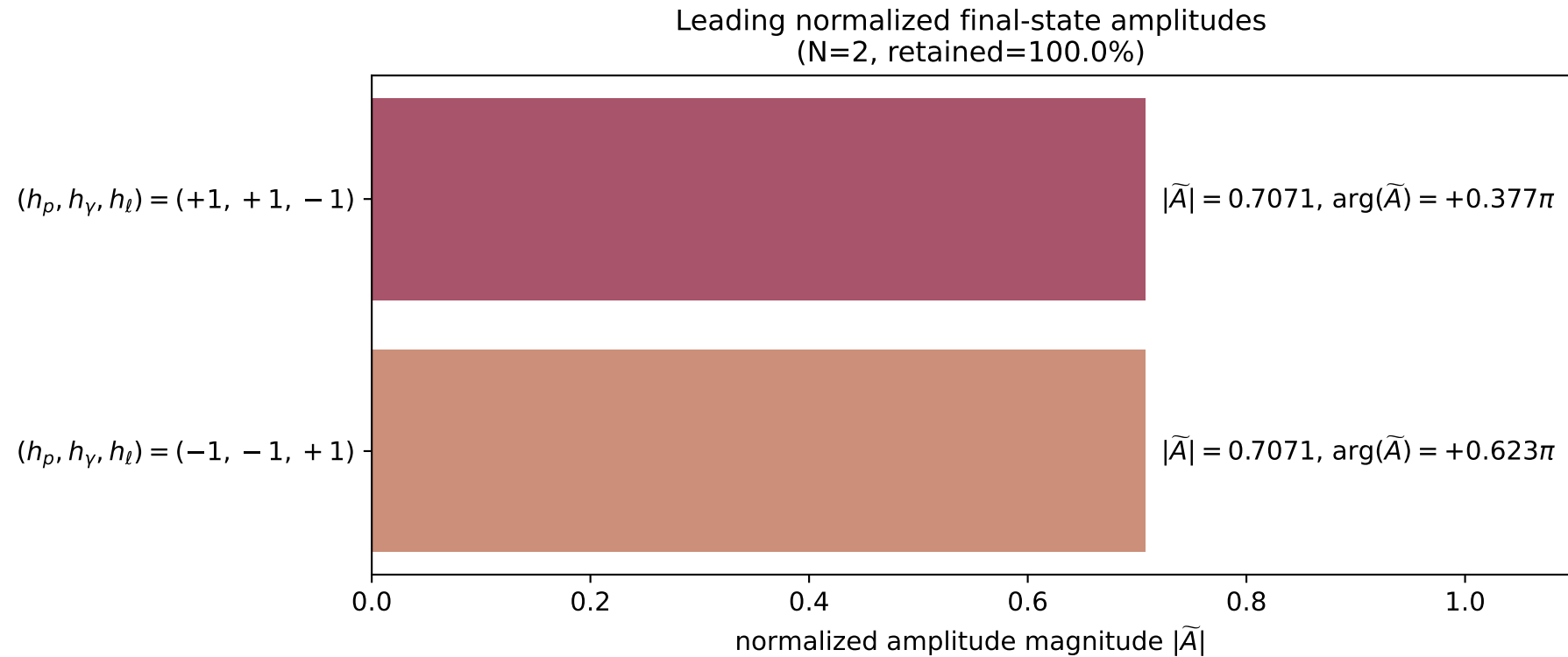


dghz_mixing_angles_polarization_02_local_minimum_78
 $d_{\text{GHZ}}=2.97007\text{e-}08$
 region: polarization_cluster_02_local_minimum_78
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0078
 lepton mixing angle: $\alpha_e=0.55713074$ rad
 incoming lepton: $0.848776|+\rangle + 0.528753|-\rangle$
 proton mixing angle: $\alpha_p=2.5844424$ rad
 incoming proton: $-0.848765|+\rangle + 0.52877|-\rangle$

$s=24.9739$, $\sqrt{s}=4.99739$, $|\vec{P}|=2.41067$, $|\vec{P}'|=0.0137879$
 $\theta_{P'}=0.000837323$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.00351513$, $\phi_\gamma=1.9584$
 $E_\gamma=2.03654$, $\theta_{\ell'}=5.70749\text{e-}06$, $\phi_{\ell'}=3.14511$
 $Q^2=19.5047$, $x_B=0.834182$, $t=-3.02706$, $W^2=4.75696$, $y=0.970439$

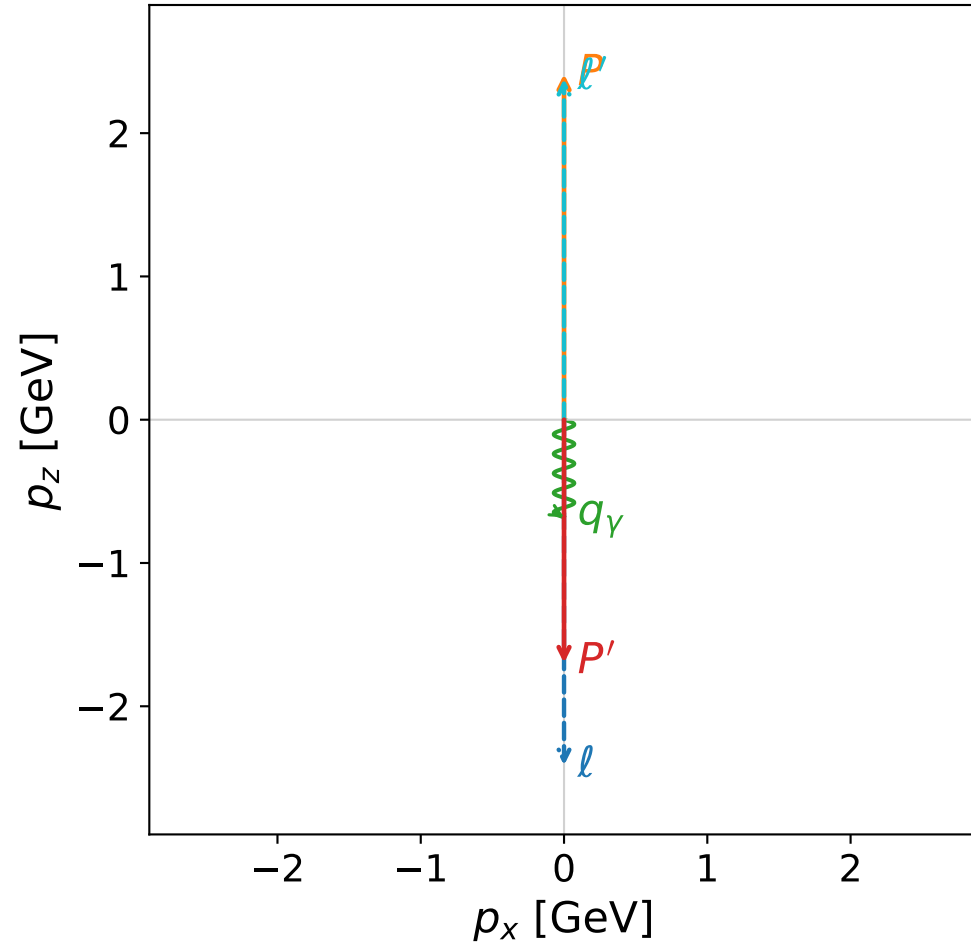
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4107$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4107$
 P $E= 2.5867$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4107$
 ℓ' $E= 2.0228$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0228$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0138$
 q_γ $E= 2.0365$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0365$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.960\text{e+}18$; $I_h=+3.517\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.565\text{e-}13$; $I_{TL}=+2.466\text{e-}07$

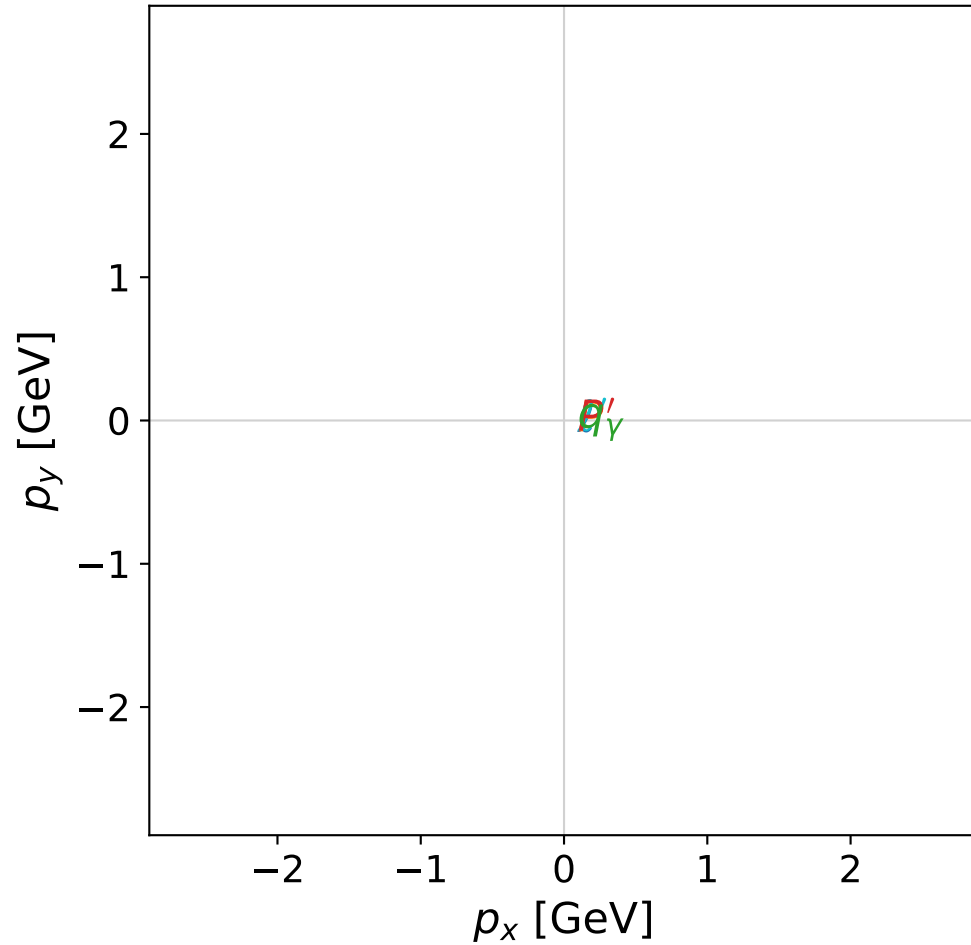


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

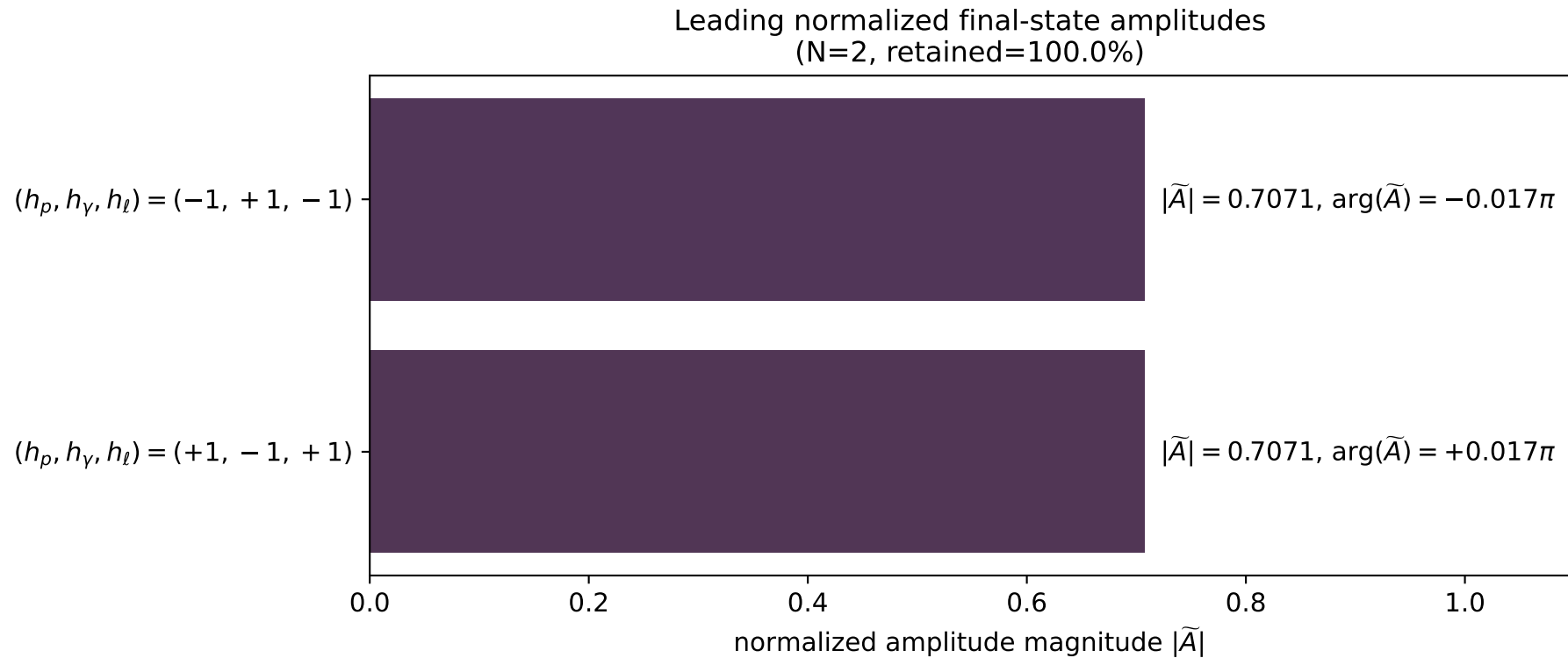


dghz_mixing_angles_polarization_02_local_minimum_86
 $d_{\text{GHZ}}=3.12213\text{e-}08$
 region: polarization_cluster_02_local_minimum_86
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0086
 lepton mixing angle: $\alpha_e=1.1624483$ rad
 incoming lepton: $0.397094|+\rangle +0.917778|-\rangle$
 proton mixing angle: $\alpha_p=1.9791355$ rad
 incoming proton: $-0.397086|+\rangle +0.917782|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.69806$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.694775$, $\phi_\gamma=0.0541782$
 $E_\gamma=0.681015$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.65694$
 $Q^2=22.9535$, $x_B=0.985852$, $t=-16.4727$, $W^2=1.20925$, $y=0.965287$

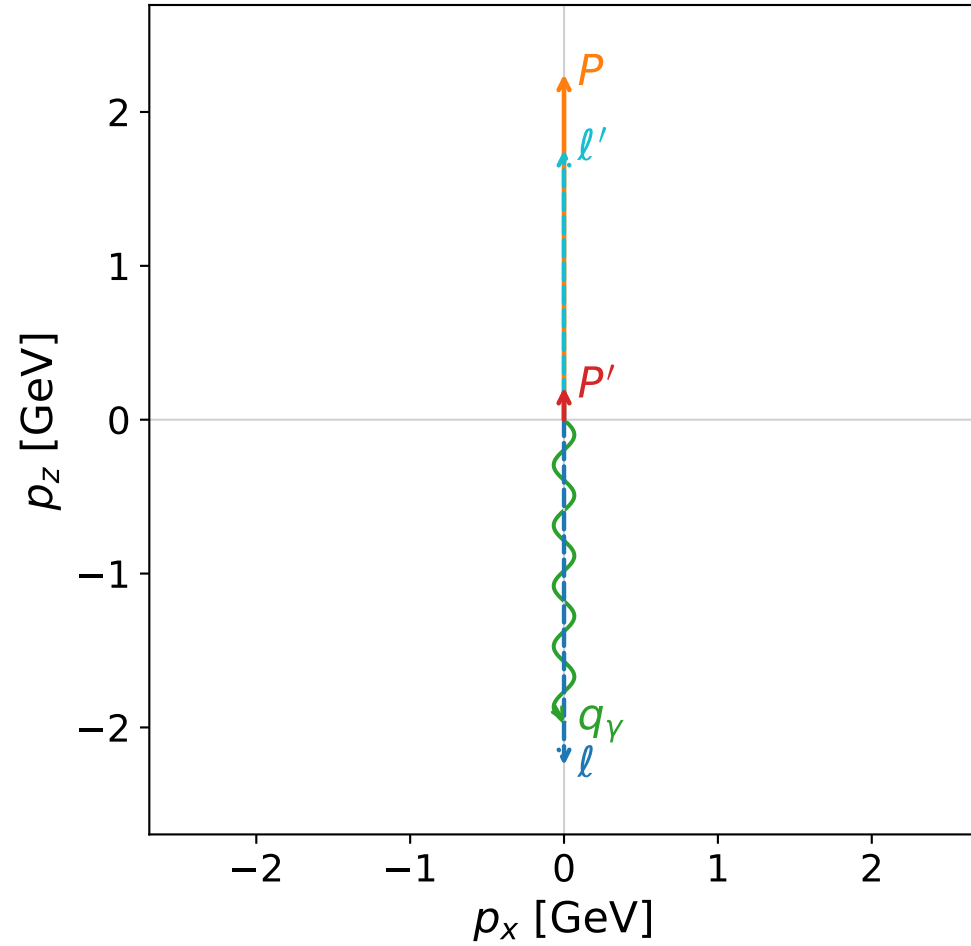
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3791$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.3791$
 P' $E= 1.9399$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6981$
 q_γ $E= 0.6810$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6810$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.663\text{e+}22$; $I_h=+2.756\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.482\text{e-}27$; $I_{TL}=-1.343\text{e-}17$

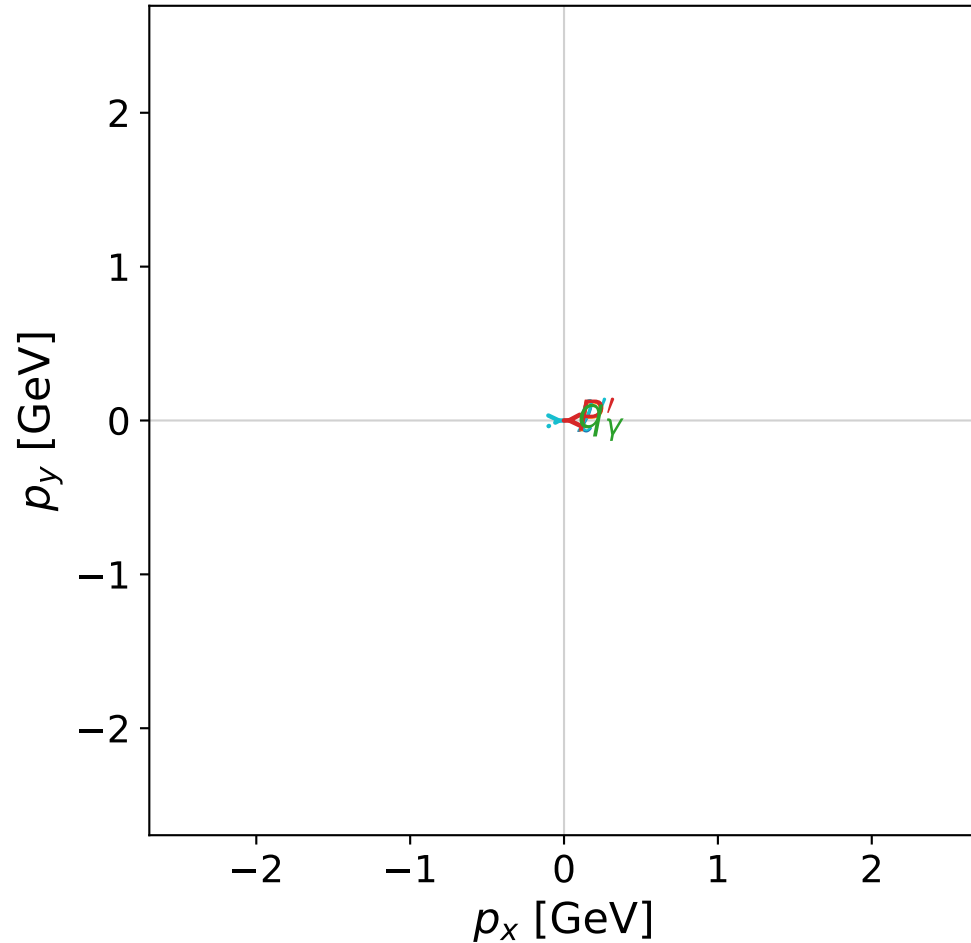


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

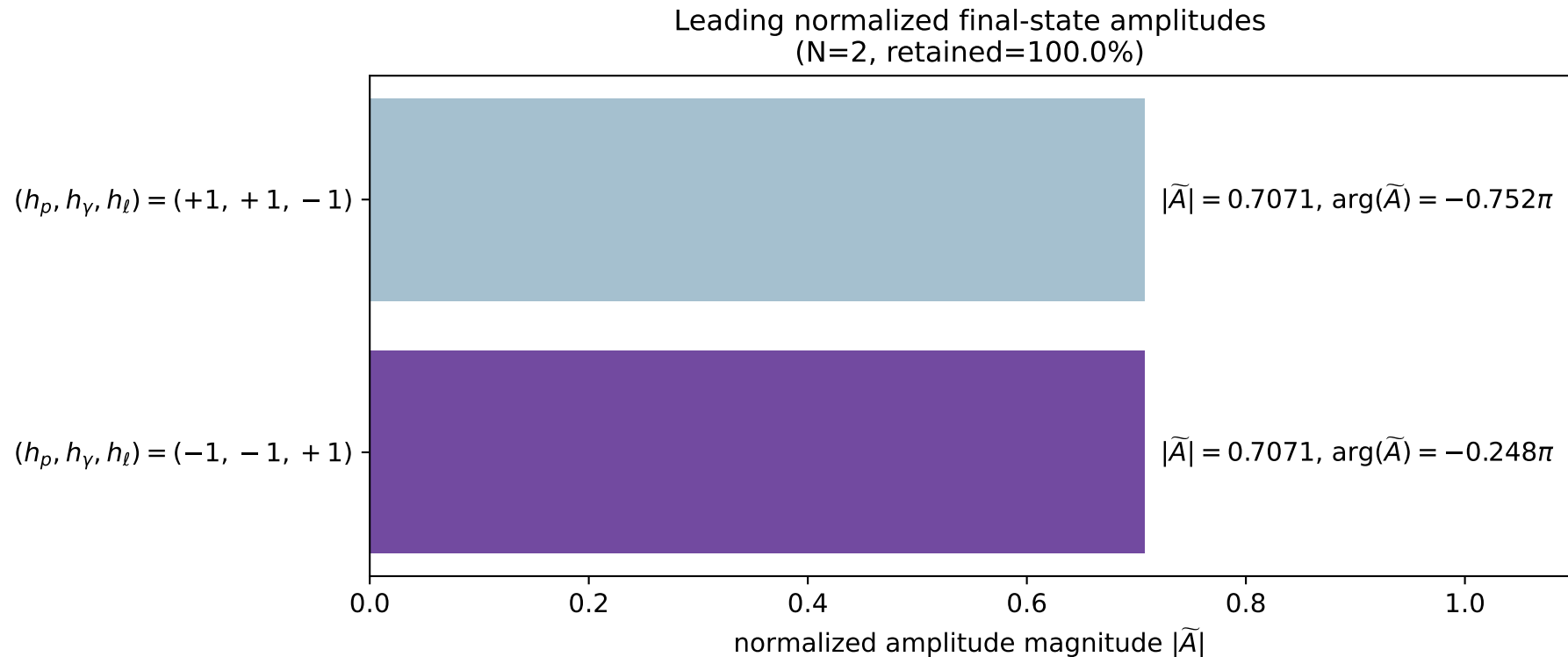


dghz_mixing_angles_polarization_02_local_minimum_90
 $d_{\text{GHZ}}=3.23249\text{e-}08$
 region: polarization_cluster_02_local_minimum_90
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0090
 lepton mixing angle: $\alpha_e=0.78715457$ rad
 incoming lepton: $0.705864|+\rangle +0.708348|-\rangle$
 proton mixing angle: $\alpha_p=2.3544342$ rad
 incoming proton: $-0.705861|+\rangle +0.70835|-\rangle$

$s=21.9038$, $\sqrt{s}=4.68015$, $|\vec{P}|=2.24608$, $|\vec{P}'|=0.210445$
 $\theta_{P'}=6.7809\text{e-}06$, $\theta_Y=3.14159$, $\phi_{P'}=3.16268$, $\phi_Y=5.50499$
 $E_Y=1.96464$, $\theta_{\ell'}=8.13445\text{e-}07$, $\phi_{\ell'}=0.0210885$
 $Q^2=15.7602$, $x_B=0.773911$, $t=-1.97479$, $W^2=5.48402$, $y=0.968628$

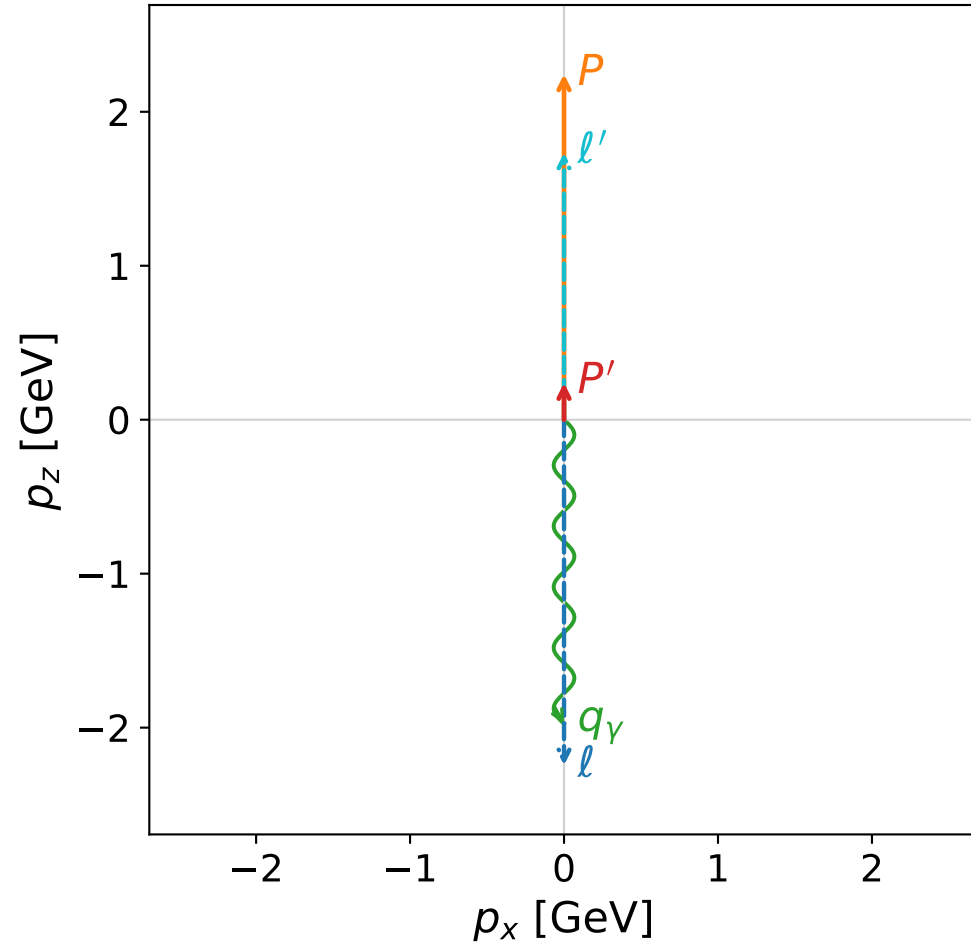
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2461$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2461$
 P $E= 2.4341$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2461$
 ℓ' $E= 1.7542$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7542$
 P' $E= 0.9613$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.2104$
 q_Y $E= 1.9646$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9646$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.092\text{e+}20$; $I_h=+3.728\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.024\text{e-}15$; $I_{TL}=+3.525\text{e-}10$

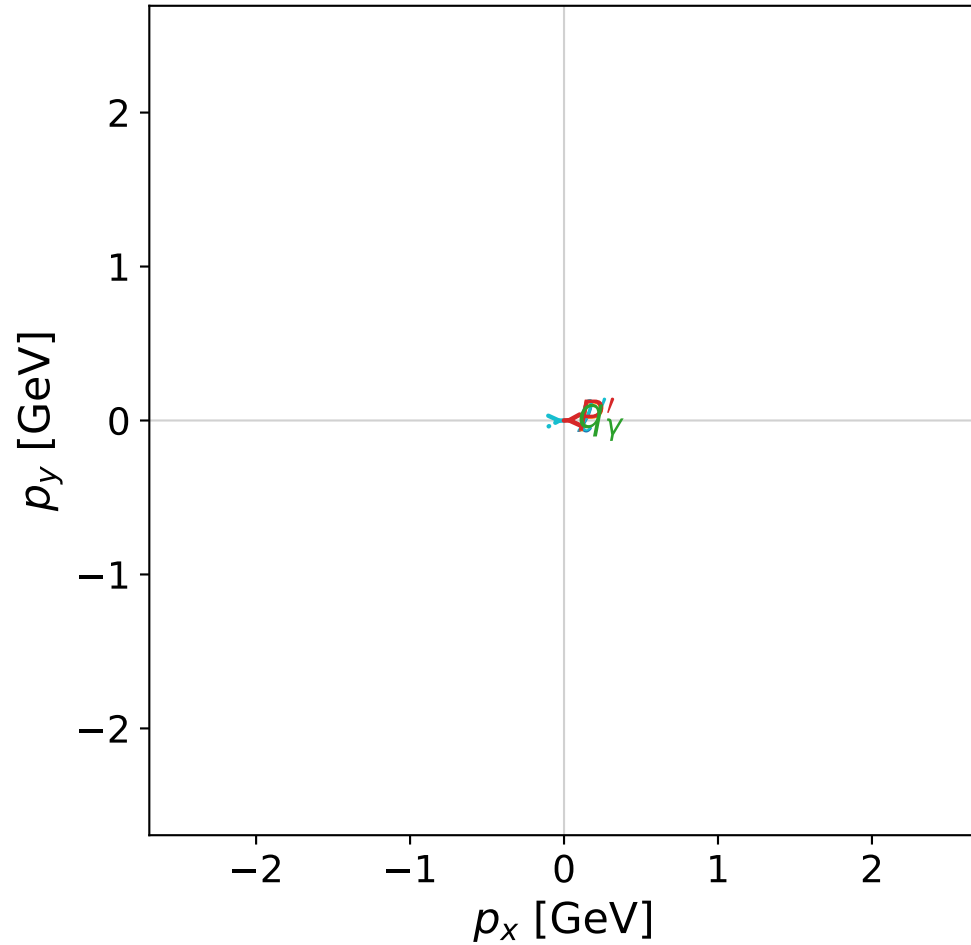


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

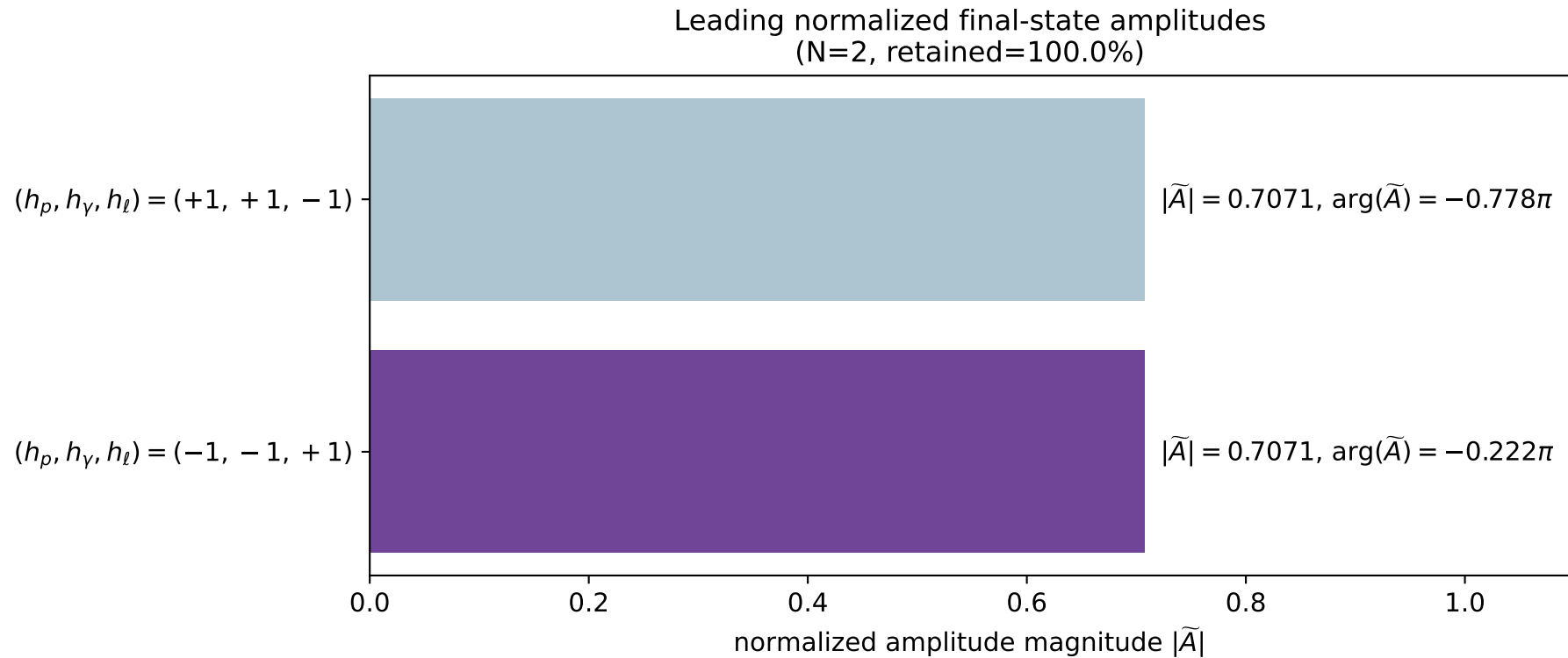


dghz_mixing_angles_polarization_02_local_minimum_94
 $d_{\text{GHZ}}=3.33046\text{e-}08$
 region: polarization_cluster_02_local_minimum_94
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0094
 lepton mixing angle: $\alpha_e=0.79992921$ rad
 incoming lepton: $0.696757|+\rangle +0.717307|-\rangle$
 proton mixing angle: $\alpha_p=2.3416514$ rad
 incoming proton: $-0.696749|+\rangle +0.717315|-\rangle$

$s=21.8786$, $\sqrt{s}=4.67746$, $|\vec{P}|=2.24468$, $|\vec{P}'|=0.239827$
 $\theta_{P'}=0.000137432$, $\theta_Y=3.14159$, $\phi_{P'}=3.17695$, $\phi_Y=5.58436$
 $E_Y=1.97455$, $\theta_{\ell'}=1.9\text{e-}05$, $\phi_{\ell'}=0.0353563$
 $Q^2=15.5756$, $x_B=0.765531$, $t=-1.87435$, $W^2=5.65037$, $y=0.968921$

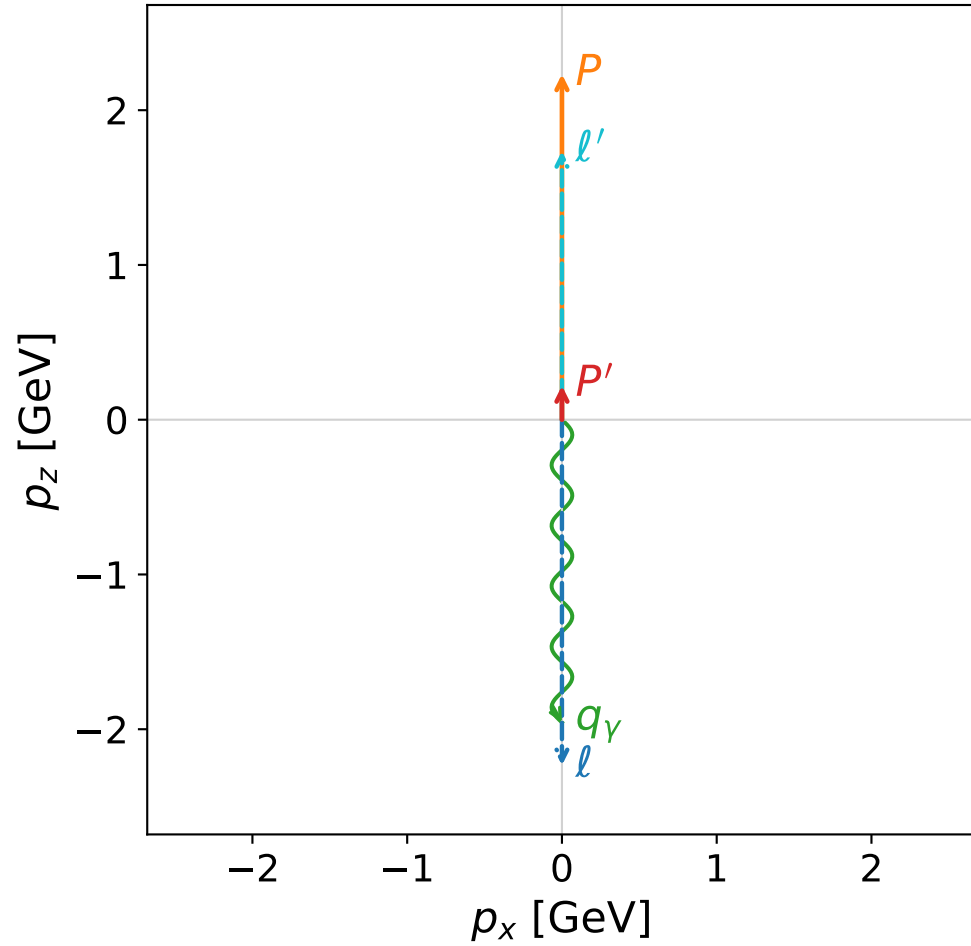
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2447$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2447$
 P $E= 2.4328$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2447$
 ℓ' $E= 1.7347$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7347$
 P' $E= 0.9682$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.2398$
 q_Y $E= 1.9746$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9746$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.005\text{e+}17$; $I_h=+7.151\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.917\text{e-}12$; $I_{TL}=+7.038\text{e-}08$

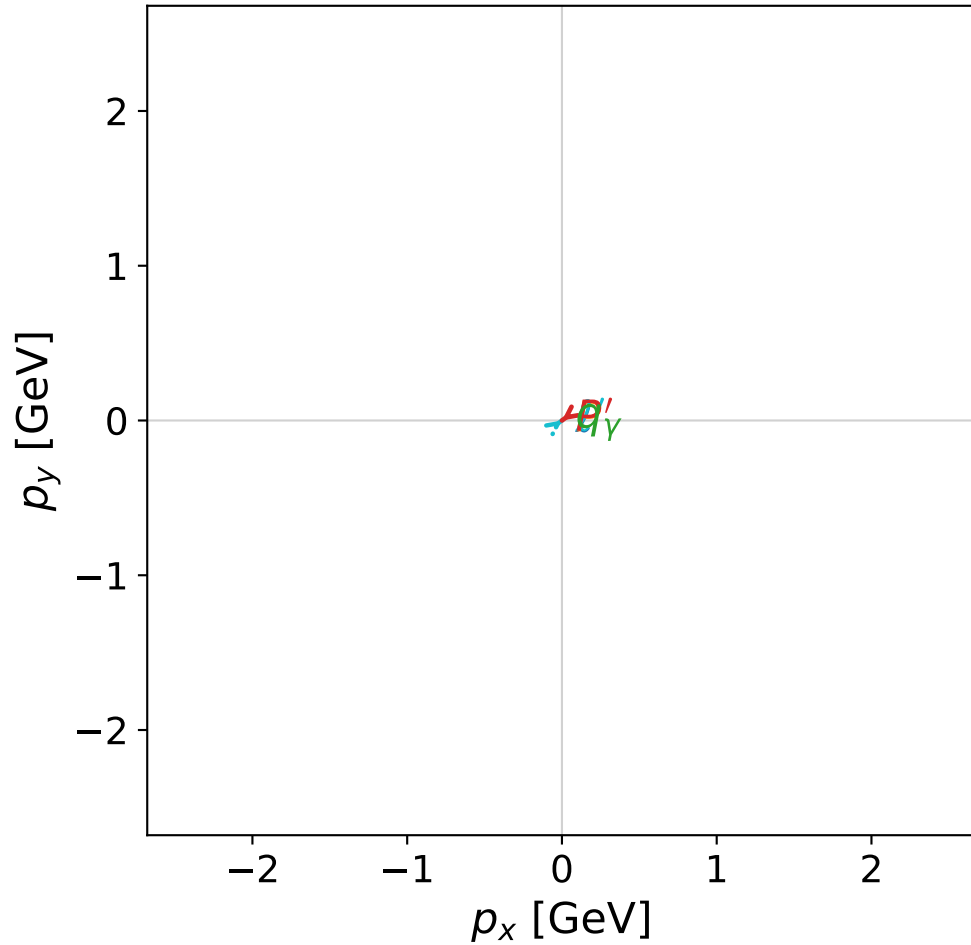


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



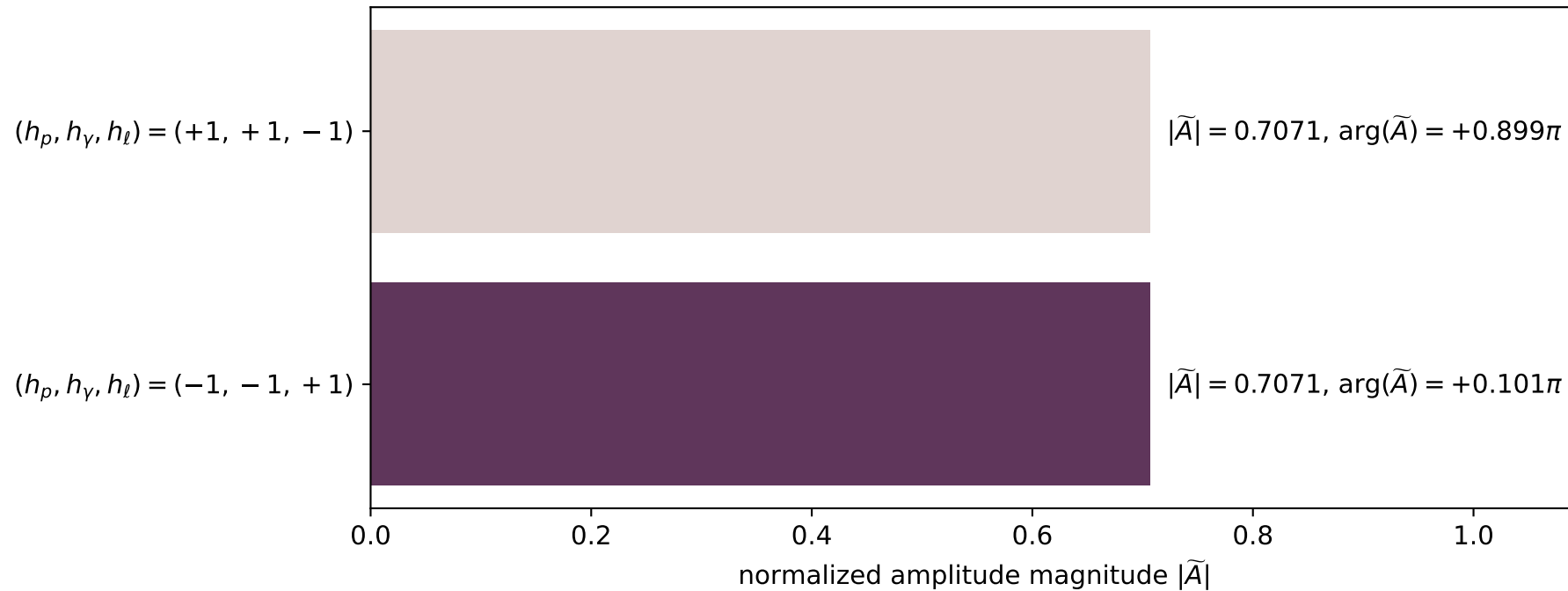
dghz_mixing_angles_polarization_02_local_minimum_96
 $d_{\text{GHZ}}=3.42638\text{e-}08$
 region: polarization_cluster_02_local_minimum_96
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0096
 lepton mixing angle: $\alpha_e=0.66918829$ rad
 incoming lepton: $0.784325|+\rangle +0.62035|-\rangle$
 proton mixing angle: $\alpha_p=2.4723833$ rad
 incoming proton: $-0.784312|+\rangle +0.620366|-\rangle$

$s=21.6714$, $\sqrt{s}=4.65526$, $|\vec{P}|=2.23313$, $|\vec{P}'|=0.219926$
 $\theta_{P'}=3.14159\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.77898$, $\phi_\gamma=0.317472$
 $E_\gamma=1.95587$, $\theta_{l'}=3.98006\text{e-}06$, $\phi_{l'}=0.637385$
 $Q^2=15.5064$, $x_B=0.770105$, $t=-1.9252$, $W^2=5.50887$, $y=0.96844$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2331$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2331$
 P $E= 2.4221$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2331$
 l' $E= 1.7359$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7359$
 P' $E= 0.9634$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.2199$
 q_γ $E= 1.9559$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9559$

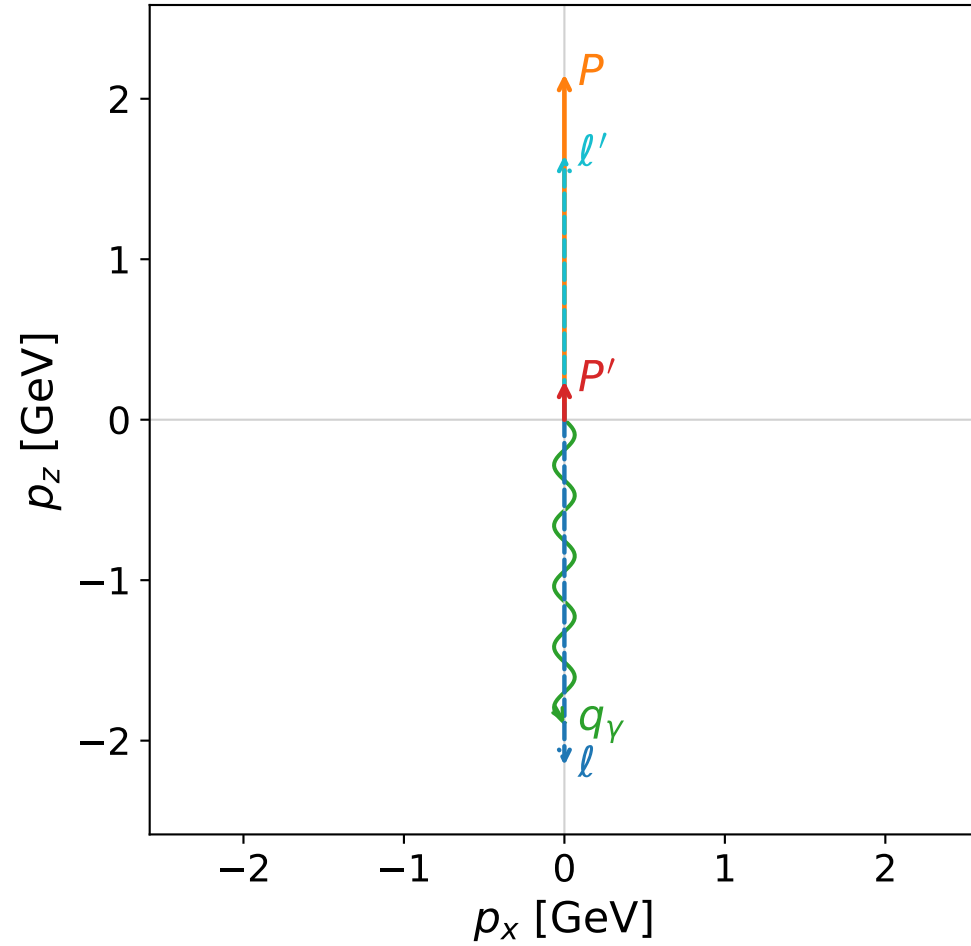
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.229\text{e+}18$; $I_h=-9.892\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.319\text{e-}13$; $I_{TL}=-9.505\text{e-}08$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

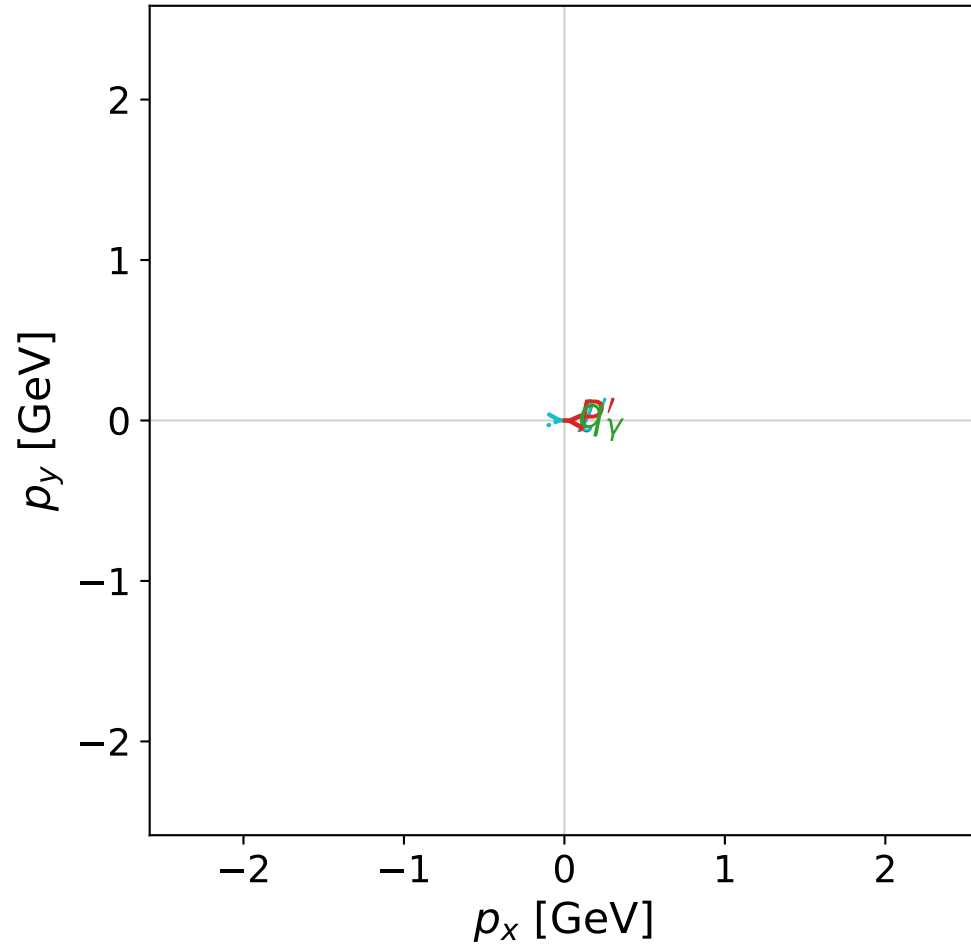


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

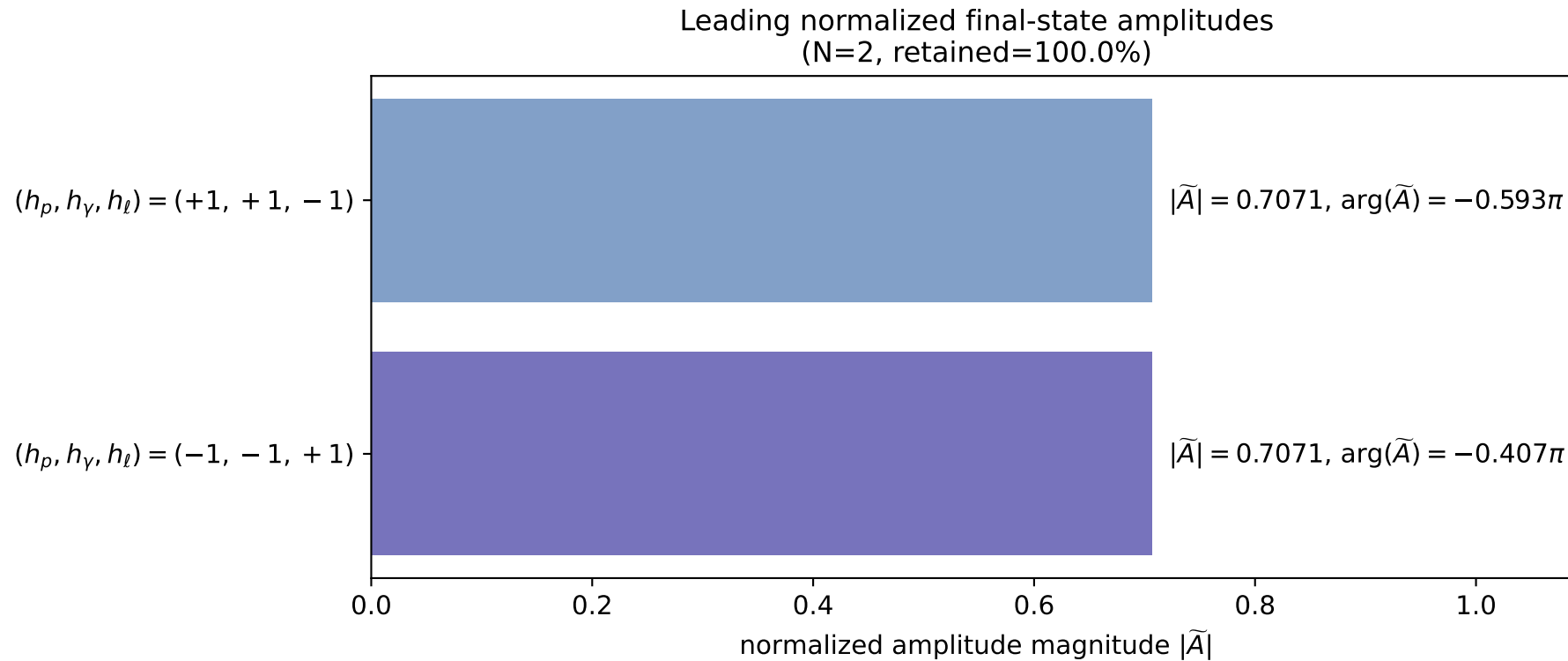


dghz_mixing_angles_polarization_02_local_minimum_99
 $d_{\text{GHZ}}=3.58348\text{e-}08$
 region: polarization_cluster_02_local_minimum_99
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0099
 lepton mixing angle: $\alpha_e=0.78633215$ rad
 incoming lepton: $0.706446|+\rangle + 0.707767|-\rangle$
 proton mixing angle: $\alpha_p=2.3552466$ rad
 incoming proton: $-0.706436|+\rangle + 0.707777|-\rangle$

$s=20.2744$, $\sqrt{s}=4.5027$, $|\vec{P}|=2.15365$, $|\vec{P}'|=0.24118$
 $\theta_{P'}=6.93359\text{e-}06$, $\theta_Y=3.14159$, $\phi_{P'}=3.0985$, $\phi_Y=5.00556$
 $E_Y=1.88769$, $\theta_{\ell'}=1.01558\text{e-}06$, $\phi_{\ell'}=6.2401$
 $Q^2=14.184$, $x_B=0.756439$, $t=-1.75164$, $W^2=5.44687$, $y=0.966822$

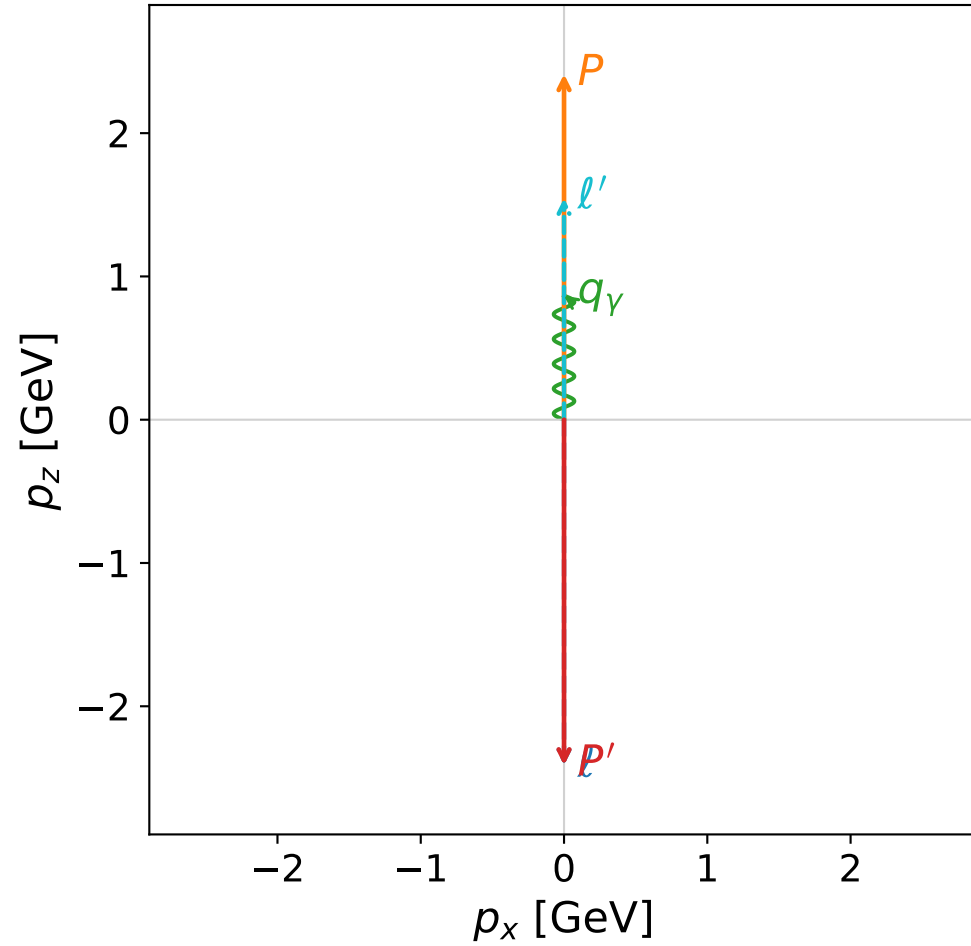
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1537$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1537$
 P $E= 2.3491$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1537$
 ℓ' $E= 1.6465$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.6465$
 P' $E= 0.9685$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.2412$
 q_Y $E= 1.8877$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8877$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.374\text{e+}19$; $I_h=+2.605\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.504\text{e-}15$; $I_{TL}=+2.580\text{e-}10$

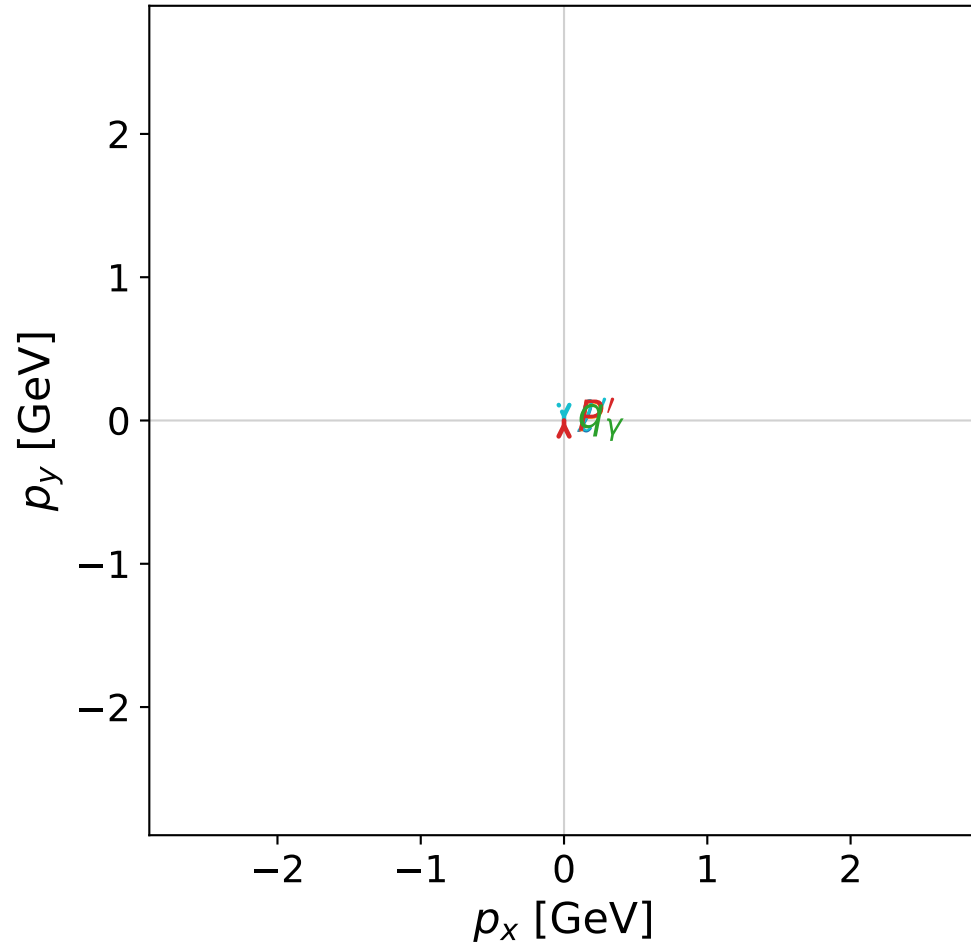


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

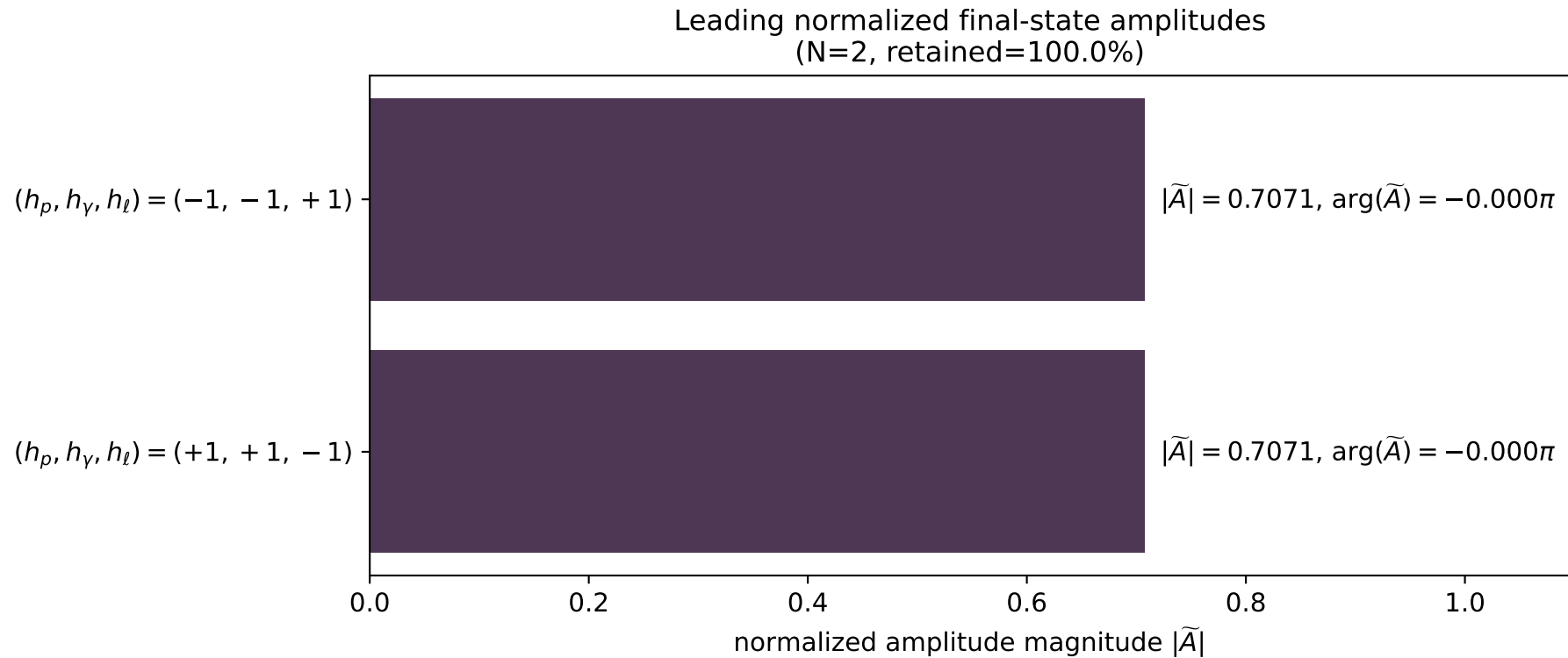


dghz_mixing_angles_polarization_02_local_minimum_100
 $d_{\text{GHZ}}=3.62681\text{e-}08$
 region: polarization_cluster_02_local_minimum_100
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0100
 lepton mixing angle: $\alpha_e=0.65339387$ rad
 incoming lepton: $0.794025|+\rangle + 0.607885|-\rangle$
 proton mixing angle: $\alpha_p=2.224186$ rad
 incoming proton: $-0.607881|+\rangle + 0.794028|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.52284$, $\phi_\gamma=1.07341$
 $E_\gamma=0.865987$, $\theta_{l'}=0$, $\phi_{l'}=4.66443$
 $Q^2=14.9162$, $x_B=0.632684$, $t=-23.2713$, $W^2=9.53971$, $y=0.977442$

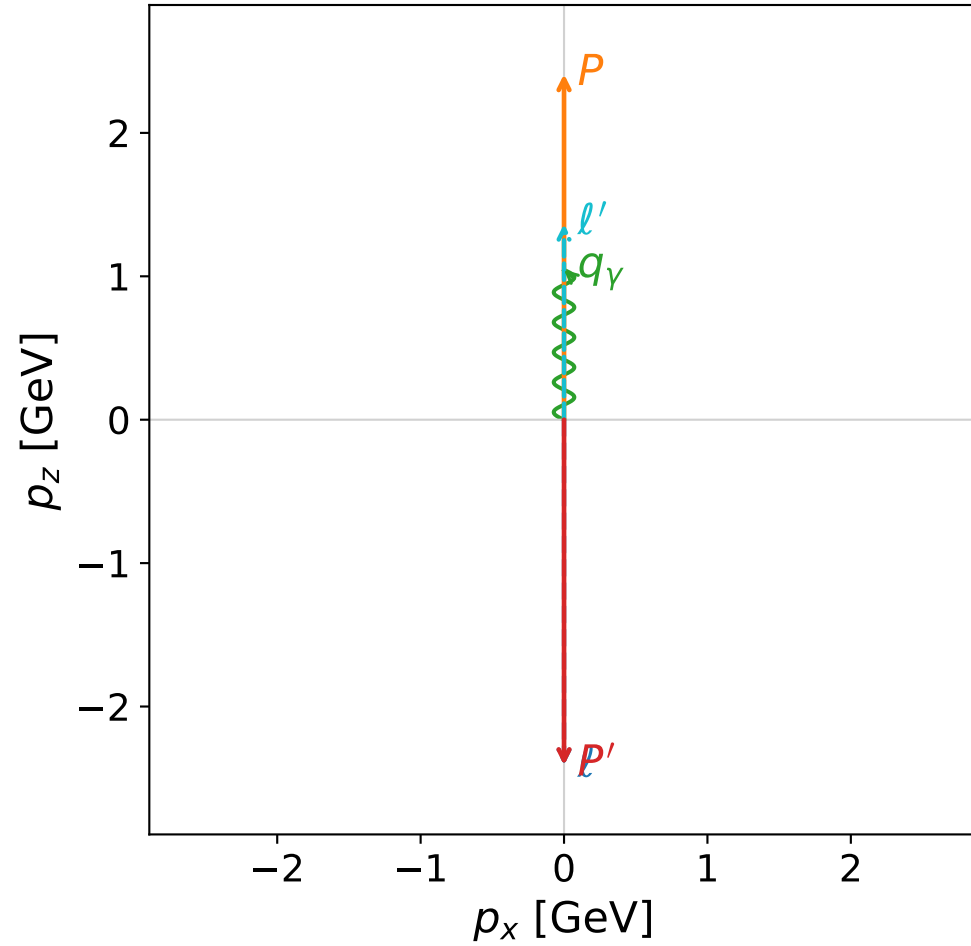
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.5460$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5460$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.8660$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8660$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.733\text{e}+23$; $I_h=-4.264\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.756\text{e-}28$; $I_{TL}=-4.264\text{e-}14$

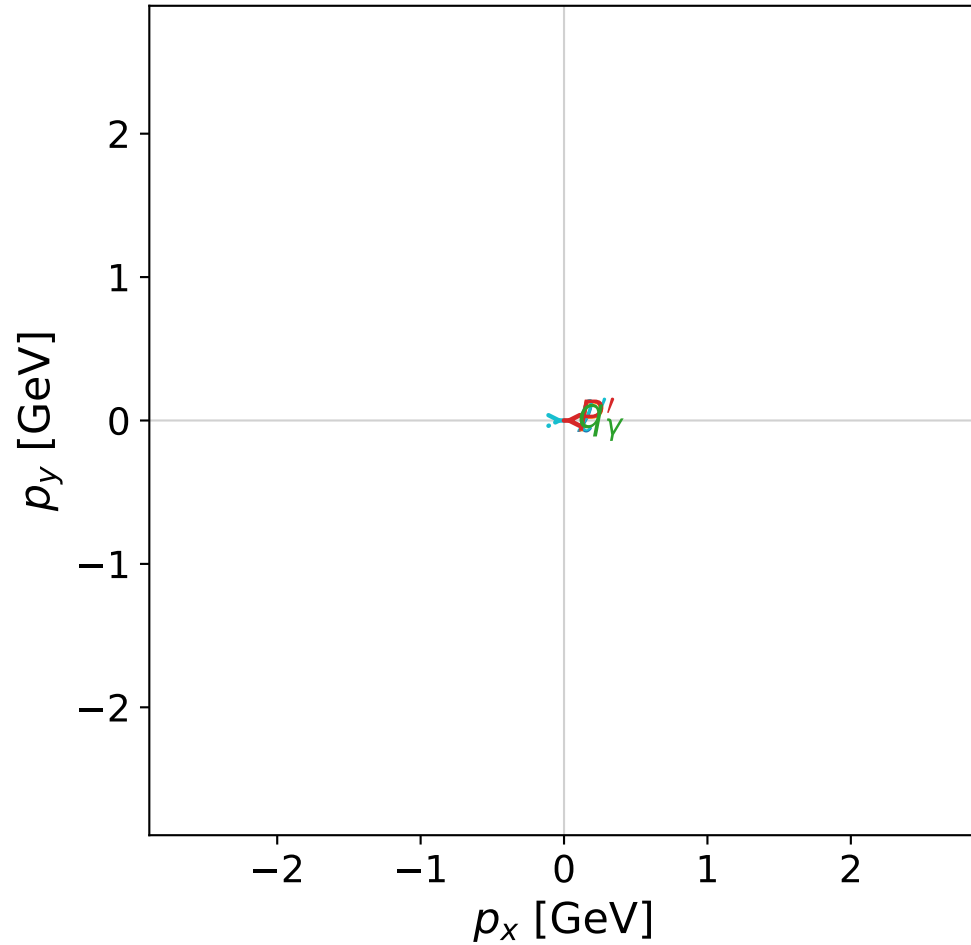


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



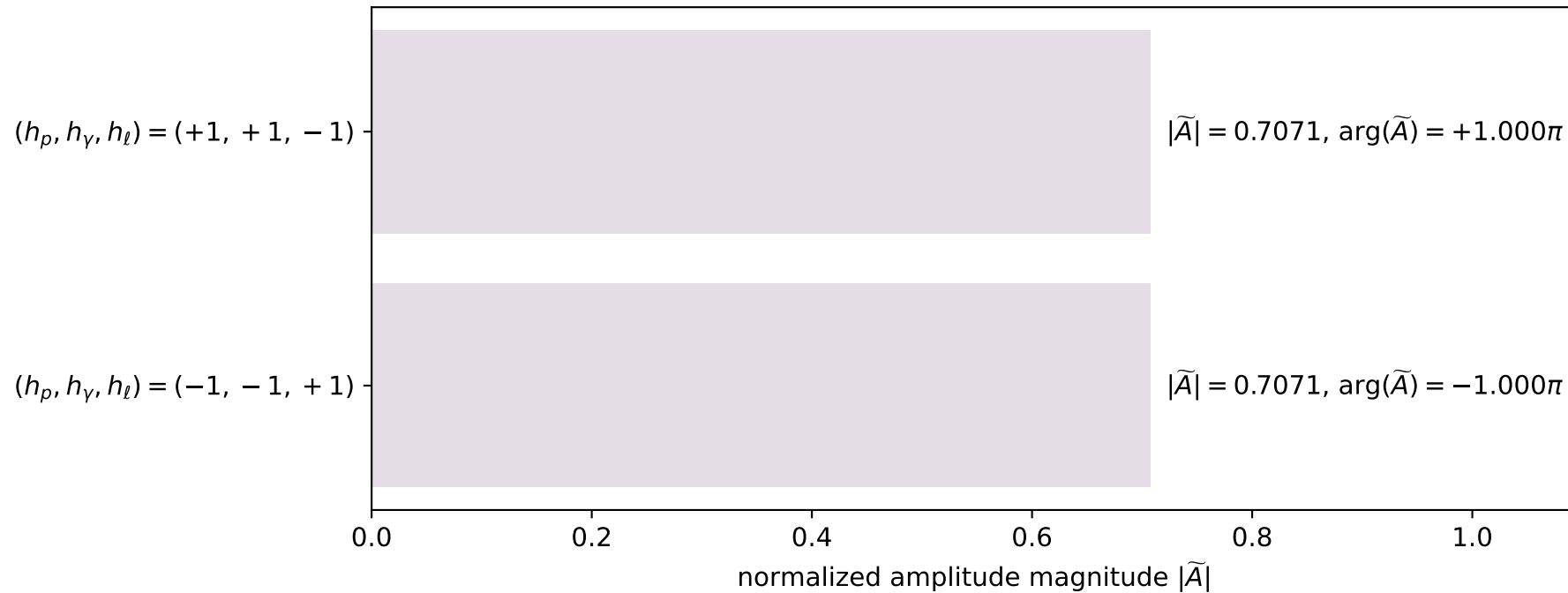
dghz_mixing_angles_polarization_02_local_minimum_110
 $d_{\text{GHZ}}=4.05095\text{e-}08$
 region: polarization_cluster_02_local_minimum_110
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0110
 lepton mixing angle: $\alpha_e=0.68998436$ rad
 incoming lepton: $0.771256|+\rangle + 0.636525|-\rangle$
 proton mixing angle: $\alpha_p=2.2608057$ rad
 incoming proton: $-0.636544|+\rangle + 0.77124|-\rangle$

$s=24.9589$, $\sqrt{s}=4.99589$, $|\vec{P}|=2.40989$, $|\vec{P}'|=2.40989$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.7054$, $\phi_\gamma=1.95022$
 $E_\gamma=1.04501$, $\theta_{l'}=0$, $\phi_{l'}=5.84699$
 $Q^2=13.1568$, $x_B=0.557533$, $t=-23.2303$, $W^2=11.3213$, $y=0.980035$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4099$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4099$
 P $E= 2.5860$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4099$
 l' $E= 1.3649$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.3649$
 P' $E= 2.5860$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4099$
 q_γ $E= 1.0450$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.0450$

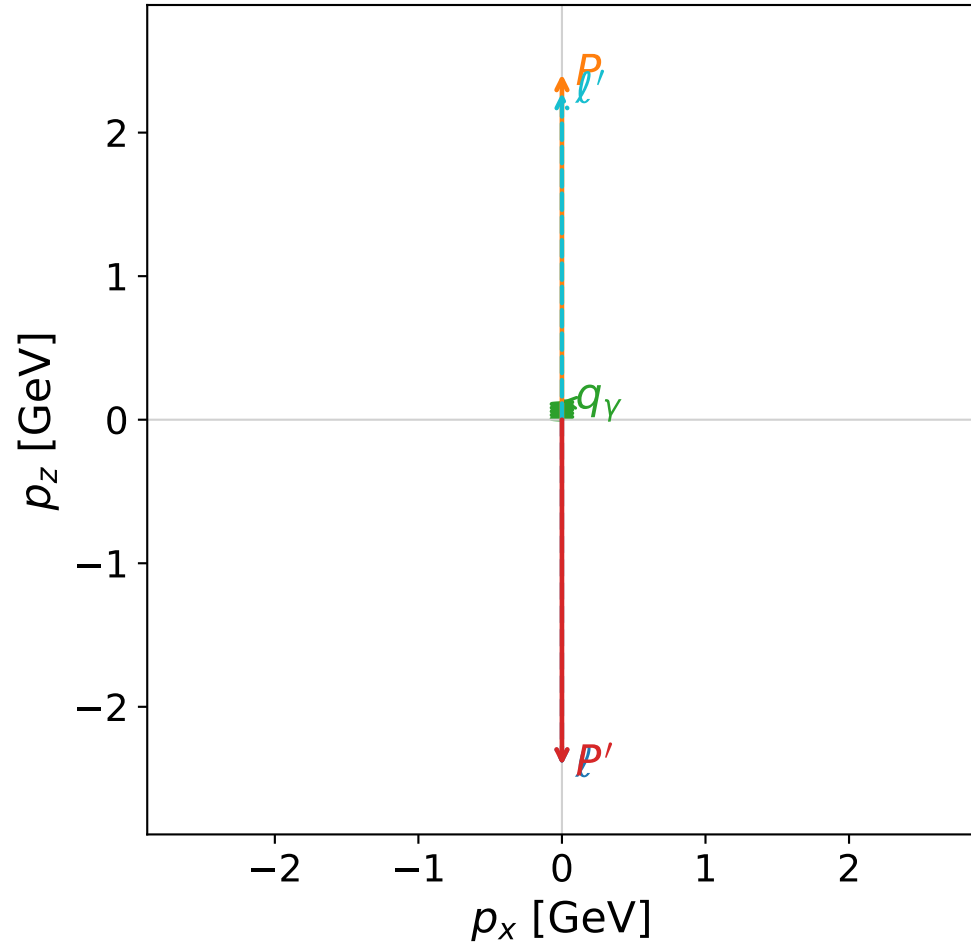
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.247\text{e}+23$; $I_h=+6.971\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.521\text{e-}27$; $I_{TL}=+6.971\text{e-}14$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

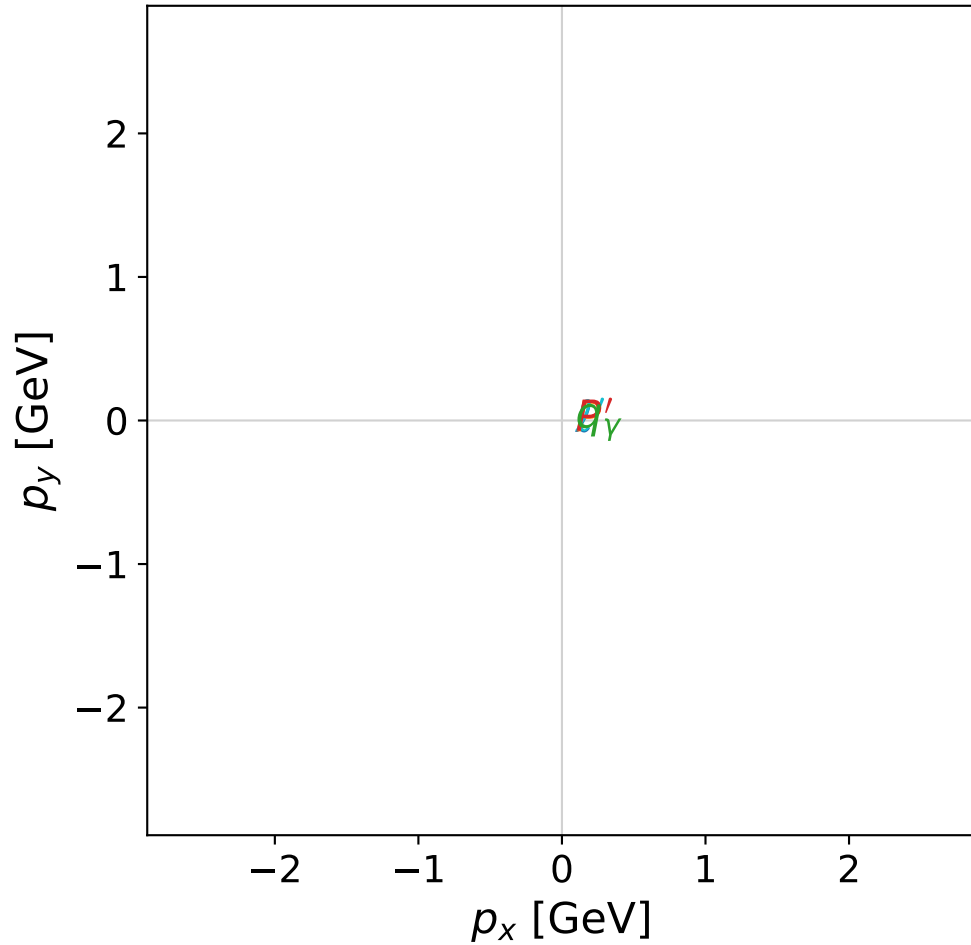


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

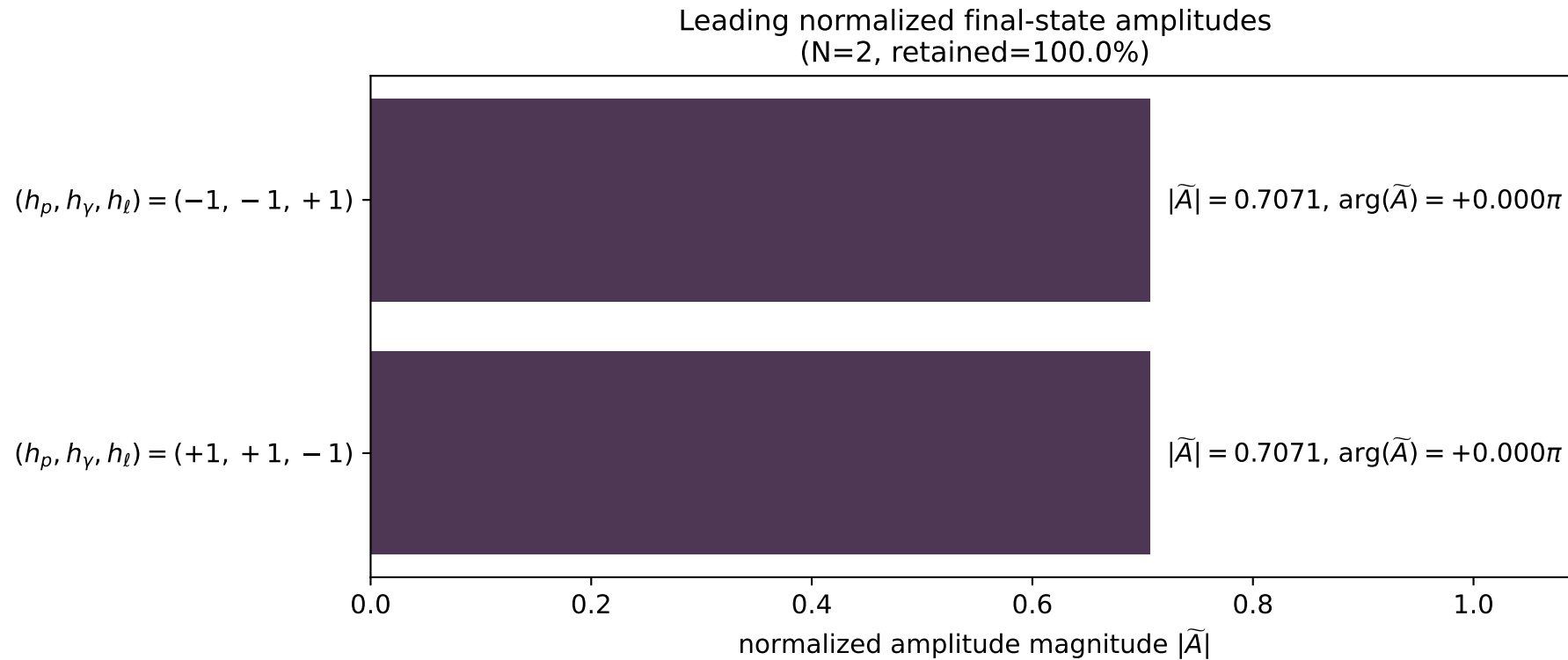


dghz_mixing_angles_polarization_02_local_minimum_116
 $d_{\text{GHZ}}=4.3521\text{e-}08$
 region: polarization_cluster_02_local_minimum_116
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0116
 lepton mixing angle: $\alpha_e=0.28922643$ rad
 incoming lepton: $0.958465|+\rangle + 0.285211|-\rangle$
 proton mixing angle: $\alpha_p=1.8600136$ rad
 incoming proton: $-0.285202|+\rangle + 0.958467|-\rangle$

$s=24.913$, $\sqrt{s}=4.99129$, $|\vec{P}|=2.40751$, $|\vec{P}'|=2.40751$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.69158$, $\phi_\gamma=0.269509$
 $E_\gamma=0.128615$, $\theta_{l'}=0$, $\phi_{l'}=2.54999$
 $Q^2=21.9458$, $x_B=0.94473$, $t=-23.1844$, $W^2=2.16375$, $y=0.96657$

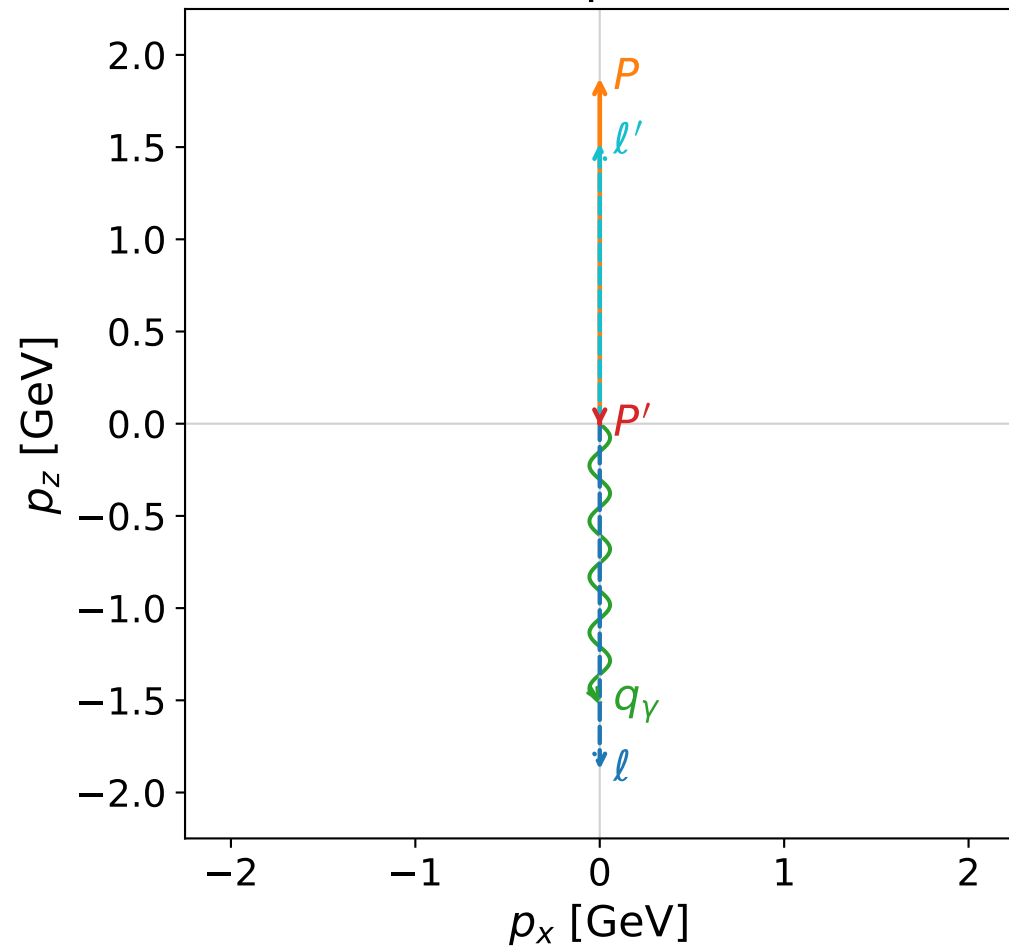
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4075$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4075$
 P $E= 2.5838$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4075$
 ℓ' $E= 2.2789$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.2789$
 P' $E= 2.5838$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4075$
 q_γ $E= 0.1286$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1286$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.795\text{e+}21$; $I_h=+1.135\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.156\text{e-}24$; $I_{TL}=+1.135\text{e-}12$

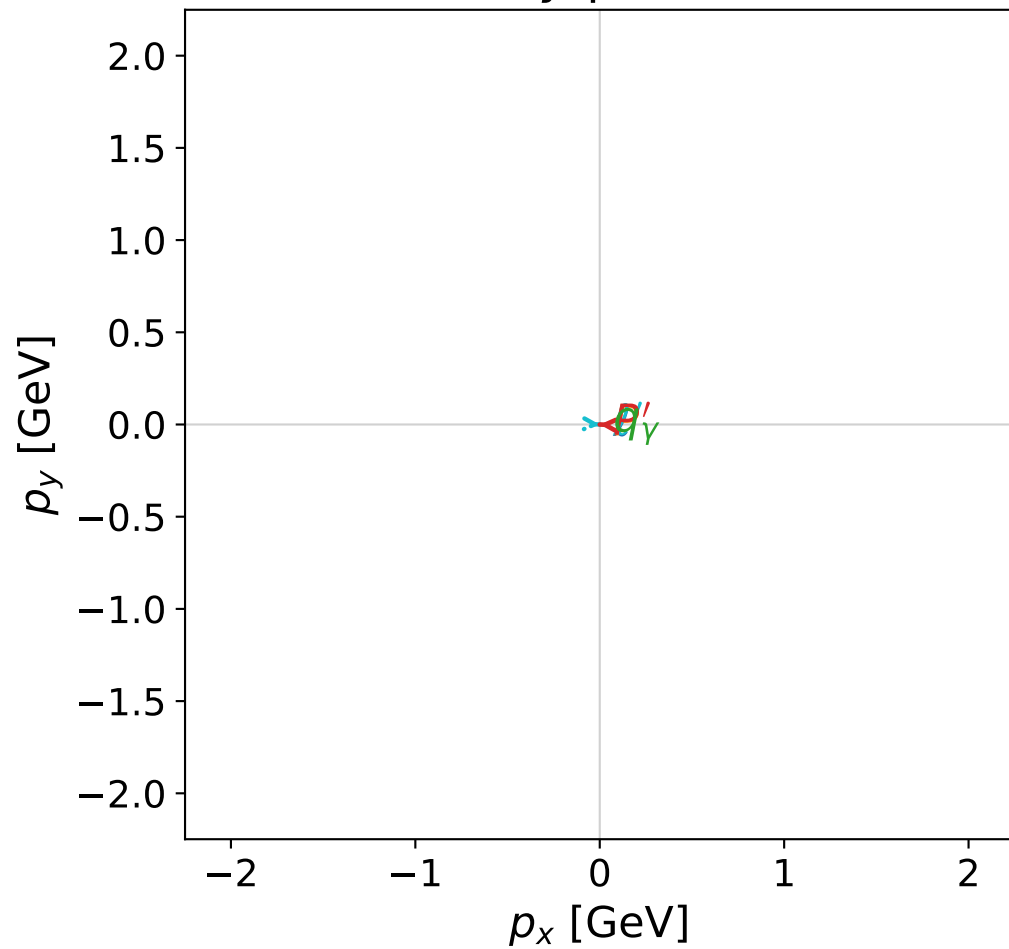


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

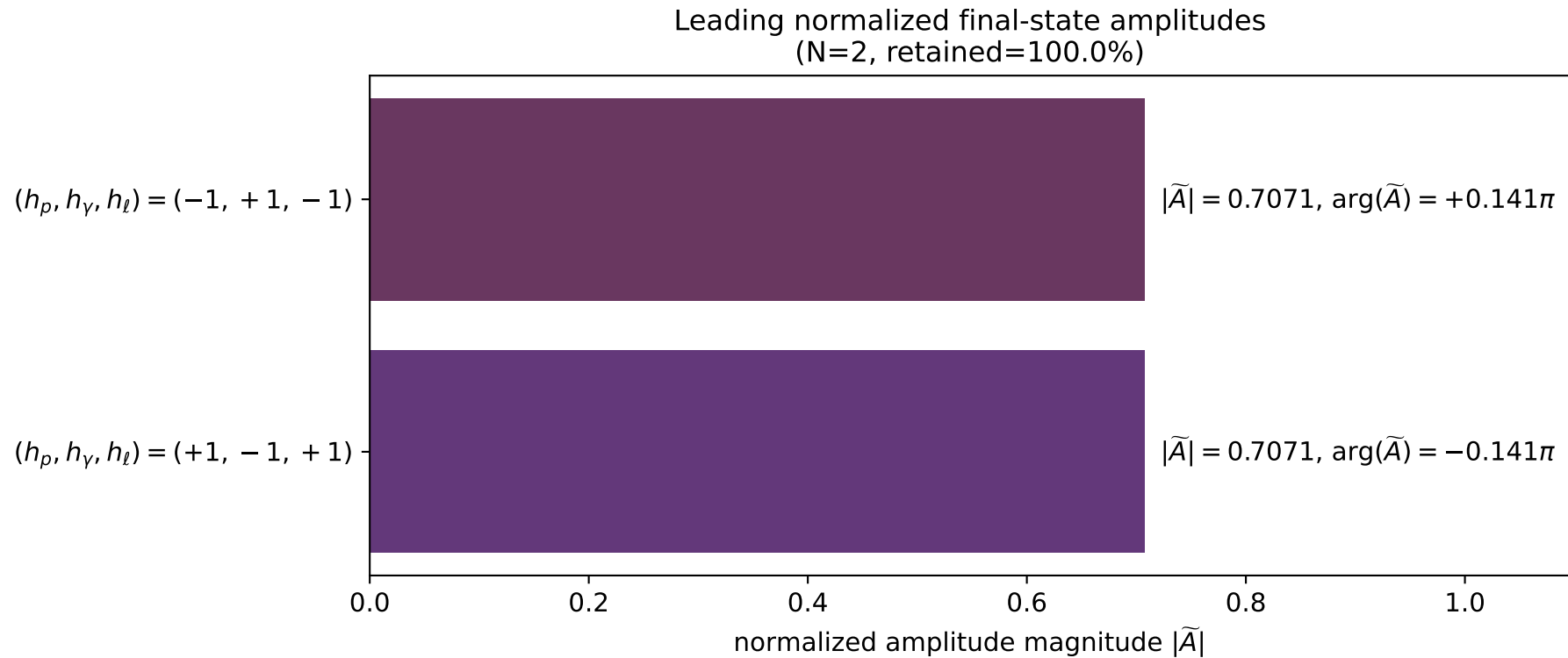


dghz_mixing_angles_polarization_02_local_minimum_120
 $d_{\text{GHZ}}=4.59538\text{e-}08$
 region: polarization_cluster_02_local_minimum_120
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0120
 lepton mixing angle: $\alpha_e=0.74588136$ rad
 incoming lepton: $0.73449|+\rangle +0.678619|-\rangle$
 proton mixing angle: $\alpha_p=2.3956994$ rad
 incoming proton: $-0.734482|+\rangle +0.678628|-\rangle$

$s=15.7593$, $\sqrt{s}=3.9698$, $|\vec{P}|=1.87408$, $|\vec{P}'|=0.00992053$
 $\theta_{p'}=3.14111$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.09611$, $\phi_\gamma=2.65443$
 $E_\gamma=1.51091$, $\theta_{\ell'}=3.16449\text{e-}06$, $\phi_{\ell'}=6.23771$
 $Q^2=11.4006$, $x_B=0.802563$, $t=-2.20928$, $W^2=3.68449$, $y=0.954693$

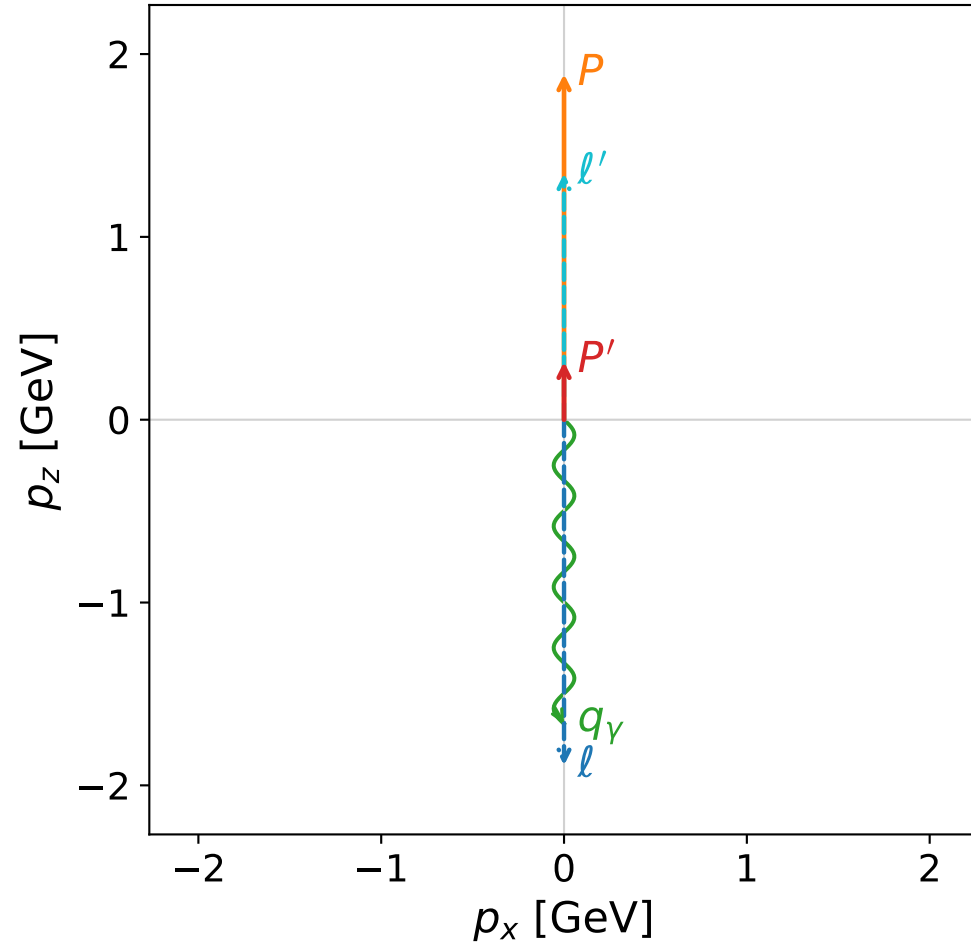
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8741$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8741$
 P $E= 2.0957$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8741$
 ℓ' $E= 1.5208$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.5208$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0099$
 q_γ $E= 1.5109$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.5109$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.644\text{e+}18$; $I_h=-4.231\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{\text{L/T}}=8.293\text{e-}14$; $I_{\text{TL}}=-3.211\text{e-}08$

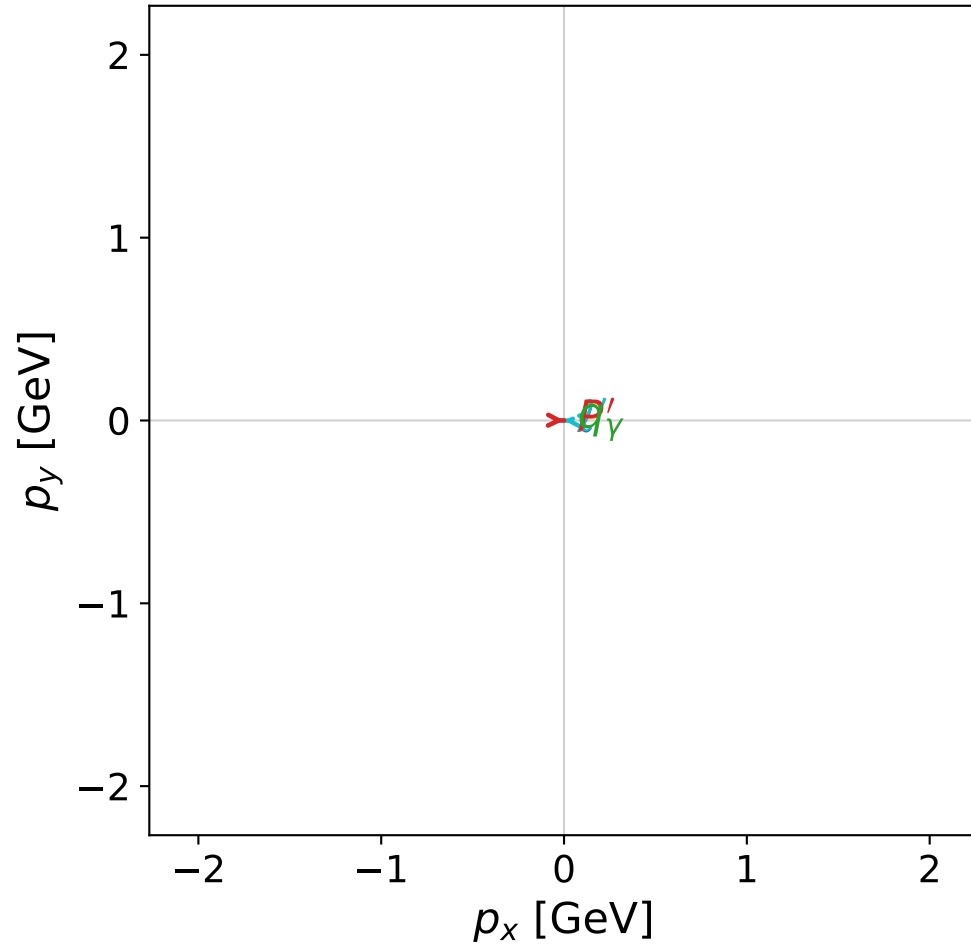


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

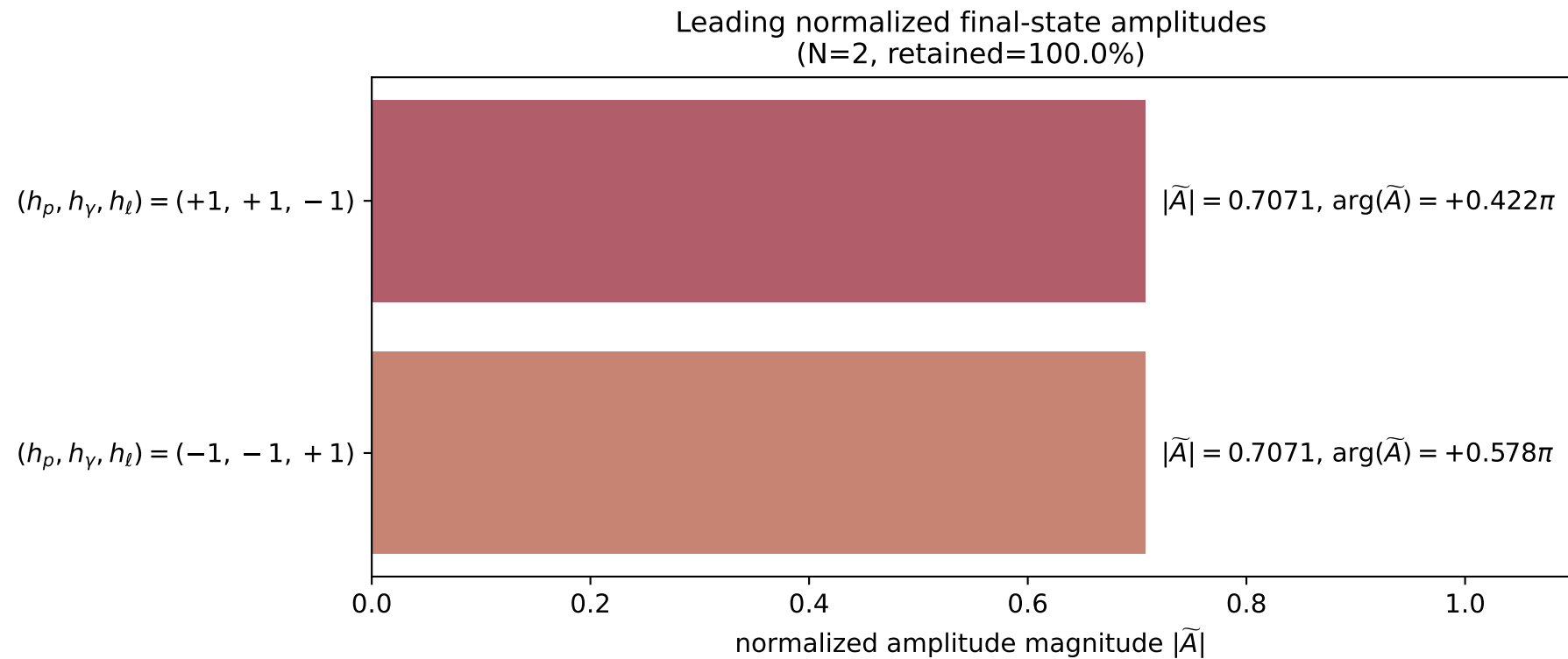


dghz_mixing_angles_polarization_02_local_minimum_127
 $d_{\text{GHZ}}=5.10304\text{e-}08$
 region: polarization_cluster_02_local_minimum_127
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0127
 lepton mixing angle: $\alpha_e=0.78216053$ rad
 incoming lepton: $0.709392|+\rangle + 0.704814|-\rangle$
 proton mixing angle: $\alpha_p=2.3594272$ rad
 incoming proton: $-0.709389|+\rangle + 0.704817|-\rangle$

$s=16.0046$, $\sqrt{s}=4.00057$, $|\vec{P}|=1.89032$, $|\vec{P}'|=0.316348$
 $\theta_{P'}=9.29374\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.26205$, $\phi_\gamma=1.81607$
 $E_\gamma=1.6635$, $\theta_{l'}=2.18242\text{e-}05$, $\phi_{l'}=3.12046$
 $Q^2=10.1862$, $x_B=0.700944$, $t=-1.22222$, $W^2=5.22577$, $y=0.960822$

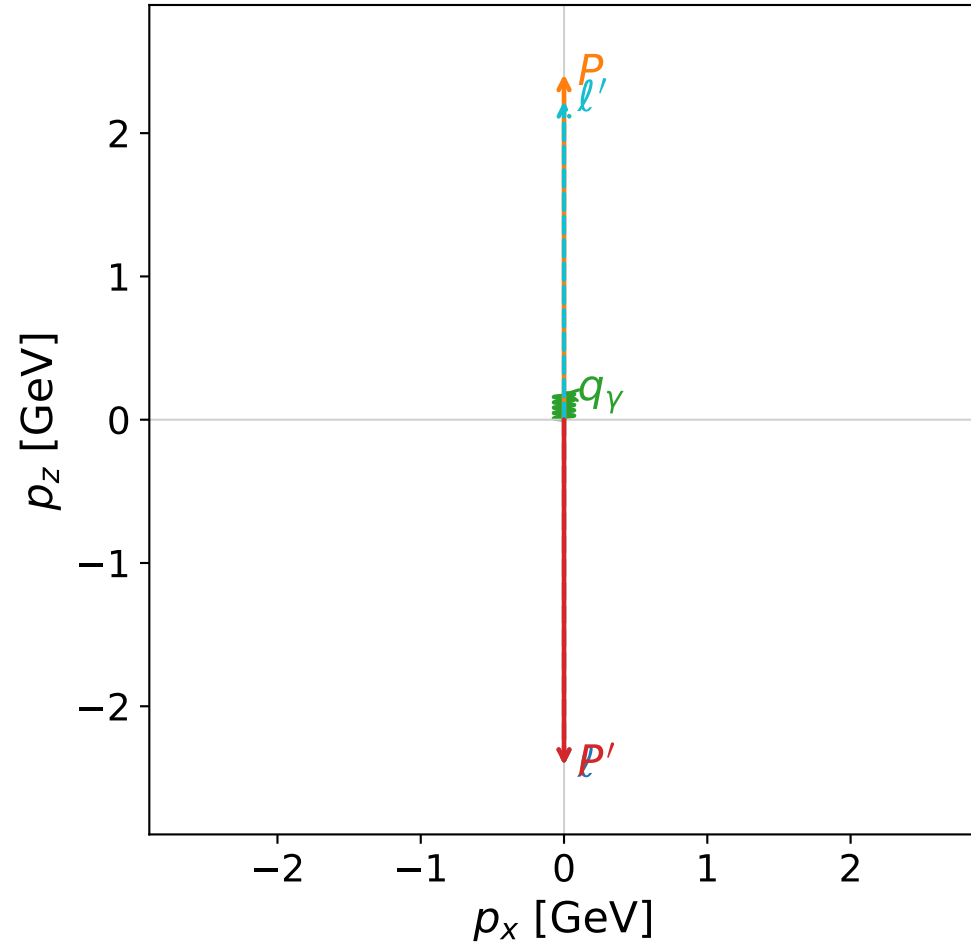
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.8903$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8903$
 P $E= 2.1102$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8903$
 l' $E= 1.3472$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.3472$
 P' $E= 0.9899$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.3163$
 q_γ $E= 1.6635$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.6635$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.010\text{e+}17$; $I_h=+2.169\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=7.592\text{e-}12$; $I_{TL}=+2.554\text{e-}08$

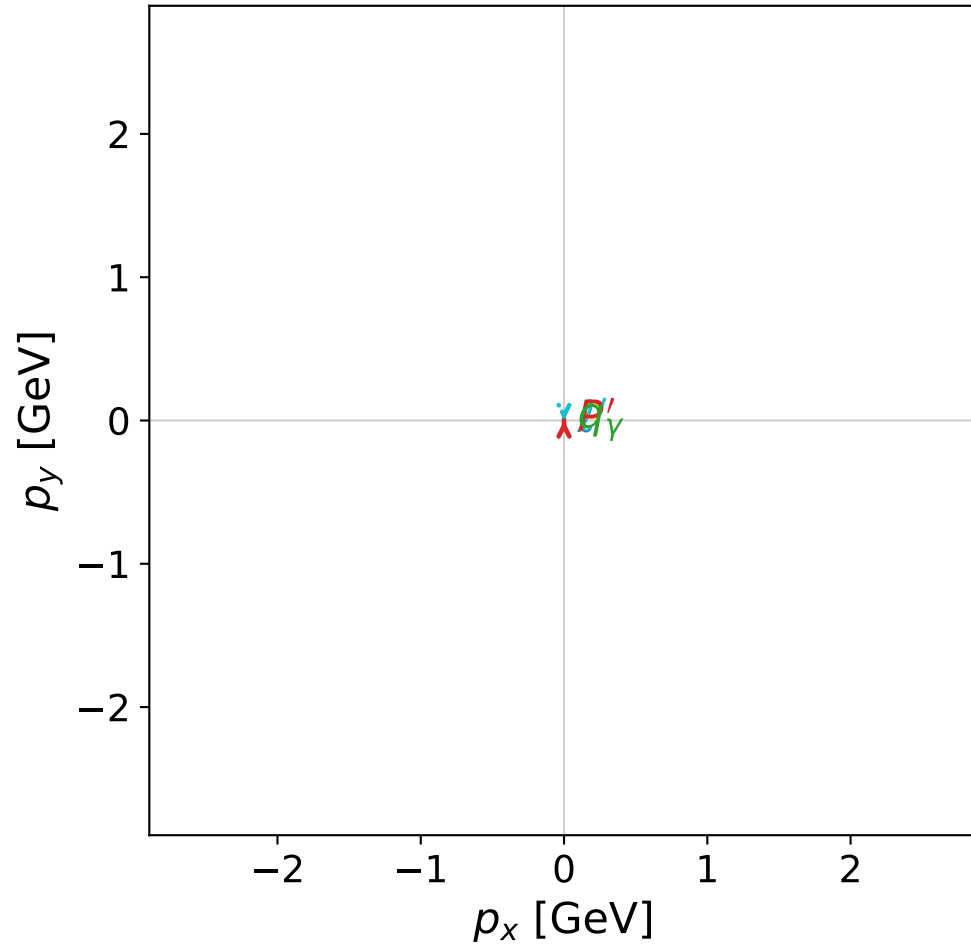


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



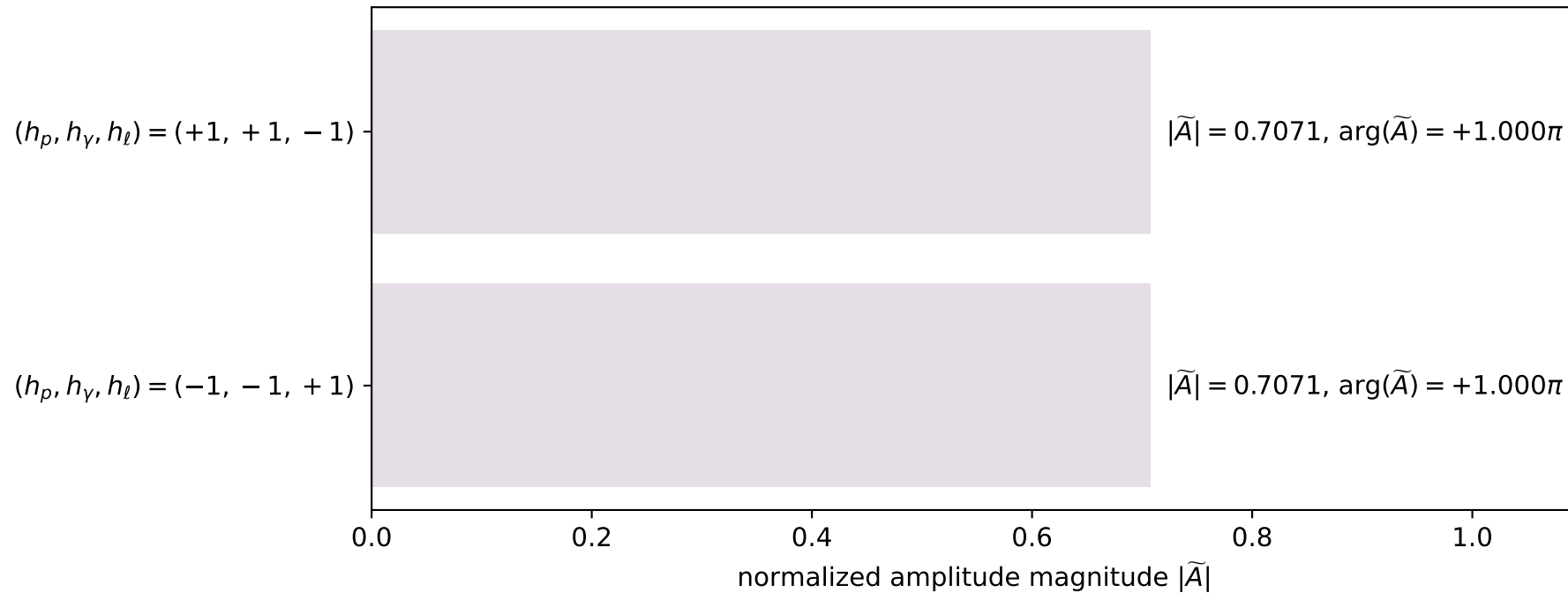
dghz_mixing_angles_polarization_02_local_minimum_136
 $d_{\text{GHZ}}=6.21675\text{e-}08$
 region: polarization_cluster_02_local_minimum_136
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0136
 lepton mixing angle: $\alpha_e=0.20100578$ rad
 incoming lepton: $0.979866|+\rangle + 0.199655|-\rangle$
 proton mixing angle: $\alpha_p=1.7718087$ rad
 incoming proton: $-0.199661|+\rangle + 0.979865|-\rangle$

$s=24.9997$, $\sqrt{s}=4.99997$, $|\vec{P}|=2.412$, $|\vec{P}'|=2.412$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=1.40635$, $\phi_Y=3.24397$
 $E_Y=0.186808$, $\theta_{l'}=0$, $\phi_{l'}=4.54794$
 $Q^2=21.4686$, $x_B=0.919952$, $t=-23.2709$, $W^2=2.74791$, $y=0.967532$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2252$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2252$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 0.1868$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1868$

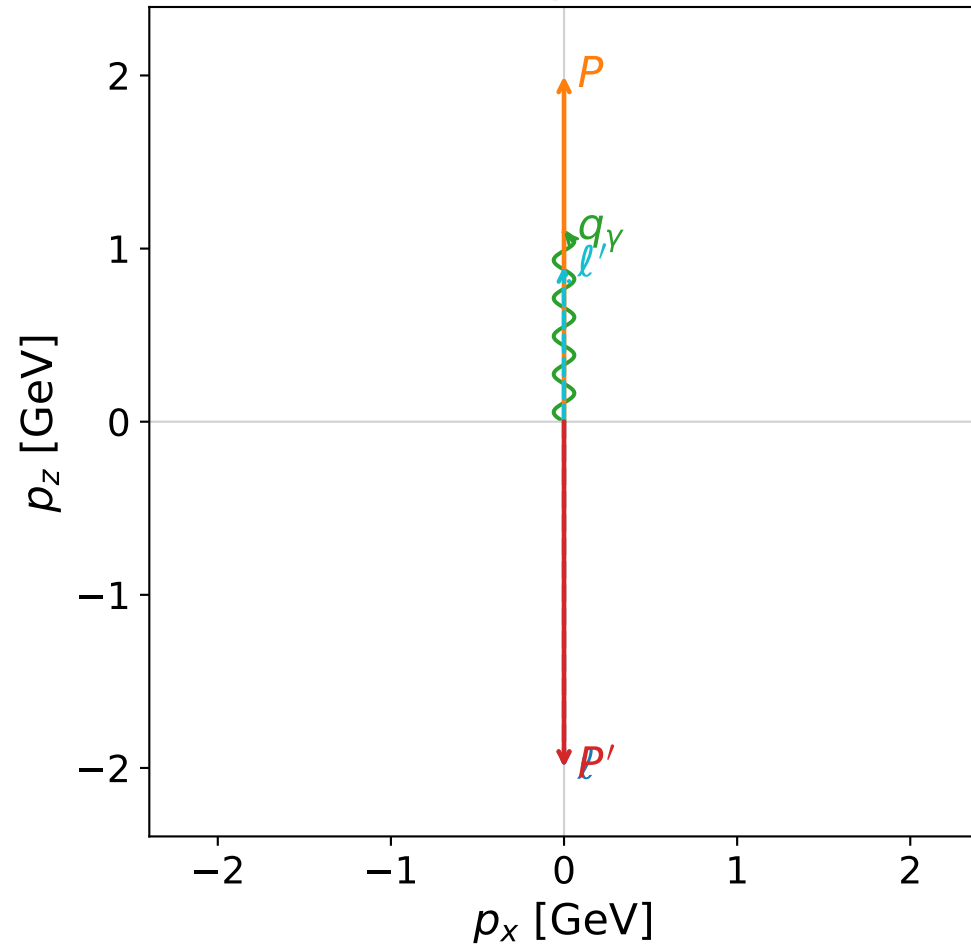
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=9.730\text{e}+21$; $I_h=+1.582\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=8.169\text{e-}26$; $I_{TL}=+1.582\text{e-}13$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

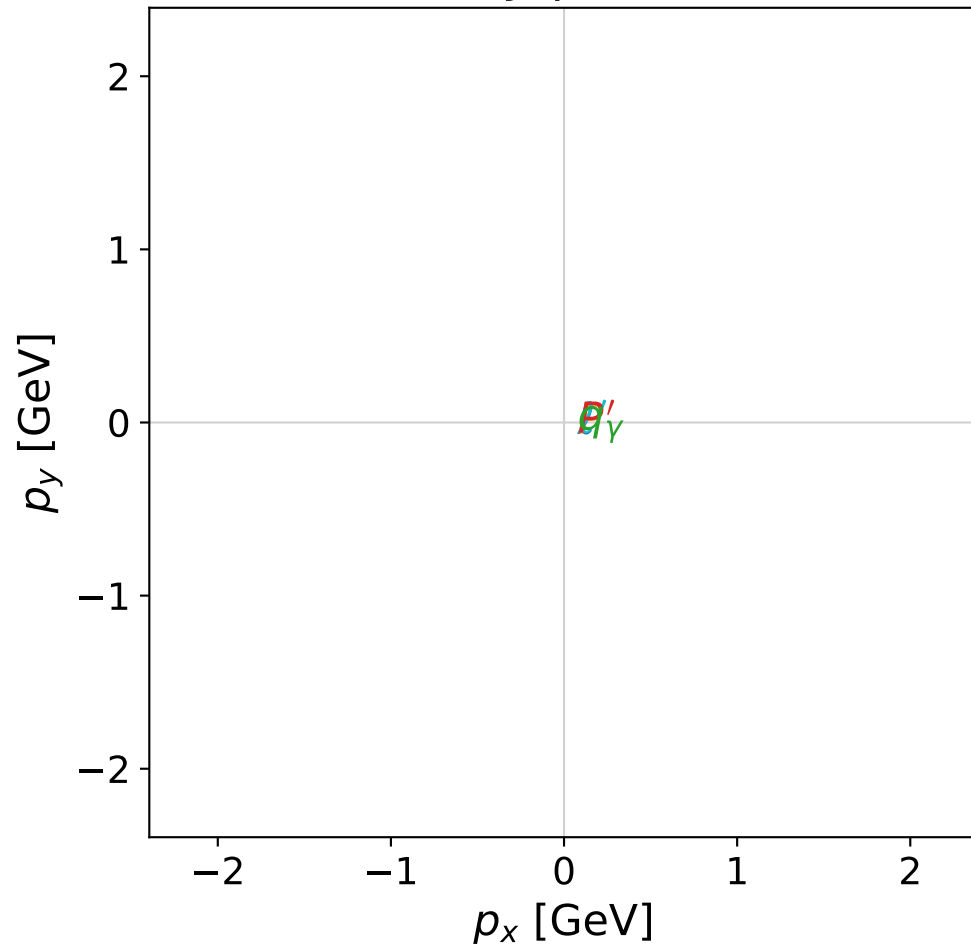


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

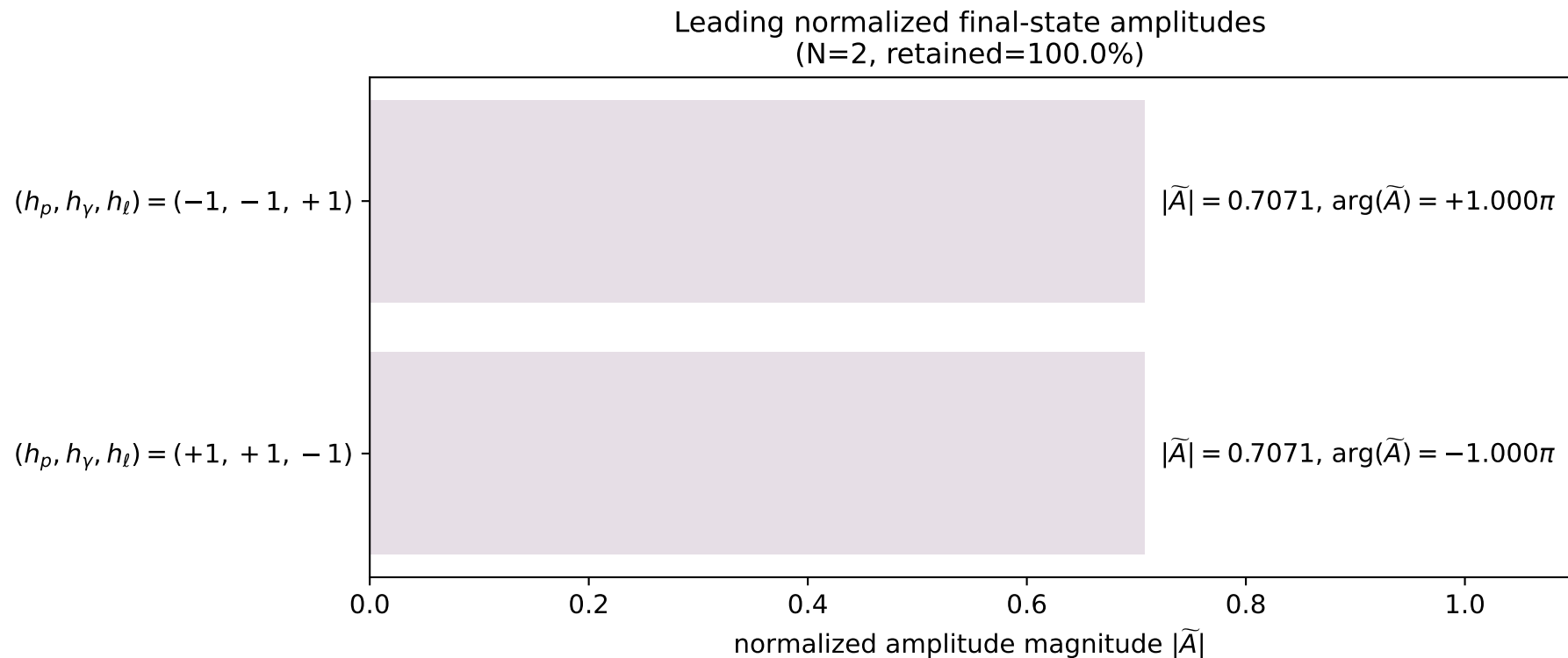


dghz_mixing_angles_polarization_02_local_minimum_142
 $d_{\text{GHZ}}=7.52972\text{e-}08$
 region: polarization_cluster_02_local_minimum_142
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0142
 lepton mixing angle: $\alpha_e=0.65625654$ rad
 incoming lepton: $0.792282|+\rangle +0.610155|-\rangle$
 proton mixing angle: $\alpha_p=2.2270306$ rad
 incoming proton: $-0.610138|+\rangle +0.792295|-\rangle$

$s=17.6556$, $\sqrt{s}=4.20186$, $|\vec{P}|=1.99623$, $|\vec{P}'|=1.99623$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=6.06891$, $\phi_Y=4.38119$
 $E_Y=1.09748$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.92732$
 $Q^2=7.17649$, $x_B=0.437607$, $t=-15.9398$, $W^2=10.1027$, $y=0.977564$

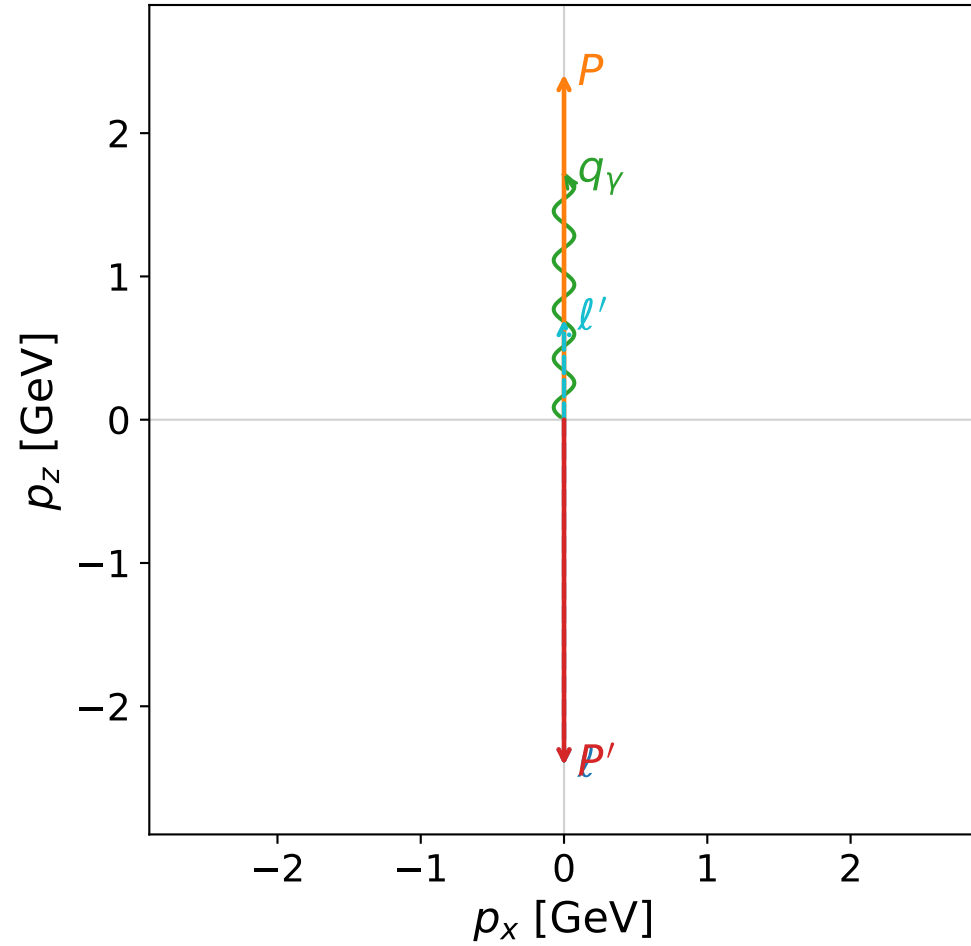
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9962$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9962$
 P $E= 2.2056$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9962$
 ℓ' $E= 0.8988$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.8988$
 P' $E= 2.2056$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9962$
 q_Y $E= 1.0975$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.0975$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.099\text{e}+24$; $I_h=+5.045\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.361\text{e-}29$; $I_{TL}=+5.046\text{e-}15$

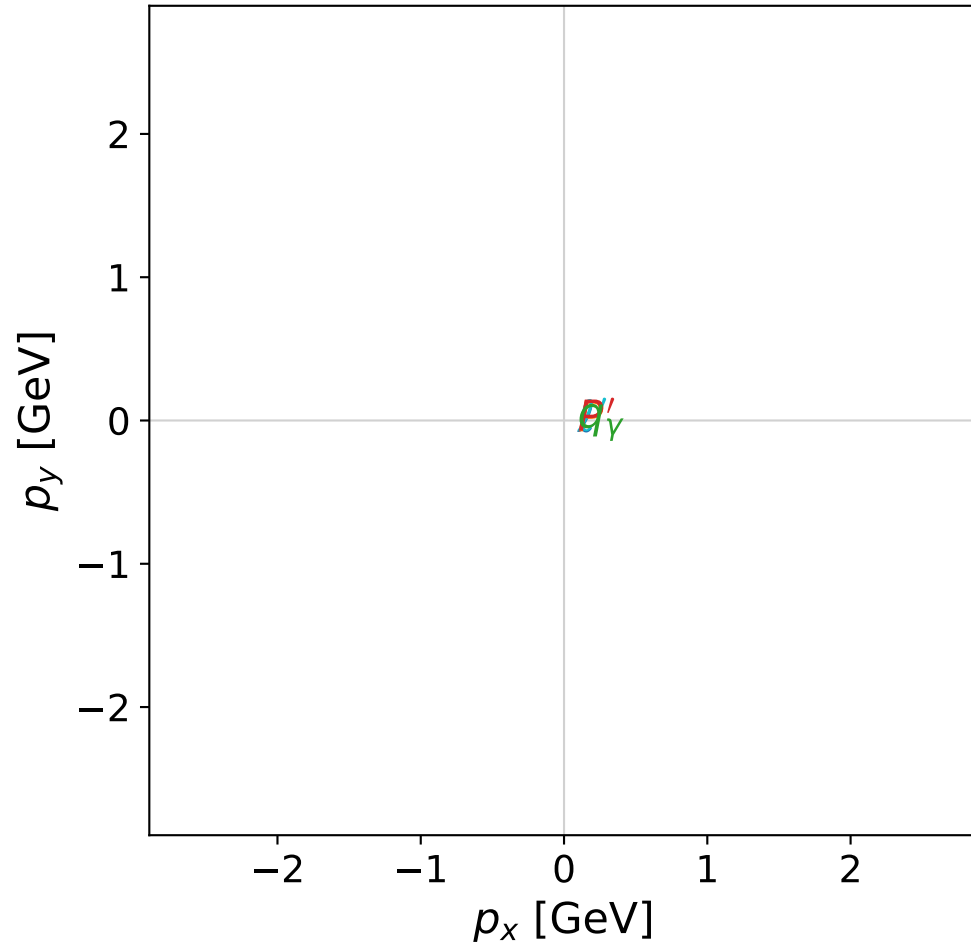


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



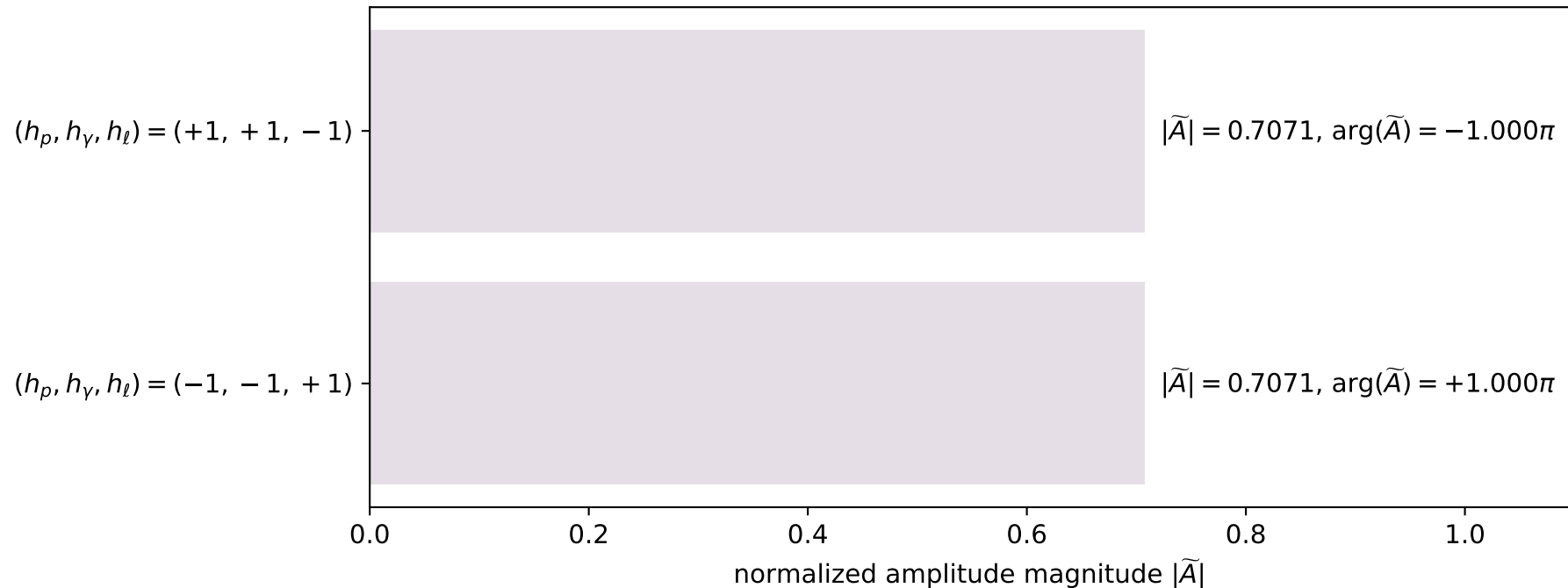
dghz_mixing_angles_polarization_02_local_minimum_144
 $d_{\text{GHZ}}=7.83262\text{e-}08$
 region: polarization_cluster_02_local_minimum_144
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0144
 lepton mixing angle: $\alpha_e=0.70738913$ rad
 incoming lepton: $0.760061|+\rangle +0.649852|-\rangle$
 proton mixing angle: $\alpha_p=2.2781902$ rad
 incoming proton: $-0.649855|+\rangle +0.760058|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.15414$, $\phi_\gamma=4.45412$
 $E_\gamma=1.71245$, $\theta_{l'}=0$, $\phi_{l'}=5.29573$
 $Q^2=6.74947$, $x_B=0.282713$, $t=-23.2713$, $W^2=18.0043$, $y=0.989793$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 0.6996$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6996$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 1.7124$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.7124$

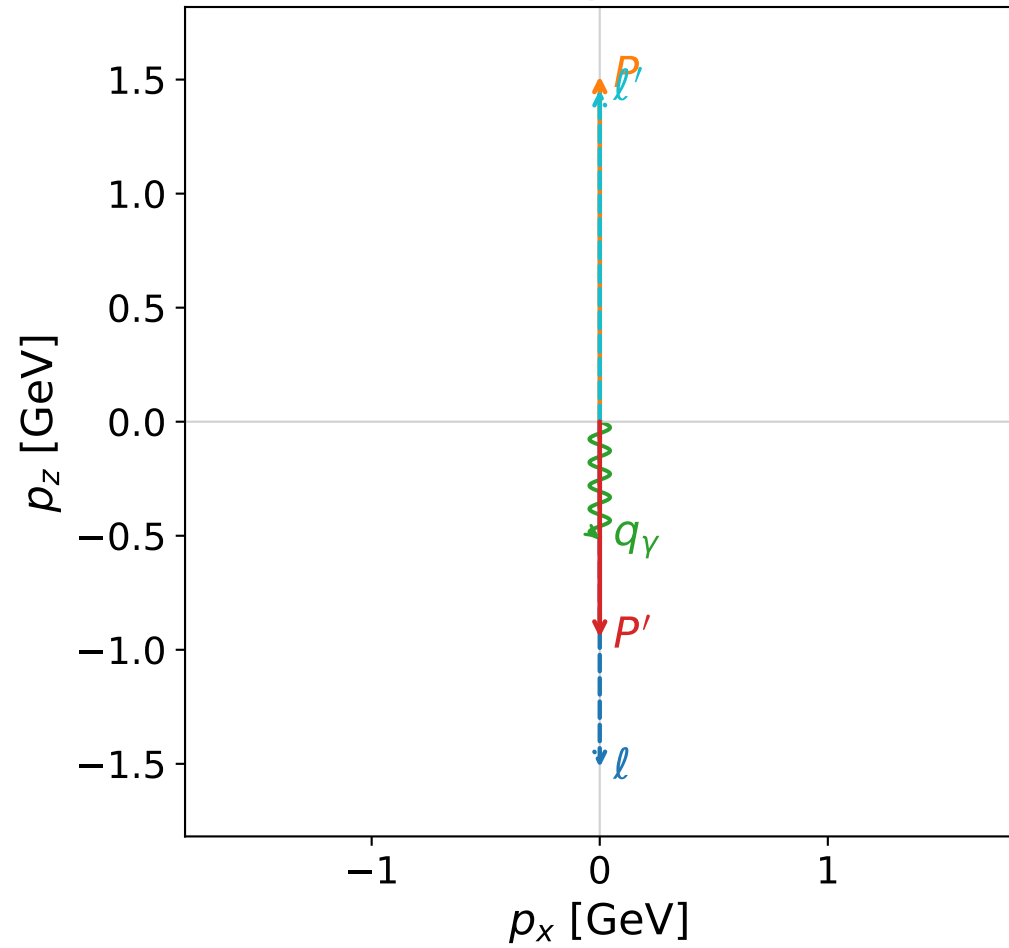
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.390\text{e}+24$; $I_h=+1.765\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.596\text{e-}28$; $I_{TL}=+1.765\text{e-}14$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

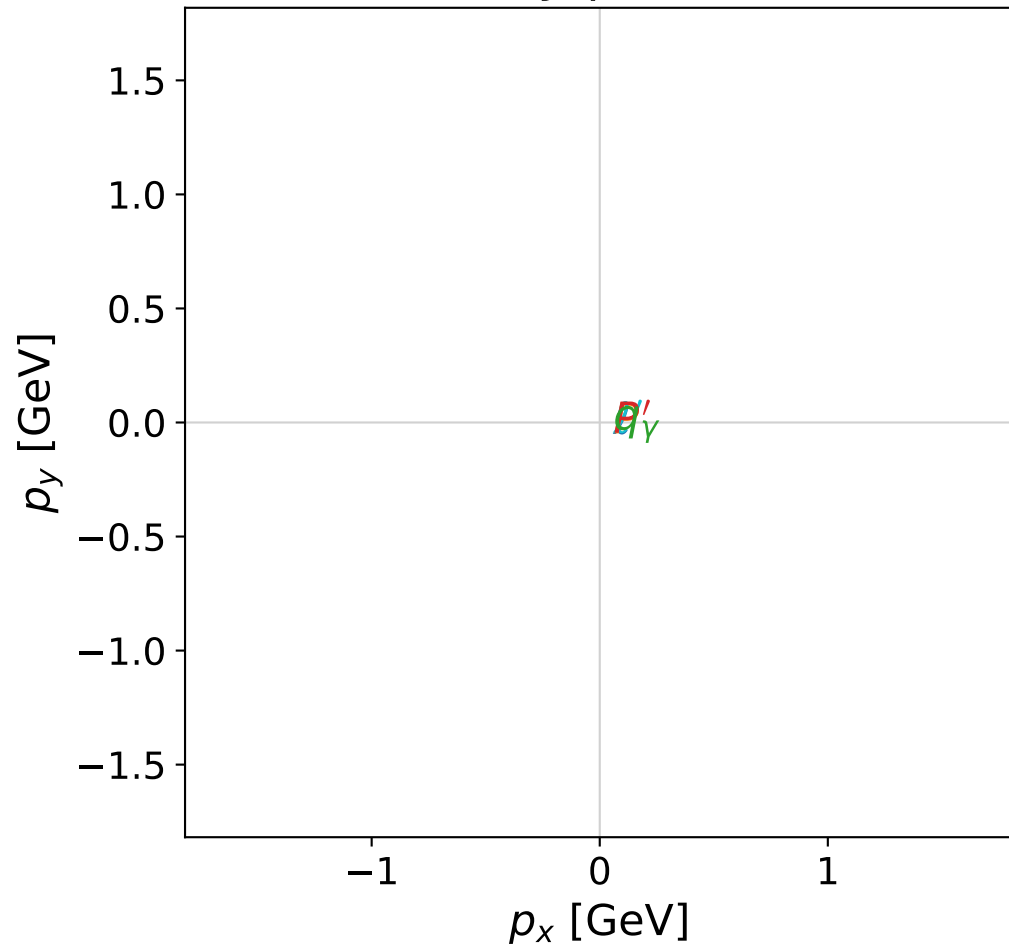


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

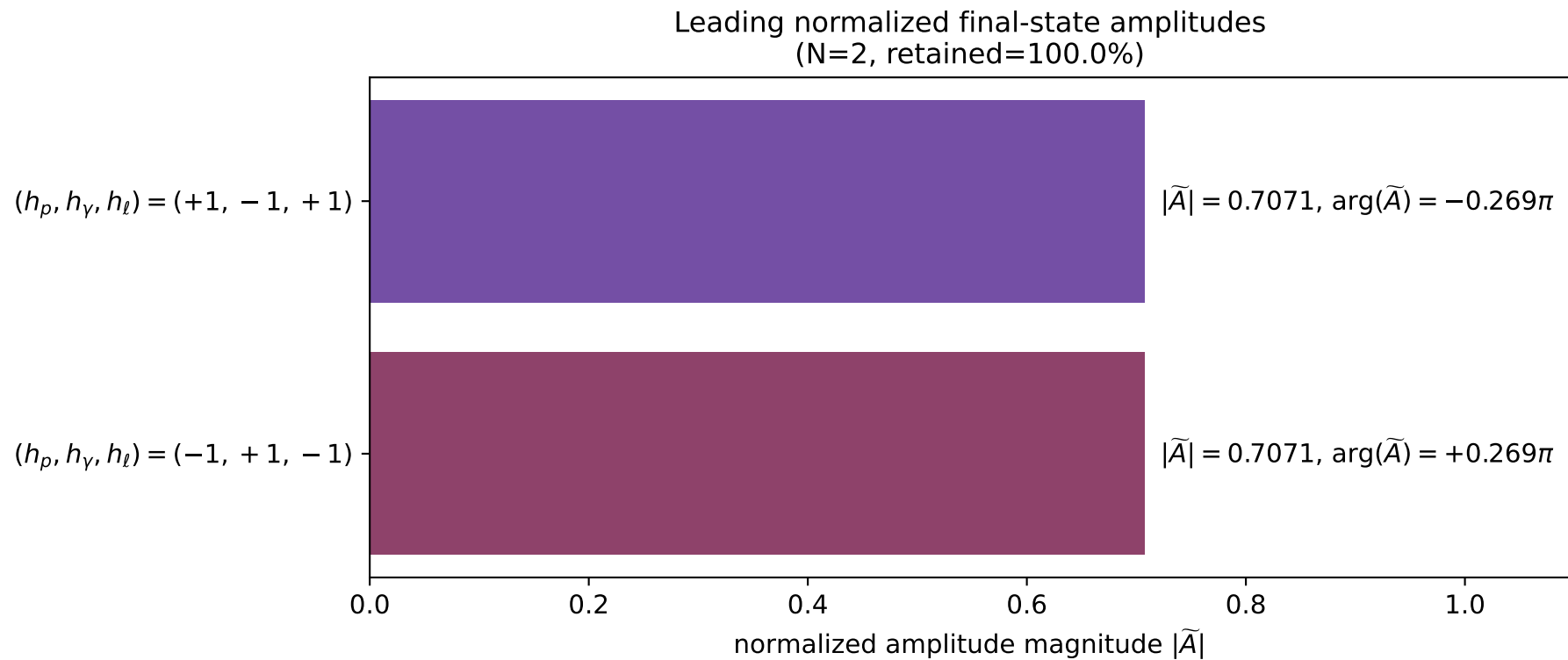


dghz_mixing_angles_polarization_02_local_minimum_147
 $d_{\{\text{rm GHZ}\}}=8.24447\text{e-}08$
 region: polarization_cluster_02_local_minimum_147
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0147
 lepton mixing angle: $\alpha_e=0.40049047$ rad
 incoming lepton: $0.92087|+\rangle +0.38987|-\rangle$
 proton mixing angle: $\alpha_p=2.7411031$ rad
 incoming proton: $-0.92087|+\rangle +0.389869|-\rangle$

$s=10.8717$, $\sqrt{s}=3.29723$, $|\vec{P}|=1.51519$, $|\vec{P}'|=0.945903$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.07214$, $\phi_\gamma=5.43934$
 $E_\gamma=0.509598$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.96521$
 $Q^2=8.82147$, $x_B=0.957283$, $t=-5.85459$, $W^2=1.27349$, $y=0.922259$

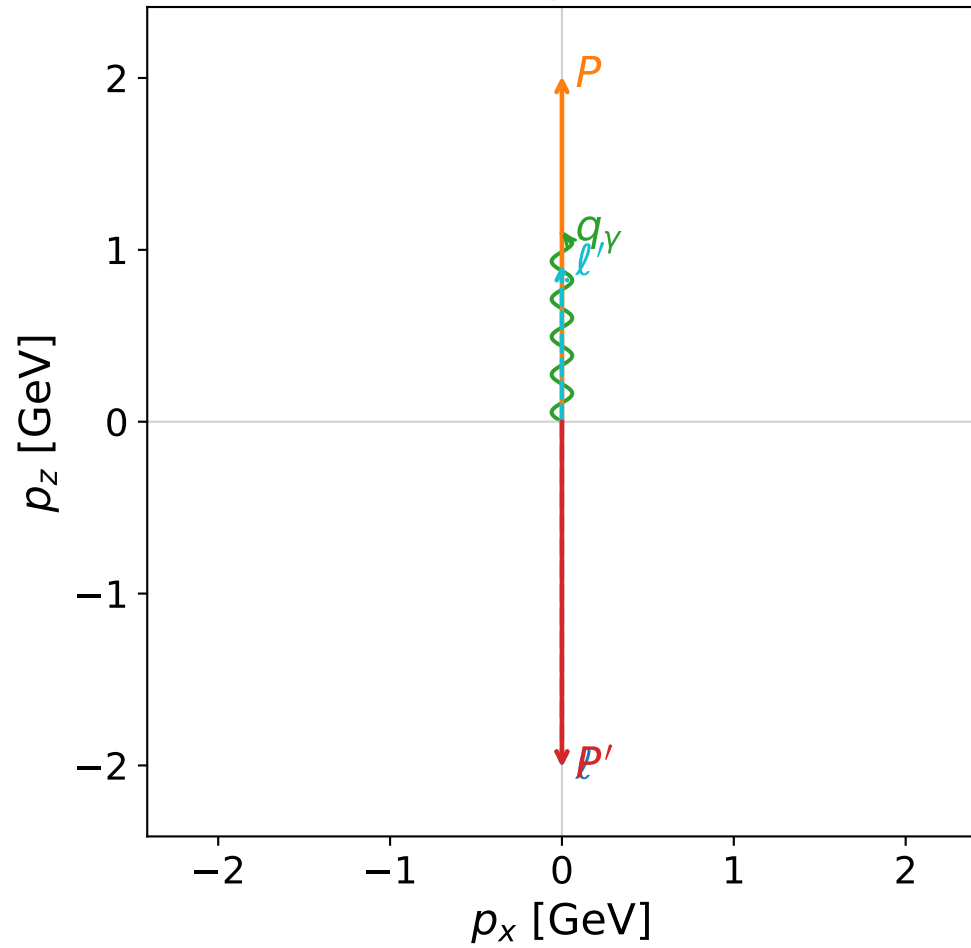
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5152$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5152$
 P $E= 1.7820$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5152$
 ℓ' $E= 1.4555$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.4555$
 P' $E= 1.3321$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.9459$
 q_γ $E= 0.5096$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.5096$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.114\text{e}+23$; $I_h=-9.232\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.515\text{e-}26$; $I_{TL}=-2.765\text{e-}17$

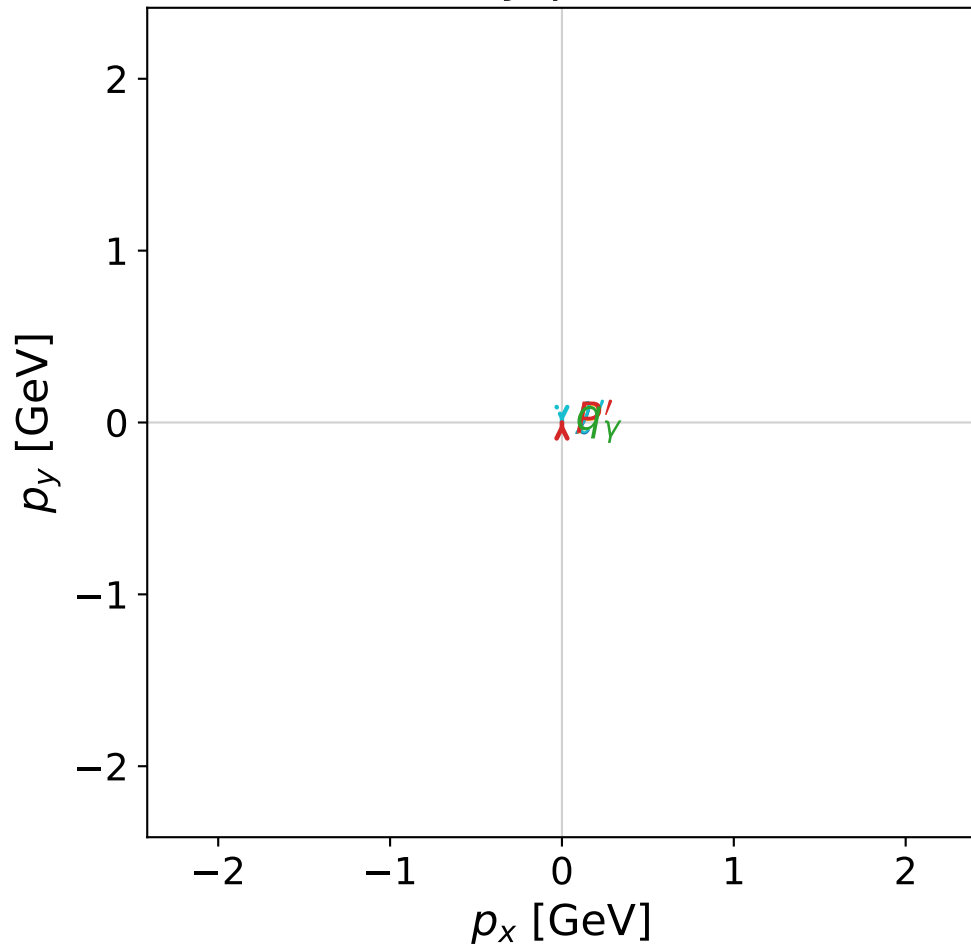


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

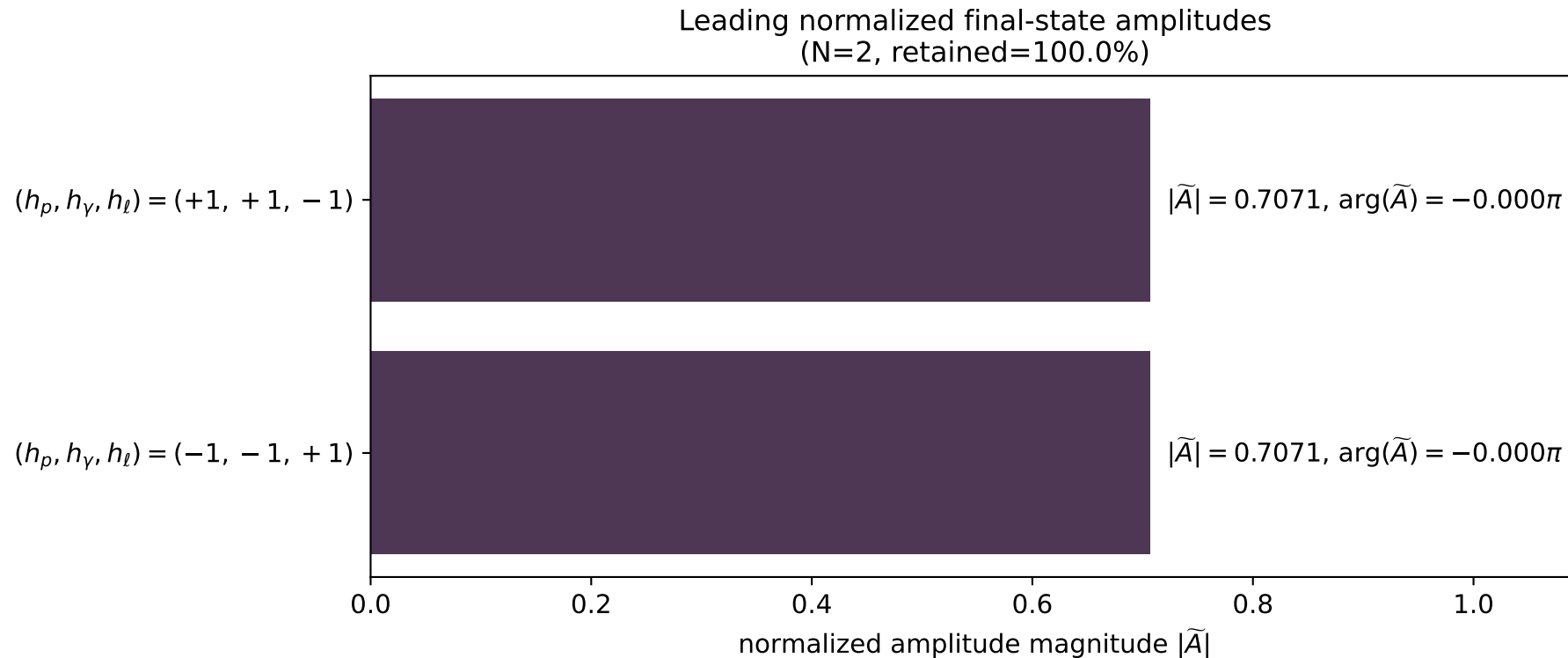


dghz_mixing_angles_polarization_02_local_minimum_148
 $d_{\text{GHZ}}=8.27454\text{e-}08$
 region: polarization_cluster_02_local_minimum_148
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0148
 lepton mixing angle: $\alpha_e=1.0537095$ rad
 incoming lepton: $0.49435|+\rangle +0.869263|-\rangle$
 proton mixing angle: $\alpha_p=2.6245189$ rad
 incoming proton: $-0.869269|+\rangle +0.494339|-\rangle$

$s=17.8907$, $\sqrt{s}=4.22973$, $|\vec{P}|=2.01086$, $|\vec{P}'|=2.01086$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.82192$, $\phi_\gamma=5.84502$
 $E_\gamma=1.09781$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.96351$
 $Q^2=7.34406$, $x_B=0.44159$, $t=-16.1742$, $W^2=10.1667$, $y=0.97767$

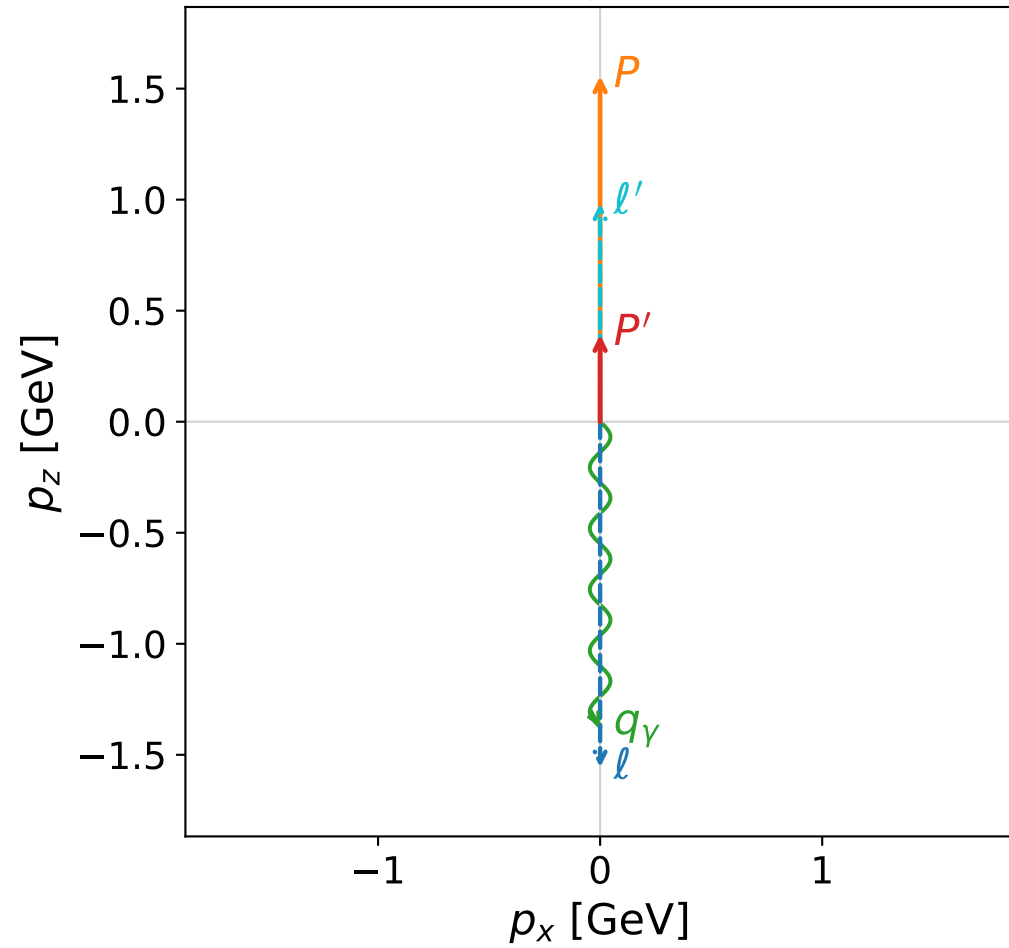
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0109$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0109$
 P $E= 2.2189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0109$
 ℓ' $E= 0.9130$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.9130$
 P' $E= 2.2189$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0109$
 q_γ $E= 1.0978$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0978$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.061\text{e+}24$; $I_h=-2.899\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.690\text{e-}28$; $I_{TL}=-2.899\text{e-}14$

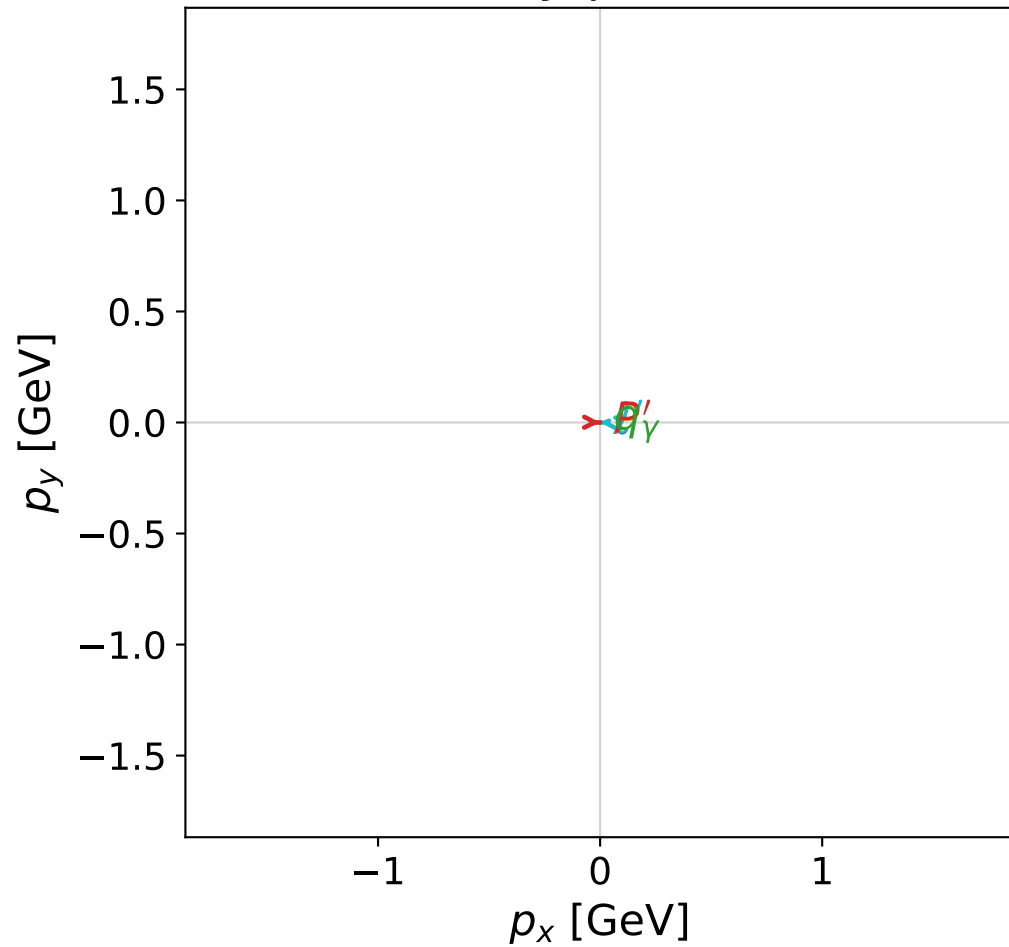


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

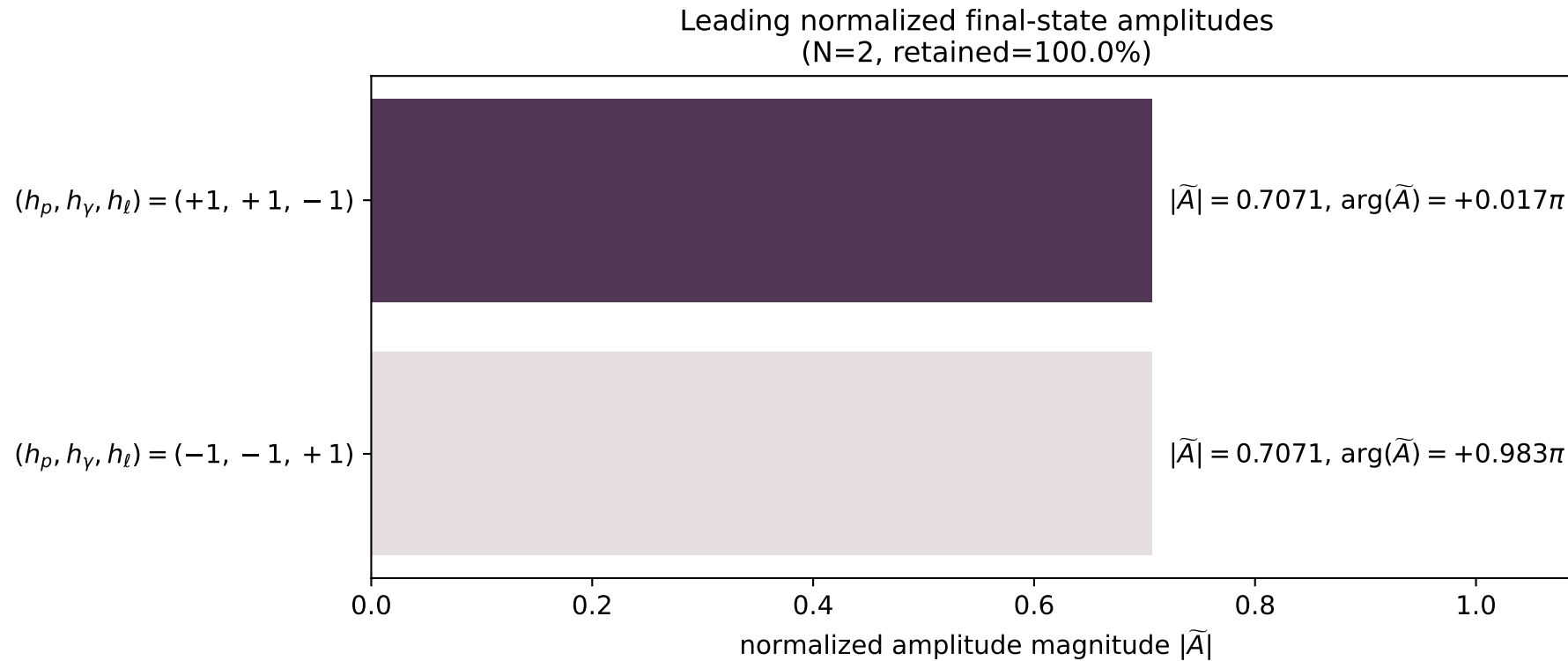


dghz_mixing_angles_polarization_02_local_minimum_150
 $d_{\text{GHZ}}=8.40814\text{e-}08$
 region: polarization_cluster_02_local_minimum_150
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0150
 lepton mixing angle: $\alpha_e=0.77196915$ rad
 incoming lepton: $0.716538|+\rangle + 0.697548|-\rangle$
 proton mixing angle: $\alpha_p=2.3696107$ rad
 incoming proton: $-0.71653|+\rangle + 0.697557|-\rangle$

$s=11.3797$, $\sqrt{s}=3.37339$, $|\vec{P}|=1.55628$, $|\vec{P}'|=0.391465$
 $\theta_{P'}=7.31453\text{e-}06$, $\theta_Y=3.14159$, $\phi_{P'}=6.26658$, $\phi_Y=3.08863$
 $E_Y=1.37422$, $\theta_{\ell'}=2.91362\text{e-}06$, $\phi_{\ell'}=3.12499$
 $Q^2=6.1178$, $x_B=0.61256$, $t=-0.715694$, $W^2=4.74931$, $y=0.951176$

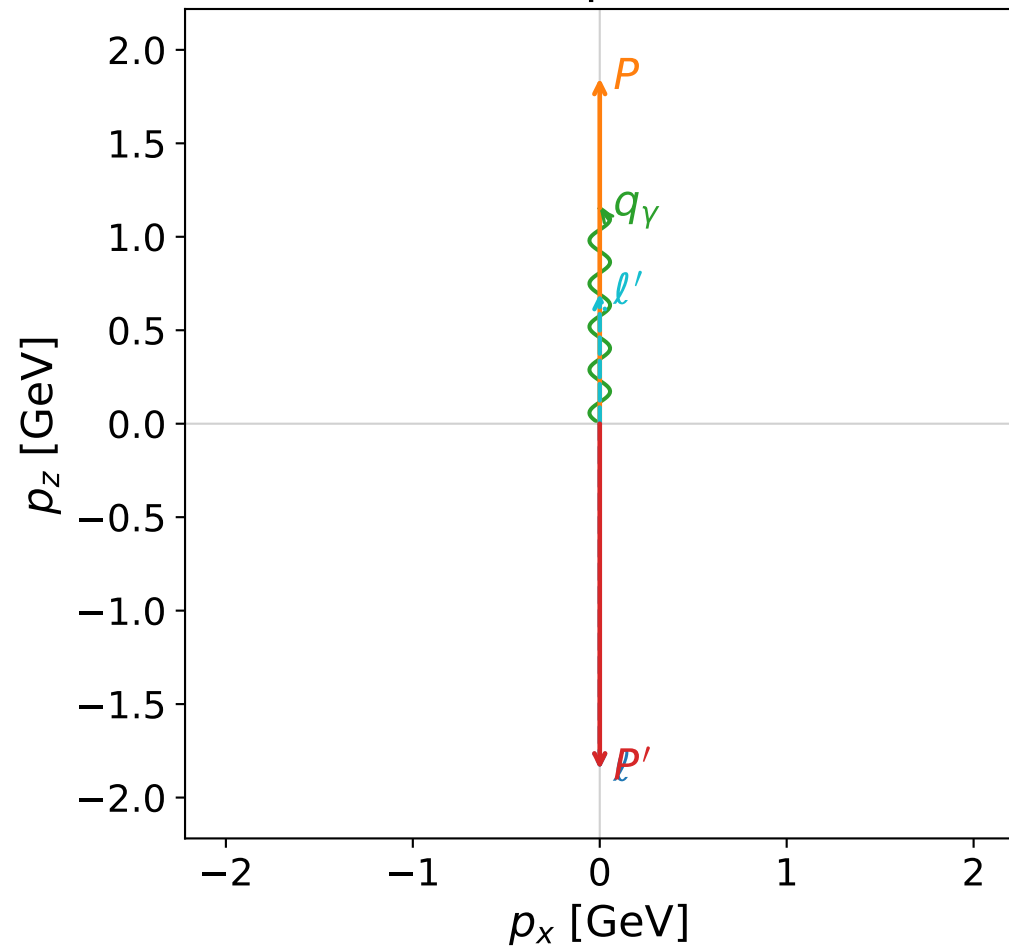
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5563$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5563$
 P $E= 1.8171$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5563$
 ℓ' $E= 0.9828$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.9828$
 P' $E= 1.0164$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.3915$
 q_Y $E= 1.3742$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.3742$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.441\text{e+}18$; $I_h=+1.454\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.881\text{e-}13$; $I_{TL}=+2.039\text{e-}08$

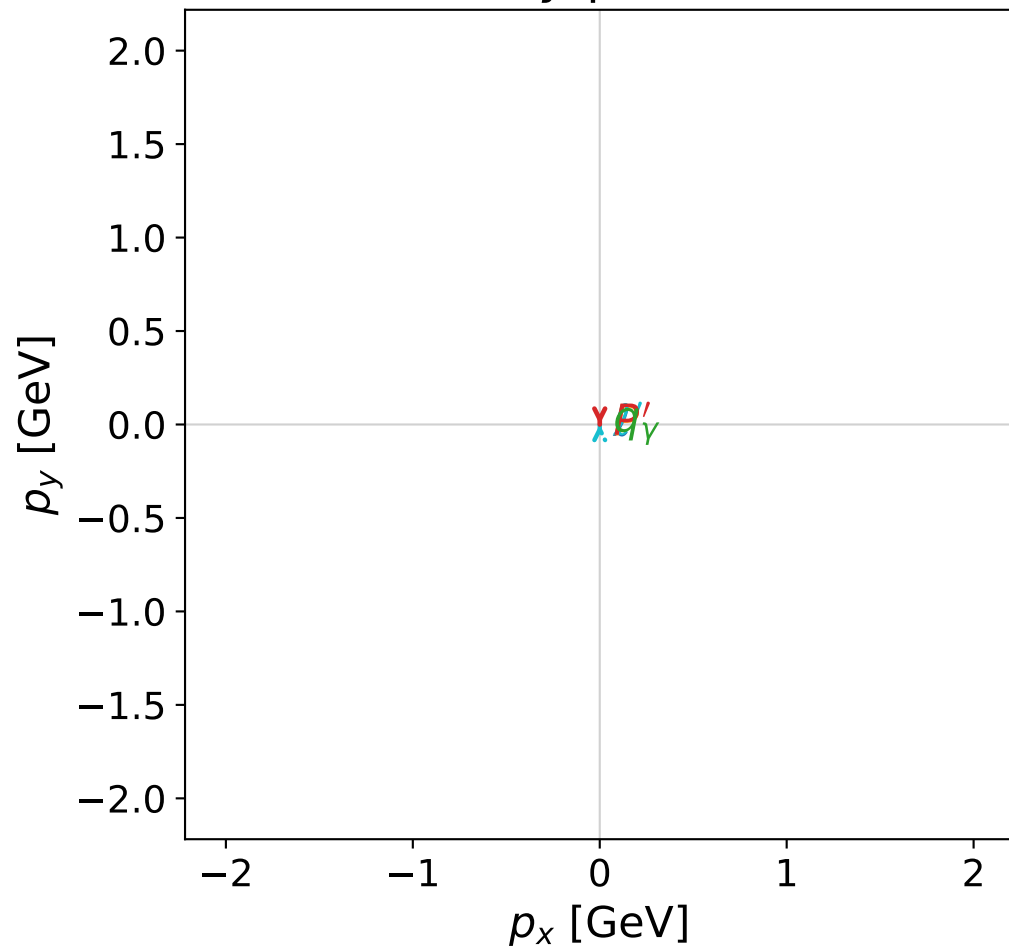


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

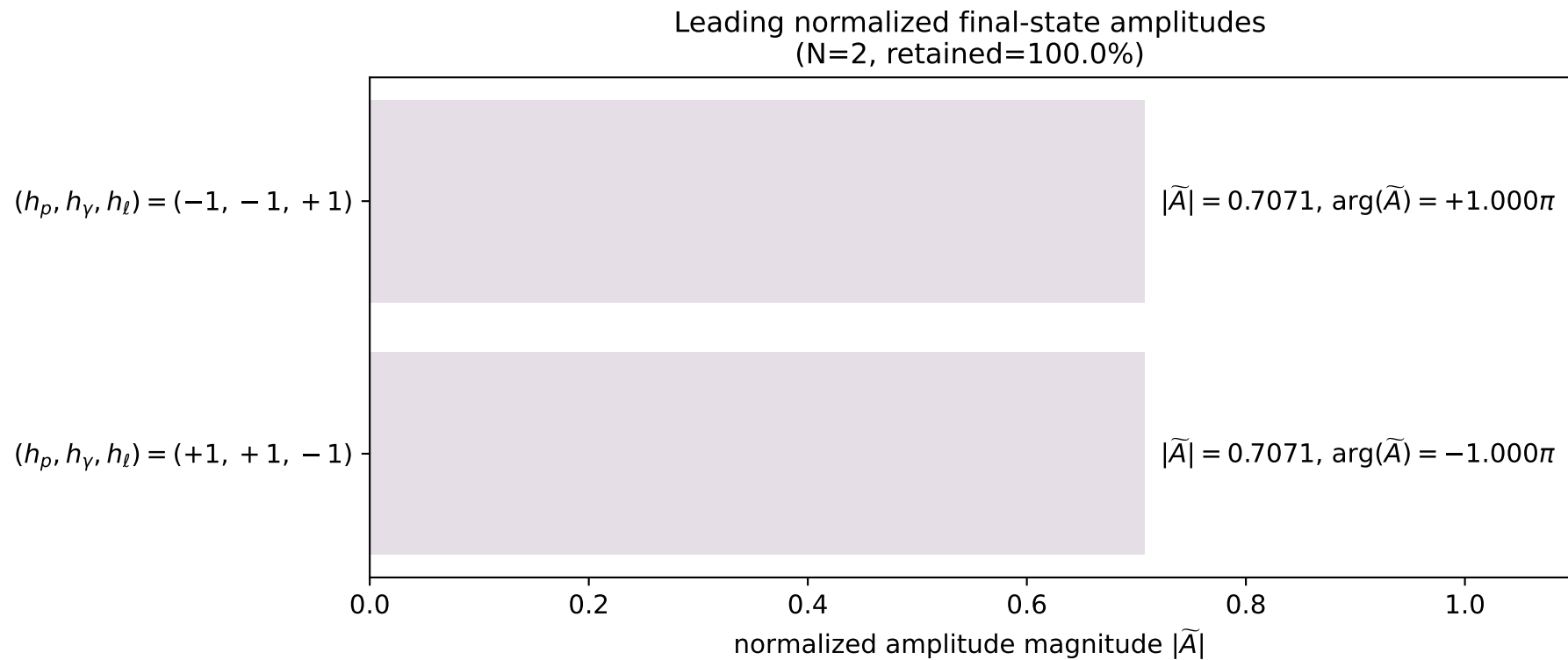


dghz_mixing_angles_polarization_02_local_minimum_157
 $d_{\text{GHZ}}=1.01492\text{e-}07$
 region: polarization_cluster_02_local_minimum_157
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0157
 lepton mixing angle: $\alpha_e=0.78801275$ rad
 incoming lepton: $0.705256|+\rangle + 0.708953|-\rangle$
 proton mixing angle: $\alpha_p=2.3588075$ rad
 incoming proton: $-0.708952|+\rangle + 0.705257|-\rangle$

$s=15.3816$, $\sqrt{s}=3.92194$, $|\vec{P}|=1.8488$, $|\vec{P}'|=1.8488$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.11449$, $\phi_\gamma=3.59494$
 $E_\gamma=1.15299$, $\theta_{l'}=0$, $\phi_{l'}=1.97289$
 $Q^2=5.14568$, $x_B=0.362639$, $t=-13.6722$, $W^2=9.92371$, $y=0.978472$

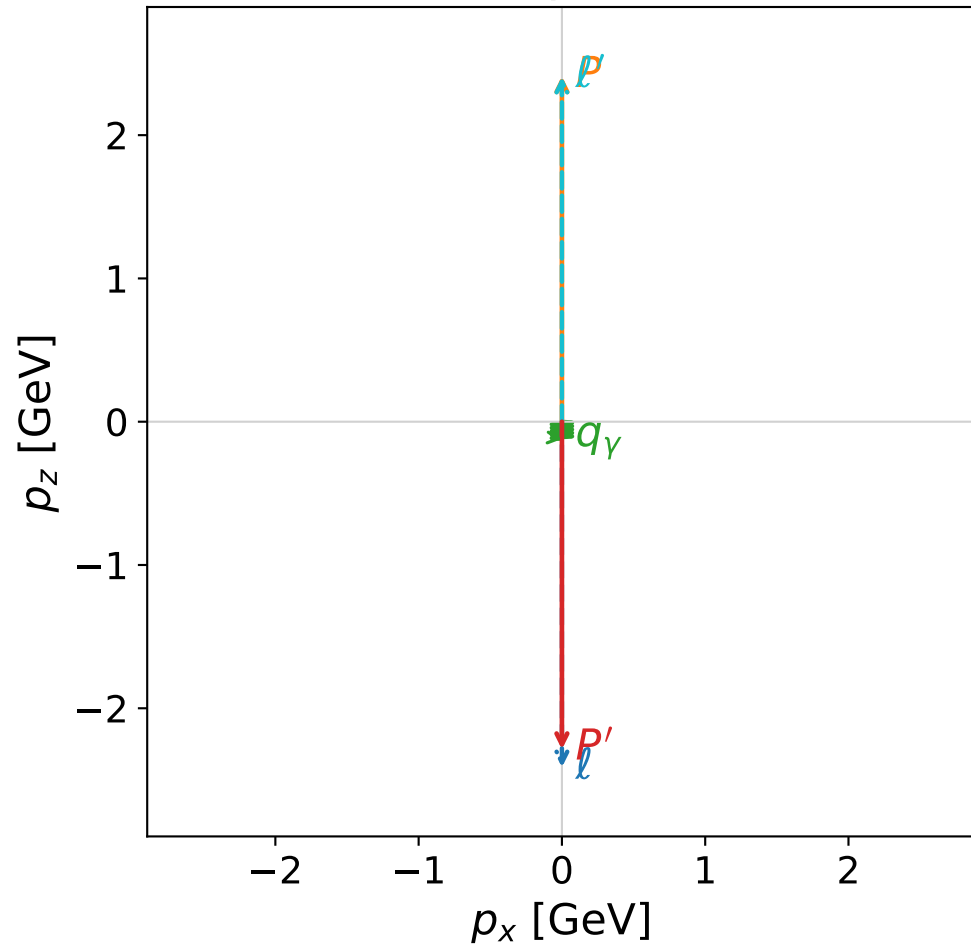
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8488$
 P $E= 2.0731$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8488$
 ℓ' $E= 0.6958$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.6958$
 P' $E= 2.0731$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8488$
 q_γ $E= 1.1530$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.1530$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.735\text{e}+24$; $I_h=+7.802\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.044\text{e-}29$; $I_{TL}=+7.802\text{e-}15$

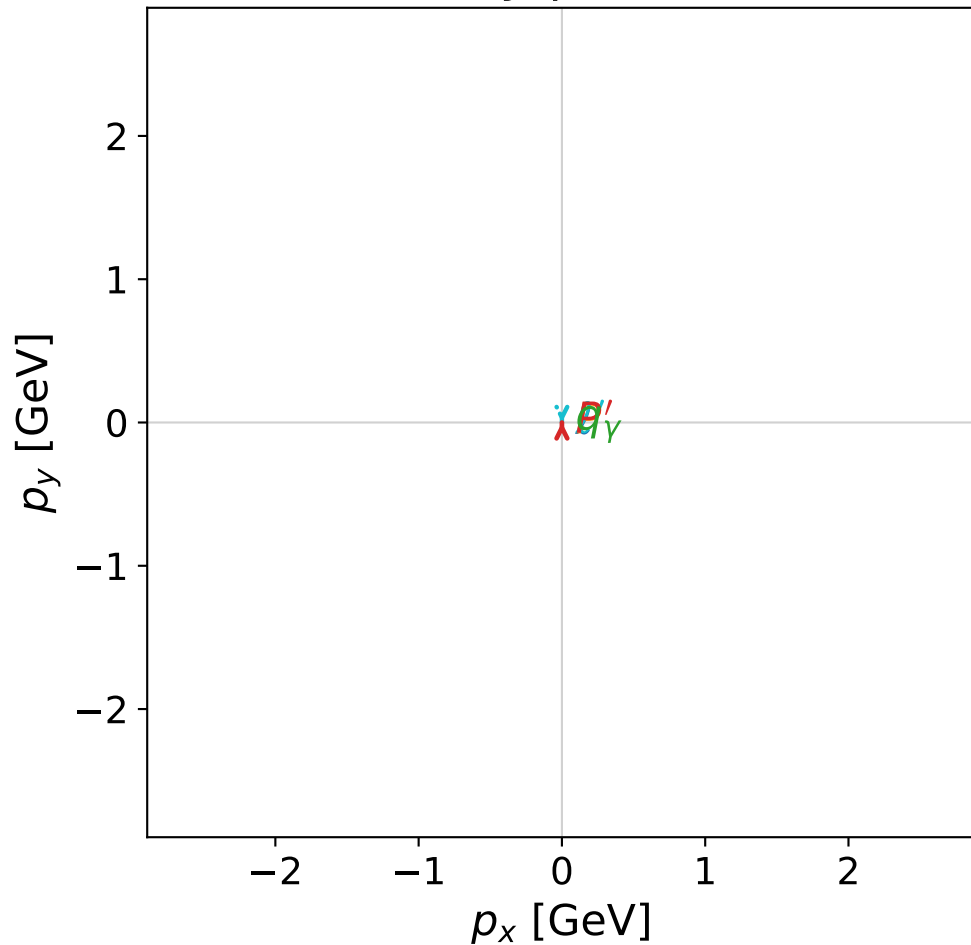


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



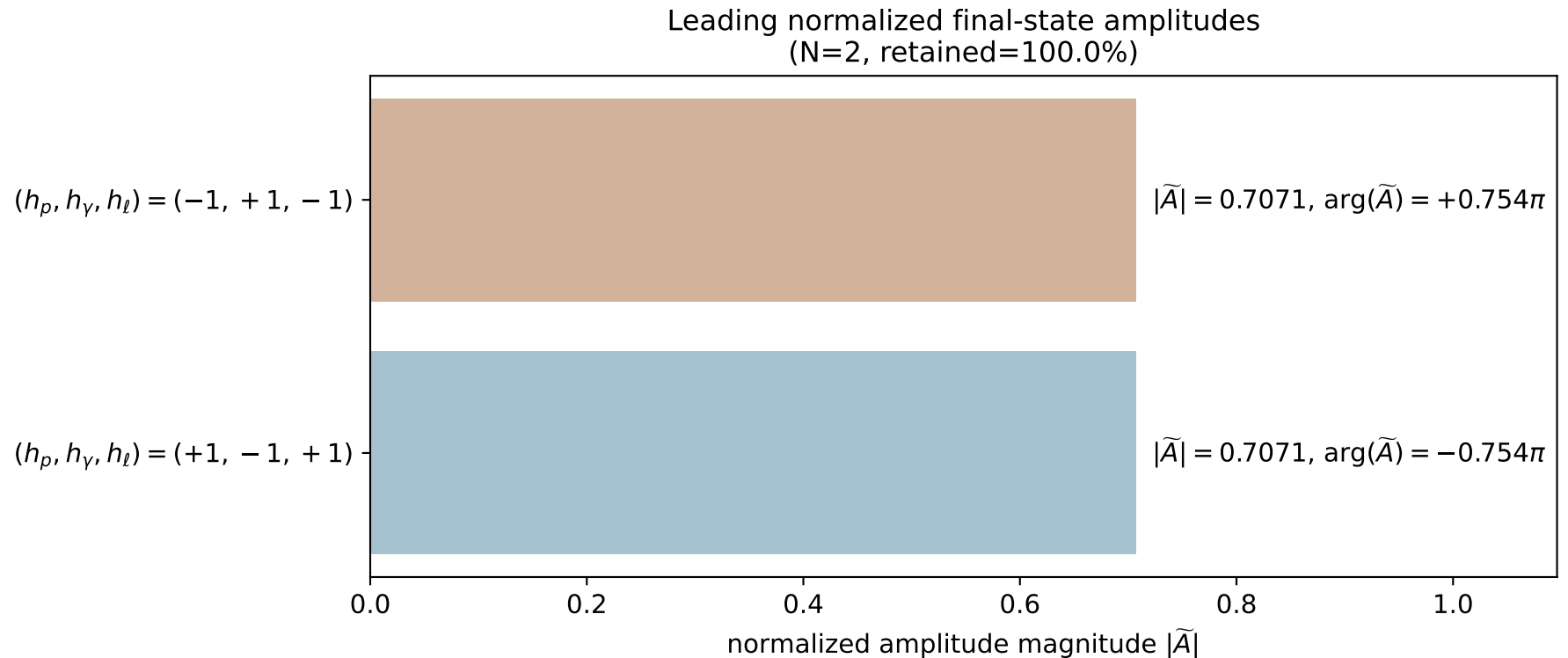
dghz_mixing_angles_polarization_02_local_minimum_159
 $d_{\{\rm GHZ\}}=1.07274\text{e-}07$
 region: polarization_cluster_02_local_minimum_159
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0159
 lepton mixing angle: $\alpha_e=0.10562157$ rad
 incoming lepton: $0.994427|+\rangle +0.105425|-\rangle$
 proton mixing angle: $\alpha_p=3.0359631$ rad
 incoming proton: $-0.994426|+\rangle +0.105433|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28696$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.69535$, $\phi_\gamma=3.91457$
 $E_\gamma=0.120596$, $\theta_{l'}=0$, $\phi_{l'}=4.88033$
 $Q^2=23.2283$, $x_B=0.998084$, $t=-22.0669$, $W^2=0.924437$, $y=0.964871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

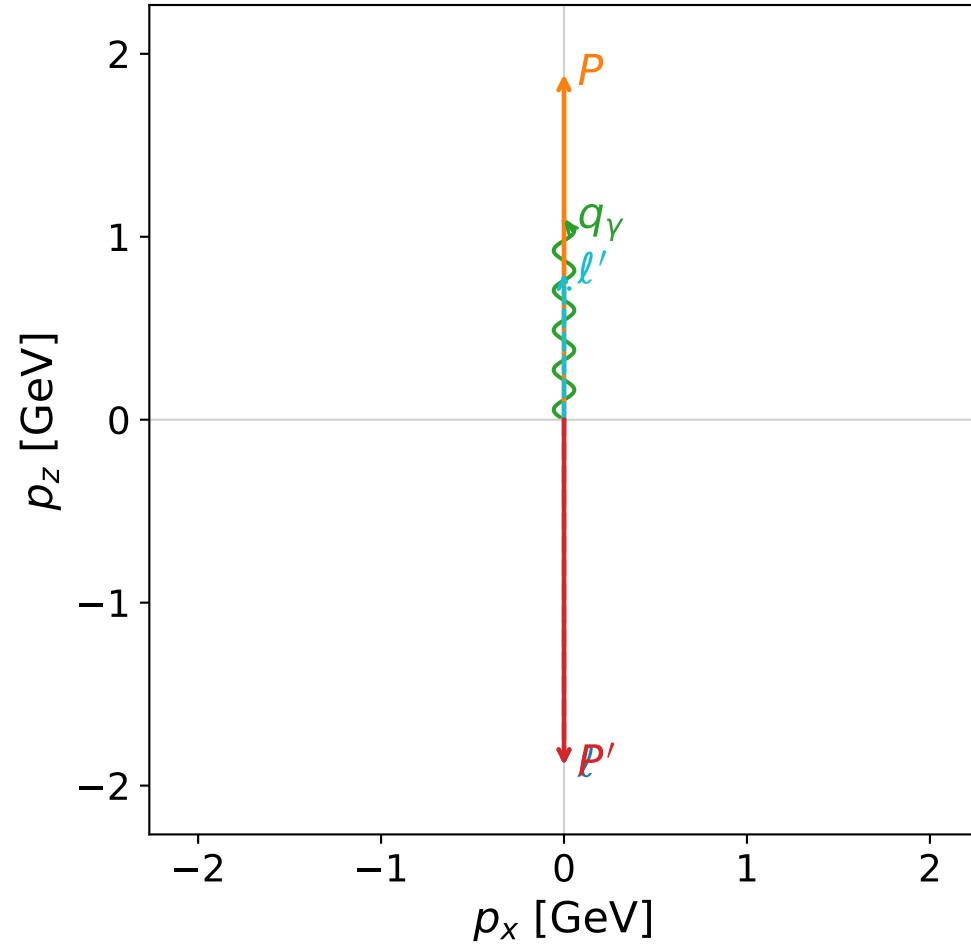
Particle	E	p_x	p_y	p_z
l	2.4120	0.0000	0.0000	-2.4120
P	2.5880	0.0000	0.0000	2.4120
l'	2.4076	0.0000	-0.0000	2.4076
P'	2.4718	-0.0000	0.0000	-2.2870
q_γ	0.1206	-0.0000	-0.0000	-0.1206

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=8.842\text{e+}19$; $I_h=+7.332\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{L/T}=1.800\text{e-}25$; $I_{TL}=-4.178\text{e-}16$

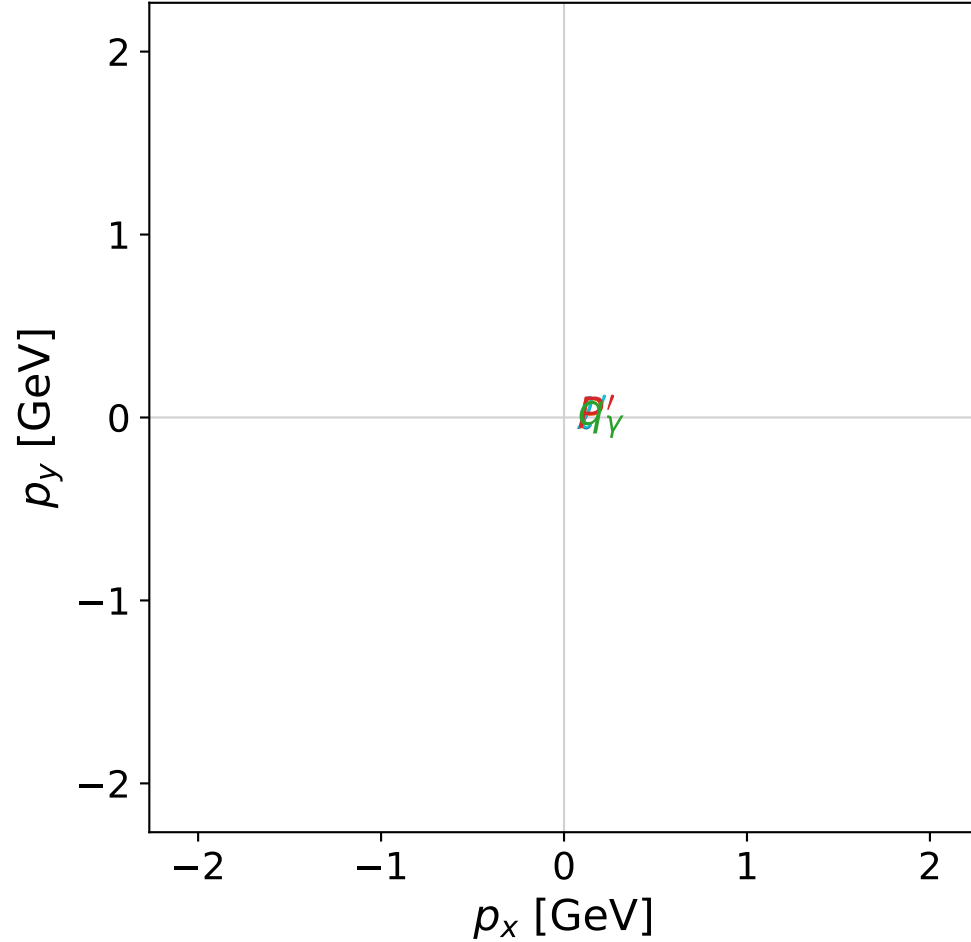


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

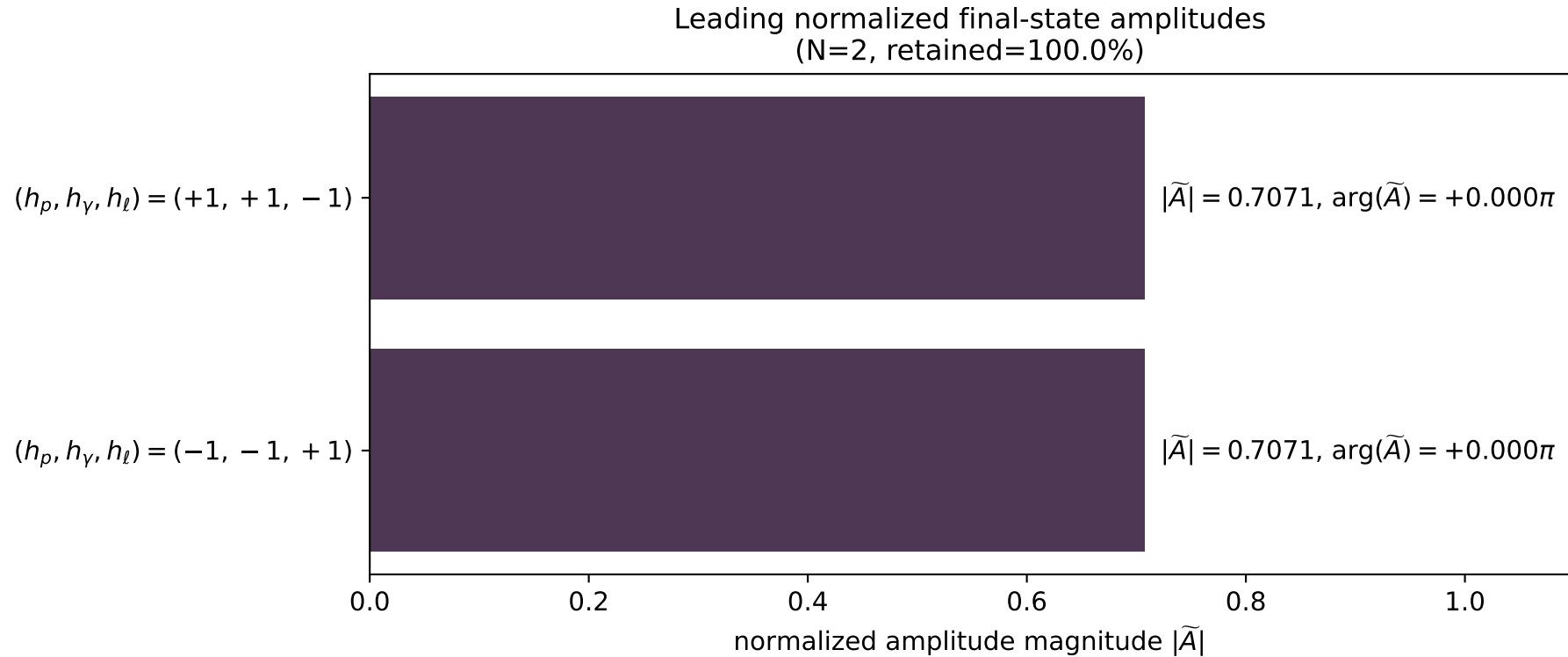


dghz_mixing_angles_polarization_02_local_minimum_160
 $d_{\text{GHZ}}=1.12987\text{e-}07$
 region: polarization_cluster_02_local_minimum_160
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0160
 lepton mixing angle: $\alpha_e=1.1362778$ rad
 incoming lepton: $0.420974|+\rangle + 0.907073|-\rangle$
 proton mixing angle: $\alpha_p=2.7070795$ rad
 incoming proton: $-0.907075|+\rangle + 0.420969|-\rangle$

$s=15.9848$, $\sqrt{s}=3.9981$, $|\vec{P}|=1.88901$, $|\vec{P}'|=1.88901$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.2262$, $\phi_\gamma=6.10955$
 $E_\gamma=1.08804$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.08461$
 $Q^2=6.05225$, $x_B=0.410256$, $t=-14.2735$, $W^2=9.57998$, $y=0.976661$

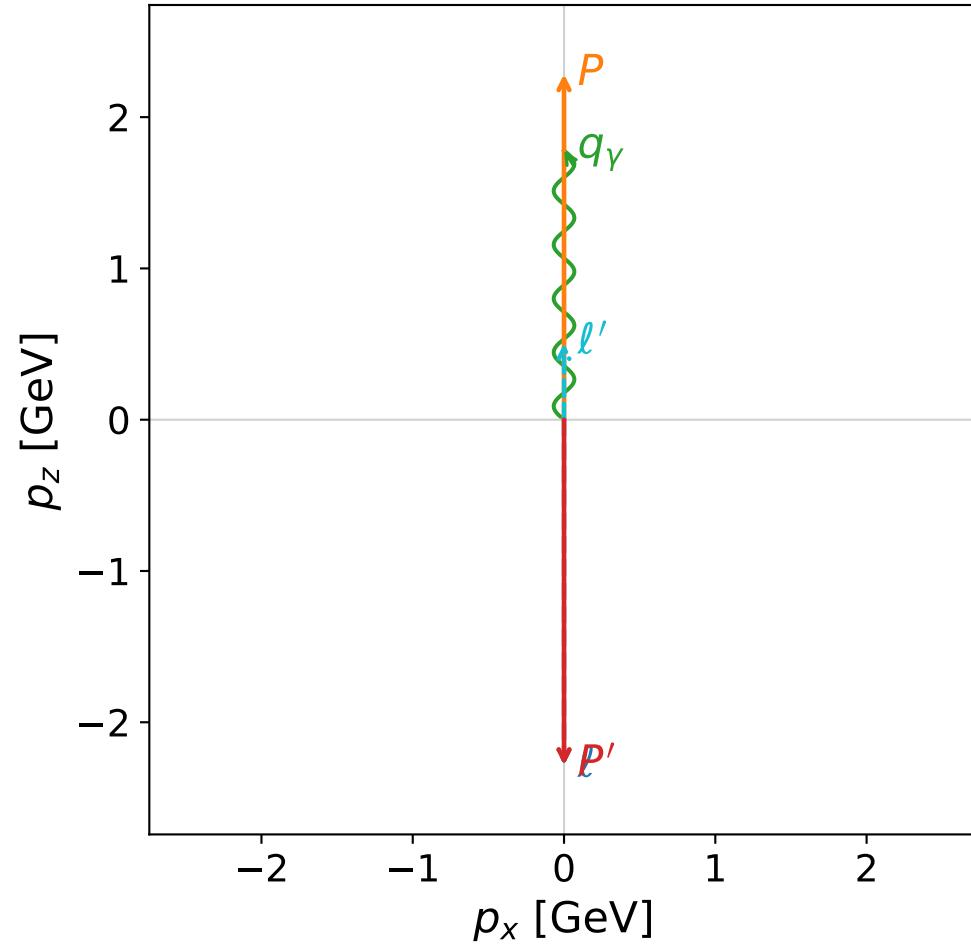
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8890$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8890$
 P $E= 2.1091$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8890$
 ℓ' $E= 0.8010$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8010$
 P' $E= 2.1091$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8890$
 q_γ $E= 1.0880$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0880$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.375\text{e}+24$; $I_h=+1.247\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.333\text{e-}28$; $I_{TL}=+1.247\text{e-}14$

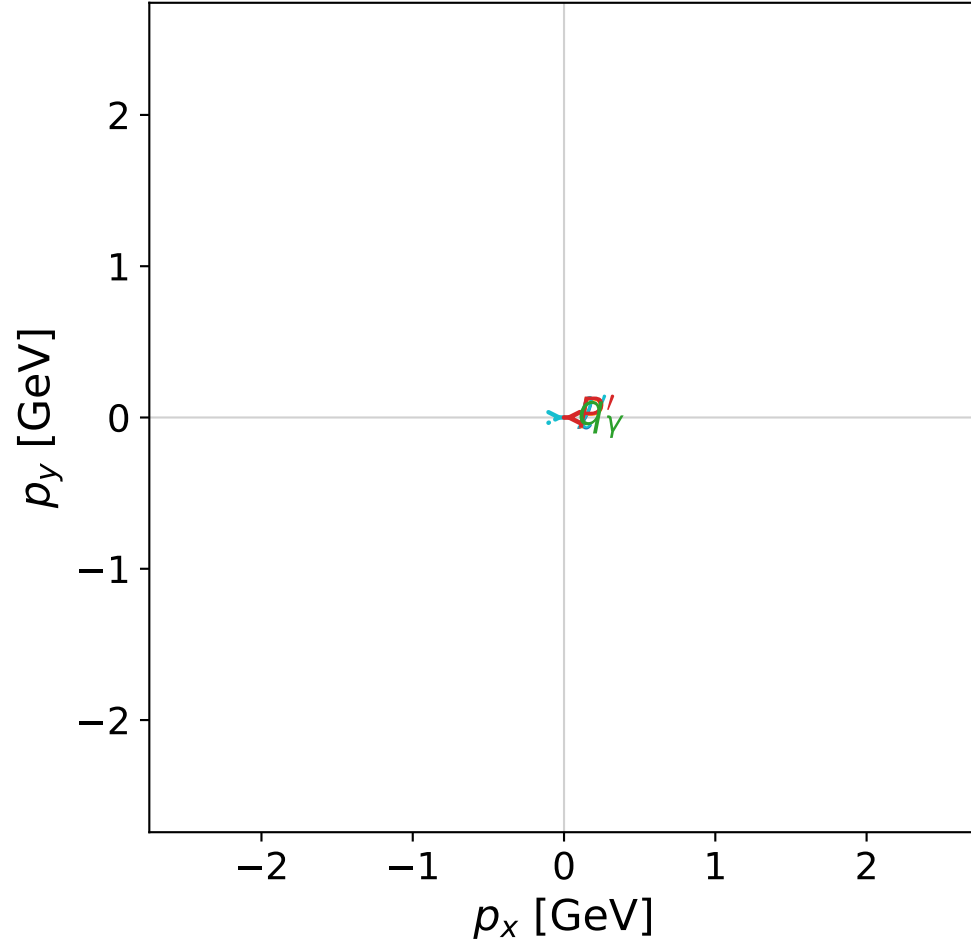


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



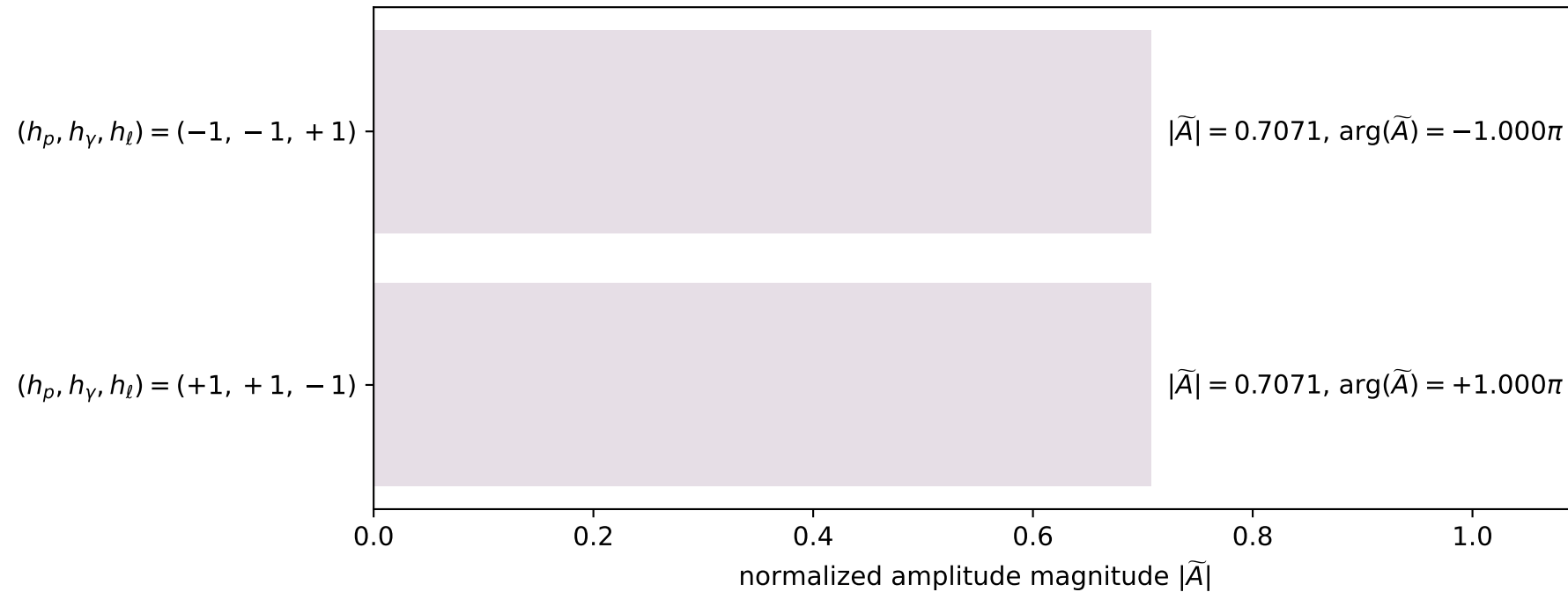
dghz_mixing_angles_polarization_02_local_minimum_164
 $d_{\text{GHZ}}=1.19686\text{e-}07$
 region: polarization_cluster_02_local_minimum_164
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0164
 lepton mixing angle: $\alpha_e=0.61981444$ rad
 incoming lepton: $0.813986|+\rangle + 0.580884|-\rangle$
 proton mixing angle: $\alpha_p=2.1905907$ rad
 incoming proton: $-0.580868|+\rangle + 0.813998|-\rangle$

$s=22.5997$, $\sqrt{s}=4.75391$, $|\vec{P}|=2.28442$, $|\vec{P}'|=2.28442$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.04512$, $\phi_\gamma=2.44102$
 $E_\gamma=1.7794$, $\theta_{l'}=0$, $\phi_{l'}=6.18671$
 $Q^2=4.61466$, $x_B=0.214308$, $t=-20.8742$, $W^2=17.7981$, $y=0.991393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2844$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2844$
 P $E= 2.4695$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2844$
 l' $E= 0.5050$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.5050$
 P' $E= 2.4695$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.2844$
 q_γ $E= 1.7794$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.7794$

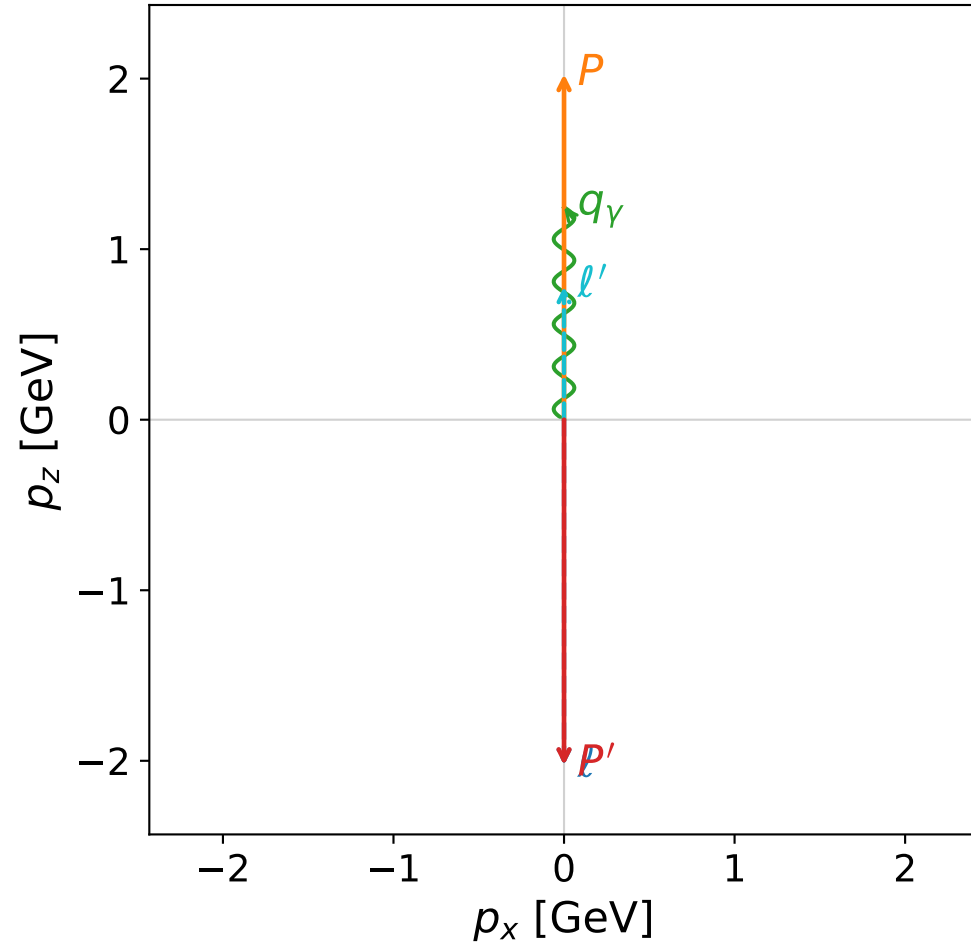
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.931\text{e+}24$; $I_h=-3.905\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=8.524\text{e-}30$; $I_{TL}=-3.906\text{e-}15$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

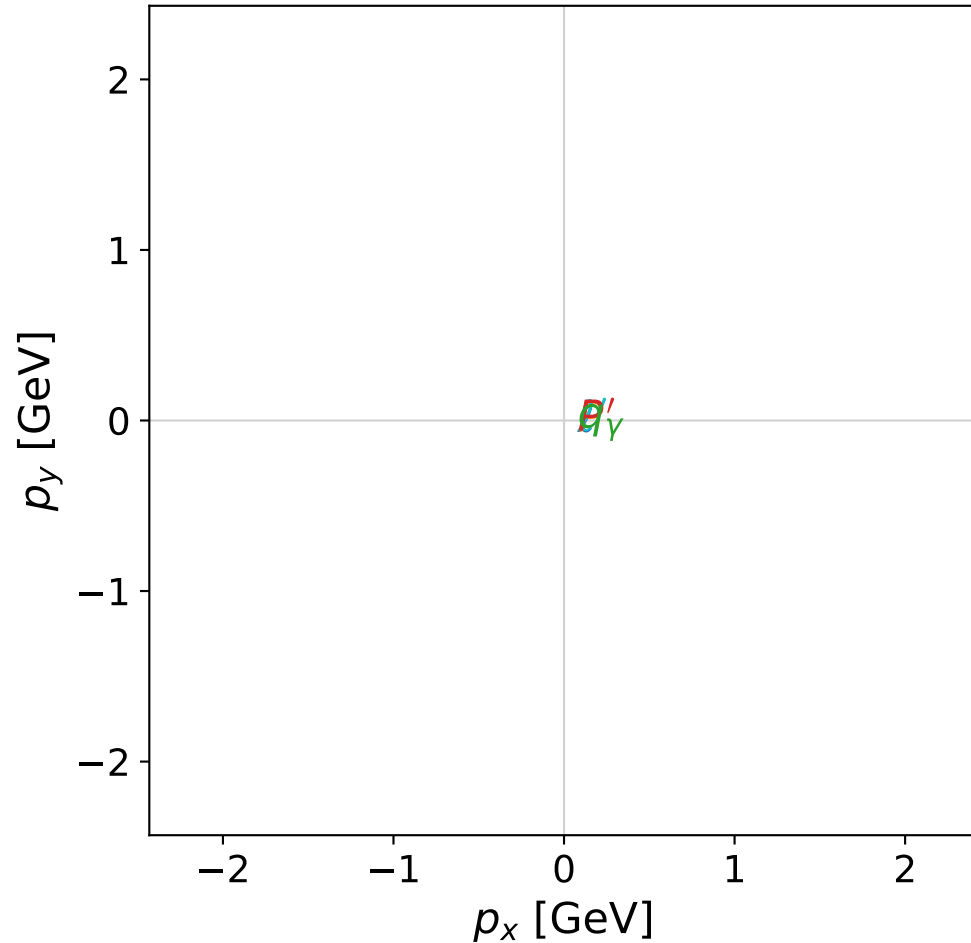


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

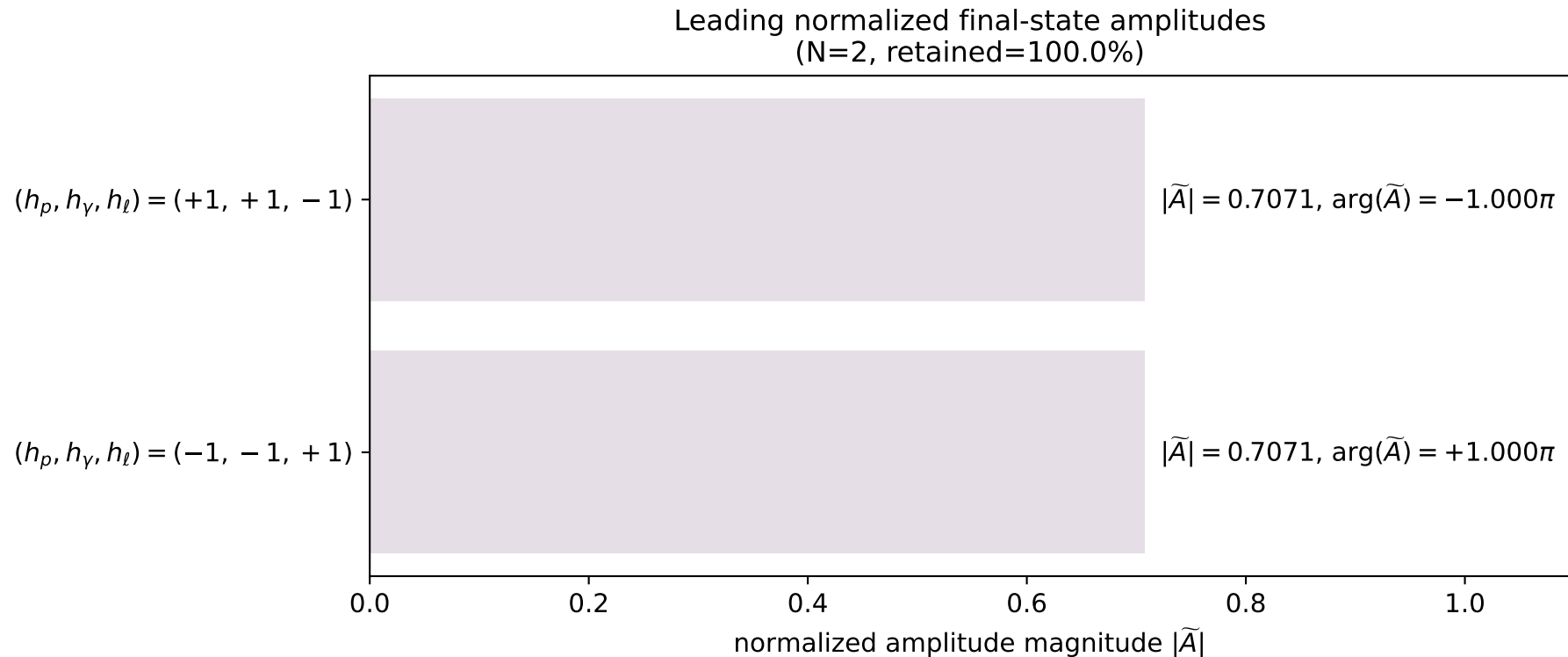


dghz_mixing_angles_polarization_02_local_minimum_167
 $d_{\text{GHZ}}=1.5068\text{e-}07$
 region: polarization_cluster_02_local_minimum_167
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0167
 lepton mixing angle: $\alpha_e=0.28996137$ rad
 incoming lepton: $0.958255|+\rangle + 0.285915|-\rangle$
 proton mixing angle: $\alpha_p=1.8607761$ rad
 incoming proton: $-0.285933|+\rangle + 0.95825|-\rangle$

$s=18.1422$, $\sqrt{s}=4.25936$, $|\vec{P}|=2.0264$, $|\vec{P}'|=2.0264$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.950105$, $\phi_\gamma=4.60864$
 $E_\gamma=1.24571$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.0917$
 $Q^2=6.32793$, $x_B=0.373554$, $t=-16.4252$, $W^2=11.4917$, $y=0.981316$

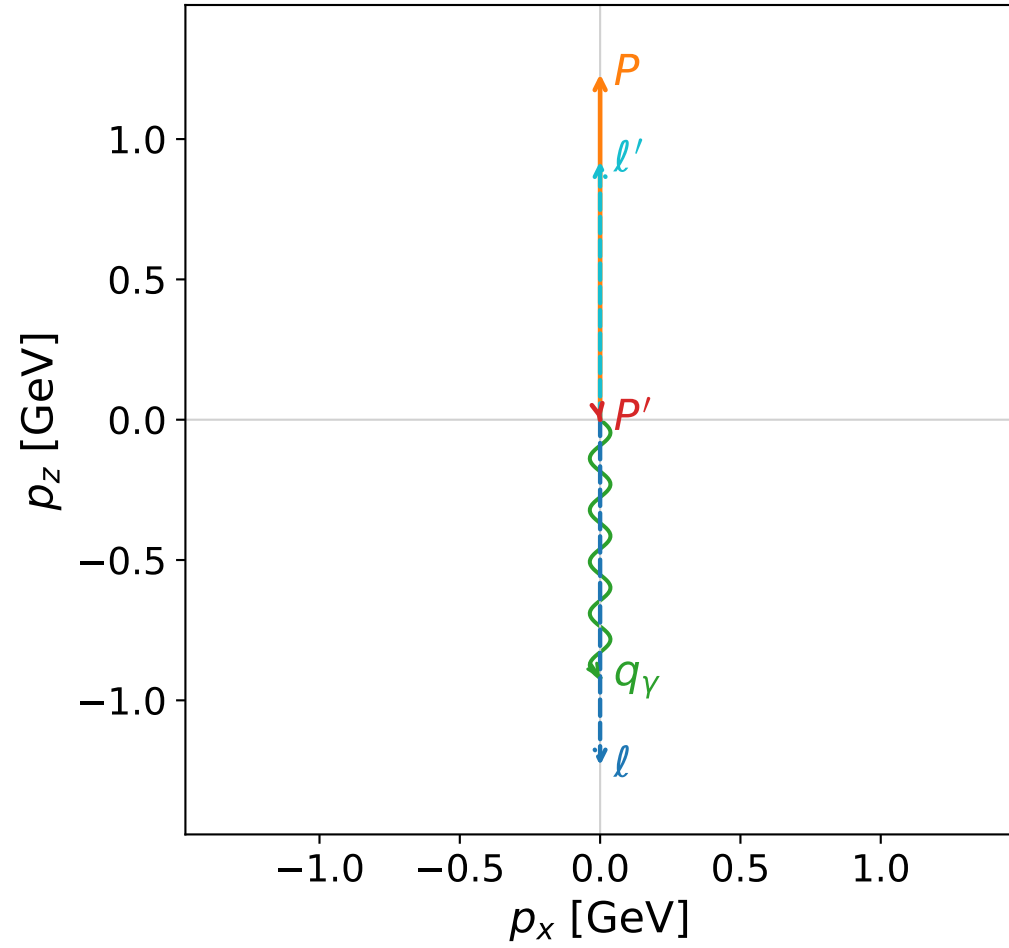
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0264$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0264$
 P $E= 2.2330$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0264$
 ℓ' $E= 0.7807$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.7807$
 P' $E= 2.2330$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0264$
 q_γ $E= 1.2457$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.2457$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.407\text{e}+24$; $I_h=+2.769\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.277\text{e-}27$; $I_{TL}=+2.769\text{e-}14$

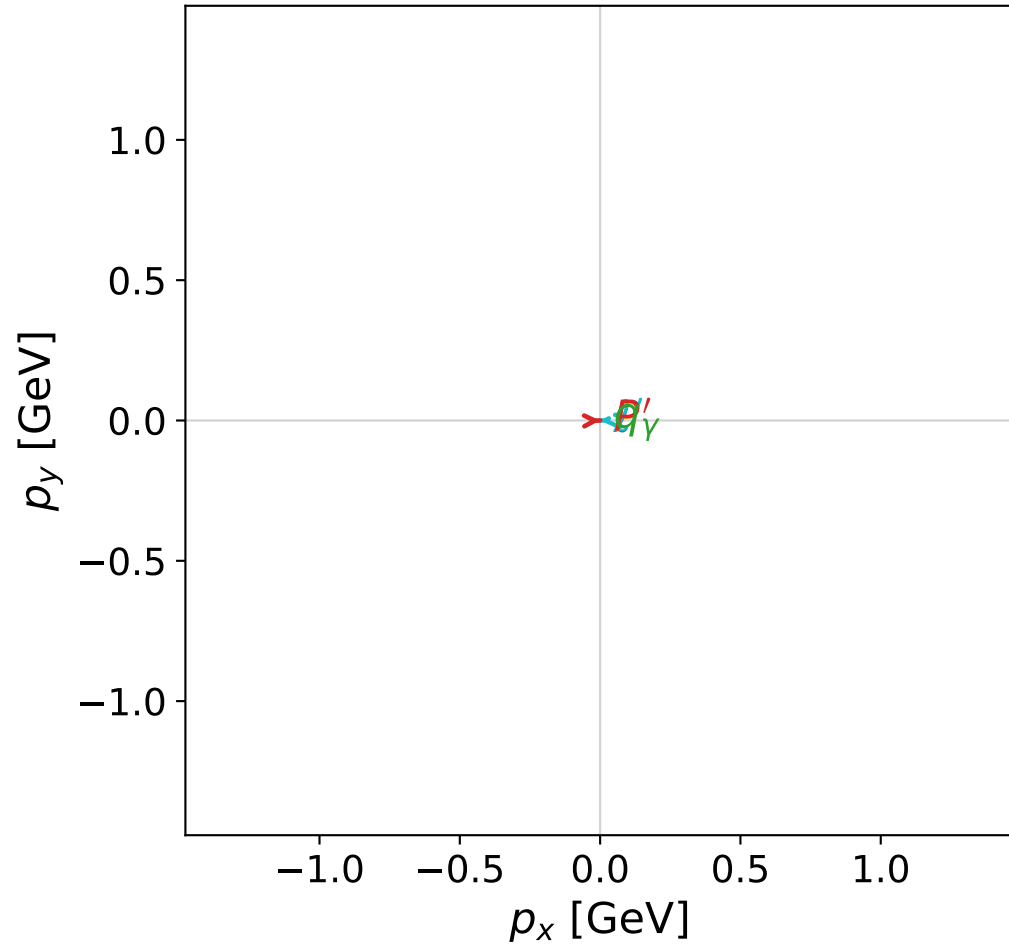


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

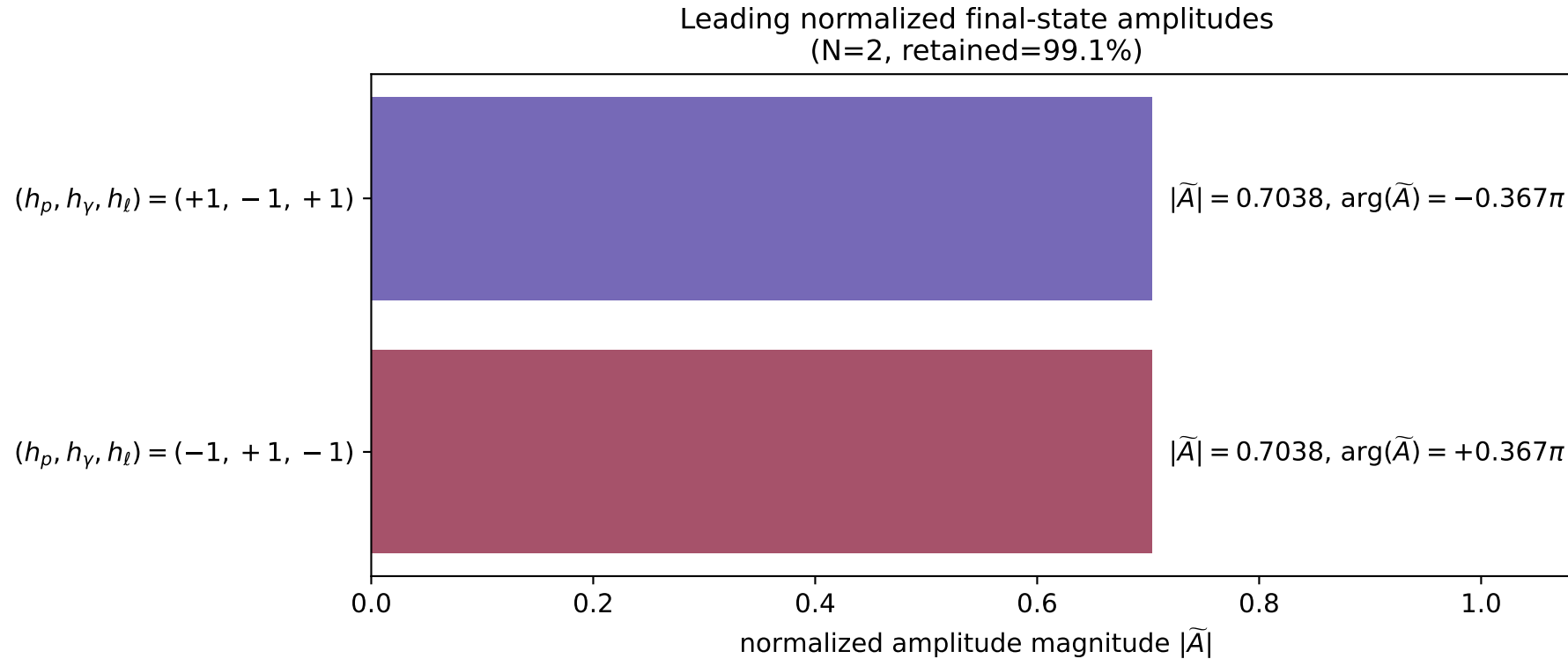


dghz_mixing_angles_polarization_02_local_minimum_168
 $d_{\text{GHZ}}=1.70137\text{e-}07$
 region: polarization_cluster_02_local_minimum_168
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0168
 lepton mixing angle: $\alpha_e=1.2036147$ rad
 incoming lepton: $0.358986|+\rangle + 0.933343|-\rangle$
 proton mixing angle: $\alpha_p=1.9380061$ rad
 incoming proton: $-0.359013|+\rangle + 0.933333|-\rangle$

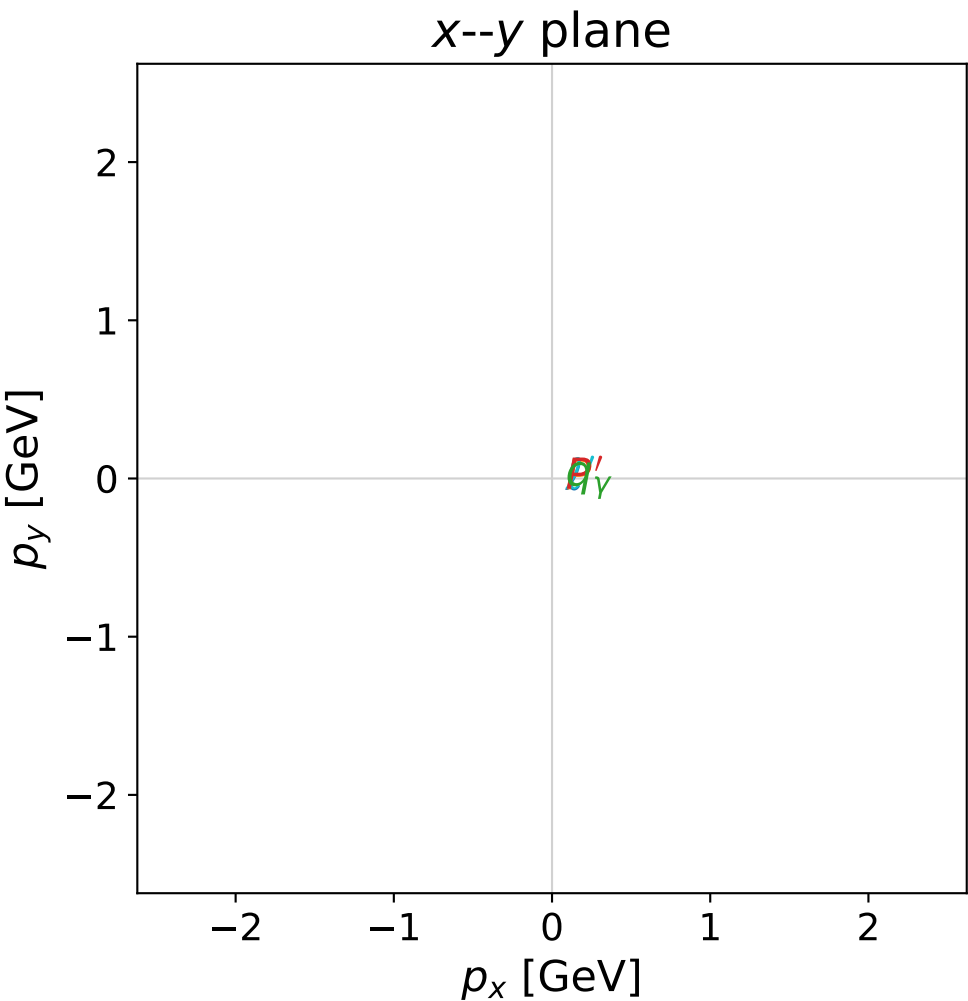
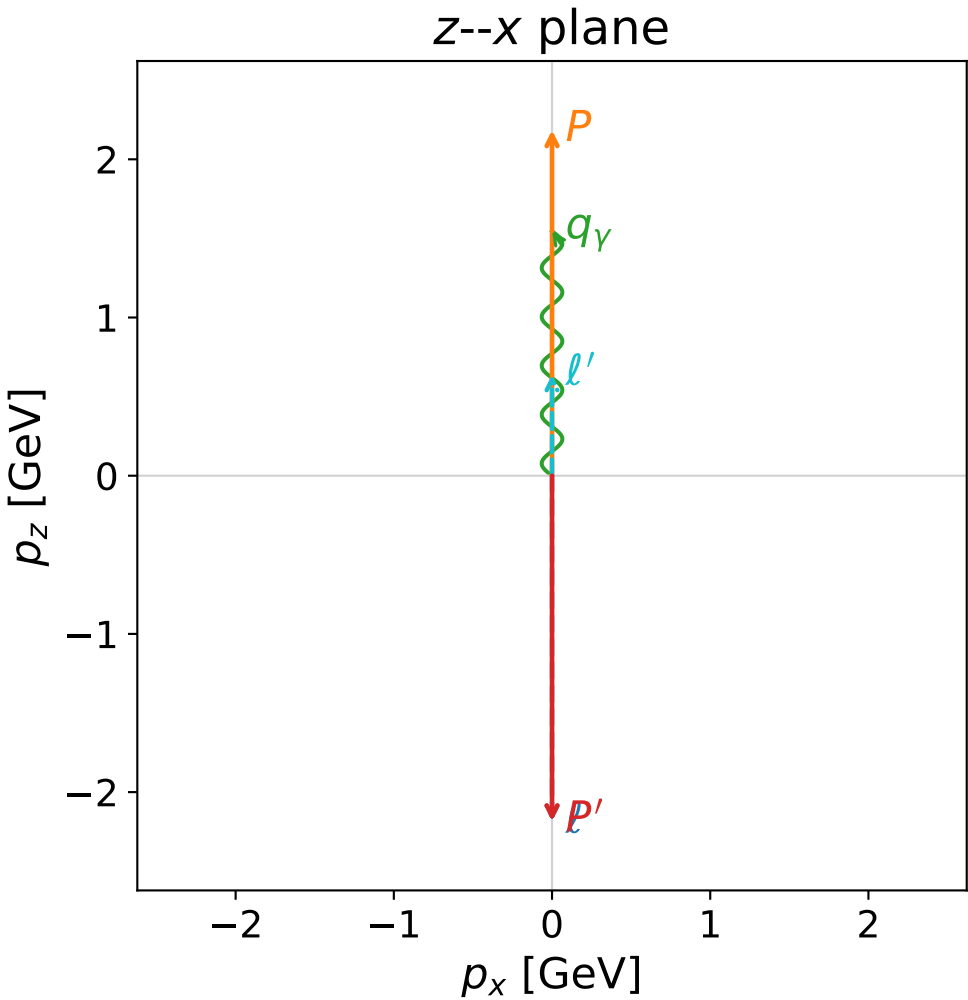
$s=7.72597$, $\sqrt{s}=2.77956$, $|\vec{P}|=1.23151$, $|\vec{P}'|=0.000244016$
 $\theta_{P'}=2.94845$, $\theta_Y=3.14159$, $\phi_{P'}=0.027548$, $\phi_Y=5.15721$
 $E_Y=0.920662$, $\theta_{\ell'}=5.08603\text{e-}05$, $\phi_{\ell'}=3.16914$
 $Q^2=4.5364$, $x_B=0.724304$, $t=-1.14505$, $W^2=2.60656$, $y=0.914842$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2315$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2315$
 P $E= 1.5481$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2315$
 ℓ' $E= 0.9209$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.9209$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0002$
 q_Y $E= 0.9207$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.9207$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.540\text{e+}15$; $I_h=-1.563\text{e-}08$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.534\text{e-}10$; $I_{TL}=-1.299\text{e-}05$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

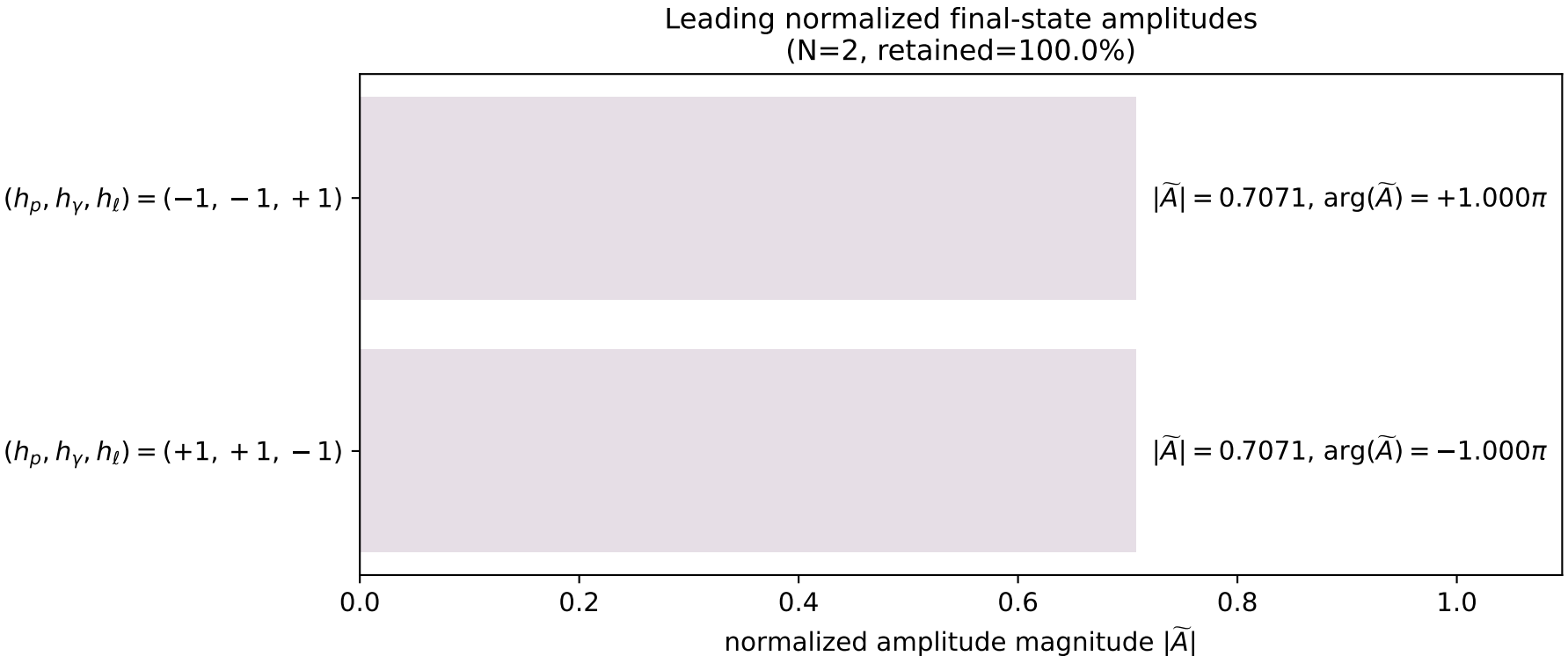


dghz_mixing_angles_polarization_02_local_minimum_171
d_{\rm GHZ}=2.04591e-07
region: polarization_cluster_02_local_minimum_171
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0171
lepton mixing angle: alpha_e=0.23742392 rad
incoming lepton: 0.971947|+> +0.2352|->
proton mixing angle: alpha_p=1.8082114 rad
incoming proton: -0.235191|+> +0.971949|->

$s=20.8112$, $\sqrt{s}=4.56193$, $|\vec{P}|=2.18453$, $|\vec{P}'|=2.18453$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.23383$, $\phi_\gamma=4.47775$
 $E_\gamma=1.54558$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.37543$
 $Q^2=5.58324$, $x_B=0.283631$, $t=-19.0887$, $W^2=14.9815$, $y=0.987634$

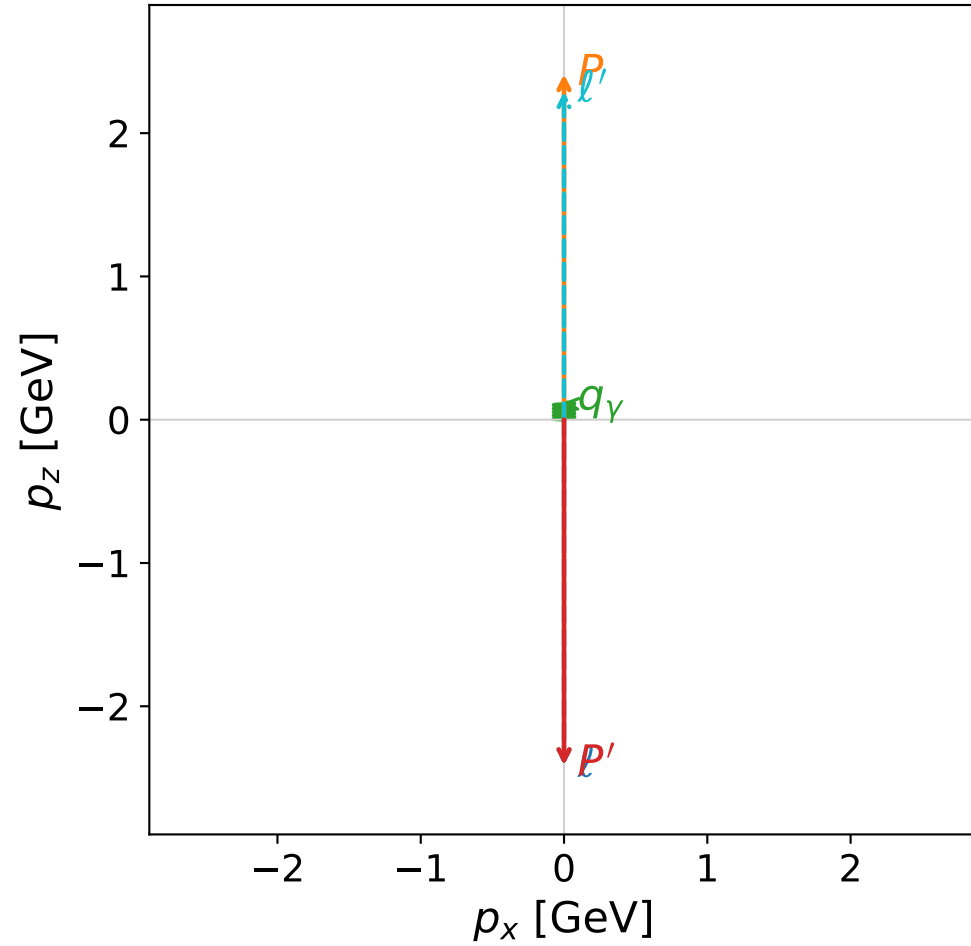
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1845$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1845$
 P $E= 2.3774$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1845$
 ℓ' $E= 0.6390$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6390$
 P' $E= 2.3774$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1845$
 q_γ $E= 1.5456$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5456$

GHZ mechanism: helicity flipping (full H G C)
helicity powers: flip 100.000%, conserve 0.000%
R_flip/conserve=1.677e+24; I_h=+7.335e-15
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=1.287e-28; I_TL=+7.329e-15

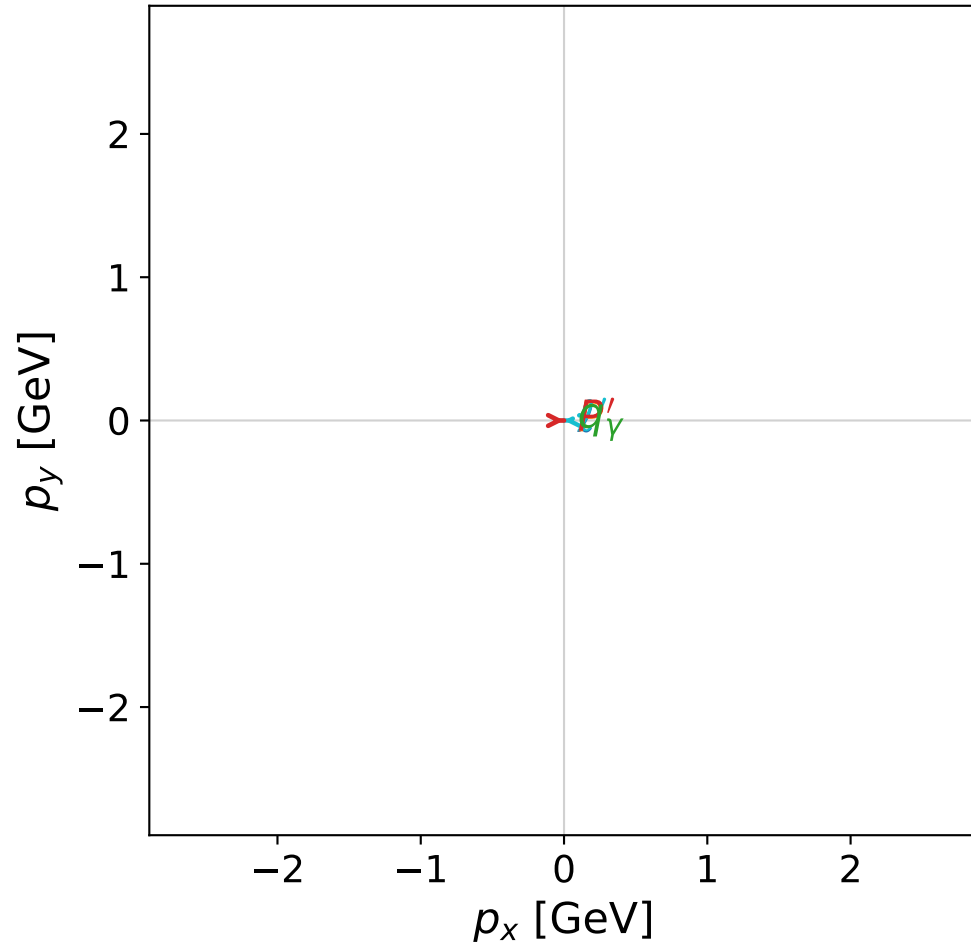


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

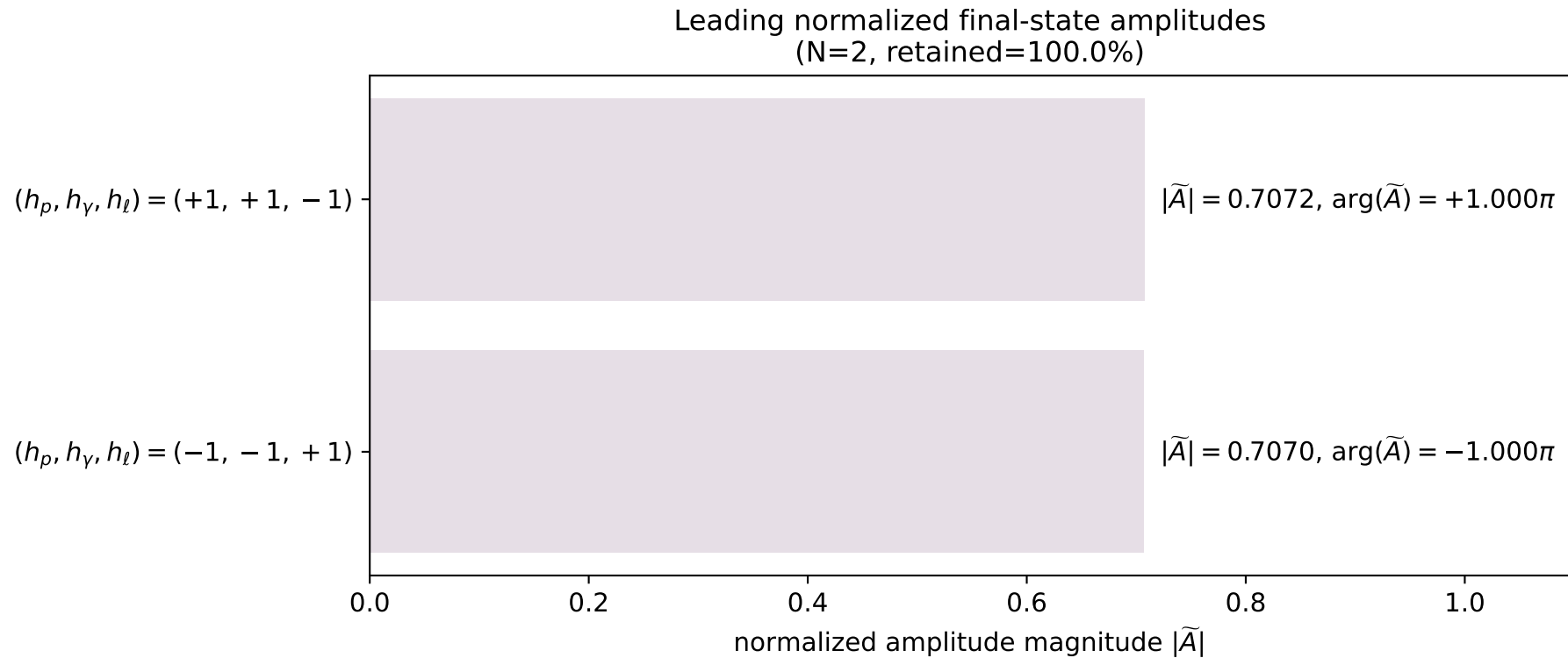


dghz_mixing_angles_polarization_02_local_minimum_175
 $d_{\text{GHZ}}=2.71276\text{e-}07$
 region: polarization_cluster_02_local_minimum_175
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0175
 lepton mixing angle: $\alpha_e=0.045198059$ rad
 incoming lepton: $0.998979|+\rangle +0.0451827|-\rangle$
 proton mixing angle: $\alpha_p=1.6160065$ rad
 incoming proton: $-0.0451948|+\rangle +0.998978|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.96908\text{e-}29$, $\phi_\gamma=1.572$
 $E_\gamma=0.120652$, $\theta_{l'}=0$, $\phi_{l'}=3.14159$
 $Q^2=22.1072$, $x_B=0.948249$, $t=-23.2713$, $W^2=2.08636$, $y=0.966567$

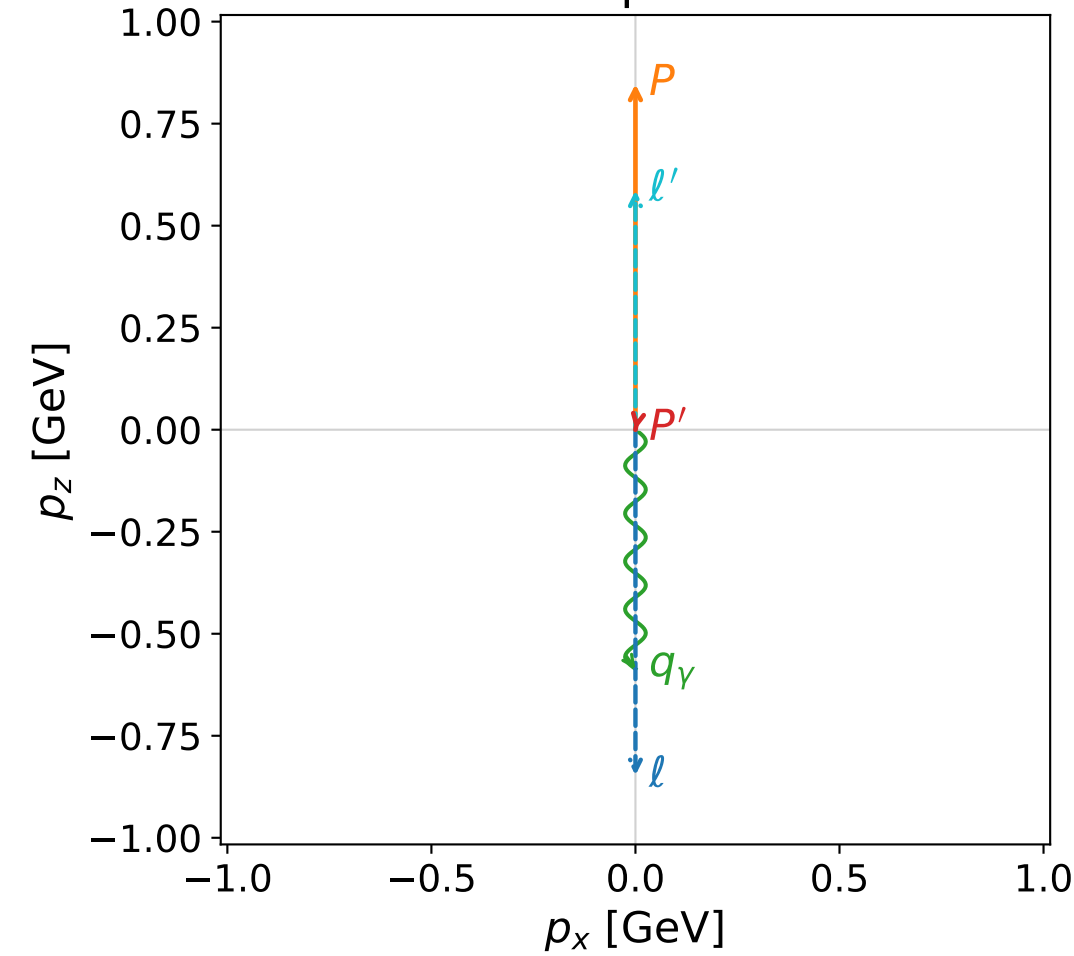
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.2914$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1207$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.1207$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.950\text{e+}21$; $I_h=+6.322\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.452\text{e-}23$; $I_{TL}=+6.336\text{e-}13$

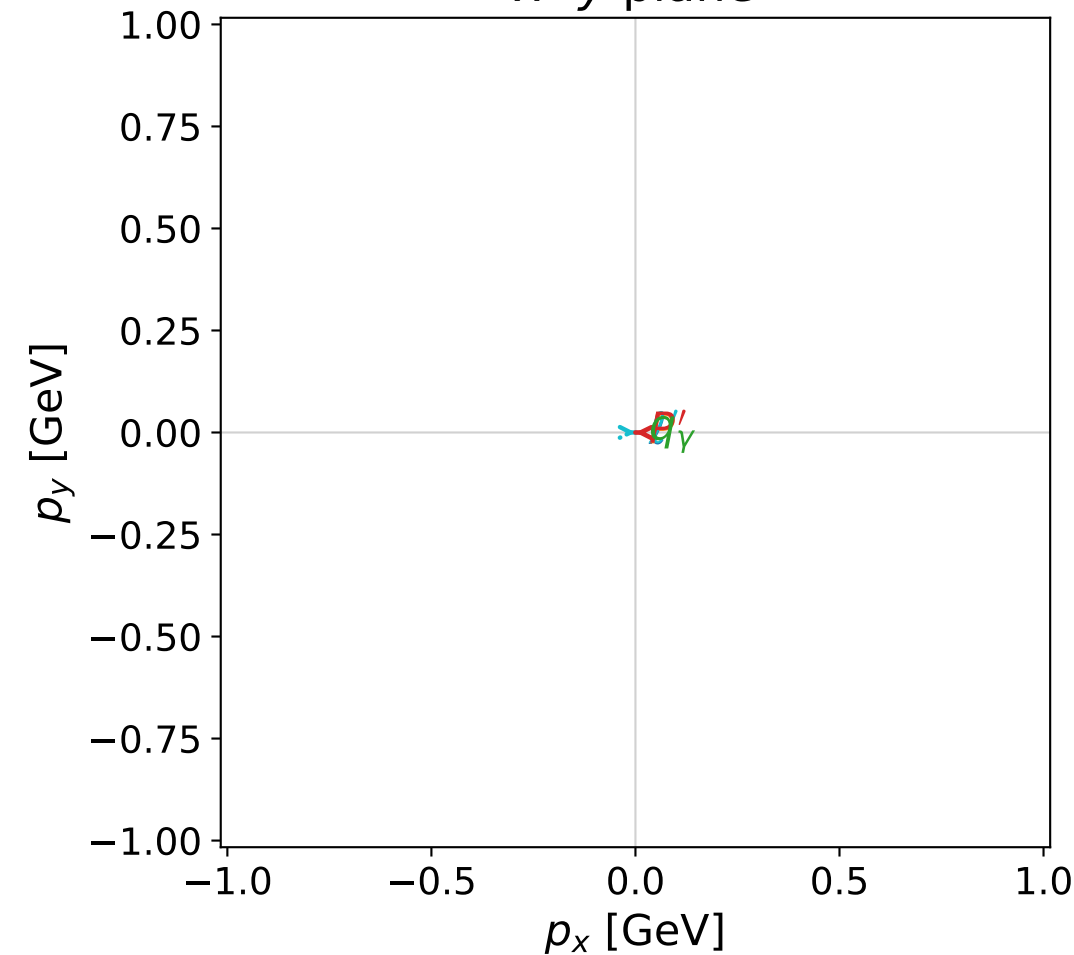


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_02_local_minimum_182
 $d_{\{\rm GHZ\}}=6.61545\text{e-}07$
 region: polarization_cluster_02_local_minimum_182
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0182
 lepton mixing angle: $\alpha_e=0.2035518$ rad
 incoming lepton: $0.979355|+\rangle +0.202149|-\rangle$
 proton mixing angle: $\alpha_p=2.9380379$ rad
 incoming proton: $-0.979354|+\rangle +0.202152|-\rangle$

$s=4.45467$, $\sqrt{s}=2.11061$, $|\vec{P}|=0.846871$, $|\vec{P}'|=1.86603\text{e-}05$
 $\theta_{p'}=2.94793$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.13457$, $\phi_\gamma=3.85199$
 $E_\gamma=0.586295$, $\theta_{\ell'}=6.12523\text{e-}06$, $\phi_{\ell'}=6.27616$
 $Q^2=1.98613$, $x_B=0.643594$, $t=-0.611116$, $W^2=1.97971$, $y=0.863258$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.8469$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.8469$
 P $E= 1.2637$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8469$
 ℓ' $E= 0.5863$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.5863$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 0.5863$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.5863$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.291\text{e+}16$; $I_h=-6.190\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{\text{L/T}}=1.588\text{e-}11$; $I_{\text{TL}}=-5.174\text{e-}06$

